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📚 Master Subject Table of Contents

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SUBJECT 01 · ANCIENT INDIA

SUBJECT 01: ANCIENT INDIAN HISTORY

प्राचीन भारत का इतिहास — दैनिक उच्च-प्राथमिकता तथ्य

Indus Valley Civilization, Rigvedic & Later Vedic Age, 16 Mahajanapadas, Mauryan Administration & Ashokan Edicts, Gupta Golden Age, and Dynastic Founders & Capitals.

1. Indus Valley Civilization (IVC) Master Matrix

Site	Location / State	River	Excavator & Year	Key Unique Findings / Exam Triggers
Harappa	Sahiwal (Punjab, Pak)	**Ravi** (Left bank)	Daya Ram Sahni (1921)	6 granaries in 2 rows, Coffin burial (R-37 & H cemetery), Sandstone torso of nude male, Copper bullock cart
Mohenjodaro (*Mound of Dead*)	Larkana (Sindh, Pak)	**Indus** (Right bank)	R.D. Banerjee (1922)	**Great Bath**, **Great Granary** (largest building of Mohenjodaro), **Bronze Dancing Girl** (lost wax technique), Steatite Bearded Priest, Pashupati seal, Piece of woven cotton
Lothal	Ahmedabad (Gujarat)	**Bhogava**	S.R. Rao (1954/57)	**Tidal Dockyard** (artificial brick basin), Rice husk, Bead factory, Fire altars, Terracotta horse figurine, Persian gulf seal, Double burial (male-female)
Kalibangan (*Black Bangles*)	Hanumangarh (Rajasthan)	**Ghaggar** (ancient Saraswati)	A. Ghosh (1953), B.B. Lal & B.K. Thapar	**Ploughed field surface** (furrows), Camel bones, Row of **7 fire altars**, Wooden drainage, Decorated floor tiles, Both citadel & lower town fortified
Chanhudaro	Sindh (Pakistan)	**Indus**	N.G. Majumdar (1931), Mackay (1935)	**Only site without a citadel**, Bead-making factory, Lipstick/cosmetics, Inkpot, Imprint of dog's paw chasing a cat
Dholavira	Kutch (Gujarat) - Khadir Bet	Luni basin (Mansar & Manhar streams)	J.P. Joshi (1967), R.S. Bisht (1990)	**Three-tier city division** (Citadel, Middle Town, Lower Town), Giant **water reservoirs/dams**, Unique **10-symbol signboard**, Sports stadium; **UNESCO WHS (2021)**
Banawali	Fatehabad (Haryana)	Saraswati / Rangoi	R.S. Bisht (1974)	**Terracotta toy model of plough**, High quantity of good quality barley, Lack of systematic grid drainage
Rakhigarhi	Hisar (Haryana)	Ghaggar-Hakra	Amarendra Nath	**Largest Harappan site in India** (and overall after recent surveys)
Ropar (Roopnagar)	Punjab (India)	**Sutlej**	Y.D. Sharma (1953)	**Dog buried with human oval pit** (post-Harappan continuation), First site excavated after independence
Surkotada	Kutch (Gujarat)	Seasonal stream	J.P. Joshi (1964)	**Horse bone remains** (disputed), Pot burials, Oval citadel
Daimabad	Ahmednagar (Maharashtra)	**Pravara** (Godavari tributary)	B.P. Bopardikar	**Southernmost IVC site**, Bronze chariot with driver, Rhino, Elephant, Buffalo

Site	Location / State	River	Excavator & Year	Key Unique Findings / Exam Triggers
Alamgirpur	Meerut (Uttar Pradesh)	**Hindon** (Yamuna tributary)	Y.D. Sharma (1958)	**Easternmost IVC site** in India; cloth impression on pottery; triangle–peacock–squirrel motifs. **Not** Rakhigarhi (Haryana = largest, not east).
Hulas	**Saharanpur** (UP; some lists tag the Baghpat belt)	Upper Ganga–Yamuna doab	**K.N. Dikshit** (ASI)	Eastern Harappan habitation: handmade + wheel-made pottery, chert blades, bone points. **UPPCS match: Hulas = Uttar Pradesh** (not Gujarat).
Mandi	**Muzaffarnagar** (UP; some lists Bijnor / Ramganga)	Doab	ASI	Late Harappan **jewellery / gold-hoard** find. **Mandi (UP) ≠ Manda (J&K)**. With Hulas, the usual UP pair that rejects **Rakhigarhi**.
Sanauli	**Baghpat** (Uttar Pradesh)	Yamuna basin	D.V. Sharma / ASI (**2018**)	About **125** graves, coffins, copper antenna swords, solid wooden-wheel **chariots**. Late Harappan / OCP — **not** a Mature grid-city with a Great Bath. Same district belt as Hulas lists, different period label.
Bargaon / Ambakheri	**Saharanpur** (Uttar Pradesh)	Doab	ASI	Late Harappan / **OCP** belt of western UP (with Hastinapur, Lal Qila, Atranjikhera). Rural doab, not a citadel city.
Manda	Jammu & Kashmir	**Chenab**	J.P. Joshi (1982)	**Northernmost IVC site** in India. Trap: **Manda ≠ Mandi**.
Sutkagen Dor	Balochistan (Pak-Iran border)	**Dasht**	Aurel Stein (1927)	**Westernmost IVC site**, Coastal trading post between Harappa & Babylon
Kunal	Haryana	Saraswati	J.S. Khatri	2 silver crowns discovered
Balu	Kaithal (**Haryana**)	-	-	Earliest evidence of garlic cultivation. Trap: **Balu ≠ UP**.
Padri	Bhavnagar belt (**Gujarat**)	Coastal Saurashtra	-	**Gujarat** match in the Balu–Manda–Padri–Hulas set. Trap: **Padri ≠ Hulas**.
Bhirrana	Fatehabad (**Haryana**)	Ghaggar	ASI / L.S. Rao	Often called among the **oldest** Harappan settlements. Not a UP site.

UP Harappan quartet: Alamgirpur (Meerut, east end) · **Hulas** (Saharanpur) · **Mandi** (Muzaffarnagar) · **Sanauli** (Baghpat, Late / OCP chariots). **Rakhigarhi is Haryana**, not UP.
UKPCS — no mature Indus city in Uttarakhand. Closest protohistoric belt: **Bahadrabad (Haridwar)** — OCP + Copper Hoards (Y.D. Sharma, 1951–53); **Bankot (Pithoragarh)** — 8 copper anthropomorphs (1989); **Malari (Chamoli)** — cist graves, 5.2 kg gold mask, Swat-style pottery. Do not list these as Harappa / Mohenjodaro-type cities.

2. Vedic Civilization — Literature, Philosophy & Geography

A. Rigvedic Rivers (Ancient vs Modern Name Match-up)

Rigvedic Ancient Name	Modern Name	Rigvedic Ancient Name	Modern Name
Vitasta	Jhelum	**Saraswati**	Ghaggar / Hakra (Most sacred: *Naditama*, *Devitama*)
Asikni	Chenab	**Drishadvati**	Chautang / Rakshi
Parushni	Ravi (Battle of 10 Kings on its bank)	**Kubha**	Kabul (Afghanistan)

Rigvedic Ancient Name	Modern Name	Rigvedic Ancient Name	Modern Name
Vipas	Beas	**Suvastu**	Swat (Afghanistan)
Sutudri	Sutlej	**Krumu**	Kurram (Afghanistan)
Sindhu	Indus (Most frequently mentioned river)	**Gomati**	Gomal (Afghanistan)
Sadaneera	Gandak (Mentioned in *Shatapatha Brahmana*)	**Marudvridha**	Maruwardwan (J&K)

B. Vedic Literature & Upanishadic Dialogues (Match Goldmine)

- **Rigveda**: 10 Mandalas, 1,028 Suktas (hymns); 3rd Mandala contains **Gayatri Mantra** (composed by Vishvamitra dedicated to solar deity Savitr); 7th Mandala describes **Dasrajan Yudha** (Battle of 10 Kings on Parushni/Ravi river won by Sudas of Bharata tribe); 9th Mandala dedicated entirely to **Soma**; 10th Mandala contains **Purusha Sukta** (first mention of 4 Varnas).
- **Samaveda**: Book of chants / melodies (*Ragas*); origin of Indian classical music; chanted by *Udgatri* priests.
- **Yajurveda**: Book of sacrificial prayers and rituals; divided into **Shukla Yajurveda** (White - poetry) and **Krishna Yajurveda** (Black - prose + verse).
- **Atharvaveda**: Charms, spells, magical formulas, everyday medicine; mentions *Sabha* and *Samiti* as twin daughters of Prajapati.
- **Key Brahmanas**:
- *Aitareya & Kaushitaki*: Rigveda.
- *Shatapatha Brahmana*: Largest and oldest Brahmana (attached to Shukla Yajurveda); story of Videgha Mathava carrying Agni across Sadaneera/Gandak.
- *Gopatha Brahmana*: Attached to Atharvaveda.
- **Famous Upanishadic Dialogues (Direct UPPSC/UKPSC Match-ups)**:
- **Mundaka Upanishad**: Sourced national motto **"Satyam Eva Jayate"** (*Truth Alone Triumphs*).
- **Katha Upanishad**: Famous philosophical dialogue between young boy **Nachiketa and Yama (God of Death)** on the nature of soul and immortality.
- **Brihadaranyaka Upanishad**: Largest Upanishad; contains celebrated philosophical debate between sage **Yajnavalkya and scholarly woman Gargi**; phrase "*Tamaso Ma Jyotirgamaya*".
- **Chhandogya Upanishad**: Contains story of **Satyakama Jabala** (boy who spoke truth about his parentage admitted by sage Gautama); dialogue between Uddalaka Aruni and son Svetaketu (*Tat Tvam Asi*).
- **Jabala Upanishad**: First explicit mention of the **Four Ashramas** (Brahmacharya, Grihastha, Vanaprastha, Sanyasa).

C. Six Vedangas (Limbs of the Vedas)

1. **Shiksha**: Phonetics and Pronunciation.
2. **Kalpa**: Ritual instructions (Shrauta Sutras, Grihya Sutras, Dharma Sutras, and **Sulba Sutras — ancient Indian Geometry & measurement!**).
3. **Vyakarana**: Grammar (Panini's *Ashtadhyayi*).
4. **Nirukta**: Etymology of Vedic words (Yaska's *Nirukta*).
5. **Chhanda**: Metrics and Poetic meters (Pingala's *Chhandasastra*).
6. **Jyotisha**: Astronomy and calculation of auspicious ritual calendar (Lagadha's *Vedanga Jyotisha*).

D. Six Schools of Indian Philosophy (Shad-Darshana)

School of Philosophy	Founder / Sage	Foundational Text	Core Philosophical Principle
1. Samkhya (*Oldest school*)	**Kapila Muni**	*Samkhya Sutra*	Dualistic realism: **Purusha** (Consciousness) and **Prakriti** (Matter); 25 basic elements

School of Philosophy	Founder / Sage	Foundational Text	Core Philosophical Principle
2. Yoga	**Patanjali**	*Yoga Sutra*	Eightfold path (**Ashtanga Yoga**: Yama, Niyama, Asana, Pranayama, Pratyahara, Dharana, Dhyana, Samadhi)
3. Nyaya	**Akshapada Gautama**	*Nyaya Sutra*	Logic, epistemology, and rigorous reasoning; 4 Pramanas (Pratyaksha, Anumana, Upamana, Shabda)
4. Vaisheshika	**Kanada (Uluka)**	*Vaisheshika Sutra*	**Atomic theory (Paramanu)**; universe composed of 9 physical substances; scientific classification
5. Purva Mimamsa	**Jaimini**	*Mimamsa Sutra*	Unquestioned authority of Vedas; ritualism, Dharma, and Vedic karma
6. Uttara Mimamsa (Vedanta)	**Badarayana (Vyasa)**	*Brahma Sutra*	Philosophy of Upanishads; Brahman is supreme reality; developed by Shankara (Advaita) & Ramanuja

E. Vedic Political Assemblies (Sabha, Samiti, Vidatha, Gana)

Assembly	Composition & Nature	Presiding Officer	Primary Functions	Women's Status & Rigvedic vs Later Vedic Evolution
Sabha (सभा)	Council of select **elders, nobles, and Brahmanas**; smaller and more elite body	**Sabhapati** / *Sabhapala*	**Judicial and advisory**: arbitrated disputes, dispensed justice, advised king on state affairs	Women attended in Early Vedic (*Sabhavati*); **strictly excluded in Later Vedic** as royal power and patriarchy grew
Samiti (समिति)	**General folk assembly of the entire tribe (*Vis*)**; broad democratic body	**Pati** or **Ishan**	**Political and electoral**: elected and re-elected the *Rajan* (King), debated peace/war, discussed national policy; Rigveda hymn (X.191) ends with prayer for common consensus	Women had **no role**; strictly male tribal gathering. In Later Vedic, kings gradually subdued Samiti's electing power
Vidatha (विदथ)	**Oldest tribal assembly** (mentioned 122 times in Rigveda vs 8 times for Sabha, 9 for Samiti)	Tribal elder / seer	**Ritual, redistributive, and military**: distributed war spoils/booty, conducted folk rituals, settled domestic customs	**Women actively participated** (*Vidathya*) and debated alongside men. **Completely disappeared in Later Vedic times**
Gana (गण)	Assembly of warrior clans / general troop gathering	**Ganapati** / *Jyeshtha*	Military deliberations and tribal defense	Minor folk council; evolved into post-Vedic oligarchic Gana-sanghas

> **Crucial Prelims Triggers:** - **Atharvaveda (7.12.1)** explicitly calls **Sabha and Samiti the twin daughters of Prajapati** (*Prajapateh duhitarau*). - **Judicial function** belonged exclusively to **Sabha** (not Samiti). - **Electoral function** (election of the King) belonged to **Samiti** (not Sabha). - **Earliest assembly: Vidatha** (Rigveda's highest mention count: 122).

F. High-Yield Vedic Terms & Vocabulary (Match & Meaning Goldmine)

Frequently tested in UPPCS & UKPCS Prelims:

Ancient Vedic Term	Literal Meaning / Context	Historical & Exam Significance
Aghanya (अघन्या)	"Not to be killed / slain"	Refers to the **Cow**; highest penalty (capital punishment or expulsion) for hurting/killing
Gavishti / Gaveshana / Goshu	"Search for cows"	**War / Battle**; cattle raids were the prime cause of inter-tribal conflicts in Rigvedic era
Gopati (गोपति)	"Protector of cattle"	Title of the **King / Tribal Chief** (*Rajan*)

Ancient Vedic Term	Literal Meaning / Context	Historical & Exam Significance
Gomat (गोमत्)	"Possessor of cattle"	A **wealthy person** (wealth was measured in cattle heads, not coins/land)
Duhitri (दुहितृ)	"One who milks the cow"	**Daughter** in the family
Goghna (गोघ्न)	"One for whom a cow is offered"	A revered **Guest** visiting the household
Godhuli (गोधूलि)	"Dust raised by returning cattle"	Evening twilight / dusk (unit of time measurement)
Gavyuti (गव्यूति)	Cattle grazing radius	Unit of **distance measurement**
Kulapa (कुलप)	Head of the *Kula* (family)	Oldest male patriarch of the family
Gramani (ग्रामणी)	Leader of the *Grama* (village)	Head of village; led local militia and ranked among the 12 Ratnins
Vispati (विशपति)	Head of the *Vis* (clan / canton)	Clan chief between village (*Grama*) and tribe (*Jana*)
Vrajapati (व्राजपति)	Master of pastures	Officer in charge of pastures and cattle camps; led village warriors
Spasa (स्पश)	"Lookout / watcher"	**Spies / Intelligence agents** employed by the King/Varuna
Jivagribh (जीवगृभ्)	"One who seizes alive"	**Police official / arresting officer**
Duta (दूत)	Messenger	Royal envoy or ambassador
Yava (यव)	Food grain	**Barley** (the primary cereal of Rigvedic Aryans)
Godhuma (गोधूम)	Cereal grain	**Wheat** (widely cultivated in Later Vedic)
Vrihi (व्रीहि)	Cereal grain	**Rice** (extensively mentioned in Later Vedic texts)
Sira / Langala (सीर / लांगल)	Agricultural tool	**Plough**; *Sita* = furrow made by plough
Kinasha (कीनाश)	Field worker	**Ploughman / peasant cultivator**
Urvara / Kshetra (उर्वरा / क्षेत्र)	Land type	Fertile, cultivated agricultural field
Ayas (अयस्)	Metal	In Rigveda: **copper / bronze**; in Later Vedic: *Krishna-ayas / Shyama-ayas* = **iron**, *Lohita-ayas* = **copper**
Pani (पणि)	Trading folk	Wealthy non-Aryan traders/merchants; notorious in hymns as cattle-thieves
Bekhanata (बेखनाट)	Lender	**Usurer / Moneylender** who charged exorbitant interest rates
Nishka / Satamana (निष्क / शतमान)	Gold ornament / unit	In Rigveda: gold necklet/bullion; in Later Vedic: standardized gold unit of value

G. The 12 Vedic Ratnins (Jewel-Bearers of Royal Administration)

Mentioned in the Shatapatha Brahmana and Taittiriya Samhita during the Ratnahavimshi ceremony of the Rajasuya Yajna (King visits each official's home to offer oblation):

#	Ratnin (Jewel-Official)	Exact Designation / Official Portfolio	Key UPPCS Exam Trap
1	**Purohita** (पुरोहित)	Chief Royal Priest and chief political adviser to Rajan	Invariable first counselor of the realm
2	**Senani** (सेनानी)	Commander-in-Chief of the military forces	Leads the army in times of war
3	**Gramani** (ग्रामणी)	Village headman / leader of the rural militia	Represents rural folk in the royal circle

#	Ratnin (Jewel-Official)	Exact Designation / Official Portfolio	Key UPPCS Exam Trap
4	**Mahishi** (महिषी)	Chief Queen (first legally wedded queen)	Participated in royal consecration rites
5	**Suta** (सूत)	Charioteer, royal court minstrel, and herald	Chronicler of heroic royal deeds
6	**Sangrahitri** (संग्रहीतृ)	**Treasurer / Custodian of royal store**	Stores revenue and tribute
7	**Bhagadugha** (भागदुघ)	**Collector of the King's share (*Bhaga*) of revenue / taxes**	**UPPCS Trap:** Collects revenue (do NOT confuse with Akshavapa)
8	**Akshavapa** (अक्षावाप)	**Superintendent of gambling & royal dice accountant**	**UPPCS Trap (2023):** Dice/gambling officer, NOT revenue collector!
9	**Govikartana** (गोविकर्तन)	Keeper of forests, game master, companion in the royal hunt	Head of livestock and hunting preserves
10	**Palagala** (पालागल)	Royal courier, messenger, and confidential king's companion	Diplomatic emissary
11	**Kshattri** (क्षात्तृ)	Royal chamberlain / master of the royal household gate	Managed inner palace affairs
12	**Takshan / Rathakara** (तक्षन् / रथकार)	Chief carpenter and chariot-builder	Highest artisan status due to importance of chariots in war

3. 16 Mahajanapadas & Their Capitals (6th Century BCE)

Mahajanapada	Capital	Modern Location	High-Yield Exam Notes
Magadha	Girivraja / Rajagriha → Pataliputra	Patna & Gaya (Bihar)	Founded by Brihadratha; rose under Bimbisara (Haryanka)
Anga	**Champa** (ancient Malini)	Bhagalpur & Munger (Bihar)	Annexed by Bimbisara; major trade port to Suvarnabhumi
Kasi	**Varanasi**	Varanasi (UP)	Between Varuna and Asi rivers; famous for cotton textiles
Vatsa	**Kausambi**	Prayagraj/Kaushambi (UP)	King **Udayana** (hero of Bhasa's *Swapnavasavadattam*); Buddha visited
Kosala	**Sravasti** (North) / Saketa (South)	Ayodhya/Faizabad & Bahraich (UP)	King **Prasenjit** (contemporary of Buddha); Buddha spent most rainy seasons here
Saurasena	**Mathura**	Mathura (UP)	King Avantiputra (disciple of Buddha); Megasthenes mentions worship of Herakles (Krishna)
Panchala	**Ahichchhatra** (North, Bareilly) & **Kampilya** (South, Farrukhabad)	Rohilkhand & Central UP	Kannauj was part of it; transition from tribal to urban center
Kuru	**Indraprastha** (near Delhi) / Hastinapur	Meerut, Delhi, Thanesar	Shifted capital to Kaushambi after Hastinapur was flooded (Nichakshu)
Matsya	**Viratnagar** (Bairat, Jaipur)	Alwar, Bharatpur, Jaipur (Raj)	Hiding place of Pandavas; King Sujata
Chedi	**Suktimati** / Sotthivati	Bundelkhand (UP/MP)	King Shishupala (killed by Krishna in Mahabharata)
Avanti	**Ujjayini** (North) & **Mahishmati** (South)	Malwa / MP	King **Pradyota** (Mahasena); Jivaka (Bimbisara's physician) sent to treat his jaundice

Mahajanapada	Capital	Modern Location	High-Yield Exam Notes
Gandhara	**Taxila** (Takshashila) & Pushkalavati	Rawalpindi / Peshawar (Pak)	Famous university; King Pushkarasarin sent embassy to Bimbisara
Kamboja	**Rajpur** / Hataka	Rajouri & Hazara (NW frontier)	Famous for fine breed of horses; mentioned by Kautilya as *Varta-Shastropajivi*
Asmaka (Assaka)	**Potana** / Potali	Banks of river **Godavari** (Telangana/Maha)	**ONLY Mahajanapada situated south of the Vindhyas**
Vajji	**Vaishali**	Muzaffarpur / Vaishali (Bihar)	Confederacy of 8 clans (Lichchhavis, Videhas, Jnatrikas); First republican state
Malla	**Kushinagar** & **Pawa**	Deoria / Gorakhpur (UP)	Republican Gana-sangha; Buddha died at Kushinagar; Mahavira died at Pawa

UKPCS Special Note: Ashoka's Major Rock Edict in Uttarakhand is at **Kalsi** (Dehradun district) on the confluence of the Yamuna and Tons rivers.

4. Ashoka's 14 Major Rock Edicts (MRE) — Complete Cheat-Sheet

Edict	Core Subject / Mandate Inscribed
Major Rock Edict I	Prohibition of animal slaughter and festive gatherings (*Samaja*).
Major Rock Edict II	Medical treatment for men and animals in his empire and border lands; mentions **Cholas, Pandyas, Satiyaputras, Keralaputras, and Antiochus II Theos of Syria**.
Major Rock Edict III	Liberal attitude towards Brahmanas and Shramanas; ordering **Yuktas, Rajukas, and Pradeshikas** to go on tour of inspection every 5 years (*Anusamyana*).
Major Rock Edict IV	Sound of Dhamma-ghosha replaced the sound of Bheri-ghosha (war drum).
Major Rock Edict V	Appointment of **Dhamma Mahamattas** (in 14th regnal year) to propagate morality.
Major Rock Edict VI	King's accessibility: reporters (**Pativedakas**) ordered to report public matters to Ashoka at all times and places.
Major Rock Edict VII	Universal religious tolerance among all sects.
Major Rock Edict VIII	Replaced royal pleasure trips (*Viharayatras*) with religious tours (**Dhammayatras**); visited Bodh Gaya in 10th regnal year.
Major Rock Edict IX	Condemned meaningless rituals and ceremonies; extolled Dhamma-mangala.
Major Rock Edict X	Denounced individual fame and glory in favour of moral attainment.
Major Rock Edict XI	Declared that the gift of Dhamma (*Dhamma-dana*) is the highest of all gifts.
Major Rock Edict XII	Directive on mutual tolerance among all religious sects; appointment of *Ithijhakha Mahamattas* (female affairs officers).

Edict	Core Subject / Mandate Inscribed
Major Rock Edict XIII	**MOST IMPORTANT EDICT**: Records the **Kalinga War (261 BCE, 8th regnal year)**, deep remorse over 1,00,000 killed and 1,50,000 captured; victory by Dhamma declared supreme; **mentions 5 contemporary Greek Kings**: 1. **Antiochus II Theos** (Syria) 2. **Ptolemy II Philadelphus** (Egypt) 3. **Antigonus Gonatas** (Macedonia) 4. **Magas** (Cyrene) 5. **Alexander** (Epirus).
Major Rock Edict XIV	Summarizes the purpose and recording of the various rock inscriptions.

5. Buddhism & Jainism Quick-Recall Matrix

A. The 4 Buddhist Councils (Memorize the Pattern!)

Council	Year	Venue	Patron King & Dynasty	President / Chairman	Landmark Accomplishment
1st	**483 BCE**	Saptaparni Cave, **Rajagriha**	**Ajatashatru** (Haryanka)	**Mahakassapa**	**Sutta Pitaka** (Ananda - Buddha's teachings) & **Vinaya Pitaka** (Upali - Monastic rules) compiled
2nd	**383 BCE**	**Vaishali**	**Kalashoka** (Shishunaga)	**Sabakami**	First schism: Split into **Sthaviravadins** (Theravadins) and **Mahasanghikas**
3rd	**250 BCE**	**Pataliputra**	**Ashoka** (Maurya)	**Moggaliputta Tissa**	**Abhidhamma Pitaka** compiled (philosophical analysis); Kathavatthu added; Missionaries sent abroad
4th	**72 CE**	Kundalvana (**Kashmir**)	**Kanishka** (Kushana)	**Vasumitra** (President), **Ashvaghosha** (Vice-President)	Final split into **Hinayana** and **Mahayana**; *Mahavibhasha Shastra* written in Sanskrit

B. Buddha's 5 Great Life Events & Symbols

1. **Janma (Birth)**: Lotus and Bull (**563 BCE**, Lumbini, Kapilavastu).
2. **Mahabhinishkramana (Great Renunciation)**: Horse (Kanthaka).
3. **Nirvana / Sambodhi (Enlightenment)**: Bodhi Tree (Pipal / Ficus religiosa) at Bodh Gaya on Niranjana (Phalgu) river.
4. **Dharmachakrapravartana (First Sermon)**: Wheel / 8-Spoked Chakra (Deer Park at Sarnath / Rishipattana).
5. **Mahaparinirvana (Death)**: Stupa / Footprints (Sal tree grove at Kushinagar, **483 BCE**, age **80**).

C. Birth and death of Buddha and Mahavira

Head	**Gautama Buddha (Siddhartha)**	**Mahavira (Vardhamana)**
Birth year	**563 BCE** (traditional; some books use a shorter chronology)	**599 BCE** (usual teaching set); some books use **540 BCE**
Birthplace	**Lumbini** (Nepal), near Kapilavastu of the **Shakya** clan	**Kundagrama** (Vaishali / Vajji belt, Bihar), **Jnatraka** clan
Parents	Father **Suddhodana**; mother **Mahamaya / Mayadevi** (Koliyan); raised by **Mahapajapati Gotami**	Father **Siddhartha**; mother **Trishala** (sister of Lichchhavi chief **Chetaka**)
Wife / child	**Yashodhara**; son **Rahul**	Jain tradition names wife **Yashoda** (a different person from Yashodhara)
Renunciation	Age **29** (*Mahabhinishkramana*; horse **Kanthaka**)	Age **30** (*grihatyaga*)

Head	Gautama Buddha (Siddhartha)	Mahavira (Vardhamana)
Awakening	Enlightenment at about **35**, **Bodh Gaya** (Vaishakha Purnima)	**Kevala jnana** at about **42**
Death year	**483 BCE** (*Mahaparinirvana*)	**527 BCE** (usual); some books use **468 BCE**
Death place	**Kushinagar** (UP, Malla republic), sala grove, age **80**	**Pavapuri / Pawa** (Bihar), age **72**
Proof of birthplace	Ashoka's **Rummindei** pillar at Lumbini	Kundagrama / Vaishali tradition; not marked by Rummindei

Logic: Buddha **563–483 BCE**, 80 years. Mahavira **599–527 BCE**, 72 years. Do not swap Lumbini with Kundagrama, or Kushinagar with Pavapuri. **Nayaputta** is an epithet of **Mahavira**, not of Buddha. If a stem uses the shorter Jain pair **540–468 BCE**, both years still sit in the 6th–5th century BCE.

D. Jain Tirthankaras & Symbols (High-Frequency Match)

- **1st Rishabhanatha (Adinatha)**: **Bull** (Emblem) | Born in Ayodhya | Mentioned in Rigveda with Arishtanemi.
- **2nd Ajitanatha**: **Elephant** | Born in Ayodhya.
- **16th Shantinaatha**: **Deer**.
- **19th Mallinatha**: **Water Jar (Kalasha)** | Shvetambaras consider Mallinatha a female.
- **22nd Neminatha (Arishtanemi)**: **Conch (Shankha)** | Cousin of Krishna.
- **23rd Parshvanatha**: **Serpent / Hooded Cobra** | Born in Varanasi (Ashvasena's son); propounded 4 vows (*Chaturyama*).
- **24th Mahavira (Vardhamana)**: **Lion** | Born **599 BCE** at Kundagrama (Vaishali); died **527 BCE** at Pavapuri (age 72); added 5th vow: **Brahmacharya**.
- **Jain Councils**:
- **1st Council (c. 300 BCE)**: Pataliputra | Presided by **Sthulabhadra** | 12 Angas compiled | Split into Shvetambara (white-clad, led by Sthulabhadra) and Digambara (sky-clad, led by Bhadrabahu).
- **2nd Council (512 CE)**: Vallabhi (Gujarat) | Presided by **Devardhi Kshamasramana** | Final compilation of 12 Angas & 12 Upangas in Ardhamagadhi Prakrit.

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6. Ancient Dynasties — From / To, Founder, Last Ruler, High-Yield

Magadha spine (do not skip a rung): **Haryanka → Shishunaga → Nanda → Maurya → Shunga → Kanva**.

Dynasty	Period (from–to)	Founder	Last ruler	Capital	Key ruler & high-yield
Haryanka	**c. 544–412 BCE**	**Bimbisara** (Shrenika; annexes **Anga**; contemporary of Buddha & Mahavira)	**Nagadasaka** (weak tail; minister Shishunaga takes the throne)	**Rajagriha / Girivraja**	**Ajatashatru** — *rathamusala*, war on Vajji, **1st Buddhist Council** (Rajagriha). **Udayin** first makes **Pataliputra** the Magadhan capital (Ajatashatru only seeds Pataligrama).
Shishunaga	**c. 412–344 BCE**	**Shishunaga** (ex-viceroy of Kashi)	**Nandivardhana / Mahanandin** (overthrown by Mahapadma Nanda)	Girivraja → **Vaishali** (second capital tradition)	Founder **takes Avanti / Malwa** (ends Pradyota). **Kalashoka / Kakavarna** — **2nd Buddhist Council** at Vaishali (schism). Did **not** found Pataliputra.

Dynasty	Period (from–to)	Founder	Last ruler	Capital	Key ruler & high-yield
Nanda	**c. 344–321 BCE**	**Mahapadma Nanda** (*Ekarat*, *Sarvakshatrantaka*, *Ugrasena*, *Aparoparashurama*)	**Dhana Nanda** (Greek **Agrammes / Xandrames**; commander Bhaddasala)	**Pataliputra**	Nine Nandas. Huge standing army; **Alexander never fought** them (326 BCE is still Dhana Nanda's reign). Hathigumpha later recalls a Nanda canal and a Jina image from Kalinga. Overthrown by **Chandragupta + Chanakya**.
Maurya	**c. 321–185 BCE**	**Chandragupta Maurya** (Greek *Sandrokottos*; Chanakya / Kautilya; treaty with Seleucus **c. 303 BCE**)	**Brihadratha** (killed by general **Pushyamitra Shunga**, 185 BCE)	**Pataliputra** (Ganga–Son)	**Bindusara** (*Amitraghata*). **Ashoka**: accession c. 272, coronation **268–232**; **Kalinga War 261 BCE** (MRE XIII); **3rd Council** at Pataliputra. Ashoka's death **232 ≠** end of empire — house runs to **185**.
Shunga	**185–73 BCE**	**Pushyamitra Shunga** (coup at a military review)	**Devabhuti** (killed / ousted by Vasudeva Kanva)	Pataliputra & **Vidisha**	Two **ashvamedhas** (Ayodhya / Dhanadeva record). **Agnimitra** (अग्निमित्र) in Kalidasa's (कालिदास) *Malavikagnimitra* (मालविकाग्निमित्रम्). **Patanjali** (*Mahabhashya* (महाभाष्य)) is Shunga-age, not Kanishka's court. Sanchi still grows.
Kanva	**73–28 BCE**	**Vasudeva Kanva**	**Susarman**	Pataliputra	Short Magadha coda after Shungas. Deccan **Satavahanas** rise as this house fades.
Satavahana (Andhra)	**c. 230 BCE – 3rd c. CE**	**Simuka**	Last great often **Yajna Sri Satakarni** (later Pulomavi line)	**Pratishthana (Paithan)** & Amaravati	**Gautamiputra Satakarni** — *Ekabrahmana*, *Trisamudrapitapoyavahana* (Nasik cave of mother **Gautami Balashri**); smashed Western Kshatrapas. **Lead & potin** coins; matrilineal *Gautamiputra / Vasisthiputra* titles. **Hala** — Prakrit *Gatha Saptasati*. Yajna Sri — **ship** on coins.
Kushana	**1st–3rd c. CE**	**Kujula Kadphises** (Kadphises I)	Last great **Vasudeva I** (then fragmentation)	**Purushapura (Peshawar)** & **Mathura**	**Vima Kadphises** — first Kushana **gold** (*dinar*). **Kanishka** — **Shaka era 78 CE** (some books peak c. 127); **4th Council** at Kundalvana (Kashmir); Gandhara + Mathura art; titles *Devaputra*. Do not make Kanishka a Gupta.
Gupta	**c. 320–550 CE**	**Sri Gupta** → **Ghatotkacha** (both *Maharaja*); real imperial start **Chandragupta I** (**Maharajadhiraja**, Gupta era **319/320 CE**, queen **Kumaradevi**)	Last great **Skandagupta**; later line thins (e.g. Vishnugupta)	**Pataliputra** (CG II also uses **Ujjain**)	**Samudragupta** — Prayag Prashasti by **Harisena**; *Napoleon of India*; Aryavarta uprooted, Dakshinapatha captured-and-freed. **Chandragupta II Vikramaditya** — *Sakari*, ends Western Kshatrapas (Rudrasimha III); Mehrauli iron pillar; Fa-Hien. **Kumaragupta I** — Nalanda tradition. **Skandagupta** — **Bhitari** (Ghazipur, UP) vs **Hunas**; repairs Sudarsana lake.

Dynasty	Period (from–to)	Founder	Last ruler	Capital	Key ruler & high-yield
Vakataka	**c. 250–500 CE**	**Vindhyashakti**	**Harishena** (Vatsagulma branch; Ajanta later caves)	**Vatsagulma** (Washim) / Nandivardhana	**Pravarasena I** — four *ashvamedhas*. Gupta–Vakataka tie: **Prabhavatigupta** (daughter of CG II) married **Rudrasena II** and ruled as regent.
Maukhari	**6th c. CE** (till **c. 606**)	**Harivarman**	**Grahavarman**	**Kannauj** (Kanyakubja, UP)	Grahavarman married **Rajyashri** (Harsha's sister); killed by **Deva Gupta of Malwa** + **Shashanka** of Gauda — the spark for Harsha's rise.
Pushyabhuti (Vardhana)	House earlier; **Harsha 606–647 CE**	**Pushyabhuti** (house); effective reunion king **Harshavardhana**	**Harsha** (no stable heir after **647**; Wang Xuance / Arunashva)	Thanesar → **Kannauj**	**Harsha-samvat 606**. Xuanzang (*Si-Yu-Ki*). Kannauj & Prayag assemblies. Stopped on the **Narmada** by **Pulakeshin II**. Banabhatta — *Harshacharita*. Not a Gupta emperor.
Badami Chalukya	**543–753 CE**	**Pulakeshin I**	**Kirtivarman II** (overthrown by Rashtrakuta **Dantidurga**)	**Vatapi (Badami**, Karnataka)	**Pulakeshin II** — defeats Harsha on Narmada; **Aihole / Meguti** prashasti by Jain poet **Ravikirti** (रविकीर्ति) (names **Kalidasa** (कालिदास) & **Bharavi** (भारवि)). Clash with Pallavas (Narasimhavarman I sacks Vatapi — *Vatapikonda*).
Pallava of Kanchi	**c. 575–897 CE**	**Simhavishnu** (imperial line)	**Aparajita** (defeated by Chola **Aditya I**)	**Kanchipuram**	**Mahendravarman I** — *Mattavilasa Prahasana*; rock-cut start. **Narasimhavarman I Mahamalla** — defeats Pulakeshin II; **Mamallapuram** rathas / shore temple age.

Logic: Maurya **321–185**, not 321–232. Gupta **320–550**, not "ends with Chandragupta II." Harsha **606–647** is Pushyabhuti, not Gupta. Shunga **185** is the Mauryan end, not Kalinga (261).

7. Epigraphy & Inscriptions (The Match-the-Following Goldmine)

Inscription	Place / State	King / Dynasty	Author / Key Fact
Prayag Prashasti (Allahabad Pillar)	Prayagraj (UP)	**Samudragupta** (Gupta)	Composed by court poet **Harisena** (हरिषेण) in *Champu* (चम्पू) (verse + prose) style in Sanskrit. Mentions *Digvijaya*, *Dakshinapatha* (12 kings - *Grahana-Mokshanugraha*), *Aryavarta* (9 kings exterminated). Also has inscriptions of Ashoka, Jahangir, Birbal!
Hathigumpha Inscription	Udayagiri hills, Bhubaneswar (Odisha)	**Kharavela** (Chedi/Mahameghavahana)	In Prakrit, Brahmi script. Mentions: canal dug by Nandaraja 300 yrs earlier brought to Kalinga, invasion of Magadha (*Bahisatimita* defeated), first epigraphic mention of **Bharatavarsha**.
Junagarh Rock Inscription	Girnar, Junagadh (Gujarat)	**Rudradaman I** (Western Kshatrapa, 150 CE)	**First chaste, elaborate Sanskrit inscription**. Records history of **Sudarshana Lake**: built under Chandragupta Maurya (Pushyagupta), canal added by Ashoka (Tushaspha), repaired by Rudradaman (Suvisakha) **without levying extra tax/Vishti**, later restored by Skandagupta (Chakrapalita)!

Inscription	Place / State	King / Dynasty	Author / Key Fact
Aihole Inscription	Bagalkot (Karnataka)	**Pulakeshin II** (Badami Chalukya)	Composed by Jain poet **Ravikirti** (रविकीर्ति) at Meguti temple. Mentions Pulakeshin II defeating Harshavardhana on the banks of Narmada; mentions poets **Kalidasa** (कालिदास) and **Bharavi** (भारवि).
Mehrauli Iron Pillar	Qutb Complex, Delhi	**Chandra** (Chandragupta II Vikramaditya)	Rustless iron metallurgy; defeated Vanga confederacy and crossed 7 mouths of Sindhu to defeat Vahlikas. Dedicated to Lord Vishnu (*Vishnupada*).
Nasik Cave Inscription	Nasik (Maharashtra)	**Gautamiputra Satakarni** (Satavahana)	Issued by his mother **Gautami Balashri**. Titles: *Ekabrahmana*, *Kshatriya-darpa-man-mardana*, *Trisamudrapitapoyavahana* (whose horses drank water of three seas).
Besnagar (Vidisha) Pillar	Vidisha (Madhya Pradesh)	**Bhagabhadra** (Shunga)	Erected by **Heliodorus** (Greek ambassador of King Antialcidas of Taxila); calls himself *Bhagavata* and dedicates Garuda-dhvaja to *Devadeva Vasudeva*.
Eran Inscription (510 CE)	Sagar (Madhya Pradesh)	**Bhanugupta** (Gupta)	**First epigraphic / dated evidence of Sati**: Goparaja (general) died fighting Hunas; his wife immolated herself on his funeral pyre.
Uttiramerur Inscription	Kanchipuram (Tamil Nadu)	**Parantaka I** (Chola, 10th c.)	In-depth details of **Chola village self-government**, Kudavolai (pot-ticket lot) system, and committee qualification rules for *Variyam*.
Rummindei Pillar Inscription	Lumbini (Nepal)	**Ashoka** (Maurya)	Commemorates Ashoka's visit to Buddha's birthplace (20th regnal year). **Tax concessions**: Land revenue (*Bhaga*) reduced from ⅙th to **⅛th**, and religious tribute (*Bali*) completely abolished.
Bhabru (Bairat) Edict	Rajasthan	**Ashoka** (Maurya)	Ashoka explicitly affirms faith in the **Buddha, Dhamma, and Sangha**; calls himself *Piyadassi Raja Magadha*.
Maski, Gurjara, Nittur, Udegolam	Karnataka & MP	**Ashoka** (Maurya)	Inscriptions where the personal name **"Ashoka"** is explicitly written (instead of just *Devanampiya Piyadassi*).
Sohgaura (Gorakhpur) & Mahasthangarh (Bogura)	UP & Bangladesh	Maurya Period	Copper plate & limestone plaque mentioning **famine relief measures** and state granary maintenance.

8. Ancient Ports & Ancient Coins

A. Major Ancient Indian Ports (Direct Match Questions)

- **Western Coast**:
- **Bharuch / Barygaza** (Narmada estuary, Gujarat): Most active port for trade with Roman Empire.
- **Sopara (Surparaka)** (near Mumbai, Maharashtra): Ashokan edict site; major trade center.
- **Kalyan (Calliena)** (Thane, Maharashtra).
- **Muziris / Kodungallur** (Periyar river mouth, Kerala): Roman trade hub; exported black pepper (**"Yavanapriya"**); temple of Augustus built here.
- **Nelcynda & Tyndis** (Kerala coast).
- **Eastern Coast**:
- **Tamralipti / Tamluk** (Midnapore, West Bengal - Hooghly river): Premier port for trade with Southeast Asia, China, and Sri Lanka; Fa-Hien sailed back to China from here.

- **Arikamedu (Podouke)** (near Puducherry): Roman trading post; excavated by Mortimer Wheeler; Roman amphorae, Arretine ware, and Roman coins found.
- **Poompuhar / Kaveripattinam** (Kaveri mouth, Tamil Nadu): Capital port of the Early Cholas (celebrated in *Silappadikaram*).
- **Korkai** (Tamil Nadu): Famous pearl-fishery port of the Pandyas.

B. Ancient Coins & Currency Units

- **Nishka / Satamana**: Gold ornaments/weights in Vedic period; later evolved into currency terms.
- **Karshapana / Pana**: Standard **silver punch-marked coins** (bearing symbols of sun, tree, hill, peacock) in Mauryan and Mahajanapada era.
- **First Gold Coins in India**: Issued by **Indo-Greeks (2nd c. BCE)**; also first to issue coins with portraits and legends of kings.
- **Largest Number of Pure Gold Coins**: Issued by **Kushanas** (Vima Kadphises and Kanishka - *Dinar* based on Roman denarius weight standard).
- **Largest Quantity of Gold Coins**: Issued by **Guptas** (*Dinar* for gold; *Rupaka* for silver, started by Chandragupta II after defeating Shakas).
- **Lead and Potin Coins**: Distinctive currency issued by the **Satavahanas**.
- **Cowries (Kaudi)**: Common medium of exchange for daily petty transactions (recorded by Fa-Hien in Gupta era).

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9. Ancient Indian Administrative Officers Master Matrix

A. Mauryan Central Administration (Arthashastra & Edicts)

- **Central Tirthas (18 Supreme Functionaries)**: Top cabinet-level officials including *Mantri* (Prime Minister), *Purohita* (Chief Priest), *Senapati* (Army Chief), *Yuvaraja* (Crown Prince), and the two supreme fiscal heads:
- **Samaharta (समाहर्ता)**: **Collector General of Revenue** — responsible for preparing the annual state budget, assessing revenues, and overseeing tax collection across the entire empire (rural, urban, mines, forests).
- **Sannidhata (सन्निधाता)**: **Chief Treasury Officer & Custodian of Royal Storehouses** (*Koshadhyaksha*) — responsible for the safe keeping of the state treasury, constructing granaries, treasuries, and warehouses.
- **Megasthenes' Administrative Officials in *Indica***:
- **Astynomoi (नगर अध्यक्ष)**: 30 city officials of Pataliputra divided into **6 boards of 5 members each**:
    1. *Board 1*: Industrial arts & crafts inspection.
    2. *Board 2*: Care and surveillance of foreign visitors.
    3. *Board 3*: Registration of vital statistics (births and deaths).
    4. *Board 4*: Trade, commerce, and regulation of markets.
    5. *Board 5*: Inspection of manufactured goods and distinguishing new from old.
    6. *Board 6*: Collection of the **10% sales tax** on goods sold (evasion was punishable by death).
- **Agronomoi (जिला / ग्रामीण अधिकारी)**: Rural and district officers in charge of land measurement, irrigation canals, agriculture, timber, mining, and road maintenance (installed milestones every 10 stadia).
- **Ashoka's Key Administrative Cadres**:
- **Rajuka (रज्जुक)**: District administrator in charge of land survey, revenue assessment, and judicial administration (Ashoka granted them full independence in judicial punishment).
- **Yukta (युक्त)**: Subordinate secretarial officer / district accountant assisting the Rajukas.
- **Pradeshika (प्रादेशिक)**: Head of provincial/divisional administration on regular five-year inspection tours (*Anusamyana*).
- **Dhamma Mahamatta (धम्म महामात्र)**: High-ranking moral officials instituted in Ashoka's 14th regnal year (Rock Edict V) to propagate Dhamma, prevent arbitrary imprisonment, and supervise religious harmony.

B. Kautilya's 20+ Key Adhyakshas (Department Superintendents)

Adhyaksha (Superintendent)	Department / Administrative Portfolio	Exam Keyword / Function
Sitadhyaksha (सीताध्यक्ष)	**Agriculture & Crown Lands**	Supervised state farms (*Sita* lands) and farming labour
Pautavadhyaksha (पौतवाध्यक्ष)	**Weights and Measures**	Standardized scales and measures across markets
Panyadhyaksha (पण्याध्यक्ष)	**Trade and Commerce**	Regulated supply, pricing, and distribution of state merchandise
Samsthadhyaksha (संस्थाध्यक्ष)	**Markets and Guilds**	Monitored markets, prevented cartelization and cheating
Sulkadhyaksha (शुल्काध्यक्ष)	**Tolls, Customs & Octroi**	Collected transit duties and import/export taxes
Rupadarshaka (रूपदर्शक)	**Currency & Coinage Inspector**	Examined purity of punch-marked silver coins (*Pana*)
Akshapataladhyaksha (अक्षपटलाध्यक्ष)	**Accountant General & Archives**	Maintained central accounts, ledgers, and state audit
Mudradhyaksha (मुद्राध्यक्ष)	**Passports & State Seals**	Issued identity tokens and regulated travel/emigration
Navadhyaksha (नावाध्यक्ष)	**Shipping, Ferries & Waterways**	Managed maritime trade, rivers, ports, and toll-ferries
Akaradhyaksha (आकराध्यक्ष)	**Mines and Metallurgy**	Controlled state monopoly over mineral extraction
Lohadhyaksha (लोहाध्यक्ष)	**Iron and Metal Manufacture**	Supervised manufacture of iron weapons and tools
Suradhyaksha (सुराध्यक्ष)	**Liquor and Excise**	Regulated state breweries and alcohol taverns
Sunadhyaksha (सूनाध्यक्ष)	**Slaughterhouses & Animal Care**	Prevented cruelty and controlled meat production
Ganikadhyaksha (गणिकाध्यक्ष)	**Courtesans & Entertainers**	Regulated public entertainers, dancers, and royal spies
Vivitadhyaksha (विवीताध्यक्ष)	**Pastures and Grazing Lands**	Managed grazing grounds, dug wells, maintained security
Kupyadhyaksha (कुप्याध्यक्ष)	**Forest Produce & Timber**	Managed state forests, timber, and wild elephant reserves
Koshadhyaksha (कोषाध्यक्ष)	**Imperial Treasury**	Subordinate to Sannidhata; guarded gold, gems, and bullion
Ayudhagaradhyaksha (आयुधागाराध्यक्ष)	**Armoury & Weapons Storage**	Manufactured and preserved arms and military gear
Hastyadhyaksha (हस्त्यध्यक्ष)	**War Elephant Force**	Training, feeding, and deploying royal elephant corps
Ashvadhyaksha (अश्वाध्यक्ष)	**Cavalry & Royal Stables**	Imported and maintained war horses

C. Gupta & Harsha Administrative Officers

Officer Designation	Dynasty / Period	Portfolio / Exact Role
Kumaramatya (कुमारमात्य)	Gupta	**Elite administrative cadre**; equivalent to modern IAS; served as provincial governors, ministers, generals
Mahabaladhikrita (महाबलाधिकृत)	Gupta & Harsha	**Commander-in-Chief** of the imperial army
Mahadandanayaka (महादंडनायक)	Gupta	**Chief Judicial & Police Magistrate** (Harisena held Kumaramatya, Sandhivigrahika, and Mahadandanayaka)
Sandhivigrahika (सांधिविग्रहिक)	Gupta (started under Samudragupta)	**Minister for Peace and War** (Foreign Minister)

Officer Designation	Dynasty / Period	Portfolio / Exact Role
Mahapratihara (महाप्रतिहार)	Gupta & Harsha	**Chief of Palace Guards** and Lord Chamberlain
Uparika (उपरिक)	Gupta & Harsha	**Provincial Governor**; head of a *Bhukti* (province)
Vishayapati (विषयपति)	Gupta	**Head of a District** (*Vishaya*); administered with 4-member city council (*Adhisthana-adhikarana*): 1. *Nagara-sreshthi* (Chief of guild/banker) 2. *Sarthavaha* (Caravan merchant chief) 3. *Prathama-kulika* (Chief artisan) 4. *Prathama-kayastha* (Chief scribe/writer)
Pustapala (पुस्तपाल)	Gupta	**Keeper of land records and revenue surveys**
Shaulkika (शौल्किक)	Gupta	Collector of toll and customs duties (*Shulka*)
Dandapasika / Chauroddharanika	Gupta	Police officer / investigator in charge of catching thieves and criminals
Bhandagaradhikarana	Harsha	Chief superintendent of the royal treasury and storehouse

D. Chola Local Administration & Committees (Uttiramerur Inscription)

- **Three Core Assemblies**:
- **Ur**: General assembly of common tax-paying villagers in non-Brahmana settlements.
- **Sabha / Mahasabha**: Assembly of learned Brahmanas in tax-free **Brahmadeya / Agrahara** villages.
- **Nagaram**: Council of merchants and traders in commercial towns.
- **Variyam (Executive Standing Committees of Sabha)**:
- **Samvatsara-variyam**: Annual executive governing committee.
- **Eri-variyam**: **Tank and irrigation committee** (supervised desilting and water sharing).
- **Totta-variyam**: Garden and orchard committee.
- **Pon-variyam**: Gold and treasury assessment committee.
- **Panchavara-variyam**: Standing committee for general supervision.
- **Kudavolai System (कुदवोलाई)**: Pot-ticket lot system where candidates' names on palm leaves were drawn from an earthen pot by a young boy to ensure impartial committee selection.

10. Ancient Indian Taxes & Fiscal Terms Master Matrix

Tax / Fiscal Term	Period / Civilization	Nature of Levy / Source of Collection	High-Yield Exam Lock
Bali (बलि)	Vedic → Maurya	Vedic: Voluntary offering to Rajan in Rigveda → **Compulsory tax** in Later Vedic. Maurya: Religious tribute / additional land cess (Ashoka abolished Bali at Lumbini).	Voluntary in Rigveda; compulsory later
Bhaga (भाग)	Vedic → Maurya → Gupta	**King's royal share of agricultural produce**, traditionally fixed at **⅙th (*Shadbhaga*)**. Reduced to ⅛th by Ashoka at Lumbini.	Standard ⅙th land revenue
Sita (सीता)	Mauryan	**Revenue from crown agricultural lands** managed directly by the *Sitadhyaksha*.	Crown land revenue (distinct from Bhaga)
Kara (कर)	Mauryan & Gupta	Periodical tax levied on property, or specific tax levied on fruit, flower, and vegetable orchards.	Specific / periodic property tax
Vishti (विष्टि)	Maurya & Gupta	**Forced, unpaid labor** demanded by the state from artisans and peasants for public works.	Unpaid forced labor; treated as state income in Junagarh

Tax / Fiscal Term	Period / Civilization	Nature of Levy / Source of Collection	High-Yield Exam Lock
Pranaya (प्रणय)	Mauryan	**Emergency war tax / gift of affection** levied during acute financial crisis (demanded ⅓ to ¼ of produce).	Emergency fiscal levy
Vartani (वर्तनी)	Mauryan	**Transit duty / road cess** collected on goods by frontier officers (*Antapala*).	Transit duty / toll
Shulka (शुल्क)	Vedic → Gupta	**Customs duty, octroi, and toll taxes** levied on merchandise entering ports or market towns.	Toll / customs tax
Udakabhaga (उदकभाग)	Mauryan	**Water tax / irrigation cess** levied on farmers using state-constructed canals/lakes (varying from ⅕th to ⅓rd).	Irrigation / water cess
Hiranya (हिरण्य)	Maurya & Gupta	**Tax paid in cash / gold** (as opposed to payment in agricultural kind).	Cash tax (gold/coins)
Pindakara (पिण्डकर)	Mauryan	Lump-sum tax assessed collectively on an entire village community.	Collective village tax
Bhoga (भोग)	Gupta	**Daily customary tribute** of fruits, flowers, vegetables, and firewood offered by villagers to the king.	Daily tribute in kind (fruit/firewood)
Udranga (उद्रंग)	Gupta	Land revenue tax levied on **permanent tenants / water tax**.	Tax on permanent cultivators
Uparikara (उपरिकर)	Gupta	Additional tax levied on **temporary cultivators / non-permanent tenants**.	Tax on temporary tenants
Halikakara (हलिककर)	Gupta	**Plough tax** levied per plough owned by the peasant.	Tax per plough
Kadamai / Kudimai	Chola	**Primary land revenue** payable in paddy or cash to the king.	Main Chola land revenue
Vetti / Vetti-muttavalu	Chola	**Unpaid communal forced labor** for public tanks, temples, and palaces.	Chola forced labor (equivalent to Vishti)
Amanji	South India	Gratuitous labor or personal service to royal officials.	Personal royal service
Iraiyili (इरैयिली)	Chola / South India	**Tax-free land grant** bestowed upon temples (*Devadana*) or Brahmanas (*Brahmadeya*).	Tax-exempt land
Karai (करै)	Sangam	Land tax levied on agricultural produce by Tamil kings.	Sangam land tax
Ulgu / Sungam	Sangam & Chola	Customs duties, transit fees, and octroi tolls on trade caravans.	Toll / customs
Iravu (इरवु)	Sangam	Extra forced contribution or emergency demand made by the king.	Forced emergency demand

11. Ancient Land Classifications, Social Systems & Sangam Eco-Zones

A. 5 Types of Land in Ancient India (Gupta Inscriptions & Amarakosha)

1. **Kshetra (क्षेत्र)**: Land fit for **cultivation and agriculture**.
2. **Khila (खिल)**: **Waste, fallow, or uncultivated land**.
3. **Aprahata (अप्रहत)**: **Virgin, uncultivated jungle / forest land**.
4. **Vasti (वास्तु)**: **Habitable land** suitable for building homesteads and residential houses.
5. **Gapata-Sarah (गपत-सरह)**: **Pasture land** reserved for grazing cattle.

> **Ancient Land Tenure Systems:** - **Nivi-dharma (नीवी धर्म)**: Perpetual land endowment where the recipient enjoys income in perpetuity without right to sell or transfer ownership. - **Aprada-dharma (अप्रदा धर्म)**: Grantee can enjoy the revenue but has zero right to alienate or destroy property. - **Bhumi-chhidra-nyaya (भूमि-छिद्र-न्याय)**: Customary right of first clearance: anyone who clears virgin/waste land acquires ownership rights over it.

B. The 8 Types of Hindu Marriage (Manusmriti)

- **Approved / Dharmya Marriages (Prashasta — 4 Types)**:
- **Brahma**: Father gifts decorated daughter to a learned, virtuous groom without demanding dowry (Highest & purest form).
- **Daiva**: Father gives daughter to an officiating priest during a formal Vedic sacrifice.
- **Arsha**: Groom offers a pair of cows and a bull to the bride's father to fulfill religious duties (not as bride-price).
- **Prajapatya**: Father gifts daughter with blessing: "*May both of you perform your religious duties together*".
- **Unapproved / Adharmya Marriages (Aprashasta — 4 Types)**:
- **Asura**: Groom buys the bride by paying money/wealth to her father and relatives (Bride-price marriage).
- **Gandharva**: Love marriage based on mutual passion between bride and groom without parents' consent (e.g., Dushyanta & Shakuntala).
- **Rakshasa**: Forcible abduction and marriage of a crying maiden after fighting and slaying her kinsmen.
- **Paishacha**: Ravishing a maiden while she is asleep, intoxicated, or mentally deranged (Lowest and most sinful form).

> **Varna Marriage Rules:** - **Anuloma (अनुलोम)**: Marriage of a higher varna male with a lower varna female (Permitted by Dharma texts). - **Pratiloma (प्रतिलोम)**: Marriage of a lower varna male with a higher varna female (Severely condemned; produced untouchables/Chandalas).

C. Sangam Five Landscapes (Ainthinai / 5 Tinai)

Each Tinai represents an ecological zone, characteristic deity, and primary livelihood:

Tinai (Landscape)	Eco-Zone / Topography	Presiding Deity	Primary Livelihood / Occupation
Kurinji (कुरिंजी)	**Hilly / Mountainous**	**Murugan** (Seyon - God of War)	Hunting, honey-gathering, shifting cultivation
Mullai (मुल्लै)	**Forest / Pastoral Grasslands**	**Mayon** (Vishnu / Krishna)	Cattle-rearing, pastoralism, dairy farming
Marudam (मरुदम)	**Fertile Agricultural Plains**	**Vendan** (Indra - God of Rain)	Wet-paddy agriculture, farming
Neydal (नेयदाल)	**Coastal / Littoral Belt**	**Varunan** (God of the Sea)	Fishing, boat building, salt manufacturing
Palai (पालै)	**Arid / Desert / Parched Land**	**Korravai** (Goddess of Victory/War)	Highway robbery, raiding, military plunder

SUBJECT 02 · MEDIEVAL INDIA

SUBJECT 02: MEDIEVAL INDIAN HISTORY

मध्यकालीन भारत का इतिहास — दैनिक उच्च-प्राथमिकता तथ्य

Delhi Sultanate Chronology & Departments, Mughal Administrative Hierarchy, Maratha Administration, Bhakti & Sufi Movements, and Key Chroniclers & Battles.

1. Delhi Sultanate Chronology & Dynastic Founders

Dynasty	Period	Founder	Greatest / Key Ruler	Last Ruler	Capital
Slave / Mamluk / Ilbari	**1206–1290**	**Qutbuddin Aibak** (Lakhbaksh)	**Iltutmish** (real consolidator, introduced Silver Tanka & Copper Jital, Chalisa / Turkan-i-Chahalgani), **Balban** (Blood and Iron policy, Sijda & Paibos, Nauroz festival)	Kayumars (killed by Jalaluddin)	Lahore → Delhi (Iltutmish made Delhi capital)
Khalji	**1290–1320**	**Jalaluddin Khalji**	**Alauddin Khalji** (Market regulations / Shahna-i-Mandi, Dagh & Chehra, expanded to South India under Malik Kafur)	Khusrau Khan (usurper)	Delhi (Siri Fort)
Tughlaq (*Longest ruling: 94 yrs*)	**1320–1414**	**Ghiyasuddin Tughlaq** (Ghazi Malik; built Tughlaqabad)	**Muhammad bin Tughlaq** (Token currency, capital shift to Daulatabad, Qarachil & Khurasan expeditions), **Firoz Shah Tughlaq** (Canals, cities: Jaunpur, Firozabad, Hissar)	Nasiruddin Mahmud (Timur invaded in **1398 CE** during his reign)	Delhi → Daulatabad → Delhi
Sayyid	**1414–1451**	**Khizr Khan** (governor of Multan under Timur; took title *Raiyat-i-Ala*, never called himself Sultan)	Mubarak Shah (patron of Yahya bin Ahmad Sirhindi - *Tarikh-i-Mubarak Shahi*)	Alam Shah (abdicated voluntarily)	Delhi
Lodi (*First Afghan Dynasty*)	**1451–1526**	**Bahlul Lodi** (issued Bahloli coins)	**Sikandar Lodi** (Founded **Agra in 1504**, capital in 1506; wrote poetry as *Gulrukhi*; introduced **Gaz-i-Sikandari** 39 digits)	**Ibrahim Lodi** (Defeated & killed at First Battle of Panipat, 1526)	Delhi → Agra

Logic: Sultanate spine is **Slave 1206–1290 → Khalji 1290–1320 → Tughlaq 1320–1414 → Sayyid 1414–1451 → Lodi 1451–1526**. Do not stop at the founder year.

1A. Ghaznavid & Ghurid (before Delhi)

House	Period (from–to)	Key raids / rulers	Last / end
Ghaznavid (Mahmud of Ghazni)	Raids **1000–1027 CE** (17 expeditions in Indian tradition)	**Somnath 1025**; Al-Biruni came with him; capital Ghazni	Line continues in Afghanistan; **not** the Delhi Sultanate
Ghurid (Muhammad Ghori)	Indian career **1173–1206**	**Tarain I 1191** lost to Prithviraj; **Tarain II 1192** won; **Chandawar 1194** vs Jaichand	Murdered **1206**; slave generals (Aibak, Yalduz, Qubacha, Bakhtiyar) take the north

1B. Mughal Emperors — From / To, Last, High-Yield

Imperial high point **1526–1707**; the house on paper lasts to **1857**.

Emperor	Period (from–to)	Last / how it ended	Capital	Key ruler & high-yield
Babur	**1526–1530**	Died; buried **Kabul** (not Delhi)	Agra	Panipat I (**21 Apr 1526**, *Tulughma* + *Araba*); Khanwa 1527 (title *Ghazi*); Chanderi 1528; Ghaghra 1529. *Baburnama* in **Chagatai Turkish**.
Humayun	**1530–1540** and **1555–1556**	First reign ends at **Kannauj/Bilgram 1540**; restored **Sirhind 1555**; died 1556 (Sher Mandal stairs). Tomb **Delhi**	Agra / Delhi	Lost **Chausa 1539** and Kannauj 1540 to Sher Shah. Do not swap Daurah **1532** with Dharmat **1658**.
Sur (Afghan interregnum)	**1540–1555**	**Sikandar Shah Suri** beaten at Sirhind 1555	Delhi / Sasaram	**Sher Shah 1540–1545**: GT Road, rupiya, sarkar–pargana, Sasaram tomb. Islam Shah follows. **Not** a Mughal house.
Akbar	**1556–1605**	Died; buried **Sikandra (Agra)**	Agra → **Fatehpur Sikri (1571–85)** → Lahore / Agra	Panipat II **5 Nov 1556** (Bairam Khan vs Hemu). *Zabt / Dahsala 1580–81* (Todar Mal). Abolished jizya **1564**. Ibadat Khana 1575; *Sulh-i-Kul*; Din-i-Ilahi **1582**. Allahabad fort **1583**. *Ain-i-Akbari* (Abul Fazl).
Jahangir	**1605–1627**	Died; buried **Lahore** (Shahdara)	Agra / Lahore	*Tuzuk-i-Jahangiri*. Hawkins **1608** (*English Khan*); **Sir Thomas Roe 1615–19**. Nur Jahan's party. Tomb of Itimad-ud-Daulah (Nur Jahan).
Shah Jahan	**1628–1658** (died 1666 in Agra Fort)	Deposed by Aurangzeb after Samugarh; buried **Taj / Agra**	Agra → **Shahjahanabad (Delhi)**	Taj Mahal **1631–48** (Ustad Ahmad Lahori). Red Fort & Jama Masjid. Peacock Throne. Patron of Kavindra Acharya (Banaras).
Aurangzeb (Alamgir)	**1658–1707**	Died **Ahmadnagar 1707**; buried Khuldabad	Delhi / Deccan camps	Dharmat Apr 1658 → Samugarh May 1658 → Deorai 1659. Reimposed jizya **1679**. Bijapur **1686**, Golkonda **1687**. About **21 subahs**. Shivaji coronation **1674** in his reign.
Bahadur Shah I (Muazzam)	**1707–1712**	Died; **not** the 1857 emperor	Delhi	Won **Jajau 1707** vs Azam. Khafi Khan: *Shahe Bekhabar*.
Bahadur Shah II Zafar	**1837–1857**	Last Mughal; **1857** Delhi; exiled **Rangoon**	Red Fort, Delhi	Son of **Akbar II**. Teachers Zauq & Ghalib. Do **not** merge with Bahadur Shah I.

> **Logic:** Mughal **founded 1526**, imperial end **1707** (Aurangzeb), legal end **1857** (Zafar). Humayun's gap **1540–1555** is **Sur**, not "Mughal continued."

1C. Regional houses — From / To (often asked as founder-only)

Dynasty	Period (from–to)	Founder	Last ruler / end	Capital	Key ruler & high-yield
Vijayanagara	**1336–1565** as great power; Aravidu remnant to **c. 1646**	**Harihara I & Bukka I** (Sangama; Vidyaranya tradition)	**Rama Raya** killed at **Talikota 23 Jan 1565**; later Aravidu at Penukonda / Chandragiri	**Hampi** (Tungabhadra)	Order: **Sangama → Saluva → Tuluva → Aravidu**. Peak **Krishnadeva Raya (Tuluva)** — *Amuktamalyada* (Telugu); Paes & Barbosa visited. *Nayankara / amara*. Mahanavami Dibba. Berar **absent** from the Talikota alliance.
Bahmani	**1347–1518/1527**	**Hasan Gangu = Alauddin Bahman Shah**	Splits **1518** into five Deccan Sultanates	**Gulbarga → Bidar**	**Mahmud Gawan** (prime minister; executed **1481**). Nikitin visited. Tin Darwaza = **Bidar**, not Ahmedabad.
Five Deccan Sultanates	From **c. 1518**	—	Absorbed by Mughals (Bijapur **1686**, Golkonda **1687**)	Bijapur, Golkonda, Ahmadnagar, Bidar, Berar	Adil Shahi, Qutb Shahi, Nizam Shahi, Barid Shahi, Imad Shahi. Charminar = Qutb Shahi Hyderabad.
Sharqi (Jaunpur, UP)	**c. 1394–1479/1484**	**Malik Sarwar** (*Malik-us-Sharq / Khwaja-i-Jahan*)	**Hussain Shah** falls to **Bahlul Lodi**	**Jaunpur** (eastern UP)	City founded by **Firuz Tughlaq** for cousin Jauna Khan — city ≠ dynasty. **Ibrahim Shah** — *Siraj-i-Hind*; **Atala Masjid**. Vidyapati's *Kirtilata*; Jaisi's *Padmavat* in this circle. **Only this regional capital sits in modern UP.**
Maratha Swaraj / Confederacy	**1674–1818**	**Shivaji** crowned **Raigad 1674** (Gaga Bhatt)	**Peshwa Bajirao II**; Third Anglo-Maratha War **1817–18** ends Peshwaship	Raigad → later Poona (Peshwas)	Chauth 25% + Sardeshmukhi 10%. Purandar **1665** (Jai Singh I). Panipat III **1761**.

1D. Uttarakhand medieval houses (UKPCS)

Dynasty	Period (from–to)	Founder	Last ruler / end	Capital	Key ruler & high-yield
Katyuri / Kattyuri	**c. 740–1050 CE**	**Basantandev (Basu Dev)**	Tyrant **Bir Dev / Biram Dev** overthrown; realm splits (Askot, Doti, Dwarahat, Baijnath)	**Joshimath (Kartikeyapura) → Baijnath** (Katyur valley, Bageshwar)	First historic empire of the hills. Unifier **Ishtaganadeva**. Apex **Lalitsuradeva** (Pandukeshwar plates). Bageshwar inscription of **Bhuwandev**.
Chand of Kumaon	c. 10th/11th c. legend → **1790**	**Somchand** (Jhusi/Prayag tradition) at **Champawat (Rajbunga)**	**Mahendra Chand** defeated by Gorkhas at **Hawalbagh, Jan 1790**	Champawat → **Almora (1563)** under Balo Kalyan Chand	Emblem **Garud**. **Char Aal**: Karki, Bora, Taragi, Chaudhary. Garud Gyan Chand vs Firoz Tughlaq. Baz Bahadur Chand — Taklakot / Kailash route. Tax memory: **36 Rakam 32 Kalam**.
Parmar / Panwar of Garhwal	Legend **Kanakpal 888 CE → 1804**	**Kanakpal** at **Chandpur Garhi** (Chamoli)	**Pradyumna Shah** dies vs Gorkhas at **Khudbuda / Khurbura (Dehradun), 14 May 1804**	Chandpur → Dewalgarh (1512) → **Srinagar (1517)**	**Ajay Pal** unites **52 garhs**. Rani **Karnavati** (*Nakkati Rani*, 1635). Fateh Shah grants four villages to **Guru Ram Rai** (1676). Painter **Mola Ram**.

2. Central Administrative Departments (Diwan-i-...)

Department Name	Portfolio / Nature of Office	Creator / Reformer Sultan	Key Official in Charge
Diwan-i-Wizarat	**Finance & Revenue Department** (Prime Minister)	Existed throughout; reorganized by Balban	**Wazir**
Diwan-i-Arz	**Military Department** (recruitment, inspection, muster rolls)	Created by **Ghiyasuddin Balban**	**Ariz-i-Mamalik** (Sultan remained supreme commander)
Diwan-i-Insha	**Royal Correspondence & State Chancery**	Early Sultanate	**Dabir-i-Khas**
Diwan-i-Risalat	**Religious Matters, Charities, Endowments & Foreign Appeals**	Early Sultanate	**Sadr-us-Sudur**
Diwan-i-Mustakhraj	Department for **Extracting Arrears from Revenue Officers**	Created by **Alauddin Khalji**	Mustakhraj officers
Diwan-i-Riyasat	**Market Control & Commerce Regulatory Department**	Created by **Alauddin Khalji**	**Naib-i-Riyasat** (controlled Shahna-i-Mandi)
Diwan-i-Kohi	**Agriculture Development Department** (crop loans/Sondhar, experimental farms)	Created by **Muhammad bin Tughlaq**	Amir-i-Kohi
Diwan-i-Khairat	**Charity & Public Welfare** (aid for poor, widows, orphan marriages)	Created by **Firoz Shah Tughlaq**	Royal almoners
Diwan-i-Bandagan	**Department of Royal Slaves** (~1,80,000 slaves trained in crafts)	Created by **Firoz Shah Tughlaq**	Khwajasara
Diwan-i-Ishtiaq	**Pension Department** for retired and destitute officials	Created by **Firoz Shah Tughlaq**	-
Diwan-i-Qaza	**Judicial Department** (Civil & Criminal justice)	Established early	**Qazi-ul-Quzat**
Barid-i-Mamalik	**Intelligence & Royal Postal Spies**	Reorganized by Balban & Alauddin	Barids (reporters)
Muhtasib	**Censor of Public Morals & Inspector of Weights/Measures**	Alauddin / Firoz / Aurangzeb	Muhtasib
Dar-ul-Shafa	**Free Royal Charitable Hospital**	Built by **Firoz Shah Tughlaq**	Hakims

2B. Delhi Sultanate Territorial & Local Administrative Hierarchy

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Iqta / Wilayat (Province) ⟶ Shiq (District) ⟶ Pargana (Sub-district) ⟶ Deh / Gram (Village)
[Head: Muqti / Wali]          [Head: Shiqdar]     [Head: Amil / Mushrif]     [Head: Muqaddam / Khut]
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Administrative Unit	Heading Officer / Functionary	Nature of Authority / Duties
Iqta / Wilayat (Province)	**Muqti / Wali / Naib Sultan**	Maintained law & order, commanded provincial troops, collected revenue, remitted surplus (*Fawazil*) to Centre
Shiq (District)	**Shiqdar**	Head of executive and military administration of the district
Pargana (Sub-district)	**Amil / Mushrif**	Assessed and collected agricultural revenue from villages
Pargana Accounts	**Karkun / Nawisinda**	Scribe and accountant maintaining land registers in Persian and Hindi
Pargana Treasury	**Fotadar / Khazandar**	Custodian of the local treasury and cash receipts
Deh / Gram (Village)	**Muqaddam / Mandal**	Village headman; collected land revenue from peasants

Administrative Unit	Heading Officer / Functionary	Nature of Authority / Duties
Village Landed Elite	**Khut**	Rich peasant / hereditary village landlord (Alauddin stripped their privileges)
Village Official	**Patwari**	Maintained village land survey and revenue assessment registers
Village Group Leader	**Chaudhari**	100-village leader acting as intermediary between state and villages

3. Medieval Indian Architecture & Monuments Chronology

Monument / Structure	Location	Builder / Sultan / Emperor	Architectural Landmark / Exam Highlight
Quwwat-ul-Islam Mosque	Qutb Complex, Delhi	**Qutbuddin Aibak** (1193 CE)	**First mosque built in Delhi**; constructed using materials from 27 demolished temples
Adhai Din Ka Jhonpra	Ajmer (Rajasthan)	**Qutbuddin Aibak** (1199 CE)	Built on the ruins of a Sanskrit college (*Saraswati Kanthabharana* built by Vigraharaja IV)
Qutb Minar	Mehrauli, Delhi	Started by Aibak; **completed by Iltutmish**; repaired by Firoz Tughlaq	Dedicated to Sufi saint Khwaja Qutbuddin Bakhtiyar Kaki; 72.5 m red sandstone
Tomb of Balban	Mehrauli, Delhi	**Ghiyasuddin Balban** (c. 1287)	**First true architectural arch (*True Arch*)** constructed in India
Alai Darwaza	Qutb Complex, Delhi	**Alauddin Khalji** (1311 CE)	**First true horseshoe arch and true dome in India**; exquisite red sandstone & white marble
Siri Fort & Hauz Khas	Delhi	**Alauddin Khalji**	Siri Fort built to protect Delhi against Mongol raids
Tughlaqabad Fort	Delhi	**Ghiyasuddin Tughlaq**	Sloping battered walls (*batter*) and stone masonry
Kotla Firoz Shah	Delhi	**Firoz Shah Tughlaq**	Ashokan pillar brought from Topra (Haryana) and installed atop 3-tiered pyramid
Atala Masjid	**Jaunpur (Uttar Pradesh)**	**Ibrahim Shah Sharqi** (1408 CE)	Sharqi style; sloping pylons, massive central propylon gate, no minarets
Tomb of Sikandar Lodi	Lodi Gardens, Delhi	Ibrahim Lodi (1517 CE)	**First double dome in India**; octagonal plan set within a walled garden
Sasaram Tomb	Sasaram (Bihar)	**Sher Shah Suri** (architect Aliwal Khan)	Octagonal red sandstone tomb situated in the middle of an artificial square lake
Purana Qila & Qila-i-Kuhna	Delhi	**Sher Shah Suri**	Qila-i-Kuhna mosque represents transition to refined Mughal architectural aesthetic
Humayun's Tomb	Delhi	Begum Bega / Haji Begum (architect Mirak Mirza Ghiyas, 1565)	**First grand garden-tomb in India (*Charbagh*)**; precursor and model for the Taj Mahal; white marble double dome; UNESCO WHS
Agra Fort	Agra (UP)	**Akbar** (1565–1573)	Massive red sandstone ramparts on the Yamuna; Jahangiri Mahal
Fatehpur Sikri Complex	Near Agra (UP)	**Akbar** (1571–1585)	City of victory: **Buland Darwaza** (1601, commemorating conquest of Gujarat; 54 m high), **Panch Mahal** (5-tier Buddhist-inspired pyramid), **Ibadat Khana** (1575), Tomb of Sheikh Salim Chishti (white marble lattice *Jali*)

Monument / Structure	Location	Builder / Sultan / Emperor	Architectural Landmark / Exam Highlight
Akbar's Tomb	Sikandra, Agra (UP)	Begun by Akbar; **completed by Jahangir** (1613)	Unique tier without a central dome; minarets at four corners
Tomb of Itimad-ud-Daulah	Agra (UP)	**Empress Nur Jahan** (1628)	Tomb of her father Mirza Ghiyas Beg; **first building made entirely of white marble** with extensive **Pietra Dura** (floral inlay with semi-precious stones); called **"Baby Taj"**
Taj Mahal	Agra (UP)	**Shah Jahan** (1631–1648)	Masterpiece of Indo-Islamic architecture; chief architect **Ustad Ahmad Lahori**; calligrapher Amanat Khan; UNESCO WHS
Red Fort & Jama Masjid	Delhi	**Shah Jahan** (1638–1648)	Shifted capital from Agra to **Shahjahanabad**; Diwan-i-Aam, Diwan-i-Khas ("*If there is a paradise on earth, it is this...*"), Peacock Throne (*Takht-i-Taus*)
Bibi Ka Maqbara	Aurangabad (Maharashtra)	**Aurangzeb / Prince Azam Shah** (1678)	Tomb of Dilras Banu Begum (Rabia-ud-Daurani); known as **"Dakhani Taj"** or poor man's replica of Taj
Moti Masjid (Red Fort Delhi)	Delhi	**Aurangzeb** (1659)	White marble private mosque within the Red Fort
Badshahi Mosque	Lahore (Pakistan)	**Aurangzeb** (1673)	Largest mosque of the Mughal Empire

4. Medieval Literature & Chroniclers (The Complete Exam Sheet)

Book / Work Title	Author	Language / Patron Ruler	Key Subject / Historical Value
Kitab-ul-Hind* / *Tahqiq-i-Hind	**Al-Biruni**	Arabic / Mahmud of Ghazni	Encyclopedic study of Indian religion, science, astronomy, and society
Chachnama	Translated by Ali Kufi	Persian (original Arabic)	History of the Arab conquest of Sindh by Muhammad bin Qasim (712 CE)
Taj-ul-Maasir	**Hasan Nizami**	Persian / Qutbuddin Aibak	First official chronicle of the Delhi Sultanate
Tabakat-i-Nasiri	**Minhaj-us-Siraj**	Persian / Nasiruddin Mahmud	Detailed political history from Muhammad Ghori to the Slave Dynasty
Tarikh-i-Firoz Shahi & ***Fatwa-i-Jahandari***	**Ziauddin Barani**	Persian / Firoz Shah Tughlaq	Eyewitness account of Alauddin's market reforms and Muhammad bin Tughlaq's schemes
Futuhat-i-Firoz Shahi	**Firoz Shah Tughlaq**	Persian / Self	Royal autobiography detailing his public works, charities, and orthodox reforms
Tughlaq Nama, Khazain-ul-Futuh, Qiran-us-Sadain, Nuh Sipihr, Ashiqa	**Amir Khusrau** (*Tuti-i-Hind*)	Persian / Sultanate (Balban to Ghiyasuddin Tughlaq)	*Khazain-ul-Futuh* records Alauddin's conquests; *Nuh Sipihr* praises India's climate, flora, and culture; invented **Sitar**, **Qawwali**, and **Khayal** singing
Tarikh-i-Mubarak Shahi	**Yahya bin Ahmad Sirhindi**	Persian / Mubarak Shah Sayyid	Only contemporary source for the Sayyid Dynasty
Baburnama (Tuzuk-i-Baburi)	**Babur**	**Chagatai Turkish**	Candid royal memoirs; mentions 5 Muslim rulers (Delhi, Gujarat, Bahmani, Malwa, Bengal) and 2 Hindu kings (Krishnadeva Raya & Rana Sanga); translated into Persian by Abdur Rahim Khan-i-Khanan

Book / Work Title	Author	Language / Patron Ruler	Key Subject / Historical Value
Humayun-nama	**Gulbadan Begum** (Babur's daughter)	Persian / Akbar	Intimate royal domestic account of the lives of Babur and Humayun
Akbarnama* & *Ain-i-Akbari	**Abul Fazl**	Persian / Akbar	Official history in 3 volumes: Vol 1 (Ancestry), Vol 2 (Akbar's reign), Vol 3 (*Ain-i-Akbari* - administrative gazetteer, revenue statistics)
Muntakhab-ut-Tawarikh	**Abdul Qadir Badauni**	Persian / Akbar	Critical counter-narrative of Akbar's religious policies; translated Ramayana and Mahabharata (*Razmnama*) into Persian
Tuzuk-i-Jahangiri	**Jahangir** (completed by Mutamid Khan)	Persian / Self	Royal autobiography; reveals his passion for painting, birds, flowers, and natural history
Padshahnama	**Abdul Hamid Lahori**	Persian / Shah Jahan	Official illustrated chronicle of the first two decades of Shah Jahan's reign
Shah Jahan-nama	**Inayat Khan**	Persian / Shah Jahan	History of Shah Jahan's reign
Sirr-i-Akbar* & *Majma-ul-Bahrain	**Dara Shikoh**	Persian	*Sirr-i-Akbar* is the Persian translation of **50 Upanishads**; *Majma-ul-Bahrain* (*The Mingling of Two Oceans*) harmonized Sufi and Vedantic philosophy
Alamgirnama	**Mirza Muhammad Kazim**	Persian / Aurangzeb	Official history of the first 10 years of Aurangzeb's reign (stopped by Emperor)
Muntakhab-ul-Lubab	**Khafi Khan**	Persian / Later Mughal	Secretly written history of Aurangzeb's reign and Deccan campaigns
Nuskha-i-Dilkusha	**Bhimsen Saxena**	Persian / Aurangzeb	Eyewitness account of Aurangzeb's Deccan wars written from a non-courtier perspective
Amuktamalyada	**Krishnadeva Raya**	**Telugu**	Masterpiece on statecraft and devotion of Andal (Godadevi) to Lord Vishnu

5. Medieval Administration, Taxes & Agrarian Revenue Systems

A. Delhi Sultanate Taxes & Core Economic Terms

- **Jizya**: Religious protection tax levied on non-Muslim subjects (*Zimmis*):
- First levied in India by **Muhammad bin Qasim** in Sindh (712 CE).
- First levied on **Brahmins** as a separate category by **Firoz Shah Tughlaq**.
- **Abolished by Akbar in 1564 CE** (also abolished pilgrim tax in 1563).
- **Reimposed by Aurangzeb in 1679 CE**.
- Finally abolished permanently by later Mughal Emperor **Muhammad Shah "Rangeela"** in 1720 on recommendation of Sawai Jai Singh.
- **Zakat**: Religious property tax levied exclusively on well-to-do Muslims (at 2.5% on wealth/savings) dedicated to charity.
- **Kharaj**: General land revenue tax levied on agricultural produce of lands cultivated by non-Muslims (varying between ⅕th to ½).
- **Khams**: State's share of **spoils of war / war booty**:
- Islamic Sharia rule: ⅕th to the State Treasury, ⅘th distributed among soldiers.
- **Alauddin Khalji and Muhammad bin Tughlaq reversed this rule**: took **⅘th for the State** and gave only ⅕th to the soldiers! Firoz Tughlaq restored orthodox rule.

- **Haqq-i-Sharb**: **Irrigation tax** (levied at **10% of gross produce**) introduced by **Firoz Shah Tughlaq** on peasants using state-constructed canals.
- **Chari & Ghari**: Taxes introduced by **Alauddin Khalji** — *Chari* was grazing tax on milch cattle; *Ghari* was tax on residential houses.
- **Iqta System**: Reorganized and formalized by **Iltutmish**; assignment of land revenue of a territorial tract to nobles/commanders (*Iqtadars / Muqtis / Walis*) in lieu of cash salary, responsible for maintaining troop quotas and remitting surplus (*Fawazil*) to the central treasury.
- **Fawazil**: Excess balance of revenue remitted to the imperial central treasury by an Iqtadar/Muqti after meeting local administrative and military expenses.
- **Sondhar / Taccavi**: Emergency agricultural loans given by state to farmers during famine or distress (pioneered by **Muhammad bin Tughlaq** under Diwan-i-Kohi).
- **Karkhanas**: Royal manufacturing workshops; categorized into *Rikabi* (movable/travel) and *Ratibi* (fixed allowances); Firoz Shah Tughlaq maintained 36 giant royal Karkhanas.

B. Mughal Administrative Hierarchy & Key Officers

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Subah (Province) ──────► Sarkar (District) ──────► Pargana (Sub-district) ──► Deh / Gram (Village)
[Subahdar & Diwan]       [Faujdar & Amil]          [Shiqdar & Amin]           [Muqaddam & Patwari]
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1. Central Administration Officers

Official Designation	Administrative Portfolio / Key Authority	Core Exam Triggers
Wazir / Diwan-i-Ala	**Prime Minister & Head of Revenue / Finance**	Controlled revenue assessment, imperial treasury, expenditure, and provincial Diwans
Mir Bakhshi	**Head of Military Department & Intelligence**	Maintained muster rolls, inspected troops and horses (*Dagh*), issued pay certificates (**Sarkhat**) to Mansabdars
Mir Saman / Khan-i-Saman	**Head of Imperial Household & Karkhanas**	In charge of royal court provisions, imperial stores, internal palace management, state workshops
Sadr-us-Sudur	**Chief Minister of Religious Affairs & Charities**	Administered tax-free land grants (*Madad-i-Maash / Suyurghal*), religious endowments, and Islamic law
Qazi-ul-Quzat	**Chief Justice of the Empire**	Highest judicial officer below Emperor; usually held simultaneously by Sadr-us-Sudur
Muhtasib	**Censor of Public Morals & Inspector of Weights**	Checked gambling, wine consumption, maintained correct weights/measures, verified Islamic practices
Mir Atish / Darogha-i-Topkhana	**Chief of Imperial Artillery**	Controlled manufacture, storage, and deployment of cannon topkhana; guarded imperial palace
Mir Barr	**Superintendent of Forests and River Navigation**	Maintained imperial boats, timber extraction, royal river flotilla
Waqia-Nawis / Sawani-Nigar	**Imperial News-Reporters & Intelligence Scribes**	Sent confidential intelligence reports from provinces directly to Emperor through Mir Bakhshi

2. Provincial, District & Local Officers

Unit	Officer Designation	Function & Authority
Subah (Province)	**Subahdar / Sipahsalar**	Governor of the Subah; commanded provincial army, maintained law and order
	Provincial Diwan	Independent financial head; reported directly to Central Wazir (kept check on Subahdar)

Unit	Officer Designation	Function & Authority
	Provincial Bakhshi	Maintained muster rolls of provincial mansabdars; acted as chief postal scout (*Waqia-Navis*)
	Kotwal	Police chief of the provincial capital; head of city surveillance, night patrol, and crime
Sarkar (District)	**Faujdar**	Executive and military head of Sarkar; suppressed rebellions, assisted Amil in revenue recovery
	Amil / Amal-Guzar	**Chief Revenue Collector & Assessor** of the district; assisted peasants, assessed crops
	Bitikchi	Chief scribe and accountant of Sarkar; prepared land and revenue assessment records
	Khazandar	District treasurer receiving collections from Parganas
Pargana (Sub-district)	**Shiqdar**	Executive and police officer of Pargana; maintained peace and assisted revenue collection
	Amil / Amin	Revenue assessor and collector at Pargana level
	Fotadar	Treasurer of Pargana
	Qanungo	Hereditary keeper of agrarian revenue records, land measurements, and local customs
Gram / Deh (Village)	**Muqaddam / Mandal**	Hereditary village headman; collected land revenue from peasants (received 2.5% commission)
	Patwari	Village accountant; maintained village revenue registers (*Bahi*)

C. Mughal Land Classification (Cultivation Frequency)

Categorized in the Ain-i-Akbari based on continuous cultivation:

Land Category	Frequency of Cultivation	Quality / Revenue Status
Polaj (पोलज)	Cultivated **annually and continuously** for each harvest; never allowed to lie fallow	Best quality, most fertile agricultural land; assessed at full revenue
Parauti (परौती)	Left **fallow for 1 or 2 years** to recover soil fertility naturally	Second grade; paid full Polaj rate when brought back under the plough
Chachar (चाचर)	Left **uncultivated / fallow for 3 or 4 years**	Third grade; assessed at lower progressive tax rate to encourage reclamation
Banjar (बंजर)	Uncultivated and barren for **5 or more years**	Lowest grade; charged minimal nominal cess to promote cultivation

D. Mughal Revenue Assessment Methods

System Name	Also Known As	Key Architect / Monarch	Working Mechanism / Key Exam Facts
Zabt / Ain-i-Dahsala	**Todar Mal's Bandobast**	**Raja Todar Mal** under **Akbar (1580 / 24th regnal year)**	Average produce and average market prices of the preceding **10 years (1570–1580)** calculated; state share fixed at **⅓rd (33.3%)** payable in cash. Measured land using **Tanab** (bamboo sticks connected by iron rings replacing hemp ropes) and standardized **Gaz-i-Ilahi** (41 anguls/digits = ~33 inches). Applied across core provinces (North India & Gujarat).
Ghalla-Bakhshi / Batai	Crop-sharing / Bhaoli	Traditional system across India	Physical division of the harvested crop between state and peasant: 1. *Khet-batai*: Division of standing crop in the field before harvest. 2. *Lank-batai*: Division of harvested grain tied in sheaves before thrashing. 3. *Raas-batai*: Division of cleaned grain heap on thrashing floor (most accurate).

System Name	Also Known As	Key Architect / Monarch	Working Mechanism / Key Exam Facts
Kankut	Danabandi	Traditional appraisal system	Rough estimation and appraisal of standing crop by inspection and pacing the field (*Kan* = grain/estimation, *Kut* = conjecture).
Nasaq	Contractual assessment	Ancient method	Rough lump-sum calculation based on past revenue payments without new annual measurement.

E. Mughal Land Tenure & Jagir Categories

Land Tenure Category	Nature of Ownership / Revenue Rights	Transferability / Legal Condition
Khalisa (खालिसा)	Direct imperial crown land	Revenue deposited directly into the Central Imperial Treasury (*Bait-ul-Mal*)
Tankha Jagir (तनख्वाह जागीर)	Standard revenue assignment in lieu of cash salary	**Strictly transferable** every 3–4 years to prevent local roots/feudal power
Watan Jagir (वतन जागीर)	Homeland territories of autonomous Rajput rulers	**Hereditary and non-transferable**; continued within the royal family
Mashrut Jagir (मशरुत जागीर)	Conditional jagir assigned for a specific post/duty	Subject to temporary fulfillment of specific military/administrative duties
Altamgha Jagir (अल-तमगा जागीर)	Imperial grant initiated by **Jahangir**	**Perpetual, hereditary, irrevocable grant** confirmed with royal vermillion seal
Madad-i-Maash / Suyurghal	Charitable land grant for scholars, saints, and religious divines	**Tax-free (*Inam*)** and hereditary; managed by the *Sadr-us-Sudur*
Paibaqi (पायबाकी)	Reserve land tract	Surplus revenue land kept reserved for assignment to upcoming mansabdars

F. Mansabdari System Essentials (Akbar to Aurangzeb)

- **Introduced by Akbar (1571 CE)**: Military-administrative ranking system borrowed from the Mongol decimal system; ranged from rank of 10 to 5,000 (later 7,000 for top royal princes/nobles like Raja Man Singh).
- **Dual Ranks (Zat & Sawar)**:
- **Zat (जात)**: Denoted personal rank, social status, precedence at court, and fixed the personal baseline salary.
- **Sawar (सवार)**: Denoted the actual number of cavalry troops and horses the officer was required to maintain for imperial service.
- **Three Classes of Mansabdars**:
- *Class 1*: Sawar rank = Zat rank.
- *Class 2*: Sawar rank = Half or more of Zat rank.
- *Class 3*: Sawar rank = Less than half of Zat rank.
- **Dah-Bishi Rule (10-20 Rule)**: For every 10 cavalrymen (*Sawars*), the Mansabdar had to maintain **20 horses** (to avoid exhaustion of war mounts during campaigns).
- **Dagh and Chehra**: Instituted by Alauddin Khalji, revived by Sher Shah, and formalized by Akbar:
- *Dagh*: Royal branding of war horses to prevent fraudulent substitution.
- *Chehra / Tashiha*: Descriptive roll / biometric identity of every soldier.
- **Du-aspa Sih-aspa (दो-अस्पा सिह-अस्पा)**: Introduced by **Jahangir**; allowed favored nobles to double their maintained cavalry force and salary allocation **without increasing their base Zat rank**.
- **Month-Scale System (*Mahana*)**: Introduced by **Shah Jahan**; due to divergence between estimated revenue (*Jama*) and actual realization (*Hasil*), Mansabdars were paid on 4-month, 6-month, or 8-month pay scales instead of full 12-month pay.

G. Vijayanagara & South Indian Administrative Systems

- **Nayankara System**:
- Distinctive military-tenurial system of Vijayanagara.
- King was the ultimate owner of all land; granted territories called **Amaram** to military commanders called **Nayakas / Amara-nayakas**.
- Nayakas maintained a fixed quota of foot-soldiers, horses, and elephants for the King and paid an annual tribute into the imperial treasury.
- Documented in detail by Portuguese travelers **Domingo Paes** and **Fernao Nuniz**.
- **Ayagar System**:
- Unique village autonomous administrative organization in South India.
- Every village was an autonomous unit managed by a council of **12 hereditary functionaries** called **Ayagars** (Headman/Gaud, Accountant/Karnam, Blacksmith, Carpenter, Washerman, Barber, etc.).
- Ayagars were granted hereditary tax-free plots of land called **Manyam** and received customary grain shares from the villagers.
- **Key Vijayanagara Administrative Terms**:
- **Athavana (अठवण)**: Central land revenue department.
- **Kandachara (कंदाचार)**: Military department presided over by the *Dandanayaka* (Commander-in-Chief).
- **Sist**: Standard land tax (fixed at ⅙th of gross agricultural yield based on land classification under Krishnadeva Raya).
- **Kavali**: Police duty; fees collected for village policing called *Kaval-katnam*.

6. The Medieval Battle Master Matrix (Chronological Ladder)

Year	Battle Name	Contenders	Victorious Party	Historical Impact / Prelims Takeaway
1191	**First Battle of Tarain**	Prithviraj Chauhan III vs Muhammad Ghori	**Prithviraj Chauhan**	Ghori wounded, fled to Ghazni
1192	**Second Battle of Tarain**	Prithviraj Chauhan III vs Muhammad Ghori	**Muhammad Ghori**	Laid the foundation of Muslim rule in North India
1194	**Battle of Chandawar**	Muhammad Ghori vs Jaichand (Gahadavala King of Kannauj)	**Muhammad Ghori**	Kannauj & Banaras annexed; Chandawar near Firozabad (UP)
1526 (21 Apr)	**First Battle of Panipat**	Babur vs Ibrahim Lodi	**Babur**	First effective use of **Tulughma** tactic & artillery (*Topkhana*) in North India; Mughal Empire founded
1527 (16 Mar)	**Battle of Khanwa**	Babur vs Rana Sanga of Mewar	**Babur**	Babur declared it **Jihad**; assumed title of **Ghazi**; secured Mughal foothold
1528	**Battle of Chanderi**	Babur vs Medini Rai (Malwa chief)	**Babur**	Rajput women committed Jauhar; Babur captured Malwa forts
1529	**Battle of Ghaghra**	Babur vs Combined Afghan forces under Mahmud Lodi	**Babur**	First battle fought simultaneously on **land and river (water)**
1539 (26 Jun)	**Battle of Chausa**	Sher Khan (Sher Shah) vs Humayun	**Sher Khan**	Near Buxar (Bihar); Humayun saved by water carrier Nizam; Sher Khan took title **Sher Shah**
1540 (17 May)	**Battle of Kannauj (Bilgram)**	Sher Shah Suri vs Humayun	**Sher Shah Suri**	Humayun driven into 15-year exile in Persia; **Second Afghan Empire (Sur Empire)** established

Year	Battle Name	Contenders	Victorious Party	Historical Impact / Prelims Takeaway
1555	**Battle of Sirhind**	Humayun (led by Bairam Khan) vs Sikandar Shah Suri	**Humayun**	Restored Mughal dynasty in Delhi; Humayun died next year falling down stairs of Sher Mandal library
1556 (5 Nov)	**Second Battle of Panipat**	Akbar (regent Bairam Khan) vs Hemu (Hemchandra Vikramaditya)	**Akbar / Bairam Khan**	Hemu struck in eye by stray arrow; captured and beheaded; ended Sur Afghan resistance
1565 (23 Jan)	**Battle of Talikota (Rakshasi-Tangadi / Bannihatti)**	Vijayanagara (Aliya Rama Raya) vs Combined Deccan Sultanates (Bijapur, Golconda, Ahmadnagar, Bidar; **Berar absent!**)	**Deccan Sultanates**	Vijayanagara sacked and destroyed; shifted capital to Penukonda & Chandragiri (Aravidu dynasty)
1576 (18 Jun)	**Battle of Haldighati**	Mughal Army (Raja Man Singh of Amber & Asaf Khan) vs Maharana Pratap of Mewar	**Mughal Army** (indecisive strategic outcome)	Chetak (Pratap's horse) became legendary; Pratap continued guerrilla struggle from Kumbhalgarh/Chavand
1658 (15 Apr)	**Battle of Dharmat**	Aurangzeb & Murad vs Imperial Army (Jaswant Singh of Marwar & Qasim Khan)	**Aurangzeb**	Near Ujjain (MP); opened road to Agra
1658 (29 May)	**Battle of Samugarh**	Aurangzeb vs Dara Shikoh	**Aurangzeb**	Near Agra; decisive victory; Aurangzeb imprisoned Shah Jahan in Agra Fort
1659 (Apr)	**Battle of Deorai**	Aurangzeb vs Dara Shikoh	**Aurangzeb**	Near Ajmer; Dara captured, humiliated, executed
1761 (14 Jan)	**Third Battle of Panipat**	Marathas (Sadashivrao Bhau & Vishwasrao) vs Ahmad Shah Abdali (Durrani) & Shuja-ud-Daula (Awadh) & Najib-ud-Daula	**Ahmad Shah Abdali**	Annihilated Maratha leadership; cleared way for British dominance

7. Foreign Travellers & Visiting Rulers (Match the Following Goldmine)

Foreign Traveler	Country of Origin	Period / Century	Indian Ruler / Court Visited	Famous Book / Key Observation
Al-Biruni (Abu Raihan)	Khwarizm (Uzbekistan)	11th c. (1017 CE)	Came with **Mahmud of Ghazni**	***Tahqiq-i-Hind* / *Kitab-ul-Hind***; deep study of Sanskrit, Hindu astronomy, philosophy, and caste system
Ibn Battuta	Tangier (Morocco)	14th c. (1333 CE)	**Muhammad bin Tughlaq**	***Rihla***; served as Qazi of Delhi for 8 years; sent as royal ambassador to China in 1342
Marco Polo	Venice (Italy)	13th c. (1292–94 CE)	**Pandya Kingdom** (Kayal port, King Maravarman Kulasekhara) & **Kakatiya** (Queen Rudramadevi)	*Travels of Marco Polo*; noted flourishing pearl fisheries and horse imports
Nicolo Conti	Venice (Italy)	15th c. (1420 CE)	**Deva Raya I** (Vijayanagara, Sangama Dynasty)	Described Vijayanagara fortifications, festivals of Navaratri and Deepavali
Abdur Razzaq	Herat (Persia)	15th c. (1443 CE)	**Deva Raya II** (Vijayanagara, Sangama Dynasty)	*Matla-us-Sadain*; observed: "*The city is such that eyes has not seen nor ear heard of any place resembling it upon the whole earth*"
Athanasius Nikitin	Tver (Russia)	15th c. (1470–74 CE)	**Bahmani Kingdom** (under Prime Minister Mahmud Gawan & Sultan Muhammad Shah III)	*The Journey Beyond Three Seas*; noted deep contrast between extreme luxury of nobles and wretched poverty of peasants

Foreign Traveler	Country of Origin	Period / Century	Indian Ruler / Court Visited	Famous Book / Key Observation
Duarte Barbosa	Portugal	16th c. (1518 CE)	**Krishnadeva Raya** (Vijayanagara, Tuluva Dynasty)	Detailed accounts of Vijayanagara trade, currency, customs, and Sati practice
Domingo Paes	Portugal	16th c. (1520–22 CE)	**Krishnadeva Raya** (Vijayanagara)	Described Krishnadeva Raya as *"a monarch who is gentle and wise, full of justice and equity"*; praised Mahanavami Dibba
Fernao Nuniz	Portugal	16th c. (1535–37 CE)	**Achyuta Deva Raya** (Vijayanagara, Tuluva Dynasty)	Horse trader; wrote detailed chronicle on founding of Vijayanagara and women's roles in royal court
Captain William Hawkins	England	1608–1611 CE	**Jahangir**	Reached Surat on *Hector*; spoke Turkish; Jahangir gave him title **"English Khan"** and 400 Mansab
Sir Thomas Roe	England	1615–1619 CE	**Jahangir**	Ambassador of King James I; secured imperial Farman for East India Company factories
Francisco Pelsaert	Netherlands (Dutch)	1620–1627 CE	**Jahangir**	***Remonstrantie***; detailed poverty of artisans and corruption of Mughal governors in Agra
Jean-Baptiste Tavernier	France	1640–1667 CE (6 voyages)	**Shah Jahan** & **Aurangzeb**	Professional gem merchant; famous for examining and describing the **Koh-i-Noor** diamond and Peacock Throne
Francois Bernier	France	1656–1668 CE	**Aurangzeb** (Physician to Dara Shikoh)	***Travels in the Mogul Empire***; argued absence of private property in land was cause of India's decline
Niccolao Manucci	Venice (Italy)	1653–1708 CE	**Dara Shikoh** & **Aurangzeb**	***Storia do Mogor***; served as artilleryman for Dara Shikoh; later practiced medicine in Lahore and Madras

8. Bhakti & Sufi Saints & Philosophies (Core Match Matrix)

A. Philosophers & Doctrines

Philosopher Saint	Period / Century	Philosophy / Doctrine	Core Tenet	Mathas / Center
Adi Shankaracharya	8th c. (788–820 CE)	**Advaita** (*Monism / Pure Non-dualism*)	*Brahma Satyam Jagan Mithya* (Brahman alone is real, world is illusion/Maya); Atman is identical to Brahman	Established 4 cardinal Mathas: **Joshimath (Badrikashram, North - Uttarakhand)**, Sringeri (South), Dwarka (West), Puri (East)
Ramanujacharya	11th–12th c. (1017–1137 CE)	**Vishishtadvaita** (*Qualified Non-dualism*)	Atman and world are real attributes (modes) of Brahman (Vishnu); Bhakti is path to Moksha	Srirangam (Tamil Nadu); Sri Vaishnavism
Madhvacharya	13th c. (1199–1278 CE)	**Dvaita** (*Dualism*)	Strict separation between God (Brahman) and individual Soul (Jiva); 5 eternal differences	Udupi (Karnataka); Brahma Sampradaya
Nimbarkacharya	13th c.	**Dvaitadvaita** (*Dualistic Non-dualism / Bhedabheda*)	God, soul, and world are both distinct and non-distinct; Radha-Krishna devotion	Sanaka Sampradaya (Vrindavan)

Philosopher Saint	Period / Century	Philosophy / Doctrine	Core Tenet	Mathas / Center
Vallabhacharya	15th–16th c. (1479–1531 CE)	**Shuddhadvaita** (*Pure Non-dualism*)	World is not an illusion but pure manifestation of Krishna; **Pushti Marga** (Grace of God)	Rudra Sampradaya (Varanasi / Braj)
Chaitanya Mahaprabhu	16th c. (1486–1534 CE)	**Achintya Bhedabheda** (*Inconceivable Oneness & Difference*)	Kirtana & Sankirtana movement; ecstatic love for Radha-Krishna	Nadia / Navadvip (Bengal)

A2. North Indian Bhakti saints — Birth and death (do not leave as "15th c." only)

North chain: **Namdev → Ramananda → Kabir → Nanak → Chaitanya → Surdas → Tulsidas → Dadu**.

Saint	Birth–death	Place / language	High-yield
Namdev	**c. 1270–1350**	Maharashtra; verses also in Adi Granth	Varkari; Vithoba of Pandharpur
Ramananda	14th–15th c.	Banaras (UP)	Guru of Kabir and Ravidas; Rama-bhakti in the north
Kabir	15th c. (often **1398–1518** in coaching lists)	Born **Kashi / Varanasi**; died **Maghar** (Sant Kabir Nagar, UP)	Nirguna; *Bijak* (Ramaini, Sabda, Sakhi); weaver-saint. **Maghar ≠ Mathura**.
Guru Nanak	**1469–1539**	Talwandi (Nankana Sahib); died Kartarpur	First Sikh Guru; Nirguna; companion **Mardana**
Ravidas	15th–16th c.	Banaras (UP)	Ramananda circle; Dalit saint; verses in Adi Granth
Surdas	**c. 1478–1581**	**Braj** (Mathura–Vrindavan, UP)	Saguna Krishna; *Sursagar*; disciple of **Vallabhacharya**
Tulsidas	**1532–1623**	**Awadhi** belt (Rajapur / Banaras, UP)	*Ramcharitmanas* (Awadhi); Saguna Rama. Do not swap with Surdas.
Mirabai	**c. 1498–1546**	Rajasthan (Mewar) → later Dwarka tradition	Krishna devotee; **not** Andal (Tamil Alvar)
Dadu Dayal	**1544–1603**	Rajasthan	Nirguna; Dadu Panth

> **Logic:** Tulsidas = **Awadhi Rama**. Surdas = **Braj Krishna**. Kabir died at **Maghar**, not Kashi. Nanak **1469–1539**, not a 12th-century Alvar.

B. Sufi Silsilahs (Orders) in India

Silsilah	Prominent Saints	Key Center / Khanqah	Key Feature / Attitude to State
Chishti (Most popular)	**Khwaja Moinuddin Chishti** (Ajmer - *Gharib Nawaz*), Bakhtiyar Kaki (Delhi), Baba Farid (Ajodhan - verses in Guru Granth Sahib), **Nizamuddin Auliya** (*Mehboob-i-Ilahi* - saw reigns of 7 Sultans), Nasiruddin Chirag-i-Delhi	Ajmer, Delhi, Punjab	Kept strict distance from royal court; practiced **Sama** (musical recitals) and poverty; vegetarianism
Suhrawardi	**Sheikh Bahauddin Zakariya** (Multan), Hamiduddin Nagori	Multan, Sindh, Bengal	Accepted royal patronage, land grants (*Inam*), and state offices; wealthy lifestyles

Silsilah	Prominent Saints	Key Center / Khanqah	Key Feature / Attitude to State
Qadiri	Sheikh Abdul Qadir Jilani, Shah Niamatullah, Miyan Mir (laid foundation stone of Golden Temple), **Dara Shikoh** & Jahanara	Punjab, Sindh, Agra	Moderate; Dara Shikoh authored *Majma-ul-Bahrain* (*Mingling of Two Oceans*) and translated Upanishads into Persian (*Sirr-i-Akbar*)
Naqshbandi	Khwaja Baqi Billah, **Sheikh Ahmad Sirhindi** (*Mujaddid Alf-i-Sani*)	Delhi, Sirhind	Orthodox and puritanical; strictly opposed *Sama*, opposed Akbar's liberal policies; propounded *Wahdat-ul-Shuhud*

9. Marathas & Shivaji's Administration

A. Shivaji's Ashtapradhan Council (Direct Matching Questions)

Member	Designation / Portfolio	Primary Function
1. Peshwa	**Mukhya Pradhan** (Prime Minister)	Overall general administration; represented king in his absence
2. Amatya / Mazumdar	**Finance Minister** / Auditor	Audited all public and military accounts
3. Sachiv / Shurnavis	**Home / Royal Secretary**	Royal correspondence, checked official edicts and sanity of royal accounts
4. Mantri / Waqia-Navis	**Interior Minister / Chronicler**	Recorded daily proceedings of court, intelligence & espionage
5. Senapati / Sari-i-Naubat	**Commander-in-Chief**	Recruitment, military organization, and field operations
6. Sumant / Dabir	**Foreign Minister**	Managed diplomatic relations, embassies, and foreign visitors
7. Nyayadhish	**Chief Justice**	Civil and criminal justice administration
8. Panditrao / Danadhyaksha	**High Priest / Minister for Charities**	Religious affairs, endowments, grants, and penances

> **Note:** All members except **Nyayadhish** and **Panditrao** were required to lead military commands in times of war.

B. Core Maratha Terms

- **Chauth**: Levy of **one-fourth (25%)** of land revenue extracted from neighbouring territories to protect them from Maratha raids.
- **Sardeshmukhi**: Additional **10% levy** claimed by Shivaji in his capacity as hereditary supreme head (*Sardeshmukh*) of Maharashtra.
- **Mirasdars**: Hereditary landholders.
- **Bargirs**: Regular cavalry provided with horses and weapons by the state.
- **Silahdars**: Mercenary cavalrymen who brought their own horses and arms.
- **Treaty of Purandar (1665)**: Between Shivaji and Mughal General **Raja Jai Singh I** of Amber; Shivaji surrendered 23 out of 35 forts.
- **Coronation (1674)**: At **Raigad Fort** by pandit **Gaga Bhatt** of Varanasi; assumed titles **Chhatrapati** and *Haindava Dharmoddharak*.

SUBJECT 03 · MODERN INDIA

SUBJECT 03: MODERN INDIAN HISTORY & NATIONAL MOVEMENT

आधुनिक भारत का इतिहास एवं राष्ट्रीय आन्दोलन — दैनिक उच्च-प्राथमिकता तथ्य

European Factories, British Treaties, Governor-Generals & Viceroys, Land Revenue Systems, 1857 Centres, Revolutionary Movements, INC Sessions, and Tribal/Peasant Revolts.

1. European Trading Companies & First Factories

European Power	Year of Arrival	First Permanent Factory / Center	Key Fort / Headquarters	Decisive Exit / Defeat Battle
Portuguese	**1498** (Vasco da Gama, Calicut)	**Cochin** (1503), Cannanore (1505)	**Goa** (captured 1510 by Alfonso de Albuquerque from Sultan of Bijapur; made capital 1530 by Nino da Cunha)	Retained Goa, Daman & Diu until **Operation Vijay (1961)**
Dutch (VOC)	**1605**	**Masulipatnam** (1605), Pulicat (1610)	Fort Geldria (Pulicat - minted gold Pagoda coins), Fort Gustavus (Chinsurah)	Defeated by English at **Battle of Bedara / Chinsurah (1759)** & by Travancore at **Battle of Colachel (1741)**
English (EIC)	**1608** (Charter 31 Dec 1600)	**Surat** (1613 permanent; Masulipatnam 1611 temporary)	**Fort St. George** (Madras, 1639 - Francis Day), **Bombay** (1668, dowry lease), **Fort William** (Calcutta, 1698 - Job Charnock)	Became colonial rulers; left in **August 1947**
Danes (Danish)	**1616**	**Tranquebar** (Tamil Nadu, 1620)	**Serampore** (Bengal, 1676 - famous Baptist Mission press)	Sold all Indian settlements to the British in **1845**
French	**1664** (Colbert / Louis XIV)	**Surat** (1668 - Francois Caron)	**Pondicherry** (1674 - founded by Francois Martin), Chandernagore (Bengal)	Decisively defeated at **Battle of Wandiwash (1760)** by Sir Eyre Coote

Chronological Order of Arrival: Portuguese (1498) → Dutch (1605) → English (1608) → Danes (1616) → French (1664).
Chronological Order of Companies Established: Portuguese *Estado da India* (1498) → English EIC (1600) → Dutch VOC (1602) → Danish (1616) → French (1664).

2. British Treaties Timeline (1757–1857)

Frequently tested in chronological order in UPPCS:

Year	Treaty Name	Signed Between	War / Context & Significance
1757 (Feb)	**Treaty of Alinagar**	Robert Clive and Nawab Siraj-ud-Daulah	Preceded Battle of Plassey (23 June 1757); British restored Calcutta privileges
1765 (Aug)	**Treaty of Allahabad**	Robert Clive with Mughal Emperor Shah Alam II & Nawab Shuja-ud-Daula (Awadh)	Followed **Battle of Buxar (1764)**; EIC acquired **Diwani rights** of Bengal, Bihar, and Orissa; Nawab of Awadh surrendered Kora and Allahabad to Emperor; paid 50 lakh indemnity

Year	Treaty Name	Signed Between	War / Context & Significance
1769	**Treaty of Madras**	English EIC and Haidar Ali	Concluded **First Anglo-Mysore War (1767–69)**; defensive alliance
1776	**Treaty of Purandar**	English (Calcutta Council) and Nana Fadnavis	Overrode Treaty of Surat (1775); First Anglo-Maratha War
1782	**Treaty of Salbai**	Warren Hastings and Mahadji Scindia (Marathas)	Concluded **First Anglo-Maratha War (1775–82)**; established 20 years of peace
1784	**Treaty of Mangalore**	Lord Macartney (EIC) and Tipu Sultan	Concluded **Second Anglo-Mysore War (1780–84)**; mutual restitution of conquests
1792	**Treaty of Seringapatam**	Lord Cornwallis and Tipu Sultan	Concluded **Third Anglo-Mysore War (1790–92)**; Tipu ceded **half of his kingdom** (Malabar, Dindigul, Baramahal) and paid 3.3 crore indemnity
1802	**Treaty of Bassein**	Lord Wellesley and Peshwa Baji Rao II	Concluded on New Year's Eve; Peshwa surrendered sovereignty and accepted **Subsidiary Alliance**; triggered Second Anglo-Maratha War
1803	**Treaties of Deogaon & Surji-Anjangaon**	Lord Wellesley with Bhonsle (Deogaon) and Scindia (Surji-Anjangaon)	Both accepted Subsidiary Alliance and ceded territory
1809	**Treaty of Amritsar**	Charles Metcalfe (EIC) and **Maharaja Ranjit Singh**	Fixed river **Sutlej as the boundary** of Sikh Kingdom; stopped Ranjit Singh's expansion across Sutlej
1816	**Treaty of Sagauli**	Lord Hastings and King of Nepal (Gurkhas)	**UKPCS TOP PRIORITY!** Concluded **Anglo-Nepal War (1814–16)**; Nepal ceded **Garhwal and Kumaon** to British; British resident posted at Kathmandu; boundary along Kali River
1826	**Treaty of Yandabo**	Lord Amherst and King of Ava (Burma)	Concluded **First Anglo-Burmese War (1824–26)**; Burma ceded Assam, Manipur, Arakan, and Tenasserim
1846	**Treaty of Lahore**	Lord Hardinge I and minor Maharaja Duleep Singh	Concluded **First Anglo-Sikh War (1845–46)**; Jalandhar Doab ceded; Kashmir sold to Gulab Singh under **Treaty of Amritsar (1846)**

2A. Anglo wars — From / To (do not stop at the first year)

War	Period (from–to)	GG / Company side	Last fight / treaty	Result
Carnatic Wars	**1746–1763** (three wars)	Dupleix vs English (Clive at Arcot 1751)	**Paris 1763** after Wandiwash **1760** (Eyre Coote)	French political power in India broken; Pondicherry restored but unfortified
Anglo-Mysore	**1767–1799** (four wars)	Haidar → **Tipu**	**Fourth war 1799**; Tipu dies at Seringapatam; Wellesley	Mysore under Wodeyar + Subsidiary Alliance
Anglo-Maratha	**1775–1818** (three wars)	Hastings → Wellesley → Hastings (Moira)	**Third 1817–18**; Peshwaship abolished	Maratha confederacy ends; Bajirao II pensioned at Bithur
Anglo-Nepal (Gurkha)	**1814–1816**	**Lord Hastings**	**Treaty of Sagauli 1816**	Nepal cedes **Garhwal and Kumaon**; Kali River boundary; resident at Kathmandu
Anglo-Burma	**1824–26**; 1852; 1885	Amherst → Dalhousie → Dufferin	**Yandabo 1826**; Third war annexes Upper Burma	Assam, Manipur, Arakan, Tenasserim in the first war
Anglo-Sikh	**1845–1849** (two wars)	Hardinge I → Dalhousie	**Second war 1848–49**; Punjab annexed **1849**	Lahore 1846; Kashmir sold to Gulab Singh (Amritsar 1846)

3. Land Revenue Systems Comparison

Feature	Permanent Settlement (Zamindari)	Ryotwari System	Mahalwari System
Year & Architect	**1793** — Lord Cornwallis & Sir John Shore	**1820** — Sir Thomas Munro & Alexander Read	**1822** (Reg. VII) — Holt Mackenzie; modified 1833 by **Martin Bird** ("Father of land settlement in North India")
Coverage Area	Bengal, Bihar, Odisha, Northern Circars (Madras), Banaras (UP)	Madras, Bombay, parts of Assam, Coorg	Ganga Valley, North-West Frontier, Central Provinces, Punjab, Western UP
% of British India	**~19%** of British Indian territory	**~51%** of British Indian territory (Largest!)	**~30%** of British Indian territory
Settlement Made With	**Zamindars** recognized as hereditary absolute owners of land	**Ryots (Individual peasants)** directly	**Mahal / Village Community** collectively represented by Lambardar/Talukdar
Revenue Share	Fixed permanently; State = **10/11th**, Zamindar = **1/11th**; **Sunset Law** (1794) applied	Periodically revised (20–30 yrs); high rate (50–60% of gross produce)	Periodically revised; state claimed **66%** (later 50% under Bentinck 1833) of rental value

4. Governor-Generals & Viceroys High-Yield Milestones

- **Warren Hastings (1772–1785)**:
- Abolished Dual Government in Bengal (1772).
- **Regulating Act of 1773**; became first **Governor-General of Bengal**.
- Established **Supreme Court at Calcutta (1774)** (Sir Elijah Impey first Chief Justice).
- Founded **Calcutta Madrasa (1781)**; Sir William Jones founded **Asiatic Society of Bengal (1784)**.
- Charles Wilkins translated *Bhagavad Gita* into English (1785). Only Governor-General impeached in Britain (later acquitted).
- **Lord Cornwallis (1786–1793)**:
- Introduced **Permanent Settlement of Bengal (1793)**.
- Formulated **Cornwallis Code (1793)** based on separation of revenue and judicial powers.
- Established District Thanas headed by Daroga; known as **"Father of Civil Services and Police in India"**.
- Died at Ghazipur (Uttar Pradesh) in 1805 during second tenure; his **tomb is located at Ghazipur (UP)**.
- **Lord Wellesley (1798–1805)**:
- Expanded empire via **Subsidiary Alliance System** (*Order of states: Hyderabad 1798 → Mysore 1799 → Tanjore 1799 → Awadh 1801 → Peshwa Baji Rao II 1802 → Bhonsle 1803 → Scindia 1804*).
- Defeated Tipu Sultan in **Fourth Anglo-Mysore War (1799)**.
- Founded **Fort William College (1800)** at Calcutta to train British civil servants in Indian languages.
- Called himself the **"Bengal Tiger"**.
- **Lord Hastings (1813–1823)**:
- **Anglo-Nepal War (1814–1816)**: Concluded by **Treaty of Sagauli (1816)** — Kumaon and Garhwal ceded to the British (**crucial UKPCS fact!**).
- Crushed the Pindaris; defeated the Marathas in **Third Anglo-Maratha War (1817–1818)**; abolished Peshwaship.
- **Lord William Bentinck (1828–1835)**:
- **Charter Act of 1833**: Became the **First Governor-General of India**.
- **Abolition of Sati (Regulation XVII of 1829)** with active support of Raja Ram Mohan Roy.
- Suppression of **Thuggee (1830)** led by **Colonel William Sleeman**.
- **Macaulay's Minute (1835)**: Made English the official medium of higher education in India (*Filtration Theory*).
- **Lord Dalhousie (1848–1856)**:

- **Doctrine of Lapse** (*Satara 1848 → Jaitpur & Sambalpur 1849 → Baghat 1850 → Udaipur 1852 → Jhansi 1853 → Nagpur 1854*).
- Annexed **Awadh in 1856** on grounds of "maladministration" based on **James Outram's report** (Nawab Wajid Ali Shah exiled).
- **First Railway line (1853)**: Bombay to Thane (34 km).
- **First Telegraph line (1853)**: Calcutta to Agra.
- **Wood's Despatch (1854)**: Magna Carta of English education in India.
- **Post Office Act (1854)**: Uniform 2-paisa postal rate.
- Drafted **Hindu Widows' Remarriage Act (passed 1856 as Act XV)** championed by Ishwar Chandra Vidyasagar.
- **Lord Canning (1856–1862)**:
- Last Governor-General and **First Viceroy of India** (under **GOI Act 1858**).
- Suppressed the **Revolt of 1857**; read Queen Victoria's Proclamation at **Allahabad Durbar on 1 Nov 1858**.
- Introduced the **Portfolio System** (Indian Councils Act 1861); enacted **Indian Penal Code (IPC 1860)**.
- **Lord Lytton (1876–1880)**:
- **Vernacular Press Act IX of 1878** (*Gagging Act* — aimed directly at Shishir Kumar Ghosh's *Amrita Bazar Patrika*).
- **Arms Act of 1878** (prevented Indians from possessing weapons without license).
- Held grand **Delhi Durbar of 1877** during severe Madras famine; proclaimed Queen Victoria as *Kaiser-i-Hind*.
- Lowered the maximum age limit for ICS examination from **21 to 19 years**.
- **Lord Ripon (1880–1884)** (*The Good Viceroy*):
- Repealed the infamous **Vernacular Press Act (1882)**.
- **Local Self-Government Resolution (1882) → "Father of Local Self-Government in India"**.
- **First Regular Census of India (1881)** (conducted under Census Commissioner W.W. Plowden).
- **First Factory Act (1881)** (restricted child labour below 7 years).
- Appointed **Hunter Commission on Education (1882)**.
- **Ilbert Bill Controversy (1883)**: Proposed allowing Indian judges to try European offenders; triggered racist Anglo-Indian agitation.
- **Lord Curzon (1899–1905)**:
- **Partition of Bengal**: Announced on **20 July 1905**, came into effect on **16 October 1905** (day observed as *National Mourning & Raksha Bandhan*).
- **Ancient Monuments Preservation Act (1904)**; established Archaeological Survey of India (ASI) under Sir John Marshall.
- **Indian Universities Act (1904)** (based on Thomas Raleigh Commission 1902).
- Formed **Police Commission (1902)** under Sir Andrew Fraser.

4A. Remaining Governor-Generals / Viceroys — Tenure (from–to)

Office-holder	Period (from–to)	High-yield
Lord Auckland	**1836–1842**	First Afghan War; **Tripartite Treaty 1838**
Lord Ellenborough	**1842–1844**	Ended Afghan War; annexed **Sindh 1843** (Napier)
Lord Hardinge I	**1844–1848**	First Anglo-Sikh War **1845–46**
Lord Lawrence	**1864–1869**	"Master of the Punjab"; Odisha famine / Campbell Commission
Lord Mayo	**1869–1872**	First Census **1872**; Mayo College; murdered at Port Blair
Lord Lytton	**1876–1880**	(see bullets above)
Lord Dufferin	**1884–1888**	INC founded **1885**; Third Burma War **1885**
Lord Lansdowne	**1888–1894**	Age of Consent Act **1891**; Durand Line **1893**
Lord Elgin II	**1894–1899**	Lyall Famine Commission

Office-holder	Period (from–to)	High-yield
Lord Minto II	**1905–1910**	**Morley–Minto / Indian Councils Act 1909**; separate electorates
Lord Hardinge II	**1910–1916**	Delhi capital **1911**; bomb on Chandni Chowk procession (Ras Behari Bose)
Lord Chelmsford	**1916–1921**	Lucknow Pact era; **GOI Act 1919**; Jallianwala **1919**; NCM starts
Lord Reading	**1921–1926**	Chauri Chaura aftermath; Kakori **1925**; Lee Commission
Lord Irwin	**1926–1931**	Simon Commission; Dandi; **Gandhi–Irwin Pact 5 Mar 1931**
Lord Willingdon	**1931–1936**	Second RTC fallout; Communal Award **1932**; Poona Pact
Lord Linlithgow	**1936–1944** (*longest Viceroy*)	GOI Act 1935 in force; **1937** ministries; WWII declaration; August Offer; Cripps; **Quit India 1942**
Lord Wavell	**1944–1947**	Wavell Plan / Simla **1945**; Cabinet Mission; Interim Government
Lord Mountbatten	**Mar–Aug 1947** (GG of India till **Jun 1948**)	**3 June Plan**; last Viceroy; first GG of independent India

Logic: First GG of **Bengal** = Warren Hastings **1773**. First GG of **India** = Bentinck **1833**. First **Viceroy** = Canning **1858**. Last Viceroy = Mountbatten.

5. British Committees & Commissions Ladder

A. Education Commissions Chronology

1. **Charles Wood's Despatch (1854)**: "Magna Carta of English Education in India"; recommended three-tier education, female education, vernacular at school, English at university; set up Universities of Calcutta, Bombay, and Madras (1857).
2. **Hunter Commission (1882)**: Appointed by Lord Ripon under Sir William Hunter; emphasized primary education in vernacular and expansion of private enterprise.
3. **Raleigh Commission / Indian Universities Act (1902/1904)**: Appointed by Lord Curzon under Sir Thomas Raleigh; focused on university education and stricter government control.
4. **Sadler Commission (1917)**: Appointed by Lord Chelmsford under Sir Michael Sadler to investigate Calcutta University; recommended 12-year school course (10+2 pattern) and 3-year degree.
5. **Hartog Committee (1929)**: Appointed by Simon Commission under Sir Philip Hartog; critiqued rapid expansion, emphasized quality and consolidation of primary education.
6. **Wardha Scheme of Basic Education (1937)**: Formulated by Mahatma Gandhi (*Nai Talim*); detailed scheme drafted by **Zakir Hussain Committee**; free, compulsory learning through mother tongue and crafts.
7. **Sargeant Plan of Education (1944)**: Central Advisory Board under Sir John Sargeant; aimed at universal literacy in 40 years.

B. Famine, Police, Currency & Civil Services Commissions

- **Famine Commissions**:
- **Campbell Commission (1866)**: Appointed by Sir John Lawrence after Odisha famine.
- **Strachey Commission (1880)**: Appointed by Lord Lytton under Sir Richard Strachey; formulated the **First Famine Code (1883)**.
- **Lyall Commission (1897)**: Appointed by Lord Elgin II.
- **MacDonnell Commission (1900)**: Appointed by Lord Curzon under Sir Antony MacDonnell; emphasized "moral strategy" and appointment of Famine Commissioner.
- **Police Commission**:
- **Fraser Commission (1902)**: Appointed by Lord Curzon under Sir Andrew Fraser; recommended setting up Criminal Investigation Department (CID) and direct recruitment of sub-inspectors.

- **Currency & Finance Commissions**:
- **Mansfield Commission (1886)**: Examined silver depreciation.
- **Fowler Committee (1898)**: Recommended gold standard for India.
- **Babington Smith Committee (1919)**: Post-WWI currency exchange.
- **Hilton Young Commission (1926)**: Royal Commission on Indian Currency and Finance; **recommended the establishment of a Central Bank → Reserve Bank of India (RBI Act 1934, set up 1 April 1935)**!
- **Civil Services Commissions**:
- **Aitchison Commission (1886)**: Lord Dufferin; categorized services into Imperial, Provincial, and Subordinate.
- **Islington Commission (1912)**: Recommended 25% of superior posts to Indians.
- **Lee Commission (1923)**: Recommended establishment of a statutory **Public Service Commission (set up in 1926 under Sir Ross Barker)**.

6. Revolt of 1857 — Centre, Leader & British Suppressor Matrix

Centre of Revolt	Indian Leader / Commander	British Suppressor Officer	Crucial Exam Detail
Delhi	**Bahadur Shah Zafar II** & **General Bakht Khan** (Subahdar of Bareilly artillery)	**John Nicholson** (mortally wounded), **Captain Hudson**	Hudson shot Bahadur Shah's sons at Delhi Gate; Emperor exiled to Rangoon
Kanpur	**Nana Saheb** (Dhondu Pant), **Tatya Tope** (Ramchandra Pandurang), Azimullah Khan	**Sir Colin Campbell**, Havelock	Nana fled to Nepal; Tatya Tope betrayed by Mansingh, hanged at Shivpuri (1859)
Lucknow	**Begum Hazrat Mahal**, Prince Birjis Qadr, Raja Jailal Singh	**Sir Colin Campbell**, Outram, Henry Lawrence (died at Residency)	Begum Hazrat Mahal refused British pension, fled to Nepal
Jhansi	**Rani Lakshmibai** (Manikarnika / Manu)	**Sir Hugh Rose**	Hugh Rose commented: "*Here lay the woman who was the only man among the rebels*"; fell at Gwalior (1858)
Bareilly	**Khan Bahadur Khan** (grandson of Hafiz Rahmat Khan)	**Sir Colin Campbell**, Eyres	Declared himself Nawab Nazim; later captured and hanged
Jagdishpur / Arrah (Bihar)	**Kunwar Singh** (80-yr old zamindar) & brother **Amar Singh**	**William Taylor** and **Vincent Eyre**	Only leader who was never defeated by the British in open battle
Allahabad & Banaras	**Maulvi Liaquat Ali**	**Colonel James Neill**	Neill carried out merciless public executions and atrocities
Faizabad	**Maulvi Ahmadullah Shah** (*Danka Shah*)	**Sir Colin Campbell**	Issued fatwa of Jihad against British; British announced Rs 50,000 bounty on his head
Meerut	**Kadam Singh**	-	Sepoys marched from Meerut to Delhi on 10 May 1857
Mathura	**Devi Singh**	-	Peasant mobilization in Doab
Assam	**Maniram Dewan** & Piyali Barua	-	Wanted to restore Ahom prince Kandarpeshwar Singha; hanged
Kumaon / Champawat (Uttarakhand)	**Kalu Singh Mahara**	British commissioners	Formed secret rebel group **Krantiveer**; acknowledged as **First Freedom Fighter of Uttarakhand**

UK freedom (from–to, not founder-year only): Gorkha rule in Kumaon **1790–1815** and Garhwal to **1804/1815** → British after **Sagauli 1816**. **Coolie-Begar** abolished **13–14 Jan 1921** at **Bageshwar** (Uttarayani; registers flung into the **Saryu**); led by **Badridatt Pandey** (*Kumaon Kesari*), Hargovind Pant, Chiranji Lal. Gandhi called it a bloodless revolution. Badridatt met Gandhi at **Calcutta, Dec 1918** on begar (not Kausani). Gandhi stayed at **Kausani (1929)** to write *Anasakti Yoga*.

7. Revolutionary Organizations & Conspiracy Cases (Direct Match Goldmine)

Organization / Party	Year & Place Founded	Founders / Key Leaders	Landmark Conspiracies / Activities
Mitra Mela → Abhinav Bharat	1899 → **1904**, Nasik (Maharashtra)	**Vinayak Damodar Savarkar** & Ganesh Savarkar	Secret society modelled on Young Italy; Anant Laxman Kanhere assassinated Collector **A.M.T. Jackson** in **Nasik Conspiracy Case (1909)**
Anushilan Samiti	**1902**, Calcutta & Dhaka	**Pramathanath Mitra (P. Mitra)**, Satish Chandra Basu, Barindra Ghosh, Bhupendranath Dutta	Secret revolutionary body; journal *Yugantar*; Khudiram Bose & Prafulla Chaki threw bomb at Kingsford carriage in **Muzaffarpur (1908) → Alipore Bomb Case (1908)**; Aurobindo Ghosh defended by C.R. Das
Indian Home Rule Society / India House	**1905**, London	**Shyamji Krishna Varma**	Centre for Indian students in London; journal *The Indian Sociologist*; **Madan Lal Dhingra** assassinated **Curzon Wyllie** at Imperial Institute (1909)
Paris Indian Society	1905, Paris	**Madam Bhikaji Cama**, S.R. Rana, Munchershah Burjorji Godrej	Unfurled **First Indian National Flag (Tricolour)** at the International Socialist Congress in **Stuttgart (Germany) in 1907**
Ghadar Party	**1913**, San Francisco (USA)	**Lala Har Dayal**, **Sohan Singh Bhakna** (President), Kartar Singh Sarabha, Bhai Parmanand	Originally *Pacific Coast Hindustan Association*; weekly paper *Ghadar*; Komagata Maru incident; attempted armed revolt in India during WWI
Berlin Committee for Indian Independence	1915, Berlin (Germany)	**Virendranath Chattopadhyaya**, Bhupendranath Dutta, Lala Har Dayal	Formed under the "Zimmerman Plan" with German foreign office help
Hindustan Republican Association (HRA)	**1924 (Oct)**, Kanpur (UP)	**Sachindra Nath Sanyal** (wrote ***Bandi Jiwan* - "Bible of Revolutionaries"**), Ram Prasad Bismil, Jogesh Chandra Chatterjee	Executed **Kakori Train Action (9 August 1925)**; 8-Down passenger train looted; Ram Prasad Bismil (Gorakhpur jail), Ashfaqulla Khan (Faizabad jail), Roshan Singh (Allahabad Naini jail), Rajendra Lahiri (Gonda jail) hanged
Hindustan Socialist Republican Association (HSRA)	**1928 (Sep)**, Feroz Shah Kotla (Delhi)	**Chandrashekhar Azad**, **Bhagat Singh**, Sukhdev Thapar, Bhagwati Charan Vohra (wrote *Philosophy of the Bomb*)	Adopted Socialism; assassinated **J.P. Saunders** on 17 Dec 1928; Bhagat Singh & Batukeshwar Dutt threw bombs in **Central Legislative Assembly (8 April 1929)**; **Lahore Conspiracy Case**; Chandrashekhar Azad martyred at Alfred Park, Allahabad (27 Feb 1931)
Naujawan Bharat Sabha	**1926**, Lahore	**Bhagat Singh**, Chhabil Das, Yashpal	Open mass youth front propagating anti-communalism and socialist revolution
Indian Republican Army (Chittagong)	**1930 (Apr)**, Chittagong (Bengal)	**Surya Sen ("Master Da")**, Ambika Chakrabarty, Ganesh Ghosh, Lokenath Bal, **Pritilata Waddedar**, **Kalpana Datta**	**Chittagong Armoury Raid (18 April 1930)**; proclaimed Provisional Revolutionary Government; Pritilata attacked Pahartali European Club; Surya Sen hanged in 1934

> **Till when (do not leave as founded-year only):** Abhinav Bharat peaks with Nasik **1909** then is smashed. Ghadar's armed try is **WWI / 1915**, then it fades as a party. **HRA 1924 → HSRA 1928** (same stream, new name). Chittagong group ends with Surya Sen's hanging **1934**. HSRA's open run ends with Azad's death at Alfred Park, Allahabad (**27 Feb 1931**) and the Lahore hangings (**23 Mar 1931**).

7A. Social-religious organisations — Founded / split / till

Body	Founded	Founder	Till / split / high-yield
Brahmo Sabha → Brahmo Samaj	**1828**, Calcutta	**Raja Rammohan Roy**	After Roy: **Adi Brahmo** (Debendranath Tagore) vs **Sadharan Brahmo Samaj 1878** (Keshab / Ananda Mohan Bose stream). Still a living samaj, not a "ended 1833" trap.
Young Bengal	**1820s–30s**	**Henry Vivian Derozio**	Short-lived radical students of Hindu College; Derozio dismissed **1831**
Prarthana Samaj	**1867**, Bombay	Atmaram Pandurang; later **M.G. Ranade**, R.G. Bhandarkar	Western-India cousin of Brahmo; did not become Arya Samaj
Satyashodhak Samaj	**1873**, Pune	**Jyotirao Phule**	Anti-caste; *Gulamgiri* **1873**. Savitribai on girls' schools
Arya Samaj	**10 Apr 1875**, Bombay (Lahore **1877**)	**Dayananda Saraswati**	*Satyarth Prakash* **1875**; "Back to the Vedas". Split **1893**: Gurukul (Munshi Ram / Shraddhanand) vs College (Lala Hansraj / Lajpat Rai)
Theosophical Society	**1875** New York; **Adyar 1882**	Blavatsky & Olcott; in India **Annie Besant**	HQ **Adyar (Madras)**; Besant's Home Rule **1916**
Aligarh Movement / MAO College	**1875** college; AMU **1920**	**Sir Syed Ahmad Khan**	*Tahzib-ul-Akhlaq*; United Patriotic Association. College **1875–1920**, then AMU
Ramakrishna Mission	**1897**	**Swami Vivekananda**	After Chicago **1893**; Belur Math. Still active
Servants of India Society	**1905**, Pune	**Gopal Krishna Gokhale**	Train public workers; **not** a revolutionary secret society
Muslim League	**30 Dec 1906**, Dacca	Nawab Salimullah; later Jinnah	**1906–1947** as the League of British India; Lahore Resolution **1940**
Hindu Mahasabha	**1915** (all-India form)	Madan Mohan Malaviya / others	Separate from INC; Savarkar later president

8. Women Freedom Fighters of India

- **Madam Bhikaji Cama**: "Mother of the Indian Revolution"; unfurled first Indian tricolour at Stuttgart (1907); operated *Vande Mataram* journal from Paris.
- **Sarojini Naidu**: "Nightingale of India"; first Indian woman Congress President (Kanpur 1925); led salt raid on Dharasana Salt Works (1930); first woman Governor of a state in independent India (**United Provinces / Uttar Pradesh 1947–49**).
- **Aruna Asaf Ali**: "Heroine of the 1942 Quit India Movement"; unfurled Congress tricolour flag at Gowalia Tank Maidan, Bombay on **9 August 1942** after all top leaders were arrested under Operation Zero Hour; edited underground journal *Inquilab*.
- **Usha Mehta**: Operated **Secret Underground Congress Radio** (transmitting on 42.34 meters from secret locations in Bombay) during Quit India 1942 with Dr. Ram Manohar Lohia.
- **Matangini Hazra ("Gandhi Buri")**: 73-year-old freedom fighter in Tamluk (Bengal); shot dead by British police while marching with the National Flag in 1942; kept chanting "Vande Mataram".

- **Kanaklata Barua**: 17-year-old martyr from Gohpur (Assam); shot dead leading the *Mrityu Bahini* to hoist the flag at police station during Quit India 1942.
- **Bina Das**: Shot Bengal Governor **Sir Stanley Jackson** at point-blank range during the Calcutta University convocation ceremony in **February 1932**.
- **Kalpana Datta & Pritilata Waddedar**: Prominent members of Surya Sen's Chittagong revolutionary group; Pritilata consumed cyanide to avoid arrest after attacking Pahartali European Club.
- **Rani Gaidinliu**: Naga spiritual and political leader; led **Heraka religious movement** against the British in Assam/Manipur at age 13; captured and sentenced to life; Jawaharlal Nehru met her in jail and gave her the title **"Rani"** (*Daughter of the Hills*).
- **Sucheta Kripalani**: Sang *Vande Mataram* in Constituent Assembly on 14 Aug 1947; **First woman Chief Minister of an Indian state (Uttar Pradesh, 1963–1967)**.

9. Famous Slogans & Sloganeers

Famous Slogan	Coined By / Popularized By	Context / Significance
"Swaraj is my birthright and I shall have it"	**Bal Gangadhar Tilak**	Home Rule Movement (1916)
"Inquilab Zindabad"	Coined by **Maulana Hasrat Mohani** (1921); popularized by **Bhagat Singh**	Central Assembly bombing & Lahore trial
"Do or Die" (*Karo ya Maro*)	**Mahatma Gandhi**	Gowalia Tank, Bombay — Quit India Movement (8 Aug 1942)
"Give me blood and I will give you freedom"	**Subhas Chandra Bose**	Addressing Indian National Army (INA) in Burma (1944)
"Delhi Chalo" & "Jai Hind"	**Subhas Chandra Bose**	INA battle cry and national greeting
"Back to the Vedas"	**Swami Dayananda Saraswati**	Arya Samaj doctrine (1875)
"One Caste, One Religion, One God for Mankind"	**Sri Narayana Guru**	Aravippuram movement, Kerala (*Oru Jathi, Oru Matham, Oru Daivam Manushyanu*)
"Educate, Agitate, Organise"	**Dr. B.R. Ambedkar**	Bahishkrit Hitakarini Sabha (1924)
"Sarfaroshi ki Tamanna ab hamare dil mein hai"	Written by **Bismil Azimabadi** (1921); popularized by **Ram Prasad Bismil**	Kakori revolutionaries
"Sare Jahan Se Achha Hindostan Hamara"	**Muhammad Iqbal** (1904)	Published in *Ittehad*
"Aaram Haram Hai"	**Jawaharlal Nehru**	Post-independence nation-building
"Jai Jawan, Jai Kisan"	**Lal Bahadur Shastri**	During 1965 Indo-Pak War at Ramlila Maidan, Delhi
"Satyam Eva Jayate"	Sourced from **Mundaka Upanishad**; popularized as national motto by **Pt. Madan Mohan Malaviya**	Adopted as National Motto on 26 Jan 1950

10. Landmark Indian National Congress (INC) Sessions

Year	Venue	President	Landmark Historical Outcome / Prelims Trigger
1885	**Bombay** (Gokuldas Tejpal Sanskrit College)	**W.C. Bonnerjee**	**Founding session**; 72 delegates attended; founded by retired ICS officer **A.O. Hume**

Year	Venue	President	Landmark Historical Outcome / Prelims Trigger
1886	**Calcutta**	**Dadabhai Naoroji**	First Parsi president; National Congress and National Conference merged
1887	**Madras**	**Badruddin Tyabji**	**First Muslim President**; appeal made to all communities
1888	**Allahabad**	**George Yule**	**First British / Non-Indian President**; Governor Auckland attempted to block session
1896	**Calcutta**	**Rahimtullah M. Sayani**	National song **"Vande Mataram"** sung for the **first time** on Congress platform
1905	**Banaras**	**Gopal Krishna Gokhale**	Expressed deep resentment against Partition of Bengal; approved Swadeshi & Boycott
1906	**Calcutta**	**Dadabhai Naoroji**	**"Swaraj"** explicitly adopted as Congress goal for the **first time**; four resolutions (Swadeshi, Boycott, National Education, Self-Gov)
1907	**Surat**	**Rash Behari Ghosh**	**Surat Split**: Congress split into **Moderates and Extremists**; session suspended
1911	**Calcutta**	**Bishan Narayan Dhar**	National anthem **"Jana Gana Mana"** sung for the **first time**
1916	**Lucknow**	**Ambica Charan Majumdar**	**Lucknow Pact**: Reunion of Moderates & Extremists; joint charter of demands with Muslim League (separate electorates accepted)
1917	**Calcutta**	**Annie Besant**	**First Woman President** of Indian National Congress
1920 (Sp.)	**Calcutta**	**Lala Lajpat Rai**	Non-Cooperation resolution approved
1920	**Nagpur**	**C. Vijayaraghavachariar**	Congress Constitution changed; Linguistic Provincial Congress Committees formed; goal shifted to peaceful attainment of Swaraj
1924	**Belgaum**	**Mahatma Gandhi**	**ONLY session ever presided over by Mahatma Gandhi**
1925	**Kanpur**	**Sarojini Naidu**	**First Indian Woman President** of INC
1929	**Lahore**	**Jawaharlal Nehru**	**Purna Swaraj** resolution passed; Tricolour unfurled on midnight of **31 Dec 1929** on river Ravi; **26 Jan 1930** declared First Independence Day
1931	**Karachi**	**Sardar Vallabhbhai Patel**	Passed resolutions on **Fundamental Rights** and **National Economic Programme** (drafted by J.L. Nehru); endorsed Gandhi-Irwin Pact
1936	**Lucknow**	**Jawaharlal Nehru**	Nehru declared Socialism as the goal of India
1937	**Faizpur** (Maharashtra)	**Jawaharlal Nehru**	**First Congress session held in a rural / village area**
1938	**Haripura** (Gujarat)	**Subhas Chandra Bose**	National Planning Committee set up under chairmanship of Jawaharlal Nehru
1939	**Tripuri** (Jabalpur, MP)	**Subhas Chandra Bose**	Bose defeated Gandhi's candidate **Pattabhi Sitaramayya**; resigned due to Pant resolution conflict; founded **Forward Bloc (1939)**; Rajendra Prasad completed term
1940–45	**Ramgarh** (1940)	**Maulana Abul Kalam Azad**	Longest continuous presidential tenure (no sessions held 1941–45 due to Quit India)
1946	**Meerut**	**J.B. Kripalani**	President of INC at the time of Indian Independence in **August 1947**

11. Master Chronological Ladder of the Freedom Movement (1905–1947)

Memorize this unbroken chain — tests 2 to 4 questions in EVERY UPPSC / UKPCS Prelims paper!

1. **1905 (16 Oct)**: **Partition of Bengal** comes into effect → Swadeshi Movement peaks.
2. **1906 (30 Dec)**: **All-India Muslim League** founded at Dacca by Nawab Salimullah & Aga Khan.
3. **1907 (Dec)**: **Surat Split** between Moderates and Extremists.
4. **1908**: Khudiram Bose hanged (Muzaffarpur conspiracy); Tilak sentenced to 6 years at Mandalay.
5. **1909**: **Morley-Minto Reforms (Indian Councils Act 1909)** introduces separate electorates for Muslims.
6. **1911 (Dec)**: **Delhi Durbar**: Annulment of Partition of Bengal; Capital shifted from Calcutta to Delhi.
7. **1913**: **Ghadar Party** formed in San Francisco by **Lala Har Dayal** and **Sohan Singh Bhakna**.
8. **1914**: **Komagata Maru** incident (steamer turned back from Vancouver; Budge Budge firing).
9. **1915 (9 Jan)**: **Mahatma Gandhi returns to India** from South Africa (*Pravasi Bharatiya Divas*).
10. **1916 (Apr / Sep)**: **Home Rule League** launched by **Bal Gangadhar Tilak** (Apr, Pune) and **Annie Besant** (Sep, Madras); **Lucknow Pact (Dec 1916)**.
11. **1917**: **Champaran Satyagraha** (First Civil Disobedience).
12. **1918**: **Ahmedabad Mill Strike** (First hunger strike) & **Kheda Satyagraha** (First non-cooperation).
13. **1919 (Mar)**: **Rowlatt Act** (*Black Act* — detention without trial).
14. **1919 (13 Apr)**: **Jallianwala Bagh Massacre** (Amritsar, General Dyer firing; Tagore renounces Knighthood).
15. **1919 (Dec)**: **Montagu-Chelmsford Reforms (GOI Act 1919)** introduces Dyarchy in Provinces.
16. **1920 (1 Aug)**: **Non-Cooperation Movement launched**; Bal Gangadhar Tilak passes away.
17. **1922 (4 Feb)**: **Chauri Chaura Incident** (Gorakhpur, UP; 22 policemen burnt alive in police station) → Gandhi calls off NCM on **12 Feb 1922** at Bardoli.
18. **1923 (1 Jan)**: **Swaraj Party** formed by **Chitta Ranjan Das** (President) and **Motilal Nehru** (Secretary).
19. **1924 (Oct)**: **Hindustan Republican Association (HRA)** formed at Kanpur by Sachindra Sanyal, Ram Prasad Bismil, and Jogesh Chatterjee.
20. **1925 (9 Aug)**: **Kakori Train Action** (Kakori near Lucknow; 8-Down train looted; Bismil, Ashfaqulla Khan, Roshan Singh, Rajendra Lahiri hanged).
21. **1927 (Nov)**: **Simon Commission appointed** (All-white, 7 MPs; arrived in Bombay in **Feb 1928** → "Simon Go Back").
22. **1928 (Aug)**: **Nehru Report** (Chaired by Motilal Nehru; demanded Dominion Status, rejected separate electorates).
23. **1928 (Sep)**: **HSRA** (Hindustan Socialist Republican Association) founded at Feroz Shah Kotla, Delhi under **Chandrashekhar Azad** & **Bhagat Singh**.
24. **1928 (17 Dec)**: Saunders assassinated at Lahore to avenge the death of Lala Lajpat Rai.
25. **1929 (8 Apr)**: **Central Legislative Assembly Bombing** by Bhagat Singh and Batukeshwar Dutt ("To make the deaf hear").
26. **1929 (13 Sep)**: **Jatin Das dies** after 63-day hunger strike in Lahore Central Jail.
27. **1929 (Dec)**: **Lahore Congress Session** → Purna Swaraj adopted.
28. **1930 (26 Jan)**: First Independence Day celebrated nationwide.
29. **1930 (12 Mar – 6 Apr)**: **Dandi March** (Sabarmati to Dandi, 241 miles, 78 followers; broken salt law on 6 April) → **Civil Disobedience Movement (CDM)** begins.
30. **1930 (Nov) – 1931 (Jan)**: **First Round Table Conference** (London; Congress boycotted).
31. **1931 (5 Mar)**: **Gandhi-Irwin Pact** (Delhi Pact; suspension of CDM, Congress agreed to attend 2nd RTC).
32. **1931 (23 Mar)**: **Bhagat Singh, Sukhdev, and Rajguru executed** at Lahore Central Jail.
33. **1931 (Sep – Dec)**: **Second Round Table Conference** (Gandhi attended as sole Congress representative; deadlock on minority seats).
34. **1932 (16 Aug)**: **Communal Award** announced by British PM Ramsay MacDonald (separate electorates for Depressed Classes).
35. **1932 (24 Sep)**: **Poona Pact** signed between Dr. B.R. Ambedkar and Madan Mohan Malaviya (on behalf of fasting Gandhi in Yerwada jail; abandoned separate electorates, increased reserved seats from 71 to 148).
36. **1932 (Nov – Dec)**: **Third Round Table Conference** (Congress boycotted).
37. **1935 (Aug)**: **Government of India Act 1935** (Abolished Provincial Dyarchy, introduced Provincial Autonomy; All-India Federation proposed).
38. **1937 (Feb)**: Provincial Elections held; Congress formed ministries in 8 provinces.

39. **1939 (22 Oct)**: Congress ministries **resign en masse** in protest against Viceroy Linlithgow declaring India at war without consent → Muslim League celebrates **"Day of Deliverance"** on **22 Dec 1939**.
40. **1940 (Mar)**: **Lahore Resolution** of Muslim League (demanding independent Muslim states).
41. **1940 (8 Aug)**: **August Offer** by Viceroy Linlithgow (Dominion status post-war, veto to minorities; rejected).
42. **1940 (Oct)**: **Individual Satyagraha** launched (*Delhi Chalo* Satyagraha; **1st: Vinoba Bhave, 2nd: Jawaharlal Nehru, 3rd: Brahma Dutt**).
43. **1942 (Mar)**: **Cripps Mission** arrives (Stafford Cripps; post-dated cheque on a crashing bank).
44. **1942 (8 Aug)**: **Quit India Resolution** passed at Gowalia Tank Maidan (August Kranti Maidan), Bombay; Gandhi's call: **"Do or Die"**.
45. **1942 (9 Aug)**: **Operation Zero Hour**: British arrest all top leaders; underground movement led by Aruna Asaf Ali, Jayaprakash Narayan, Ram Manohar Lohia, and Usha Mehta (Secret Congress Radio); parallel governments formed (**Ballia** under Chittu Pandey, **Tamluk** Jatiya Sarkar, **Satara** Prati Sarkar under Nana Patil).
46. **1944**: **C.R. Formula** (Rajagopalachari Formula for Congress-League cooperation).
47. **1945 (Jun)**: **Wavell Plan & Simla Conference** (Breakdown over League's demand to nominate all Muslim members).
48. **1945 (Nov)**: **INA Trials (Red Fort Trials)**: Bhulabhai Desai, Tej Bahadur Sapru, K.N. Katju, and J.L. Nehru defend Prem Sahgal, Gurbaksh Dhillon, and Shah Nawaz Khan.
49. **1946 (18 Feb)**: **Royal Indian Navy (RIN) Mutiny** on HMIS Talwar in Bombay.
50. **1946 (24 Mar)**: **Cabinet Mission** arrives (Lord Pethick-Lawrence, Sir Stafford Cripps, A.V. Alexander; rejected Pakistan; proposed Constituent Assembly).
51. **1946 (16 Aug)**: Muslim League launches **"Direct Action Day"** (Great Calcutta Killings).
52. **1946 (2 Sep)**: **Interim Government** sworn in under Jawaharlal Nehru (Muslim League joined later on 26 Oct).
53. **1946 (9 Dec)**: **First meeting of the Constituent Assembly** (Dr. Sachchidananda Sinha temporary president).
54. **1947 (20 Feb)**: Clement Attlee's declaration: British will leave India before **June 1948**.
55. **1947 (3 Jun)**: **Mountbatten Plan / June 3rd Plan** (Partition plan accepted by Congress and League).
56. **1947 (18 Jul)**: **Indian Independence Act 1947** receives British Royal Assent.
57. **1947 (15 Aug)**: India attains **Independence**.

12. Famous Books, Authors & Historiography

Book Title	Author	Key Historical Theme / Argument
***Poverty and Un-British Rule in India* (1901)**	**Dadabhai Naoroji**	Propounded the **"Drain of Wealth"** theory; calculated first per capita income (Rs 20)
***The Economic History of India* (2 Vols, 1902/04)**	**Romesh Chunder (R.C.) Dutt**	Classic critique of British colonial land revenue destruction and de-industrialization
***Anandamath* (1882)**	**Bankim Chandra Chattopadhyay**	Sanyasi Rebellion; contains national song **"Vande Mataram"**
***Hind Swaraj* (1909)**	**Mahatma Gandhi**	Written on board ship *Kildonan Castle*; critique of modern western civilization and machinery
The Story of My Experiments with Truth	**Mahatma Gandhi**	Autobiography (originally in Gujarati; translated by Mahadev Desai)
The Discovery of India* (1946)** & ***Glimpses of World History	**Jawaharlal Nehru**	Written in Ahmednagar Fort jail; philosophical and cultural journey of India
***India Wins Freedom* (1959)**	**Maulana Abul Kalam Azad**	Autobiography giving inside account of Partition negotiations
Gita Rahasya* / *The Arctic Home in the Vedas	**Bal Gangadhar Tilak**	Written in Mandalay jail (Burma); Karmayoga interpretation of Bhagavad Gita

Book Title	Author	Key Historical Theme / Argument
The Indian Struggle **(1920–1942)**	**Subhas Chandra Bose**	History of Indian freedom movement from a radical nationalist viewpoint
Annihilation of Caste **(1936)** & ***Who Were the Shudras?***	**Dr. B.R. Ambedkar**	Undelivered presidential address to Jat-Pat Todak Mandal; critique of Hindu varna system
Unhappy India **(1928)**	**Lala Lajpat Rai**	Written in reply to Katherine Mayo's derogatory book *Mother India*
India Divided **(1946)**	**Dr. Rajendra Prasad**	Exhaustive scholarly examination and refutation of the partition proposal
The Indian War of Independence 1857 **(1909)**	**Vinayak Damodar (V.D.) Savarkar**	First Indian to describe the 1857 revolt as the **"First War of Indian Independence"**
Indian Unrest **(1910)**	**Valentine Chirol**	British journalist who branded **Bal Gangadhar Tilak as the "Father of Indian Unrest"**
Gulamgiri **(1873)**	**Jyotirao Phule**	Dedicated to the American abolitionist movement against slavery
Satyarth Prakash **(1875)**	**Swami Dayananda Saraswati**	Foundational Hindi treatise of Arya Samaj ("Light of Truth")

13. Tribal & Peasant Movements Master Matrix (Chronological Ladder)

A. Tribal Uprisings (Direct Match & Chronology)

Tribal Movement / Revolt	Period / Year	Region / Hills	Key Leaders	Cause / Colonial Policy Opposed
Chuar / Paik Rebellion	**1766–1816**	Midnapore & Bankura (Bengal)	**Durjan Singh**, Jagannath Singh	Enhanced land revenue, famine, resumption of Paikan rent-free lands
Bhil Uprisings	**1818–1831**	Khandesh & Western Ghats (Maha/Gujarat)	**Sewaram**; later **Govind Guru** (Bhagat Movement at Mangarh)	British occupation of Khandesh, forest rights curtailment
Ramosi Uprising	**1822–1829**	Western Ghats (Satara, Maharashtra)	**Chittur Singh** & **Umaji Naik**	Heavy land revenue assessment and British annexation of Maratha lands
Ahom Revolt	**1828**	Assam	**Gomdhar Konwar** & Dhananjay Borgohain	Non-fulfillment of British pledge to vacate Assam after First Anglo-Burmese War
Khasi Uprising	**1829–1833**	Khasi & Jaintia Hills (Meghalaya)	**Tirot Singh** & Bar Manik	British construction of military road linking Brahmaputra & Surma valleys through Khasi territory
Kol Uprising	**1831–1832**	Chhotanagpur (Ranchi, Singhbhum, Hazaribagh)	**Buddho Bhagat**, Madara Mahato, Jhindrai Manki	Transfer of tribal lands to non-tribal outsider merchants and moneylenders (*Dikus*)
Khond Uprising	**1837–1856**	Ghumsur, Kalahandi, Chinna Kimedi (Odisha)	**Chakra Bisoi**	British banning of traditional human sacrifice ritual (**Meriah**) and high land revenue
Santhal Hool (Rebellion)	**1855–1856**	**Damin-i-Koh** (Rajmahal Hills, Bihar/Jharkhand)	**Sidhu, Kanhu, Chand, and Bhairav** (four brothers)	Oppression by *Dikus* (traders & moneylenders), police corruption; carved out Santhal Parganas district

Tribal Movement / Revolt	Period / Year	Region / Hills	Key Leaders	Cause / Colonial Policy Opposed
Munda Ulgulan (Great Tumult)	**1899–1900**	Chhotanagpur (Ranchi, Khunti, Jharkhand)	**Birsa Munda ("Dharti Aaba" — Father of Earth)**	Erosion of traditional joint landholding (**Khuntkatti system**) by landlords & missionaries; led to Chotanagpur Tenancy Act 1908
Tana Bhagat Movement	**1914**	Chhotanagpur (Jharkhand)	**Jatra Bhagat**, Turia Bhagat	Monotheistic reform movement; aligned with Gandhi's Non-Cooperation and non-violent satyagraha
Kuki Revolt	**1917–1919**	Manipur Hills	Kuki chiefs	British labour recruitment during WWI and interference in tribal self-governance
Rampa Rebellion (Manyam)	**1922–1924**	Rampa (Godavari Agency, Andhra Pradesh)	**Alluri Sitarama Raju ("Manyam Veerudu" — Hero of the Jungle)**	Protest against Madras Forest Act 1882 banning Podu (shifting cultivation); guerrilla warfare
Zeliangrong / Naga Movement	**1930s**	Manipur & Nagaland	**Haipou Jadonang** & **Rani Gaidinliu**	Heraka religious movement; opposed British rule and missionary conversion; Rani jailed until 1947

B. Peasant & Civil Movements (18th to 20th Century)

Movement / Struggle	Year / Period	Region / State	Key Leaders	Crucial Historical Fact / Outcome
Sanyasi & Fakir Rebellion	**1763–1800**	Bengal	**Majnum Shah, Chirag Ali, Bhavani Pathak, Devi Chaudhurani**	Triggered by famine of 1770 and harsh tax collection; inspired Bankim Chandra's ***Anandamath* and *Devi Chaudhurani***
Paika Rebellion	**1817**	Khurda (Odisha)	**Bakshi Jagabandhu** (military chief of King of Khurda)	Abolition of cowrie currency, rent-free service land confiscation, salt tax; recognized as early uprising
Wahabi Movement	**1820s–1870s**	Rae Bareli (UP) → Patna (Bihar)	**Syed Ahmad Barelvi**	Islamic revivalist anti-British movement; established operational base at Sittana (NW Frontier)
Faraizi Movement	**1838–1857**	Faridpur (East Bengal)	**Haji Shariatullah** & son **Dudu Miyan**	Peasant mobilization against British and Hindu zamindars; Dudu Miyan proclaimed land belongs to God
Indigo Revolt (Neel Bidroha)	**1859–1860**	Nadia & Jessore (Bengal)	**Digambar Biswas** & **Bishnu Charan Biswas**	Peasants refused to sow indigo under coercive Ryoti system; chronicled in Dinabandhu Mitra's play ***Neel Darpan***; backed by Harish Chandra Mukherjee's *Hindu Patriot*; Indigo Commission set up in 1860
Pabna Agrarian League	**1873–1876**	Pabna (East Bengal)	**Ishan Chandra Roy**, Sambhunath Pal, Khoodi Mollah	Peaceful legal struggle against illegal zamindari exactions; slogan: "*We want to be the ryots of Her Majesty the Queen*"; led to Bengal Tenancy Act 1885
Deccan Riots	**1875**	Pune, Ahmednagar (Maharashtra)	Local ryots	Social boycott & burning of debt-bonds against Gujarati & Marwari moneylenders; led to **Deccan Agriculturists' Relief Act 1879**

Movement / Struggle	Year / Period	Region / State	Key Leaders	Crucial Historical Fact / Outcome
Champaran Satyagraha	**1917**	Champaran (Bihar)	**Mahatma Gandhi**, Raj Kumar Shukla, Rajendra Prasad, J.B. Kripalani	First Civil Disobedience in India; protested against oppressive **Tinkathia system** (mandatory 3/20th indigo cultivation); Tinkathia abolished, 25% compensation refunded
Kheda Satyagraha	**1918**	Kheda (Gujarat)	**Mahatma Gandhi** & **Vallabhbhai Patel**	First Non-Cooperation; crop failure entitled peasants to remission under revenue code; revenue collection suspended
UP Kisan Sabha	**1918 (Feb)**	United Provinces (UP)	**Gauri Shankar Mishra, Indra Narayan Dwivedi**, supported by **Madan Mohan Malaviya**	Mobilized peasantry against arbitrary ejectments and illegal cesses (*Nazrana*)
Awadh Kisan Sabha	**1920 (Oct)**	Pratapgarh (UP)	**Baba Ramchandra**, Jawaharlal Nehru, Mata Badal Pande	Mass peasant gathering at Ayodhya; opposed Bedakhli (eviction) and forced labour
Eka Movement (Unity Movement)	**1921–1922**	Northern Awadh (Hardoi, Bahraich, Sitapur - UP)	**Madari Pasi**	Radical grassroots peasant movement; refused to pay more than recorded rent; resisted eviction
Moplah (Mapilla) Rebellion	**1921**	Malabar (Kerala)	**Variyankunnath Kunjahammed Haji**, Ali Musliyar	Muslim tenant cultivators revolted against oppressive upper-caste Hindu Janmis (landlords) and British police; merged with Khilafat; turned communal
Bardoli Satyagraha	**1928**	Surat (Gujarat)	**Vallabhbhai Patel**	Massive non-violent tax strike against arbitrary **22% enhancement** of land revenue; revenue reduced to 6.03%; **women of Bardoli gave Vallabhbhai the title "Sardar"**
All India Kisan Sabha (AIKS)	**April 1936**, Lucknow (UP)	**Swami Sahajanand Saraswati** (President), **N.G. Ranga** (General Secretary)	First national peasant federation; session held alongside Lucknow Congress; Kisan Manifesto drafted; Indulal Yagnik edited *Kisan Bulletin*	
Tebhaga Movement	**1946–1947**	North Bengal (Dinajpur, Rangpur)	**Kisan Sabha (Bengal)**, Charu Majumdar, Kansari Halder	Sharecroppers (*Bargadars*) demanded retention of **two-thirds (⅔rd)** of harvest instead of traditional half (½); based on Floud Commission recommendation
Telangana Peasant Struggle	**1946–1951**	Hyderabad State (Telangana)	Communist Party of India (Sundarayya, Ravi Narayan Reddy)	Armed peasant insurrection against feudal landlords (*Doras*) and the oppressive **Vetti** (forced labour) system of the Nizam and Razakars

14. Newspapers, Journals & Their Editors / Founders

Newspaper / Journal Title	Language	Year & Place	Founder / Chief Editor	Landmark Historical Significance
***Bengal Gazette* (*Hicky's Gazette*)**	English	**1780**, Calcutta	**James Augustus Hicky**	**FIRST NEWSPAPER PRINTED IN INDIA**; seized by British in 1782 for attacking Warren Hastings
Sambad Kaumudi	Bengali	**1821**, Calcutta	**Raja Rammohan Roy**	Campaigned actively for the abolition of Sati
Mirat-ul-Akhbar	**Persian**	**1822**, Calcutta	**Raja Rammohan Roy**	**First Persian journal in India**; shut down in protest against John Adam's Licensing Regulations 1823
Banga-Duta	Multi-lingual	1822, Calcutta	**Raja Rammohan Roy** & Dwarakanath Tagore	Published in 4 languages: English, Bengali, Persian, Hindi
***Rast Goftar* (*Truth Teller*)**	Gujarati	**1851**, Bombay	**Dadabhai Naoroji** & Naoroji Furdunji	Voice of the Parsi social and religious reform movement
Som Prakash	Bengali	**1858**, Calcutta	**Ishwar Chandra Vidyasagar** & Dwarkanath Vidyabhushan	Championed indigo cultivators; first newspaper targeted under Lytton's **Vernacular Press Act 1878**
The Hindu	English	**1878**, Madras	**G. Subramaniya Aiyer**, M. Veeraraghavachariar	Started as a weekly by the *Triplicane Six*; premier national daily of South India
Swadesamitran	Tamil	1882, Madras	**G. Subramaniya Aiyer**	First Tamil language daily newspaper in India
Kesari* & *The Mahratta	Marathi (*Kesari*) & English (*Mahratta*)	**1881**, Pune	**Bal Gangadhar Tilak** (initially Agarkar edited Kesari, Kelkar edited Mahratta)	Mouthpiece of militant nationalism; Tilak arrested for sedition in 1897 and 1908 based on Kesari articles
The Bengalee	English	1879, Calcutta	**Surendranath Banerjee**	Leading moderate organ; fiercely opposed Partition of Bengal 1905
Amrita Bazar Patrika	Bengali → English	**1868**, Jessore/Calcutta	**Sisir Kumar Ghosh** & **Motilal Ghosh**	**Overnight switched language from Bengali to English** to escape the Vernacular Press Act 1878!
***Sudharak* (*Reformer*)**	English & Marathi	1888, Pune	**Gopal Ganesh Agarkar** (assisted by G.K. Gokhale)	Radical social reform journal advocating rationalism and women's rights
Voice of India	English	1883, Bombay	**Dadabhai Naoroji**	Propounded economic drain of wealth
Bande Mataram	English	**1905**, Calcutta	**Bipin Chandra Pal** (founder), edited by **Sri Aurobindo Ghosh**	Revolutionary nationalist organ during the Swadeshi Movement
Yugantar	Bengali	**1906**, Calcutta	**Barindra Kumar Ghosh** & **Bhupendranath Dutta**	Revolutionary journal linked to Anushilan Samiti; advocated armed uprising
The Indian Sociologist	English	**1905**, London	**Shyamji Krishna Varma**	Revolutionary monthly from India House London

Newspaper / Journal Title	Language	Year & Place	Founder / Chief Editor	Landmark Historical Significance
***Bande Mataram* (Geneva/Paris)**	English	1909, Paris	**Madam Bhikaji Cama**	Published from exile; smuggled into India
Al-Hilal* & *Al-Balagh	Urdu	**1912 / 1915**, Calcutta	**Maulana Abul Kalam Azad**	Advocated Hindu-Muslim unity and composite nationalism; banned under Press Act 1914
***Comrade* (English) & *Hamdard* (Urdu)**	English & Urdu	1911, Calcutta / Delhi	**Maulana Mohammad Ali Jauhar**	Promoted pan-Islamic and anti-colonial causes during Khilafat movement
Pratap	Hindi	**1913**, Kanpur (UP)	**Ganesh Shankar Vidyarthi**	Famous revolutionary Hindi weekly; highlighted Bijolia peasant movement and freedom struggle
New India* & *Commonweal	English	**1914 / 1915**, Madras	**Annie Besant**	Mouthpieces of the **Home Rule League** movement
Young India* & *Navjivan	English & Gujarati	**1919**, Ahmedabad	**Mahatma Gandhi**	Primary journals during the Non-Cooperation and Khilafat era
Harijan (also *Harijan Bandhu, Harijan Sevak*)	English, Gujarati, Hindi	**1933**, Poona	**Mahatma Gandhi**	Launched after Poona Pact; dedicated to anti-untouchability work and Harijan upliftment
Indian Opinion	Multi-lingual	**1903**, South Africa	**Mahatma Gandhi** & Mansukhlal Nazar	Printed at Phoenix Settlement; organ of passive resistance in South Africa
The Independent	English	**1919**, Allahabad (UP)	**Motilal Nehru**	Founded to counter the moderate *Pioneer* of Allahabad
National Herald	English	**1938**, Lucknow (UP)	**Jawaharlal Nehru**	Leading nationalist daily of the United Provinces
***Mooknayak* (*Leader of the Voiceless*)**	Marathi	**1920**, Bombay	**Dr. B.R. Ambedkar** (aided by Shahu Maharaj of Kolhapur)	Fortnightly championing Dalit rights
Bahishkrit Bharat	Marathi	**1927**, Bombay	**Dr. B.R. Ambedkar**	Organ of the Bahishkrit Hitakarini Sabha; crusade for Mahad Satyagraha
Karmyogi	Hindi	Allahabad (UP)	**Pandit Sundarlal**	Nationalist publication in UP
***Kudi Arasu* (*Republic*)**	Tamil	**1925**, Erode	**E.V. Ramasamy Naicker (Periyar)**	Mouthpiece of the **Self-Respect Movement** in Tamil Nadu

SUBJECT 04 · ART & CULTURE

SUBJECT 04: INDIAN ART & CULTURE

भारतीय कला एवं संस्कृति — दैनिक उच्च-प्राथमिकता तथ्य

8 Classical Dances, Classical Languages, Temple Architecture (Nagara, Dravida, Vesara), UNESCO World Heritage Sites, GI Tags, Folk Paintings, and Cultural Institutions.

1. The 8 Classical Dances of India (Recognized by Sangeet Natak Akademi)

Dance Form	State of Origin	Key Pioneers & Revivalists	Notable Exponents	Distinctive Features, Repertoire & Exam Keywords
Bharatanatyam	**Tamil Nadu**	**E. Krishna Iyer**, **Rukmini Devi Arundale** (founder of Kalakshetra Foundation)	Yamini Krishnamurthy, Padma Subrahmanyam, Alarmel Valli, Mrinalini Sarabhai	Regarded as the **oldest classical dance** of India, originating from ancient temple dancer (*Devadasi*) traditions where it was historically known as *Sadir* or *Dasi Attam*. It symbolizes the element of fire (*Agni*) and features the characteristic half-sitting stance termed **Araimandi**. A complete traditional recital follows the structured sequence: *Alarippu* → *Jatiswaram* → *Shabdam* → *Varnam* → *Padam* → *Tillana*. It is performed in the *Ekaharya* format, wherein a single performer assumes and portrays multiple roles.
Kathak	**Uttar Pradesh**	**Lady Leela Sokhey** (Madame Menaka), **Nawab Wajid Ali Shah** (patron of Lucknow Gharana)	**Pt. Birju Maharaj**, Sitara Devi, Shambhu Maharaj, Lachhu Maharaj, Damayanti Joshi	The name derives from the ancient aphorism *"Katha Kahe So Kathak Kahave"* (one who recites stories is a Kathak). The dance is characterized by intricate and swift rhythmic footwork (**Tatkar**), dazzling pirouettes (**Chakkars**), expressive gait narratives (*Gat Nikas*), and dynamic rhythmic contests (*Jugalbandi*) with the tabla. Its major gharanas include **Lucknow** (famed for grace, *Bhava*, and subtle mime), **Jaipur** (distinguished by fiery footwork, speed, and *Veer rasa*), **Banaras**, and **Raigarh**.
Kathakali	**Kerala**	**Vallathol Narayana Menon** (founder of Kerala Kalamandalam)	Kalamandalam Gopi, Guru Kunchu Kurup, Raman Kutty Nair	A grand visual dance-drama enacting epic narratives from the Ramayana and Mahabharata. It is celebrated for its elaborate facial makeup (*Chutti*) and ornate towering headgear. Character types are defined by strict symbolic facial colors: **Pacha** (green makeup, depicting divine heroes and noble kings), **Kathi** (knife-shaped red streaks on green with a white nose-knob, representing arrogant royalty and villains), **Thadi** (bearded characters: red/black for evil demons, white for pious figures like Hanuman), and **Minukku** (warm yellow/flesh tone, representing virtuous women and sages).

Dance Form	State of Origin	Key Pioneers & Revivalists	Notable Exponents	Distinctive Features, Repertoire & Exam Keywords
Kuchipudi	**Andhra Pradesh**	**Siddhendra Yogi** (creator of *Bhamakalapam*)	Yamini Krishnamurthy, Raja & Radha Reddy, Swapnasundari	Named after its village of origin, Kuchelapuram in Krishna district, this form developed as a dance-drama tradition combining spoken dialogue, classical singing, and dance. Its most celebrated virtuoso feature is the **Tarangam**, in which the dancer deftly balances upon the raised rim of a brass plate while carrying a brass water pitcher (*Kundi*) balanced steadily atop the head.
Odissi	**Odisha**	**Kelucharan Mohapatra**, Pankaj Charan Das, Deba Prasad Das	Sonal Mansingh, Sanjukta Panigrahi, Madhavi Mudgal	Deeply rooted in temple rituals, its lineage traces back to female temple dancers (*Maharis*) and juvenile male performers (*Gotipuas*). The performance repertoire is heavily inspired by Jayadeva's devotional masterpiece, the *Gita Govinda*. It is distinguished by two fundamental body postures: the feminine, soft, three-bend deflective stance (**Tribhanga**) and the firm, square, masculine stance (**Chowk**).
Sattriya	**Assam**	**Mahapurusha Srimanta Sankardeva** (15th century)	Bhabananda Barbayan, Jatin Goswami, Ghanakanta Bora	Created as an instrument of the Neo-Vaishnavite Bhakti movement, this sacred dance form was nurtured for over five centuries inside monastic sanctuaries known as **Sattras**. Its musical and theatrical foundation rests upon *Borgeet* (devotional songs) and *Ankiya Nat* (one-act devotional plays). The Sangeet Natak Akademi officially recognized Sattriya as India's 8th classical dance in **November 2000**.
Manipuri	**Manipur**	**King Bhagyachandra** (Ningthou Ching-Thang Khomba), **Rabindranath Tagore** (introduced at Santiniketan)	Guru Bipin Singh, Jhaveri Sisters (Nayna, Suvarna, Darshana, Ranjana), Charu Mathur	Centered upon the immortal, tender love of Radha and Krishna through devotional **Raslila** performances. It is distinguished by soft, undulating, lyrical movements in which the dancer's feet never strike the floor sharply. Female dancers wear an ornate, stiff cylindrical skirt known as the **Potloi** or *Kumin*. Performances are traditionally complemented by the energetic, acrobatic drum dance called **Pung Cholom**.
Mohiniyattam	**Kerala**	**Maharaja Swathi Thirunal** of Travancore, **Vallathol Narayana Menon**	Sunanda Nair, Bharati Shivaji, Kanak Rele, Gopika Varma	Literally meaning the "Dance of the Enchantress", this dance draws mythological inspiration from Lord Vishnu's Mohini incarnation. It is performed strictly as a solo feminine recital embodying pure **Lasya** (delicate, wave-like, swaying body movements resembling palm fronds). Performers are clad in a signature pristine off-white handloom saree embellished with a shimmering golden brocade border, known as the **Kasavu saree**.

2. Classical Languages of India (Chronology & Criteria)

A. Chronological Order of Recognition

1. **Tamil** (Declared in **2004**) — Conferred status as India's first Classical Language.
2. **Sanskrit** (Declared in **2005**).
3. **Telugu** (Declared in **2008**).
4. **Kannada** (Declared in **2008**).
5. **Malayalam** (Declared in **2013**).

6. **Odia** (Declared in **2014**).
7. **Marathi** (Approved by the Union Cabinet on **3 October 2024**).
8. **Pali** (Approved on **3 October 2024**).
9. **Prakrit** (Approved on **3 October 2024**).
10. **Assamese** (Approved on **3 October 2024**).
11. **Bengali** (Approved on **3 October 2024**).

(India officially recognizes 11 Classical Languages following the landmark Union Cabinet decision of October 2024.)

B. Eligibility Criteria (Ministry of Culture)

To qualify for status as a Classical Language, a language must satisfy the following four criteria formulated by the Ministry of Culture: - **High Antiquity:** Recorded history and early literary texts spanning a historical antiquity of **1500 to 2000 years**. - **Valuable Heritage:** A treasured corpus of ancient literature and texts cherished as invaluable heritage by generations of speakers. - **Original Literary Tradition:** A literary heritage that is demonstrably authentic and original, not borrowed from another speech community. - **Linguistic Distinctiveness:** A distinct divergence between the classical language and its modern forms or vernacular offshoots, acknowledging potential historical discontinuity across later evolutionary stages.

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3. Temple Architecture Styles & Architectural Anatomy

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│                  NAGARA vs DRAVIDA ARCHITECTURE              │
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│ NAGARA (North Indian Style)      │ DRAVIDA (South Indian Style)  │
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│ • Curvilinear tower: Shikhara    │ • Stepped-pyramidal tower: Vimana  │
│ • Crowning Amalaka & Kalasha     │ • Crowning octagonal/dome Stupika  │
│ • Built on a raised plinth: Jagati│ • Enclosed by elaborate boundary wall│
│ • Monumental gateways absent     │ • Towering ornamental gates: Gopuram│
│ • Temple water tanks not mandatory│ • Temple water tank (Kalyani) inside│
│ • Sub-types: Rekha-Deula, Phamsana│ • Multi-pillared assembly Mandapas  │
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A. Core Architectural Vocabulary

- **Garbhagriha (Sanctum Sanctorum):** The innermost, unadorned sacred chamber housing the principal deity, designed to focus divine energy.
- **Mandapa:** A pillared hall situated in front of the sanctum, serving as a gathering space for devotees during rituals.
- **Antarala (Vestibule):** An intermediate foyer or transit vestibule connecting the Garbhagriha with the main Mandapa.
- **Shikhara (North) / Vimana (South):** The monumental mountain-like spire rising skyward directly over the Garbhagriha.
- **Amalaka:** A large, fluted circular stone disc resembling a myrobalan fruit, installed at the apex of a Nagara Shikhara.
- **Kalasha:** A pot- or pitcher-shaped decorative finial placed directly atop the Amalaka as the highest point of the temple.
- **Gopuram:** A towering, monumental gateway providing entrance through the perimeter walls of South Indian Dravidian temple complexes.
- **Panchayatana Style:** An architectural layout where the main central shrine is symmetrically flanked by **four smaller subsidiary shrines** at the corners (prominent examples include the Dashavatara Temple at Deogarh, and the Kandariya Mahadeva and Lakshmana temples at Khajuraho).

B. Masterpiece Temples, Dynasties & Architectural Highlights

Temple Name	Location	Builder Dynasty & Ruler	Architectural Style & Primary Exam Significance
Kailasa Temple (Cave 16)	Ellora (Maharashtra)	**Krishna I** (Rashtrakuta Dynasty, 8th century CE)	Celebrated as the **world's largest monolithic rock-cut monument**, excavated vertically from top to bottom out of a single volcanic basalt cliff face.

Temple Name	Location	Builder Dynasty & Ruler	Architectural Style & Primary Exam Significance
Shore Temple	Mamallapuram (Tamil Nadu)	**Narasimhavarman II Rajasimha** (Pallava Dynasty)	One of the earliest free-standing structural stone temples in Southern India, built on the edge of the Bay of Bengal; designated a UNESCO World Heritage Site.
Brihadeshwara (Rajarajeshwara)	Thanjavur (Tamil Nadu)	**Rajaraja I Chola** (consecrated in 1010 CE)	Represents the zenith of Dravida architecture, crowned by a massive 66-metre pyramidal Vimana and an 80-tonne monolithic granite Stupika; renowned for its shadow engineering.
Gangaikonda Cholapuram	Ariyalur (Tamil Nadu)	**Rajendra I Chola** (1035 CE)	Commissioned to commemorate Rajendra I's victorious military campaign to the holy River Ganga; features a graceful, concave-curving Vimana profile.
Sun Temple	Konark (Odisha)	**Narasimhadeva I** (Eastern Ganga Dynasty, 13th century CE)	Designed in the form of an immense stone chariot dedicated to the Sun God Surya, complete with 24 carved stone wheels and 7 galloping horses; historically termed the **"Black Pagoda"**.
Jagannath Temple	Puri (Odisha)	**Anantavarman Chodaganga** (Eastern Ganga Dynasty)	A quintessential monument of the Kalinga Nagara style (Rekha-Deula); celebrated for the world-famous Ratha Yatra and historically known as the **"White Pagoda"**.
Lingaraja Temple	Bhubaneswar (Odisha)	Somavamshi Dynasty (King Yayati I)	Regarded as the most complete and magnificent specimen of Kalinga architecture, showcasing all four classic halls: Deula, Jagamohana, Natamandapa, and Bhogamandapa.
Sun Temple	Modhera (Gujarat)	**Bhima I** (Solanki / Chaulukya Dynasty, 1026 CE)	Built in the ornate Maru-Gurjara style; renowned for its grand stepped water tank (**Surya Kund**) adorned with 108 miniature shrines along its terraces.
Kandariya Mahadeva Temple	Khajuraho (Madhya Pradesh)	**Dhangadeva / Vidyadhara** (Chandela Dynasty, 10th–11th century CE)	The largest and most elaborate Hindu temple at Khajuraho; erected on a Panchayatana plan with a soaring Shikhara accompanied by 84 subsidiary spires (*Urushringas*).
Dilwara Jain Temples	Mount Abu (Rajasthan)	**Vimal Shah** (Vimal Vasahi, 1031 CE) & **Tejpala / Vastupala** (Luna Vasahi, 1230 CE)	Outstanding examples of Solanki temple craft, internationally famous for their luminous, translucent white marble ceiling carvings and filigree stonework.
Hoysaleswara Temple	Halebidu (Karnataka)	King Vishnuvardhana (Hoysala Dynasty)	Carved from soft chloritic schist (soapstone) on a stellate (star-shaped) plinth, covered in intricate, continuous narrative friezes; inscribed as a UNESCO World Heritage Site in 2023.
Chennakeshava Temple	Belur (Karnataka)	King Vishnuvardhana (Hoysala Dynasty, 1117 CE)	Built to commemorate Hoysala military triumphs; celebrated for its star-shaped sanctum and masterfully sculpted *Madanika* bracket figures; inscribed as a UNESCO World Heritage Site in 2023.
Dashavatara Temple	Deogarh (Lalitpur district, UP)	Gupta Period (6th century CE)	Stands as the **earliest surviving Panchayatana stone temple in Northern India**, famous for its classical relief depiction of Sheshashayi Vishnu reclining on the serpent Shesha.
Bhitargaon Temple	Kanpur district (UP)	Gupta Period (5th century CE; Gupta imperial span c. 320–550 CE)	Recognized as the **oldest surviving terracotta and brick temple** in India featuring a genuine radiating structural arch.

Dynastic Spans for Key Temple Builders: Gupta Empire (c. 320–550 CE: Deogarh, Bhitargaon); Imperial Pallavas (c. 575–897 CE: Shore Temple under Rajasimha); Rashtrakutas (c. 753–973 CE: Kailasa Temple under Krishna I); Imperial Cholas (c. 850–1279 CE: Rajaraja I in 1010 CE, Rajendra I in 1035 CE); Chandelas of Jejakabhukti (c. 9th–13th century CE: Khajuraho); Eastern Gangas (c. 11th–15th century CE: Konark, Puri); Hoysalas (c. 11th–14th century CE: Belur in 1117 CE, Halebidu); Solankis / Chaulukyas of Gujarat (c. 10th–13th century CE: Modhera in 1026 CE).

4. Fairs and Festivals of India, Uttar Pradesh & Uttarakhand

A. Major National Cultural Festivals

- **Hornbill Festival (Nagaland):** Held annually from 1 to 10 December at the Naga Heritage Village in Kisama; acclaimed as the "Festival of Festivals", it showcases the collective traditions of all 16 recognized Naga tribes.
- **Bihu (Assam):** Celebrated across three distinct agricultural cycles: *Rongali (Bohag) Bihu* in spring (April) marking the Assamese New Year, *Kongali (Kati) Bihu* in autumn marking crop protection, and *Bhogali (Magh) Bihu* in winter marking the post-harvest feast.
- **Chapchar Kut (Mizoram):** Celebrated with great joy in spring after the completion of jhum (slash-and-burn) clearing operations; famously features the traditional *Cheraw* bamboo dance.
- **Losar (Ladakh, Sikkim, Arunachal Pradesh):** The traditional Buddhist and Tibetan New Year festival, observed with monastic mask dances (*Cham*) and elaborate symbolic offerings.
- **Pushkar Fair (Ajmer, Rajasthan):** One of the world's largest livestock and camel trading fairs, organized during Kartik Purnima along the holy banks of Lake Pushkar.
- **Thrissur Pooram (Kerala):** Staged at the Vadakkunnathan Temple in Thrissur; renowned for competitive temple displays featuring caparisoned elephants, dynamic parasol exchanges (*Kudamattam*), and majestic traditional percussion ensembles (*Panchavadyam*).
- **Wangala Festival (Garo Hills, Meghalaya):** Known as the "100 Drum Festival", this post-harvest celebration is observed by the Garo tribe in honor of Saljong, the Sun God of fertility.

B. Uttar Pradesh Fairs & Festivals (High-Yield State Core)

- **Kumbh Mela & Magh Mela (Prayagraj):** Organized at the sacred Triveni Sangam (confluence of the Ganga, Yamuna, and mythical Saraswati); the Maha Kumbh takes place every 12 years, with the annual Magh Mela held during the Hindu month of Magha.
- **Bateshwar Fair (Bateshwar, Agra):** A renowned historic cattle and livestock fair hosted along the Yamuna riverbank, famous for its picturesque riverside cluster of 101 Shiva temples.
- **Nauchandi Mela (Meerut):** A celebrated symbol of communal fraternity held after Holi, where pilgrims pay homage concurrently to the shrine of Nav Chandi Devi and the dargah of Sufi saint Hazrat Bale Miyan.
- **Shakumbhari Devi Mela (Saharanpur):** Organized twice every year during the auspicious Chaitra and Ashwin Navratris at the historic Shakumbhari Devi Peeth in the Shivalik foothills.
- **Devi Patan Mela (Tulsipur, Balrampur):** A major annual fair held at the sacred Devi Patan Shaktipeeth, deeply revered across the Indo-Nepal Terai region.
- **Dadri Mela (Ballia):** Regarded as India's second-largest cattle fair, historically established and named in commemoration of the revered sage Dardar Muni.
- **Govardhan Parikrama & Lathmar Holi (Mathura, Barsana & Nandgaon):** Centuries-old Braj traditions featuring circumambulation of the sacred Govardhan Hill and the joyous playful stick-and-shield Holi celebrations between the women of Barsana and the men of Nandgaon.
- **Taj Mahotsav (Agra):** A vibrant 10-day cultural extravaganza organized each February at Shilpgram near the Taj Mahal, highlighting Mughal crafts, culinary arts, and folk performances.
- **Sulh-i-Kul Festival (Agra):** Inspired by Emperor Akbar’s imperial principle of universal peace and religious tolerance (*Sulh-i-Kul*), this festival serves as a cornerstone of interfaith unity.

C. Uttarakhand Fairs & Festivals (UKPCS Priority)

- **Nanda Devi Raj Jat (Chamoli, Uttarakhand):** Organized **once every 12 years**, this epic 280-kilometre barefoot pilgrimage spans from Nauti village to the sacred lake of Homkund at the base of Mount Trishul, led by a sacred four-horned ram (**Chasingha Khadu**).
- **Ramman (Saloor-Dungra village, Joshimath, Chamoli):** A ritualistic religious festival and folk theatre dedicated to the tutelary deity **Bhumiyal Devta**; features masked dances utilizing traditional painted wooden masks and has been inscribed on UNESCO's Intangible Cultural Heritage list.
- **Bagwal Mela / Stone-Throwing Fair (Devidhura, Champawat):** Celebrated annually on Raksha Bandhan at the Maa Varahi Devi Temple, where warriors of four local clans (*Khams*) traditionally engage in stone-hurling ritual combat.
- **Purnagiri Mela (Tanakpur, Champawat):** Attracts hundreds of thousands of devotees during Chaitra Navratri to the Purnagiri Shaktipeeth perched above the Sharda (Kali) River.
- **Uttarayani Mela (Bageshwar):** Celebrated on Makar Sankranti at the holy confluence of the Saryu and Gomati rivers; carries immense modern historical importance as the site where registers of the oppressive Coolie-Begar forced labor system were dumped into the river in 1921.
- **Jaiti Mela (Almora):** A traditional cultural fair of the Kumaon region, featuring spirited performances of the circular *Jhora* dance and traditional folk ballads.

5. UNESCO Tangible World Heritage Sites in India (45 Total Sites)

India currently possesses 45 inscribed UNESCO World Heritage Sites following the historic inscription of Sarnath in July 2026.

- **The Sole Mixed Heritage Site: Khangchendzonga National Park** (Sikkim, inscribed in 2016 for combined natural and cultural criteria).
- **Latest Landmark Inscriptions:**
- **#45: Ancient Buddhist Site of Sarnath (Uttar Pradesh)** — Inscribed in **July 2026** at the 48th World Heritage Committee session in Busan. This serial cultural site encompasses the monumental **Chaukhandi Stupa** together with the excavated core remains including the Dhamek Stupa and Ashokan pillar sanctuary. It represents **Uttar Pradesh's 4th cultural World Heritage Site**.
- **#44: Maratha Military Landscapes of India** (Inscribed in **2025**; covers 12 historic fortifications: 11 in Maharashtra and Gingee Fort in Tamil Nadu).
- **#43: Moidams – Mound-Burial System of the Ahom Dynasty** (Charaideo district, Assam, inscribed in **2024**).
- **#42: Sacred Ensembles of the Hoysalas** (Karnataka — Belur, Halebidu, and Somnathpura, inscribed in **2023**).
- **#41: Santiniketan** (Birbhum district, West Bengal, inscribed in **2023**).
- **#40: Dholavira: A Harappan City** (Kutch district, Gujarat, inscribed in **2021**).
- **#39: Kakatiya Rudreshwara / Ramappa Temple** (Palampet, Mulugu district, Telangana, inscribed in **2021**).
- **Uttar Pradesh Tangible Cultural World Heritage Sites (4 Sites):**
- **Taj Mahal** (Agra — inscribed in 1983)
- **Agra Fort** (Agra — inscribed in 1983)
- **Fatehpur Sikri** (Agra — inscribed in 1986)
- **Ancient Buddhist Site of Sarnath** (Varanasi district — inscribed in **2026**)
- **Uttarakhand Tangible Natural World Heritage Site (1 Serial Property):**
- **Nanda Devi and Valley of Flowers National Parks** (Chamoli district, inscribed in 1988, expanded in 2005).

> **Important Distinction:** In examination questions, do not quote obsolete counts of 42 or 43 sites, nor limit Uttar Pradesh to only 3 tangible sites following the 2026 inscription of Sarnath. Furthermore, note that the inclusion of **Deepavali** (December 2025) belongs to UNESCO's **Intangible Cultural Heritage (ICH)** list, which is entirely distinct from physical World Heritage monuments.

6. Important Geographical Indications (GI Tags) of UP & UK

State	Product Name & Craft	Category	Distinctive Origin District & Significance
Uttar Pradesh	**Banaras Brocades and Sarees**	Handicraft	Varanasi; renowned globally for rich gold and silver (*Zari*) brocade handloom weaving.
	Bhadohi Handmade Carpet	Handicraft	Bhadohi (Mirzapur-Bhadohi carpet manufacturing belt); premier woollen hand-knotted carpets.
	Lucknow Chikan Craft	Handicraft	Lucknow; centuries-old delicate white-on-white hand embroidery patronized by Awadh Nawabs.
	Kannauj Perfume (Attar)	Manufactured	Kannauj; acclaimed as the "Perfume Capital of India", distilling natural floral attar in copper stills.
	Moradabad Metal Craft	Handicraft	Moradabad; widely celebrated as *Peetal Nagari* (Brass City) for ornate engraved brassware.
	Firozabad Glassware	Handicraft	Firozabad; known as *Suhag Nagari*, the historic hub of India's glass bangle and glassware industry.
	Aligarh Locks	Manufactured	Aligarh; traditionally designated as *Tala Nagari* (City of Locks) for precision brass and iron locks.
	Gorakhpur Terracotta	Handicraft	Gorakhpur; traditional reddish clay craftsmanship producing distinctive elephant and horse figurines.
	Kalanamak Rice	Agriculture	Siddhartha Nagar and surrounding Terai districts; aromatic black-husked rice associated with Lord Buddha.
	Mahoba Gaura Patthar Hastashilp	Handicraft	Mahoba (Bundelkhand); delicate carving in soft, radiant white pyrophyllite stone.
Uttarakhand	**Berinag Tea**	Agriculture	Pithoragarh district; exquisite, aromatic orthodox Himalayan tea made from wild and cultivated bush leaves.
	Almora Bal Mithai	Food Stuff	Almora; iconic roasted fudge-like sweet made from condensed milk (*khoya*), coated with white sugar globules.
	Uttarakhand Bhotia Dan	Handicraft	Pithoragarh and Chamoli; dense, hand-knotted traditional woollen floor rugs woven by Bhotia tribal women.
	Munsyari White Rajma	Agriculture	Munsyari (Pithoragarh); high-altitude organic kidney beans prized for delicate texture and swift cooking.
	Uttarakhand Ringal Craft	Handicraft	Chamoli and Garhwal hills; flexible Himalayan dwarf bamboo (*Ringal*) woven into baskets, mats, and wares.
	Uttarakhand Aipan Art	Handicraft	Kumaon region; sacred ritual geometric floor and wall art painted with white rice paste over red ochre.
	Uttarakhand Bichhu Buti (Nettle Fabric)	Textile	Chamoli district; eco-friendly natural fiber extracted from wild Himalayan stinging nettle (*Girardinia diversifolia*).
	Jhangora & Manduwa	Agriculture	Mountain districts of Uttarakhand; highly nutritious traditional hill millets (Barnyard Millet and Finger Millet).

7. Folk Paintings of India — State-Wise Master Guide

Traditional Painting Form	State / Region	Key Features, Media & Cultural Significance
Sanjhi	**Uttar Pradesh** (Mathura–Vrindavan / Braj)	An exquisite traditional stencilled **paper-cutting art** practiced along the Yamuna ghats of Mathura and Vrindavan. Traditionally fashioned during Krishna festivals and Pitru Paksha; recognized with an official GI tag.
Aipan	**Uttarakhand** (Kumaon)	A sacred ritual floor and wall art created using white rice-flour paste (*Biswar*) upon an ochre-red background (*Geru*), featuring complex geometric and floral protective motifs. Holds a GI tag.
Madhubani (Mithila Painting)	**Bihar** (Mithila region)	Practiced traditionally by women using natural pigments and brushes made of twigs and nibs; characterized by vivid double-line borders and stylized motifs of deities, nature, and the wedding chamber (*Kohbar*). Holds a GI tag.
Warli	**Maharashtra** (Thane / Palghar)	An ancient tribal art executed on mud-and-cowdung plastered walls using white pigment made of rice paste and water, depicting community dancing, harvesting, and reverence for the mother goddess *Palaghata*.
Pattachitra	**Odisha** (also practiced in West Bengal)	A traditional scroll painting executed on treated cloth (*Patta*) or dried palm leaves using natural mineral colors, illustrating episodes from the life of Lord Jagannath, Krishna, and Vaishnava legends.
Phad	**Rajasthan** (Bhilwara / Shahpura)	Immense narrative cloth scroll paintings depicting the heroic exploits of folk deities such as Pabuji and Devnarayan, traditionally recited and sung by traveling priest-bards (*Bhopas*).
Gond	**Madhya Pradesh** (Mandla / Dindori)	Celebrated tribal art of the Gond community, characterized by rich textures composed of fine dots, dashes, and rhythmic lines, depicting flora, fauna, and their supreme forest divinity *Bada Dev*.
Kalamkari	**Andhra Pradesh**	Traditional hand-painted and block-printed textile art using natural vegetable dyes and a sharpened bamboo reed (*kalam*). Maintained through two prominent traditions: **Srikalahasti** (freehand temple hangings depicting epics) and **Machilipatnam** (block-printed trade designs).
Pithora	**Gujarat & Madhya Pradesh** (Rathwa & Bhil tribes)	Sacred ritual wall paintings executed inside tribal homes to invoke prosperity and fulfill vows, centered on the consecrated depiction of horses honoring the deity *Pithora Baba*.
Kalighat	**West Bengal** (Kolkata)	A 19th-century urban folk painting style that developed around the Kalighat Kali Temple, renowned for sweeping brush lines and watercolor washes depicting Hindu divinities and satirical caricatures of the colonial elite.
Phulkari	**Punjab**	Traditional folk floral embroidery executed on coarse dyed cotton cloth (*khaddar*) using vibrant untwisted silk threads (*pat*), traditionally gifted during weddings.
Mandana	**Rajasthan & Malwa**	Auspicious geometric floor and wall designs drawn with white chalk (*khadiya*) against red clay plaster to welcome deities and mark domestic festivals.

> **Key Exam Distinction:** Do not confuse **Sanjhi** (indigenous to the Braj region of Uttar Pradesh), **Aipan** (traditional to Kumaon in Uttarakhand), and **Madhubani** (native to Bihar). Each holds a distinct geographical and cultural identity.

8. All 16 UNESCO Intangible Cultural Heritage (ICH) Elements of India

Inscribed on the Representative List of the Intangible Cultural Heritage of Humanity (Ministry of Culture & UNESCO):

#	Year Inscribed	Intangible Cultural Heritage Element	Region / Community	Primary Ritual & Cultural Significance
1	**2008**	**Koodiyattam (Kutiyattam)**	**Kerala**	India's oldest living Sanskrit temple theatre tradition, performed in consecrated theatre halls called *Koothambalams* with the accompaniment of copper *Mizhavu* drums.
2	**2008**	**Ramlila**	**Uttar Pradesh & North India**	The traditional dramatic enactment of the Ramayana during Dussehra, epitomized by the world-famous month-long open-air Ramnagar Ramlila of Varanasi supported by the Kashi Naresh.
3	**2008**	**Tradition of Vedic Chanting**	Pan-India	The uninterrupted oral recitation of sacred Vedic hymns handed down over millennia through sophisticated mnemonic recitation techniques such as *Jatapatha* and *Ghanapatha*.
4	**2009**	**Ramman**	**Saloor-Dungra village, Chamoli (Uttarakhand)**	A ritual festival and sacred masked theatre held in honor of the tutelary deity **Bhumiyal Devta**, where villagers perform epic episodes wearing traditional wooden masks.
5	**2010**	**Chhau Dance**	**Jharkhand, West Bengal, Odisha**	A semi-classical martial dance tradition that encompasses three distinct regional streams: **Seraikella** (Jharkhand, masked), **Purulia** (West Bengal, large theatrical masks), and **Mayurbhanj** (Odisha, performed without masks).
6	**2010**	**Kalbelia Folk Songs and Dances**	**Rajasthan**	A sensuous, serpent-like folk dance performed by women of the nomadic Kalbelia snake-charmer community, rhythmically driven by the *Poongi* (been) and *Dhaf*.
7	**2010**	**Mudiyettu**	**Kerala**	A sacred ritual dance-drama performed in Kali temples, dramatizing the mythological battle between the goddess Kali and the demon Darika; preceded by elaborate floor paintings called *Kalam*.
8	**2012**	**Buddhist Chanting of Ladakh**	**Ladakh (Trans-Himalaya)**	The solemn rhythmic chanting of sacred Buddhist texts and philosophical teachings by Mahayana monks (Lamas) of the Gelukpa, Drukpa, and Kagyu monastic lineages.
9	**2013**	**Sankirtana**	**Manipur**	A sacred ritual performance encompassing devotional singing, drumming (*Pung*), and graceful dance, performed by Meitei Vaishnavites to mark life-cycle ceremonies and religious festivals.
10	**2014**	**Traditional Brass and Copper Craft of Utensil Making**	**Jandiala Guru, Amritsar (Punjab)**	The artisanal metal-working technique of hammering heated brass and copper plates into household vessels, practiced traditionally by the hereditary **Thathera** community.
11	**2016**	**Yoga**	Pan-India (Global)	The ancient Indian physical, mental, and spiritual discipline founded upon Patanjali's Ashtanga Yoga; commemorated internationally each year on **21 June**.
12	**2016**	**Nawrouz (Nowruz)**	Parsi & Kashmiri communities	The ancient Persian New Year festival celebrated annually on the vernal equinox (**21 March**), signifying renewal, peace, and the rebirth of nature.
13	**2017**	**Kumbh Mela**	**Prayagraj (UP), Haridwar (UK), Ujjain (MP), Nashik (Maharashtra)**	The largest peaceful religious gathering of pilgrims on Earth, rotating across four sacred river sanctuaries (Ganga, Godavari, and Kshipra) over twelve-year cycles.

#	Year Inscribed	Intangible Cultural Heritage Element	Region / Community	Primary Ritual & Cultural Significance
14	**2021**	**Durga Puja in Kolkata**	**Kolkata (West Bengal)**	A monumental 10-day autumn festival celebrating the triumph of Goddess Durga over Mahishasura, renowned for extraordinary public art installations, community pandals, and syncretic celebrations.
15	**2023**	**Garba of Gujarat**	**Gujarat**	A devotional, circular community dance performed during the nine nights of Navratri around a perforated earthen lamp (*Garbha Deep*) or an image of Goddess Amba.
16	**2025 (Dec)**	**Deepavali (Festival of Lights)**	Pan-India	Inscribed at the 20th session of the Intergovernmental Committee in December 2025 as India's 16th ICH element, honoring the universal celebration of light over darkness and wisdom over ignorance.

9. Traditional Folk Theatres & Puppetry of India

A. Major Folk Theatres

Folk Theatre Form	State / Region	Key Features, Dramaturgy & Musical Accompaniment
Nautanki	**Uttar Pradesh**	Highly popular operatic folk musical theatre of Northern India; evolved two renowned styles: **Hathras** (emphasizing classical singing) and **Kanpur** (noted for crisp theatrical dialogue). Plays feature verses in *Doha, Chaubola*, and *Chhand* meters accompanied by the thunderous **Nakkara** (kettledrum).
Bhand Pather	**Jammu & Kashmir**	Traditional satirical folk theatre practiced by rural farming communities; blends witty social criticism, parodies, and slapstick comedy with music played on the *Surnai* (shehnai-like reed) and *Dhol*.
Swang / Saang	**Haryana & Western UP**	Dialogue-driven musical folk drama written in rhymed verse; pioneered by celebrated playwrights such as Pt. Deep Chand and Kishan Lal Bhat; staged on open platforms surrounded by spectators.
Rasleela & Ramlila	**Uttar Pradesh** (Mathura–Vrindavan & Ayodhya / Varanasi)	Devotional dance-dramas dramatizing the divine childhood pastimes of Lord Krishna in Braj and the heroic life journey of Lord Rama across Awadh and Varanasi.
Bhavai	**Gujarat & Rajasthan**	Fast-paced socio-religious folk theatre initiated by the 14th-century poet Asaita Thakar; features male actors balancing up to eight earthen pots on their heads while playing the long brass trumpet known as the **Bhungal**.
Jatra	**West Bengal & Odisha**	Open-air musical melodrama that emerged during the Bhakti movement under Chaitanya Mahaprabhu; characterized by grand musical prologues, impassioned dialogues, and high dramatic intensity.
Yakshagana	**Karnataka** (Coastal)	Spectacular traditional theatre based on episodes from the Hindu epics; distinguished by elaborate costumes, towering decorated headdresses (**Muguta**), vivid facial cosmetics, and high athletic leaps.
Therukoothu	**Tamil Nadu**	Traditional rural street theatre performed in village squares, historically centered on the annual festival worship of Draupadi Amman.
Koodiyattam	**Kerala**	Classical Sanskrit temple drama recognized as a UNESCO ICH masterpiece; features stylized hand gestures (*Mudras*) and highly sophisticated eye movements accompanied by *Mizhavu* drums.

Folk Theatre Form	State / Region	Key Features, Dramaturgy & Musical Accompaniment
Ankiya Nat / Bhaona	**Assam**	One-act Vaishnavite devotional plays created by **Srimanta Sankardeva**; composed in the literary language *Brajavali* and traditionally staged in monastic assembly halls (**Namghars**).
Tamasha	**Maharashtra**	Energetic, fast-paced secular folk theatre that evolved in military camps; combines satirical comedy, witty banter, and lively *Lavani* dance accompanied by the *Dholki*.
Dashavatar	**Konkan / Goa / Maharashtra**	Traditional temple folk theatre depicting the ten divine incarnations of Lord Vishnu, widely performed during local harvest festivities.

B. Traditional Indian Puppetry (Four Master Traditions)

Puppetry Form	Traditional Name	State of Origin	Key Construction & Manipulation Features
String Puppets	**Kathputli**	**Rajasthan**	Carved from a single block of mango wood and dressed in vibrant medieval Rajasthani garments; manipulated with strings, while the puppeteer produces sharp whistling voices using a reed whistle called a *Boli*.
	Kundhei	**Odisha**	Made of lightweight wood with long flowing skirts and no articulated legs; movements are closely attuned to the rhythms of Odissi music.
	Gombeyatta	**Karnataka**	Highly stylized wooden puppets modeled directly upon the elaborate costumes, headdresses, and characters of **Yakshagana** theatre.
	Bommalattam	**Tamil Nadu**	Combines mechanisms of **both rod and string puppetry**; represents the largest, heaviest, and most mechanically complex puppets in India, operated via a cloth ring worn on the puppeteer's head.
Shadow Puppets	**Tholu Bommalata**	**Andhra Pradesh**	Translucent leather puppets painted in glowing vegetable colors on both sides; projected against a backlit white cloth screen.
	Togalu Gombeyatta	**Karnataka**	Small, delicate leather shadow figures depicting epic scenes, operated behind an illuminated white screen.
	Ravanachhaya	**Odisha**	Made from opaque deerskin; **casts purely black-and-white shadows** on the screen without color projections and has no articulated joints, producing dramatic silhouetted forms.
Glove Puppets	**Pavakoothu**	**Kerala**	Worn directly over the puppeteer’s hand; intricately modeled with carved wooden heads and painted costumes inspired directly by **Kathakali** makeup and crowns.
Rod Puppets	**Putul Nach**	**West Bengal**	Heavy wooden dolls sculpted with shoulder joints; manipulated from beneath the stage by puppeteers holding sturdy control rods.
	Yampuri	**Bihar**	Carved entirely from single pieces of wood without joints; specifically dramatizes moral tales set in the court of Yama, the God of Death.

10. Martial Arts of India

Martial Art Form	State of Origin	Key Weapons, Principles & Distinctive Features
Kalaripayattu	**Kerala**	Celebrated as the **oldest living martial art in the world** (mythologically attributed to Sage Parashurama); practiced within a sacred subterranean earthen arena (*Kalari*). Employs strikes, grappling, and traditional weapons including the deadly flexible whip-sword (**Urumi**), staff, and buckler.

Martial Art Form	State of Origin	Key Weapons, Principles & Distinctive Features
Silambam	**Tamil Nadu**	An ancient staff-fencing martial art referenced in Sangam literature such as the *Silappadikaram*; utilizes a 1.68-metre rapid-spinning **bamboo staff** (*Silambam*) to deflect attacks and strike vulnerable targets.
Thang-Ta & Sarit Sarak	**Manipur**	The traditional combat system of Meitei warriors; **Thang-Ta** emphasizes armed warfare (**Thang** = sword, **Ta** = spear), while **Sarit Sarak** comprises unarmed hand-to-hand combat and evasion.
Gatka	**Punjab**	Traditional Sikh martial art codified under the patronage of the sixth Sikh Guru, **Guru Hargobind Ji**; practitioners utilize wooden training swords (*Gatka*) and leather target shields, prominently demonstrated during the Hola Mohalla festival.
Pari-Khanda	**Bihar**	Created historically by Rajput warriors; utilizes the shield (**Pari**) and two-edged sword (**Khanda**); its footwork patterns and athletic steps form the foundational physical vocabulary of **Chhau dance**.
Thoda	**Himachal Pradesh**	A traditional martial sport rooted in **archery** that traces its lineage to the Mahabharata; practiced during Baisakhi in the Kullu and Shimla hills, where competing archers aim blunt arrows strictly at their opponents' legs below the knees.
Mallakhamb	**Madhya Pradesh & Maharashtra**	An extraordinary traditional gymnastic martial discipline in which athletes perform dynamic aerial yoga postures and acrobatic maneuvers on a vertical polished teak pole or a hanging cotton rope; declared the **State Sport of Madhya Pradesh**.
Musti-Yuddha	**Varanasi (Uttar Pradesh)**	Ancient traditional unarmed Indian pugilism and bare-knuckle boxing utilizing punches, elbow strikes, and grappling maneuvers.
Mardani Khel	**Maharashtra** (Kolhapur)	A historic Maratha armed martial art emphasizing agility, swordplay with the flexible gauntlet-sword (**Patta**), and defense using the bamboo staff (*Kathi*).
Kuttu Varisai	**Tamil Nadu**	Traditional Dravidian unarmed martial art focusing on rapid footwork, devastating animal-style strikes, and knowledge of vital pressure points (**Varma Kalai**).

11. Hindustani Classical Music Gharanas & Leading Exponents

Gharana	Founders & Historical Pioneers	Key Stylistic Characteristics & Aesthetics	Renowned Maestros & Legendary Exponents
Gwalior Gharana	**Nathan Pir Bakhsh**, **Hassu Khan**, **Haddu Khan**	Acknowledged as the **oldest and foundational Khayal gharana**; characterized by an unadorned, noble, open-throated singing style (*Aakaar*), balanced exposition of the raga, and systematic note expansion.	Vishnu Digambar Paluskar, Pt. Omkarnath Thakur, D.V. Paluskar, Malini Rajurkar, Neela Bhagwat
Agra Gharana	**Haji Sujan Khan**, **Khuda Bakhsh**; revitalized by **Ustad Faiyaz Khan**	Famously known as the **"Rangeela Gharana"**; marked by a robust, deep, resonant voice culture that masterfully infuses classical Khayal with ancient Dhrupad-Dhamar *Nom-Tom* alap elements.	Ustad Faiyaz Khan, Vilayat Hussain Khan, Shrikrishna Narayan Ratanjankar, Khadim Hussain Khan
Kirana Gharana	**Ustad Abdul Karim Khan**, **Ustad Abdul Wahid Khan** (originating from Kirana, Shamli district, UP)	Distinguished by deep emotional resonance, supreme tonal purity (*Sur*), and leisurely, contemplative note-by-note melodic evolution (**Vilambit Khayal**).	**Pt. Bhimsen Joshi**, Gangubai Hangal, Sawai Gandharva, Prabha Atre, Hirabai Barodekar

Gharana	Founders & Historical Pioneers	Key Stylistic Characteristics & Aesthetics	Renowned Maestros & Legendary Exponents
Jaipur-Atrauli Gharana	**Ustad Alladiya Khan**	Celebrated for complex, rare, and compound ragas (*Jod Ragas*); distinguished by intricate, spiral rhythmic phrasing (*Taans*) executed in an offbeat, asymmetrical swing across challenging talas.	**Kishori Amonkar**, Kesarbai Kerkar, Mogubai Kurdikar, Mallikarjun Mansur, Shruti Sadolikar
Patiala Gharana	**Ustad Fateh Ali Khan** and **Ustad Ali Bakhsh Khan** ("Alia-Fattu")	Known for dynamic, lightning-fast cascade passages (*Taans*), expressive emotional embellishments, and seamless mastery over both classical Khayal and romantic semiclassical **Thumri**.	**Ustad Bade Ghulam Ali Khan**, Begum Akhtar, Parveen Sultana, Jagjit Singh, Ajoy Chakrabarty
Banaras Gharana (Vocal & Instrumental)	**Pt. Ram Sahai** (Tabla pioneer), **Thakur Prasad** (Kathak pioneer)	Celebrated worldwide for its unparalleled mastery over the **Tabla**, **Sarangi**, and **Shehnai**, as well as soul-stirring semiclassical vocal forms including **Thumri, Dadra, Chaiti, and Kajri**.	**Ustad Bismillah Khan** (Shehnai — Bharat Ratna), **Pt. Kishan Maharaj** (Tabla), **Pt. Samta Prasad / Gudai Maharaj** (Tabla), **Girija Devi** (Queen of Thumri), Siddheshwari Devi, Rajan & Sajan Mishra

SUBJECT 05 · GEOGRAPHY

SUBJECT 05: GEOGRAPHY (INDIA & WORLD)

भारत एवं विश्व का भूगोल — दैनिक उच्च-प्राथमिकता तथ्य

Mountain Passes, Highest Peaks, River Drainage & Dams, Major Crops & Producing States, Tribes of India, Minerals & Mining, Ocean Currents, Global Straits, and Planetary Winds.

1. Indian Mountain Passes (State-Wise Cheat-Sheet)

State / UT	Mountain Pass	Connects / Geographic Significance
Ladakh	**Karakoram Pass**	Ancient caravan route between India and Yarkand (Tarim Basin, China)
	Zoji La	Connects **Srinagar with Kargil and Leh**; Great Himalayas; Zojila Tunnel
	Khardung La	Gateway to the Shyok and Nubra Valleys
	Chang La	Connects Leh to Pangong Tso lake
	Fotu La	Highest point on the Srinagar-Leh highway (Zaskar Range)
	Umling La	**World's highest motorable pass/road** (19,024 ft) built by BRO
Jammu & Kashmir	**Banihal Pass**	Connects **Jammu with Srinagar**; Jawahar Tunnel under Pir Panjal Range
	Pir Panjal Pass	Traditional route from Jammu to Srinagar valley (*Peer Ki Gali*)
	Burzil Pass	Connects Srinagar with Gilgit; ancient gateway to Central Asia
Himachal Pradesh	**Rohtang Pass**	Connects Kullu Valley with Lahaul and Spiti Valleys; **Atal Tunnel** (9.02 km) under Rohtang
	Shipki La	**Sutlej River enters India** from Tibet through this pass; Zaskar Range
	Bara-lacha La	Connects **Mandi/Manali to Leh** (Zaskar Range); origin of Chandra & Bhaga rivers
	Debsa Pass	Connects Spiti and Parvati Valleys
Uttarakhand (UKPCS Top Priority!)	**Lipulekh Pass**	Tri-junction of **India, Nepal, and Tibet**; primary route for **Kailash Mansarovar Yatra** (Pithoragarh)
	Mana Pass (Dungri La)	Connects Uttarakhand with Tibet; north of Badrinath (Chamoli)
	Niti Pass	Connects Chamoli (Uttarakhand) with Tibet
	Muling La	High mountain pass in Uttarkashi north of Gangotri connecting with Tibet
	Traill's Pass	Connects **Pithoragarh with Bageshwar** (Pindari Glacier to Milam Valley)
	Sin La	Connects Darma and Byans valleys in Pithoragarh
Sikkim	**Nathu La**	Strategic border post on India-China border; Old Silk Route branch into Chumbi Valley; reopened for trade in 2006
	Jelep La	Connects **Sikkim with Lhasa (Tibet)** passing through the Chumbi Valley
Arunachal Pradesh	**Bom Di La**	Connects Arunachal Pradesh (Kameng) with Lhasa (Tibet)

State / UT	Mountain Pass	Connects / Geographic Significance
	Diphu Pass	Strategic pass at the **tri-junction of India, China, and Myanmar**
	Bum La	High mountain pass near Tawang on India-China McMahon line
	Dihang Pass	Connects Arunachal Pradesh with Myanmar
	Pangsau Pass	Historic Ledo Road (Stilwell Road) connecting Assam to Myanmar

Western Ghats Gaps (Strict North-to-South Chronology)

1. **Thalghat (Kasara Ghat)**: Connects **Mumbai to Nashik / Nagpur / North India** (National Highway 160 & railway line).
2. **Bhorghat**: Connects **Mumbai to Pune / Bengaluru / South India** (Mumbai-Pune Expressway).
3. **Palghat (Palakkad Gap)**: Located between **Nilgiri Hills (North) and Anaimalai Hills (South)**; connects **Coimbatore (Tamil Nadu) to Palakkad / Kochi (Kerala)**.
4. **Shenkottai Pass**: Located between **Cardamom Hills and Agasthyamalai Hills**; connects **Kollam (Kerala) to Madurai / Tenkasi (Tamil Nadu)**.

2. Highest Peaks of Major Ranges in India

Mountain Range / Region	Highest Peak	Elevation (m)	Location / Crucial Takeaway
Trans-Himalayas (Karakoram)	**K2 (Godwin-Austen)**	**8,611 m**	**Second highest peak in the world**; highest point in Indian territory (POK)
Great Himalayas (India)	**Kanchenjunga**	**8,586 m**	**Highest peak in undisputed Indian territory**; located on Sikkim-Nepal border
Garhwal Himalayas (Uttarakhand)	**Nanda Devi**	**7,816 m**	**Highest peak located ENTIRELY within India**; Chamoli, UNESCO WHS
Garhwal (Second Highest)	**Kamet**	**7,756 m**	Chamoli district, Uttarakhand
Aravalli Range	**Guru Shikhar**	**1,722 m**	Mount Abu (Sirohi district, Rajasthan); oldest fold mountains in India
Satpura Range	**Dhupgarh**	**1,350 m**	Near Pachmarhi (Hoshangabad/Narmadapuram, Madhya Pradesh)
Vindhya Range	**Sadbhavna Shikhar (Kalumar)**	**752 m**	Damoh district, Madhya Pradesh
Western Ghats (Sahyadri)	**Anamudi**	**2,695 m**	Anaimalai Hills (Idukki, Kerala) — **HIGHEST PEAK OF PENINSULAR / SOUTH INDIA**
Nilgiri Hills	**Doda Betta**	**2,637 m**	Udhagamandalam / Ooty (Tamil Nadu); meeting point of Western & Eastern Ghats
Eastern Ghats	**Jindhagada Peak / Arma Konda**	**1,690 m**	Araku Valley (Visakhapatnam, Andhra Pradesh); Mahendragiri (1,501 m, Odisha) often given in older PCS options
Chhotanagpur Plateau	**Parasnath Peak (Sammed Shikhar)**	**1,366 m**	Giridih (Jharkhand); sacred Jain pilgrimage site (20 Tirthankaras attained Moksha)
Andaman & Nicobar Islands	**Saddle Peak**	**732 m**	North Andaman Island (Saddle Peak National Park)
Great Nicobar Island	**Mount Thullier**	**642 m**	Southernmost peak in India
Naga Hills (Purvanchal)	**Saramati Peak**	**3,826 m**	Indo-Myanmar border, Nagaland

Mountain Range / Region	Highest Peak	Elevation (m)	Location / Crucial Takeaway
Mizo Hills (Lushai Hills)	**Phawngpui (Blue Mountain)**	**2,157 m**	Mizoram

3. Panch Prayag & Indian River Drainage System

A. The Panch Prayag of Uttarakhand (Order along Alaknanda River)

Memorize downstream order from source towards plains: 1. **Vishnuprayag**: **Alaknanda** + **Dhauliganga** (Chamoli) 2. **Nandaprayag**: **Alaknanda** + **Nandakini** (Chamoli) 3. **Karnaprayag**: **Alaknanda** + **Pindar** (Chamoli) 4. **Rudraprayag**: **Alaknanda** + **Mandakini** (Rudraprayag) 5. **Devprayag**: **Alaknanda** + **Bhagirathi** → Official **GANGA** begins here! (Tehri Garhwal)

B. Ganga River Tributaries (Left vs Right Bank)

- **Left Bank Tributaries (West to East)**:
- **Ramganga** (originates in Dudhatoli range, Pauri Garhwal; flows through Jim Corbett NP; meets Ganga at Kannauj)
- **Gomati** (originates in plains at **Gomat Taal / Fulhar Lake, Pilibhit, UP** — *only Himalayan/Ganga tributary originating in the plains!*; Lucknow & Jaunpur on its banks; joins Ganga near Saidpur, Ghazipur)
- **Ghaghra / Karnali** (originates from Mapchachungo glacier near Mansarovar; joined by Sarda/Kali; **largest tributary by water volume**; joins Ganga at Chhapra, Bihar)
- **Gandak** (originates in Nepal Himalayas; joins Ganga at Sonpur / Hajipur, Bihar)
- **Kosi** (*Sorrow of Bihar*; 7 streams / Saptakoshi; notorious for course-shifting; joins at Kursela, Katihar)
- **Mahananda** (originates in Darjeeling hills; **last left-bank tributary of Ganga in India**).
- **Right Bank Tributaries**:
- **Yamuna** (longest tributary, 1376 km; Yamunotri glacier, Bandarpunch peak; joins at Prayagraj Triveni Sangam; its tributaries: Chambal, Sind, Betwa, Ken, Tons, Hindon)
- **Son** (originates from **Amarkantak Plateau, MP**; flows north-eastwards through Sonbhadra UP; joins Ganga near Danapur/Patna, Bihar)
- **Punpun** (originates in Palamu, Jharkhand; joins Ganga at Fatuha near Patna).

C. Peninsular Drainage: West-Flowing vs East-Flowing

- **West-Flowing (into Arabian Sea — form Estuaries, NO deltas)**:
- **Narmada** (1312 km): Amarkantak plateau (MP); flows through **Rift Valley** between Vindhyas (North) and Satpuras (South); forms **Dhuandhar Falls** at Marble Rocks (Bhedaghat, Jabalpur); Sardar Sarovar Dam.
- **Tapi / Tapti** (724 km): Multai (Betul district, MP); flows in rift valley south of Satpuras; Ukai Dam & Kakrapar Project (Gujarat).
- **Sabarmati**: Dhebar Lake (Aravalli Range, Udaipur); flows through Ahmedabad and Gandhinagar into Gulf of Khambhat.
- **Mahi**: Vindhyan hills (Dhar, MP); **CUTS THE TROPIC OF CANCER TWICE!**
- **Luni**: Originates near Ajmer (Aravallis) as Sagarmati; **ends in the marshy Rann of Kutch (Inland drainage, never reaches the sea)**; sweet water up to Balotra, then saline.
- **Sharavati**: Shimoga (Karnataka); forms famous **Jog Falls (Gersoppa / Mahatma Gandhi Falls)**.
- **Periyar & Bharathappuzha (Ponnani)**: Kerala.
- **East-Flowing (into Bay of Bengal — form large Deltas)**:
- **Mahanadi** (851 km): Sihawa hills (Dhamtari, Chhattisgarh); flows through Odisha; Hirakud Dam.
- **Godavari** (1465 km — **Dakshin Ganga / Vriddha Ganga**): Trimbakeshwar (Nashik, Maharashtra); **longest river in Peninsular India**; Tributaries: Pravara, Manjra, Penganga, Wainganga, Wardha, **Pranhita** (largest), Indravati, Sabari.
- **Krishna** (1401 km): Mahabaleshwar (Maharashtra); Tributaries: Koyna, Ghataprabha, Malaprabha, Bhima, **Tungabhadra** (largest), Musi (Hyderabad on its banks), Munneru.

- **Kaveri** (800 km — **Ganga of the South**): Talakaveri (Brahmagiri Range, Coorg, Karnataka); flows year-round (perennial due to SW & NE monsoons); creates river islands: **Srirangapatna, Shivanasamudra, and Srirangam**; Tributaries: Hemavati, Harangi, Shimsha, Arkavathi (left); Lakshmantirtha, Kabini, Bhavani, Noyyal, Amaravati (right).

4. Multipurpose River Valley Projects, Dams & Waterfalls

Project / Dam / Waterfall	River	State	Exam Crucial Fact
Tehri Dam	**Bhagirathi + Bhilangana**	**Uttarakhand**	**Highest dam in India** (260.5 m); rock and earth-fill embankment
Bhakra Nangal Dam	**Sutlej**	Himachal Pradesh / Punjab	Highest straight gravity dam in India; reservoir is **Gobind Sagar**
Hirakud Dam	**Mahanadi**	Odisha (Sambalpur)	**Longest earthen dam in India / world** (main dam 4.8 km, total 25.8 km)
Sardar Sarovar Dam	**Narmada**	Gujarat (Navagam)	Beneficiary states: Gujarat, MP, Maharashtra, Rajasthan; Statue of Unity nearby
Nagarjuna Sagar Dam	**Krishna**	Andhra Pradesh / Telangana border	World's largest masonry dam; creates Nagarjunasagar reservoir
Rihand Project (Govind Ballabh Pant Sagar)	**Rihand** (tributary of Son)	**Sonbhadra (Uttar Pradesh)**	**Largest man-made artificial reservoir / lake in India by volume**
Mata Tila & Rajghat Dam	**Betwa**	Lalitpur (Uttar Pradesh / MP)	Joint venture between UP and MP
Kunchikal Falls	**Varahi**	Karnataka (Shimoga)	Highest tiered waterfall in India (455 m)
Jog Falls (Gersoppa)	**Sharavati**	Karnataka (Shimoga)	Highest untiered plunge waterfall (253 m)
Dhuandhar Falls	**Narmada**	Bhedaghat, Jabalpur (MP)	"Smoke cascade" cutting through marble gorge
Chitrakote Falls	**Indravati**	Bastar (Chhattisgarh)	Known as the **"Niagara of India"** (widest horseshoe waterfall in India)
Hundru Falls	**Subarnarekha**	Ranchi (Jharkhand)	98 m drop on Chotanagpur plateau
Nohkalikai Falls	Rain-fed stream	Cherrapunji (Meghalaya)	Tallest plunge waterfall in India (340 m)

5. Major Agricultural Crops & Top Producing States

Crop Category	Specific Crop	Top Producing State (1st Rank)	Other Key States / Prelims Note
Foodgrains	**Total Foodgrains**	**Uttar Pradesh**	Followed by Madhya Pradesh, Punjab
	Wheat	**Uttar Pradesh**	Followed by MP, Punjab (Punjab highest per hectare yield)
	Rice	**West Bengal**	Followed by Uttar Pradesh, Punjab
	Maize	**Karnataka**	Followed by Madhya Pradesh, Maharashtra
	Coarse Cereals (Nutri-cereals)	**Rajasthan**	Followed by Karnataka, Maharashtra
	Total Pulses	**Madhya Pradesh**	Followed by Maharashtra, Rajasthan
	Gram (Chana)	**Maharashtra / MP**	Accounts for ~40% of total pulses
Oilseeds	**Total Oilseeds**	**Rajasthan**	Followed by Maharashtra, Madhya Pradesh

Crop Category	Specific Crop	Top Producing State (1st Rank)	Other Key States / Prelims Note
	Rapeseed & Mustard	**Rajasthan**	(>45% of national production)
	Soyabean	**Maharashtra / MP**	MP traditionally called *Soyabean State*
	Groundnut	**Gujarat**	Followed by Rajasthan, Tamil Nadu
Cash / Commercial	**Sugarcane**	**Uttar Pradesh**	Followed by Maharashtra, Karnataka (Maha highest sugar recovery)
	Cotton	**Gujarat**	Followed by Maharashtra, Telangana
	Jute & Mesta	**West Bengal**	(>75% of national production - Golden Fibre)
	Tea	**Assam**	Followed by West Bengal (Darjeeling - 1st GI tag 2004), Tamil Nadu
	Coffee	**Karnataka**	(>70% of national production - Arabica & Robusta in Kodagu/Chikkamagaluru)
	Natural Rubber	**Kerala**	(>75% of national production)
	Tobacco	**Gujarat / Andhra Pradesh**	CTRI located at Rajahmundry (AP)

6. Major Tribes of India & State-Wise Matrix (Direct Match-Ups)

State / UT	Prominent Indigenous Tribes	Key Ethnographic / Exam Fact
Uttarakhand (UKPCS Top Priority!)	**Tharu, Jaunsari, Bhotia, Buksa, Raji (Van Rawat)**	**Tharu**: Largest tribe in UK & UP; historically celebrate Diwali as mourning (*Shok Parva*); live in Terai (US Nagar). **Jaunsari**: Second largest; polyandry tradition; Dehradun (Chakrata). **Bhotia**: Transhumance (*Ritu Pravas*); seasonal migration to bugyals; Indo-Tibetan trade. **Buksa**: Claim Rajput descent (Pawar); Terai region. **Raji**: Smallest tribe (*Van Rawat* / kings of forest); cave dwellings.
Uttar Pradesh	**Tharu, Gond, Kharwar, Sahariya, Parahiya, Baiga, Agariya, Chero**	**Sonbhadra** has highest ST population & diversity in UP; Sahariya in Lalitpur; Agariya (iron smelters) in Sonbhadra.
Madhya Pradesh	**Bhil, Gond, Baiga, Sahariya, Kol, Korku, Bharia**	**Highest ST population in India**; Bhils largest tribe of India; *Bhagoria* festival; *Pithora* painting.
Rajasthan	**Bhil, Meena, Garasia, Sahariya, Damor**	Meenas are the most populous tribe in Rajasthan; Sahariya in Baran.
Odisha	**Santhal, Bonda, Dongria Kondh, Juang, Saora**	Largest number of **Particularly Vulnerable Tribal Groups (PVTGs: 13 out of 75 in India)**; Dongria Kondh worship Niyamgiri hills.
Tamil Nadu	**Toda, Badaga, Kota, Irula, Kurumba**	**Toda**: Inhabit Nilgiri Hills; pastoral tribe centered on water buffaloes; practice fraternal polyandry; barrel-shaped huts.
Andhra Pradesh / Telangana	**Chenchu, Yanadi, Koya, Lambada**	Chenchu: Primitive hunter-gatherers in Nallamala forest / Srisailam.
Sikkim	**Lepcha, Bhutia, Limbu**	Lepchas are indigenous inhabitants; practice sacred groves.
Himachal Pradesh	**Gaddi, Gujjar, Kinnaura, Pangwala**	Gaddis: Pastoral transhumance shepherds of Dhauladhar range.

State / UT	Prominent Indigenous Tribes	Key Ethnographic / Exam Fact
Andaman & Nicobar Islands	**Great Andamanese, Onge, Jarawa, Sentinelese, Shompen, Nicobarese**	Negrito stock (Jarawa, Sentinelese, Onge - North Sentinel isolated); Shompen are Mongoloid stock.

7. Major Sea Ports & National Waterways of India

A. 13 Major Ports of India

- **Western Coast**:
- **Deendayal Port (Kandla)** (Gulf of Kutch, Gujarat): Tidal port; free trade zone (SEZ); handles crude oil & grains.
- **Mumbai Port** (Maharashtra): Largest natural port and harbour in India.
- **Jawaharlal Nehru Port (JNPT / Nhava Sheva)** (Navi Mumbai): **Largest container port in India**; fully mechanized.
- **Mormugao Port** (Goa - Zuari estuary): Premier port for iron ore exports from Goa.
- **New Mangalore Port** (Karnataka): Handles iron ore exports from Kudremukh.
- **Cochin Port** (Kerala - Willingdon Island in Vembanad Lake): Natural harbour; International Container Transshipment Terminal (Vallarpadam).
- **Eastern Coast**:
- **V.O. Chidambaranar Port (Tuticorin)** (Tamil Nadu - Gulf of Mannar): Artificial deep-sea port; pearl fishery.
- **Chennai Port** (Tamil Nadu): **Oldest artificial harbour on the east coast**; second busiest.
- **Kamarajar Port (Ennore)** (Tamil Nadu): **First corporatized port in India** (public company).
- **Visakhapatnam Port** (Andhra Pradesh): **Deepest landlocked and protected port in India** (Dolphin's Nose hill protects from storms).
- **Paradip Port** (Odisha - Mahanadi delta): Deep-water port exporting iron ore to Japan.
- **Syama Prasad Mookerjee Port (Kolkata–Haldia)** (West Bengal): **Only riverine major port in India** (on Hooghly River); twin dock system (Kolkata & Haldia).
- **Island**:
- **Port Blair Port** (Andaman & Nicobar Islands).

B. Core National Waterways (NW) of India

- **NW-1**: **Ganga–Bhagirathi–Hooghly River System** from **Prayagraj to Haldia** (1,620 km) — **Longest National Waterway in India**; passes through UP, Bihar, Jharkhand, WB.
- **NW-2**: **Brahmaputra River** from **Sadiya to Dhubri** (891 km) in Assam.
- **NW-3**: **West Coast Canal** from **Kottapuram to Kollam** (205 km) in Kerala (includes Champakara & Udyogmandal canals).
- **NW-4**: **Krishna–Godavari Rivers & Canals** from Kakinada to Puducherry (1,078 km).
- **NW-5**: **Brahmani River, Deltaic channels of Mahanadi & East Coast Canal** (588 km) in Odisha & WB.
- **NW-16**: **Barak River** from Lakhipur to Bhanga (121 km) in Assam.

8. Census 2011 Essential Fact Card (India vs UP vs UK)

Demographic Indicator	All-India	Uttar Pradesh	Uttarakhand
Total Population	**121.08 Crore** (17.5% of world)	**19.98 Crore** (16.51% of India - Rank 1)	**1.01 Crore** (0.83% of India - Rank 20)
Decadal Growth Rate (2001–11)	**17.70%** (Highest: Meghalaya 27.95%; Lowest/Negative: Nagaland -0.58%)	**20.22%** (Highest: Gautam Buddha Nagar 49.1%; Lowest: Kanpur Nagar 9.9%)	**18.81%** (Highest: US Nagar 33.4%; Negative: Almora -1.28%, Pauri Garhwal -1.41%)

Demographic Indicator	All-India	Uttar Pradesh	Uttarakhand
Population Density	**382 per sq km** (Highest: Bihar 1106; Lowest: Arunachal 17)	**829 per sq km** (Rank 4 among states) • Highest: **Ghaziabad** (3,971) • Lowest: **Lalitpur** (242)	**189 per sq km** (Rank 25) • Highest: **Haridwar** (801) • Lowest: **Uttarkashi** (41)
Overall Sex Ratio	**943 females / 1000 males** (Highest: Kerala 1084; Lowest: Haryana 879)	**912 females / 1000 males** • Highest: **Jaunpur** (1,024), Azamgarh (1,019), Deoria (1,017) • Lowest: **Gautam Buddha Nagar** (851)	**963 females / 1000 males** • Highest: **Almora** (1,139), Rudraprayag (1,114) • Lowest: **Haridwar** (880), Dehradun (902)
Child Sex Ratio (0–6 yrs)	**919** (Highest: Mizoram 970; Lowest: Haryana 834)	**902** • Highest: **Balrampur** (950) • Lowest: **Baghpat** (841)	**890** • Highest: **Pithoragarh** (916) • Lowest: **Pithoragarh / Champawat**
Literacy Rate (Overall)	**73.0%** (Male: 80.9%, Female: 64.6%; Gap: 16.3%)	**67.7%** (Male: 77.3%, Female: 57.2%; Gap: 20.1%) • Highest: **Gautam Buddha Nagar** (80.1%) • Lowest: **Shravasti** (46.7%)	**78.82%** (Male: 87.4%, Female: 70.0%; Gap: 17.4%) • Highest: **Dehradun** (84.25%) • Lowest: **Haridwar** (73.43%)
SC Population %	**16.6%** (Highest %: Punjab 31.9%; Highest No.: UP)	**20.7%** of state pop (Highest No.: Sitapur; Highest %: Kaushambi 34.7%; Lowest: Baghpat)	**18.8%** of state pop (Highest %: Bageshwar; Lowest %: Dehradun)
ST Population %	**8.6%** (Highest %: Lakshadweep/Mizoram; Highest No.: MP; Nil in Punjab, Haryana)	**0.6%** of state pop (Highest No. & %: **Sonbhadra**; Lowest: Baghpat; Tharu is largest tribe)	**2.9%** of state pop (Highest %: **Udham Singh Nagar**; Tharu & Jaunsari are largest)

9. Minerals of Uttar Pradesh & Uttarakhand (state-site map)

National mineral belts stay in the Minerals chapter. Daily sheet needs the **UP / UK places** that otherwise vanish (same gap as Hulas on the Indus table).

Mineral / resource	Uttar Pradesh	Uttarakhand	Trap
Limestone / cement	**Sonbhadra** (Churk, Dalla, Ghurma); also Banda–Mirzapur belt	**Dehradun**, Tehri, Pithoragarh, Pauri belts	Do not give Rajasthan's Makrana marble as a UP fact
Coal	**Singrauli coalfield** (Sonbhadra; shared with MP)	Negligible	Jharia / Raniganj are **Jharkhand–WB**, not UP
Silica sand / glass	**Shankargarh (Prayagraj)**, Banda; feeds **Firozabad** glass	—	Firozabad = glass craft; sand is Prayagraj belt
Bauxite	**Banda, Vindhyan** pockets (small)	—	Big bauxite = Odisha / Gujarat, not UP
Andalusite / sillimanite	**Sonbhadra / Mirzapur** (refractory)	—	
Magnesite	—	**Jhiroli / Bageshwar** (Kumaon) — India's important magnesite belt	Magnesite ≠ magnesium metal plant trap
Gypsum	—	**Dehradun / Nainital** foothill pockets	Rajasthan remains India's gypsum giant

Mineral / resource	Uttar Pradesh	Uttarakhand	Trap
Steatite / soapstone	—	**Almora / Pithoragarh** (also used in local craft)	
Copper (old)	—	**Askot (Pithoragarh)** — historic Kumaon copper	Khetri copper is **Rajasthan**
Rock phosphate	—	**Mussoorie / Durmala–Maldeota** (Dehradun)	
Building stone / granite	Mirzapur–Sonbhadra; Lalitpur	Granite–quartzite of Lesser Himalaya	
Energy water	**Rihand / Govind Ballabh Pant Sagar** (Sonbhadra) — largest man-made lake by volume	**Tehri** (Bhagirathi–Bhilangana) — highest dam	Hydel ≠ mineral ore

Logic: Sonbhadra is UP's mineral hub (coal + limestone + refractories). **Bageshwar magnesite** and **Askot copper** are the UK pair. Do not park Rajasthan or Jharkhand fields in these two states.

10. Major Minerals of India & Famous Mines Master Matrix

Mineral / Resource	Leading Producing State(s)	Famous Mines / Belts (All-India Match Goldmine)	High-Yield Prelims Exam Triggers
Iron Ore (Hematite & Magnetite)	**Odisha** (1st in production & reserves), Chhattisgarh, Karnataka, Jharkhand	• **Odisha**: Mayurbhanj (Gorumahisani, Badampahar), Keonjhar (Joda), Sundargarh • **Chhattisgarh**: **Bailadila** (Bastar/Dantewada — high-grade hematite exported to Japan via Visakhapatnam Port), Dalli-Rajhara (Durg — feeds Bhilai Steel Plant) • **Karnataka**: **Kudremukh** (Chikkamagaluru — horse-face shaped; slurry pipeline to Mangalore), Bababudan Hills, Bellary-Hospet (Sandur) • **Jharkhand**: **Noamundi**, Gua, Kiriburu (West Singhbhum) • **Goa**: Marmagao belt	• India is rich in **Hematite** (~70%) and **Magnetite** (~30%). • Dharwar and Cuddapah rock systems are primary sources. • *Trap:* Bailadila is in **Chhattisgarh**, NOT MP or Odisha!
Bauxite (Aluminium Ore)	**Odisha** (>50% of national production), Gujarat, Jharkhand	• **Odisha**: **Panchpatmali** (Koraput — India's largest deposit; feeds NALCO refinery at Damanjodi), Kalahandi, Rayagada, Bolangir • **Jharkhand**: **Lohardaga**, Gumla, Latehar • **Gujarat**: Jamnagar, Junagadh, Kutch • **Madhya Pradesh**: Amarkantak plateau (Shahdol), Balaghat • **Maharashtra**: Kolhapur, Ratnagiri	• Associated with **laterite rocks** of tertiary age formed via intense leaching. • NALCO (Damanjodi & Angul) and HINDALCO (Renukoot, UP) are major processors.
Copper	**Madhya Pradesh** (1st in production), Rajasthan (1st in reserves), Jharkhand	• **Madhya Pradesh**: **Malanjkhand** (Balaghat district — largest open-cast copper mine in Asia; accounts for ~50% of Indian production) • **Rajasthan**: **Khetri**, Chandmari, Kolihan (Jhunjhunu), Kho-Dariba (Alwar), Delwara (Udaipur) • **Jharkhand**: **Mosabani**, **Ghatshila**, Rakha (Singhbhum Copper Belt) • **Uttarakhand**: Askot (Pithoragarh — old workings)	• Hindustan Copper Limited (HCL, est. 1967) holds public monopoly. • India is heavily import-dependent for copper ore (~80% imported).

Mineral / Resource	Leading Producing State(s)	Famous Mines / Belts (All-India Match Goldmine)	High-Yield Prelims Exam Triggers
Manganese	**Madhya Pradesh** (1st in production), Maharashtra, Odisha	• **Madhya Pradesh**: **Balaghat** (Bharveli underground mine), Chhindwara • **Maharashtra**: Nagpur, Bhandara, Ratnagiri • **Odisha**: Sundargarh, Keonjhar, Koraput • **Karnataka**: Shimoga, Bellary	• Vital ferro-alloy: ~10 kg manganese is needed to manufacture **1 tonne of steel**. • MOIL (Manganese Ore India Limited) headquartered in Nagpur.
Coal	**Jharkhand** (1st in reserves), **Odisha / Chhattisgarh** (1st in production)	• **Jharkhand**: **Jharia** (Dhanbad — **richest and only storehouse of prime coking coal in India**), **Bokaro**, Giridih, North/South Karanpura, Ramgarh • **West Bengal**: **Raniganj** (Asansol/Bardhaman — **oldest commercial coalfield in India, started 1774 by Summer & Heatly**) • **Chhattisgarh**: **Korba** (Hasdeo valley), Tatapani-Ramkola, Chirimiri • **Odisha**: **Talcher** (Brahmani valley — feeds NTPC super thermal plant), Ib valley • **Madhya Pradesh**: **Singrauli** (Jhingurdah seam — **second thickest coal seam in the world**; shared with UP Sonbhadra), Pench-Kanhan • **Telangana**: Singareni (Godavari valley — SCCL) • **Tamil Nadu**: **Neyveli** (Cuddalore — premier deposit of **Lignite / Brown Coal** in India) • **Assam**: Makum, Nazira, Ledo (Tertiary coal; high sulphur content)	• **Gondwana Coal** (~98% of India's reserves; 250 million yrs old; low moisture, low sulphur, high ash, non-coking/coking). • **Tertiary Coal** (~2% reserves; 15–60 million yrs old; NE states & Jammu; high sulphur & volatile matter).
Petroleum & Crude Oil	**Offshore (Bombay High)** (1st in production), Rajasthan, Gujarat, Assam	• **Offshore**: **Bombay High** (1974 discovery by ONGC vessel *Sagar Samrat*; 160 km off Mumbai), Bassein, Ravva (Krishna-Godavari) • **Rajasthan**: **Barmer Basin** (Mangala, Bhagyam, Aishwarya fields — discovered by Cairn/Vedanta; largest onshore producer) • **Gujarat**: **Ankleshwar** (*Fountain of Prosperity* named by Nehru), Kalol, Mehsana, Aliabet Island (Gulf of Khambhat) • **Assam**: **Digboi** (**oldest producing oilfield & refinery in India and Asia, 1901**), Naharkatiya, Moran-Hugrijan, Surma Valley	• Refineries: **Jamnagar (Gujarat)** is the world's largest grass-root refining complex (private - Reliance). • Indian Oil's Digboi is the oldest operational refinery.
Gold	**Karnataka** (>90% of national production), Andhra Pradesh	• **Karnataka**: **Kolar Gold Fields (KGF)** (shut down in 2001 due to depth & costs), **Hutti Gold Mines** (Raichur — **only active primary gold producing mine in India today**) • **Andhra Pradesh**: Ramagiri (Anantapur) • **Jharkhand**: Alluvial placer gold in Subarnarekha river sands	• Hutti produces ~99% of India's indigenous mined gold.
Diamond	**Madhya Pradesh** (holds >90% of diamond resources)	• **Madhya Pradesh**: **Panna** (**Majhgawan pipe mine** operated by NMDC — **the ONLY mechanized active diamond mine in Asia/India**) • Historic: **Kollur Mine** (Guntur/Golconda, Andhra Pradesh — origin of the Koh-i-Noor, Hope Diamond, and Daria-i-Noor)	• Kimberly pipe formations in Vindhyan conglomerates.

Mineral / Resource	Leading Producing State(s)	Famous Mines / Belts (All-India Match Goldmine)	High-Yield Prelims Exam Triggers
Uranium	**Jharkhand, Andhra Pradesh**	• **Jharkhand**: **Jaduguda** (East Singhbhum — India's first uranium mine, 1967), Bhatin, Narwapahar, Turamdih, Banduhurang (all run by UCIL) • **Andhra Pradesh**: **Tummalapalle** (Kadapa/YSR district — massive global-scale reserves discovered in 2011) • **Meghalaya**: Domiasiat / Mawthabah (Kasi Hills - sandstone deposit) • **Rajasthan**: Rohil (Sikar)	• Operated under Department of Atomic Energy (DAE) via Uranium Corporation of India Ltd (UCIL, Jaduguda).
Thorium	**Kerala, Odisha, Tamil Nadu**	• **Kerala**: **Monazite beach sands** of Chavara, Kollam, and Alappuzha • **Odisha**: Ganjam coast (Chhatrapur beach sands) • **Tamil Nadu**: Manavalakurichi (Kanyakumari)	• Derived from **Monazite sand** (contains thorium phosphate, cerium, and rare earths). • Indian Rare Earths Limited (IREL) processes beach sand minerals. India holds the world's largest thorium reserves.
Lead & Zinc / Silver	**Rajasthan** (near 100% monopoly)	• **Rajasthan**: **Zawar Mines** (Udaipur — ancient & modern lead-zinc-silver), **Rampura-Agucha** (Bhilwara — world-class open-pit zinc deposit), Rajpura-Dariba (Rajsamand), Sindesar Khurd	• Hindustan Zinc Limited (HZL, Vedanta group). • Galena (lead ore) and Sphalerite (zinc ore). Silver is recovered as a by-product.
Mica (Ruby Mica / Muscovite)	**Andhra Pradesh** (1st in production), Rajasthan, Jharkhand	• **Andhra Pradesh**: Nellore Mica Belt (Gudur) • **Jharkhand**: **Koderma** (historically called the "*Mica Capital of the World*"), Giridih, Hazaribagh • **Rajasthan**: Bhilwara, Ajmer, Jaipur	• India was historic world leader in sheet mica (Muscovite / Ruby mica). • Essential electrical insulator resistant to high voltages.

11. Ocean Currents & Planetary Local Winds of the World

A. Major Ocean Currents (Warm vs Cold Master Matrix)

Rule of Thumb: Currents moving from Equator to Poles = Warm; Currents moving from Poles to Equator = Cold.

Ocean Basin	Warm Ocean Currents	Cold Ocean Currents	High-Yield Prelims Exam Takeaways
Atlantic Ocean	• **Gulf Stream** (powerful warm current) • **North Atlantic Drift** (keeps NW European ports ice-free year-round) • **Florida Current** • **Antilles Current** • **Brazil Current** (South Atlantic) • **Guinea Current** (West Africa)	• **Labrador Current** (flows from Arctic past Greenland) • **Canary Current** (North Africa coast) • **Benguela Current** (South-West Africa coast; creates **Namib & Kalahari Deserts**) • **Falkland / Malvinas Current** (Patagonia coast) • **East Greenland Current**	• **Grand Banks of Newfoundland**: Confluence of warm **Gulf Stream** and cold **Labrador Current** produces dense fog and creates world's richest cod-fishing ground. • **Sargasso Sea**: Clockwise North Atlantic gyre creates a calm, landless sea surrounded by currents, filled with *Sargassum* seaweed.

Ocean Basin	Warm Ocean Currents	Cold Ocean Currents	High-Yield Prelims Exam Takeaways
Pacific Ocean	• **Kuroshio (Japan Current)** (warm counterpart of Gulf Stream) • **North Pacific Drift** • **East Australian Current** (flows south along eastern Australia) • **Alaska Current** (keeps Alaskan coast ice-free) • **Tsushima Current** (Sea of Japan)	• **Oyashio (Kuril Current)** (cold current from Bering Strait) • **California Current** (West coast of USA) • **Peru / Humboldt Current** (flows along Chile & Peru; high upwelling, rich in anchovies; suppressed during **El Niño**) • **Okhotsk Current**	• Meeting of warm **Kuroshio** and cold **Oyashio** creates dense fog and prime fishing grounds off Japan. • **Atacama Desert** (driest non-polar desert) created by the desiccating effect of the cold **Humboldt/Peru Current**.
Indian Ocean	• **Mozambique Current** (East Africa) • **Agulhas Current** (South Africa) • **South-West Monsoon Drift** (summer)	• **West Australian Current** (flows along West Australia; creates **Great Sandy/Victoria Deserts**) • **Somali Current** (reverses direction seasonally; cold upwelling in summer)	• Indian Ocean currents completely **reverse their direction of flow twice a year** due to the seasonal reversal of the Monsoon winds.
Circumpolar	—	• **West Wind Drift (Antarctic Circumpolar Current)**	• **Largest ocean current by water volume**; circles Antarctica unbroken from West to East under roaring westerlies.

B. Famous Local Winds of the World (Hot vs Cold & Nicknames)

Local Wind	Nature (Hot / Cold)	Geographic Region	Distinctive Nickname / Exam Trigger
Chinook	**Warm & Dry** (Katabatic)	Rocky Mountains (USA & Canada)	**"Snow Eater"** — descends eastern slopes of Rockies, melting winter snow rapidly and opening pastures for cattle.
Foehn (Föhn)	**Warm & Dry**	Northern slopes of the **Alps (Switzerland)**	Accelerates melting of alpine snow, aids grape ripening in Swiss valleys (*the climber's wind*).
Harmattan	**Warm, Dry & Dusty**	Sahara Desert blowing south-west to the **Gulf of Guinea (West Africa)**	**"The Doctor"** — provides welcome relief from the sultry, humid coastal tropical heat and kills pathogens.
Sirocco	**Hot, Dry & Dust-laden**	Sahara Desert blowing north across Mediterranean into **Southern Europe (Italy/Sicily)**	Causes **"Blood Rain"** (red Sahara dust particles falling with condensation rain in Italy and Greece).
Khamsin	**Hot & Dusty**	**Egypt**	Blows from the southern desert for approximately 50 days (*Khamsin* = fifty in Arabic).
Gibli	**Hot & Dry**	**Libya**	Sahara desert wind blowing north towards Libyan coast.
Brickfielder	**Hot & Dusty**	**Southern Australia**	Blows from the central Australian desert towards coastal cities (Melbourne/Sydney).
Black Roller	**Hot & Dust-storm**	Great Plains of **North America / USA**	Intense summer dust squall darkening the sky.
Zonda	**Hot & Dry** (Foehn-type)	Eastern slopes of the **Andes (Argentina)**	Blows down Argentine pampas from the Andes.
Santa Ana	**Hot, Dry & Gusty**	Southern **California (USA)**	Blows from Mojave desert down coastal mountain canyons; notoriously triggers catastrophic wildfires.

Local Wind	Nature (Hot / Cold)	Geographic Region	Distinctive Nickname / Exam Trigger
Loo	**Scorching Hot & Dry**	Northern and North-Western Plains of **India (UP, Bihar, Punjab, Rajasthan)**	Blows during May–June afternoons; causes severe heatstrokes.
Nor'wester / Kalbaishakhi	**Hot-season thunderstorm**	**West Bengal, Odisha, Assam** (*Bordoisila*)	Violent afternoon squall in Baisakh (April–May); beneficial for early tea, jute, and aus paddy crops.
Mistral	**Extremely Cold & Dry**	Blows from Alps down the **Rhone Valley into Gulf of Lion (France & Spain)**	Winter/spring gale-force cold wind; trains often disrupted; houses in Provence face south without north windows.
Bora	**Extremely Cold, Gusty & Dry**	Blows from Dinaric Alps down towards the **Adriatic Sea (Italy, Croatia)**	Fierce northeastern catabatic katabatic wind reaching hurricane speeds in winter.
Blizzard	**Freezing Cold with Blinding Snow**	**Canada, Northern USA, Siberia, Polar regions**	Gale-force winds with sub-zero temperatures and zero visibility.
Buran / Purga	**Bitterly Cold & Fierce Blizzard**	**Siberia & Central Asian steppes (Russia)**	Buran brings icy blizzard conditions; Purga is tundra version.
Pampero	**Cold & Violent Wind**	**Pampas of Argentina and Uruguay**	Cold polar squall blowing from southwest across the pampas.

12. Structure of Atmosphere & Key Meteorological Phenomena

Atmospheric Layer	Height Range	Temperature Trend	Landmark Phenomenon / Exam Triggers
1. Troposphere	**0 to 18 km** (Equator: ~18 km; Poles: ~8 km)	**Decreases with altitude** (Lapse Rate: **6.5°C per km** or **1°C per 165 m**)	• Thickest at Equator due to strong vertical convectional currents. • Contains **~75% to 80% of total atmospheric mass** and **almost ALL atmospheric water vapour and dust particles**. • **ALL weather phenomena** (clouds, rainfall, lightning, thunder, cyclones, fog) occur EXCLUSIVELY in this layer. • Upper boundary is the **Tropopause** (temperature drops to -80°C over Equator and -45°C over Poles).
2. Stratosphere	**From Tropopause up to 50 km**	**Increases with altitude** (from -60°C up to 0°C at Stratopause)	• **Contains the Ozone Layer (Ozonosphere)** between **15 km and 35 km** (absorbs harmful solar ultraviolet UV-B and UV-C radiation). • Completely free from clouds and turbulent weather; ideal condition for flying **commercial supersonic jet aircraft**. • High-altitude **Subtropical Jet Streams** blow near the tropopause. • Rare iridescent **Nacreous Clouds (Mother-of-Pearl clouds)** form in polar winter stratosphere.
3. Mesosphere	**50 km to 80 km**	**Decreases rapidly with altitude** (reaches **-100°C at Mesopause**)	• **COLDEST LAYER OF THE ATMOSPHERE** (-100°C to -130°C). • **Meteors burn up** in this layer upon entering from space due to friction with thin gases. • Rare **Noctilucent Clouds** (night-shining polar clouds composed of ice crystals on meteor dust) appear here.

Atmospheric Layer	Height Range	Temperature Trend	Landmark Phenomenon / Exam Triggers
4. Thermosphere / Ionosphere	**80 km to 400+ km**	**Increases rapidly with altitude** (exceeds 1,500°C due to intense solar radiation absorption)	• Contains electrically charged ionized gas particles (**Ions**). • **Reflects terrestrial radio waves back to Earth**, enabling wireless global radio communication: - **D-layer (60–90 km)**: Reflects low-frequency radio waves, absorbs high-frequency; disappears at night. - **E-layer / Kennelly-Heaviside Layer (90–130 km)**: Reflects medium-frequency radio waves. - **F-layer / Appleton-Barnett Layer (140–400 km)**: Reflects high-frequency short radio waves. • **Auroras** (polar light displays caused by solar wind hitting charged atmospheric gases): - **Aurora Borealis** (Northern Lights, Arctic) - **Aurora Australis** (Southern Lights, Antarctic) • **International Space Station (ISS)** orbits in this layer (~400 km).
5. Exosphere	**>400 km to outer space** (~10,000 km)	Extremely high temperature, but imperceptible due to ultra-low density	• Outermost rarefied fringe of Earth's atmosphere merging into interplanetary space. • Dominated by light gases: **Hydrogen and Helium**. • Artificial communication and weather satellites orbit in higher exosphere.

13. Frontiers, Latitudinal/Longitudinal Framework & Territorial Limits of India

Locational / Border Indicator	Geographic Fact / Value	States / Locations Involved	Crucial Prelims Exam Triggers
Latitudinal Extent	**8°4' N to 37°6' N**	Mainland southern tip to northern tip	Total span = ~29°2'; Entirely in Northern & Eastern Hemispheres
Longitudinal Extent	**68°7' E to 97°25' E**	Westernmost point to easternmost point	Total span = ~29°18'; ~2 hours time lag between Arunachal & Gujarat
North–South Distance	**3,214 km**	Indira Col to Kanyakumari	Greater than East–West distance by 281 km
East–West Distance	**2,933 km**	Kibithu to Guhar Moti	
Total Land Frontier	**15,106.7 km**	16 States & 2 UTs share land borders	India shares land borders with 7 sovereign countries
Total Coastline	**7,516.6 km**	Mainland = 6,100 km; Island territories = 1,416.6 km	9 Coastal States + 4 Coastal UTs
Northernmost Point	**Indira Col** (37°6' N)	Siachen Glacier (Ladakh)	Near trijunction of India, Pakistan, and China
Southernmost Point (Overall)	**Indira Point (Pygmalion / Parsons Point)** (6°45' N)	Great Nicobar Island	Submerged partially during 2004 Tsunami; ~150 km from Sumatra (Indonesia)
Southernmost Point (Mainland)	**Cape Comorin (Kanyakumari)** (8°4' N)	Tamil Nadu	Confluence of Arabian Sea, Bay of Bengal, and Indian Ocean
Westernmost Point	**Guhar Moti / Sir Creek** (68°7' E)	Kutch district, Gujarat	Sir Creek is a 96 km tidal estuary disputed between India & Pak

Locational / Border Indicator	Geographic Fact / Value	States / Locations Involved	Crucial Prelims Exam Triggers
Easternmost Point	**Kibithu** (97°25' E)	Anjaw district, Arunachal Pradesh	Lohit River enters India near Kibithu
Tropic of Cancer (23°30' N)	Passes through **8 Indian States**	**Gujarat → Rajasthan → Madhya Pradesh → Chhattisgarh → Jharkhand → West Bengal → Tripura → Mizoram** (West to East)	• **Ranchi** is the closest state capital to Tropic of Cancer • **Mahi River** cuts the Tropic of Cancer TWICE! • Does NOT pass through Odisha, Bihar, or UP
Standard Meridian of India (82°30' E)	**IST = GMT + 5 hours 30 mins**	Passes through **5 States**: **Uttar Pradesh** (Mirzapur / Shankargarh / Naini, Prayagraj) → **Madhya Pradesh** → **Chhattisgarh** → **Odisha** → **Andhra Pradesh**	• Tropic of Cancer and 82°30' E meridian intersect in **Koriya / Surajpur district (Chhattisgarh)**

Land Frontiers by Length (Strict Descending Order)

1. **Bangladesh (4,096.7 km)** — **Longest border**; 5 States: West Bengal (longest border with BD), Assam, Meghalaya, Tripura (surrounded on 3 sides by BD; Zero Line), Mizoram.
2. **China (3,488 km)** — 1 UT (Ladakh) + 4 States: Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh; McMahon Line (1914).
3. **Pakistan (3,323 km)** — 2 UTs (J&K, Ladakh) + 3 States: Gujarat, Rajasthan (longest border with Pak), Punjab; Radcliffe Line (17 August 1947).
4. **Nepal (1,751 km)** — 5 States: Uttarakhand, Uttar Pradesh (longest border with Nepal), Bihar, West Bengal, Sikkim; Kali River forms western border.
5. **Myanmar (1,643 km)** — 4 States: Arunachal Pradesh (longest), Nagaland, Manipur, Mizoram; Purvanchal hills form watershed.
6. **Bhutan (699 km)** — 4 States: Assam (longest), West Bengal, Sikkim, Arunachal Pradesh.
7. **Afghanistan (106 km)** — **Shortest border**; Ladakh (POK - Wakhan corridor); Durand Line (1896).

Coastal States by Coastline Length (Descending)

- **Gujarat (1,214.7 km)** > **Andhra Pradesh (973.7 km)** > **Tamil Nadu (906.9 km)** > **Maharashtra (652.6 km)** > **Kerala (569.7 km)** > **Odisha (476.4 km)** > **Karnataka (280 km)** > **Goa (101 km — shortest among states)** > **West Bengal (157.5 km)**.
- **UTs**: **Andaman & Nicobar Islands (1,962 km — longest overall territory)** > Lakshadweep (132 km) > Puducherry (47.6 km) > Daman & Diu (42.5 km).

Maritime Channels & Straits Around India

- **8° Channel**: Separates **Minicoy (India)** from **Maldives**.
- **9° Channel**: Separates **Minicoy Island** from the **main Lakshadweep archipelago** (Kalpeni/Suheli).
- **10° Channel**: Separates **Little Andaman** from **Car Nicobar** (Andaman Islands from Nicobar Islands; ~150 km wide).
- **Duncan Passage**: Separates **South Andaman (Rutland Island)** from **Little Andaman**.
- **Sombrero Channel**: Separates **Car Nicobar** from **Central Nicobar (Nancowry)**.
- **Great / Grand Channel**: Separates **Great Nicobar (Indira Point)** from **Sumatra (Aceh, Indonesia)**.
- **Palk Strait & Adam's Bridge (Rama Setu)**: Separates **Tamil Nadu (Pamban Island / Dhanushkodi)** from **Sri Lanka (Mannar Island / Talaimannar)**; connects Palk Bay with Gulf of Mannar.
- **Coco Channel**: Separates **North Andaman (Landfall Island)** from **Coco Islands (Myanmar)**.

14. Peninsular & Himalayan Hill Sequences (North to South & West to East)

A. Trans-Himalayan & Himalayan Ranges (North to South Order)

1. **Karakoram Range** (K2 / Godwin-Austen 8,611 m; Siachen, Baltoro, Biafo, Hispar glaciers)
2. **Ladakh Range** (Khardung La pass; Leh lies between Ladakh and Zaskar ranges)
3. **Zaskar Range** (Kamet peak, Nanda Devi; Shipki La, Lipulekh passes)
4. **Great Himalayas (Himadri)** (Everest, Kanchenjunga, Makalu, Dhaulagiri, Nanga Parbat, Namcha Barwa)
5. **Lesser / Middle Himalayas (Himachal)** (Pir Panjal in J&K/HP, Dhauladhar in HP, Mussoorie & Nag Tibba in Uttarakhand, Mahabharat Lekh in Nepal)
6. **Shiwalik Range (Outer Himalayas)** (Youngest foothill range; flat-bottomed structural valleys called **Duns** in west e.g., Dehradun, and **Duars** in east).

B. Western Ghats (Sahyadri) Hill Sequence (Strict North to South)

1. **Northern Sahyadri**: **Kalsubai** (1,646 m — highest peak of Maharashtra / Northern Sahyadri), Salher, Mahabaleshwar (source of Krishna river), Harishchandra Range.
2. **Central Sahyadri**: **Mullayanagiri** (1,930 m — highest peak of Karnataka), **Kudremukh** (1,892 m), Baba Budan Hills (first coffee planted in India).
3. **Nilgiri Hills ("Blue Mountains")**: **Doda Betta** (2,637 m); junction meeting point of Western and Eastern Ghats; Nilgiri Tahr and Toda pastoral tribe.
4. *[Palghat / Palakkad Gap separates Nilgiris from Anaimalai Hills]*
5. **Anaimalai Hills ("Elephant Hills")**: **Anamudi** (2,695 m — **HIGHEST PEAK OF PENINSULAR / SOUTH INDIA**; Idukki, Kerala).
6. **Palani Hills**: Eastward spur from Anaimalai into Tamil Nadu; famous hill station **Kodaikanal** located around artificial star-shaped lake.
7. **Cardamom Hills (Yela Mala)**: Southernmost hill range of mainland India (Kerala-Tamil Nadu border); Periyar National Park.
8. *[Shenkottai Pass separates Cardamom Hills from Agasthyamalai]*
9. **Agasthyamalai Hills**: Southernmost biosphere reserve of Western Ghats (Kerala-TN border).

C. Eastern Ghats & Peninsular Discontinuous Hills (Strict North to South)

1. **Garhjat Hills**: Northern Odisha (Mayurbhanj / Keonjhar iron-ore belt).
2. **Nallamala Hills**: Andhra Pradesh (between Krishna and Pennar rivers; Srisailam dam and tiger reserve).
3. **Erramala Hills**: Kurnool district, Andhra Pradesh.
4. **Velikonda Range**: Eastern Andhra Pradesh (Nellore/Kadapa border).
5. **Palkonda Range**: Southern Andhra Pradesh (Tirupati sacred hills / Tirumala located on Seshachalam-Palkonda range).
6. **Nagari Hills**: Chittoor district, Andhra Pradesh.
7. **Javadi Hills**: Northern Tamil Nadu (Vellore / Tiruvannamalai; famous for blue-grey granites and sandalwood).
8. **Shevaroy Hills**: Salem district, Tamil Nadu; famous hill resort **Yercaud**; rich in bauxite deposits.
9. **Panchaimalai Hills**: Central Tamil Nadu (Tiruchirappalli).
10. **Sirumalai Hills**: Dindigul district, Tamil Nadu.
11. **Varushanad Hills**: Southern Tamil Nadu (source of Vaigai River).

D. Central Highlands & Transverse Order (North to South)

- **Vindhya Range** (Sandstone scarplands: Kaimur, Bhander, Bharner)
  ↓ *[Narmada River flows west in Rift Valley]*
- **Satpura Range** (Horst block mountain: **Rajpipla Hills** in west → **Mahadeo Hills / Dhupgarh 1,350 m** in middle → **Maikal Range / Amarkantak 1,065 m** in east)
  ↓ *[Tapi River flows west in Rift Valley south of Satpura]*
- **Ajanta Range & Satmala Hills** (Maharashtra; entirely within single state).

E. North-East Hills & Valleys

- **Arunachal Foothills (West to East)**: **Dafla Hills** → **Miri Hills** → **Abor Hills** → **Mishmi Hills** (Daphla-Miri-Abor-Mishmi named after local tribes).
- **Purvanchal Eastern Border Ranges (North to South)**: **Patkai Bum** (Arunachal-Myanmar border) → **Naga Hills** (Saramati Peak 3,826 m) → **Manipur Hills** → **Mizo Hills / Lushai Hills** (Phawngpui / Blue Mountain 2,157 m).
- **Meghalaya Plateau (West to East)**: **Garo Hills** (Nokrek Biosphere Reserve 1,412 m) → **Khasi Hills** (Shillong Peak 1,961 m; Mawsynram and Cherrapunji) → **Jaintia Hills** → **Mikir Hills** → **Barail Range** (connects Meghalaya plateau to Naga hills).

15. Major Lakes & Inland Wetlands of India Master Matrix

Lake Name	State / UT	Type / Origin	High-Yield Prelims Exam Triggers
Chilika Lake	**Odisha** (Puri, Khordha, Ganjam)	**Brackish coastal lagoon**	• **Largest brackish water lagoon in India** (fed by Daya River) • Designated **1st Indian Ramsar site (1981)** along with Keoladeo NP • Habitat of the endangered **Irrawaddy Dolphin** • Nalabana Island declared Bird Sanctuary
Vembanad Lake	**Kerala** (Kottayam, Alappuzha, Ernakulam)	**Brackish lagoon / backwater**	• **LONGEST LAKE IN INDIA** (96.5 km); largest in Kerala • Famous **Nehru Trophy Boat Race (Vallam Kali)** held on Punnamada Lake • Willingdon Island (Kochi Port) man-made in it • Kumarakom Bird Sanctuary on its eastern shore
Sasthamkotta Lake	**Kerala** (Kollam)	**Freshwater lake**	• Known as the **"Queen of Lakes"**; largest freshwater lake in Kerala • Contains unique larva *Cavaborus* which cleanses and purifies lake water
Ashtamudi Lake	**Kerala** (Kollam)	**Brackish estuarine lagoon**	• Eight-armed (*Ashta-mudi*) palm-shaped lake; gateway to Kerala backwaters; Ramsar site
Kolleru Lake	**Andhra Pradesh** (Eluru/Krishna)	**Freshwater lake**	• Located between **Krishna and Godavari deltas**; fed by Budameru and Tammileru streams • Atapaka Bird Sanctuary (pelican paradise); Ramsar site
Pulicat Lake	**Andhra Pradesh (96%) & Tamil Nadu (4%)**	**Brackish water coastal lagoon**	• **Second largest brackish lagoon in India** • Separated from Bay of Bengal by the barrier island of **Sriharikota** (ISRO Satish Dhawan Space Centre) • Famous annual **Flamingo Festival**
Loktak Lake	**Manipur** (Moirang, Bishnupur)	**Freshwater lake**	• **Largest freshwater lake in North-East India** • Famous for heterogeneous mass of vegetation and soil called **Phumdis** • Hosts **Keibul Lamjao National Park** — **WORLD'S ONLY FLOATING NATIONAL PARK** • Exclusive natural habitat of endangered **Sangai (Brow-antlered / Dancing deer)**
Lonar Lake	**Maharashtra** (Buldhana district)	**Meteorite Impact Crater Lake**	• Created by hyper-velocity meteorite impact during Pleistocene epoch in Deccan basalt • Highly saline and alkaline (pH ~10.5); turns pink due to haloarchaea microbes • Notified National Geo-heritage Monument; Ramsar site (2020)

Lake Name	State / UT	Type / Origin	High-Yield Prelims Exam Triggers
Wular Lake	**Jammu & Kashmir** (Bandipora)	**Tectonic Freshwater Lake**	• **LARGEST FRESHWATER LAKE IN INDIA** • Formed by tectonic activity; fed and drained by **Jhelum River** • **Tulbul Navigation Project** (disputed with Pakistan) located at its mouth
Dal Lake	**Jammu & Kashmir** (Srinagar)	Freshwater flood-plain lake	• "Jewel in the crown of Kashmir"; famous for **Shikaras** and floating gardens (*Radh*)
Sambhar Lake	**Rajasthan** (Jaipur, Nagaur, Ajmer)	**Inland Saline Playa Lake**	• **LARGEST INLAND SALINE LAKE IN INDIA**; represents an endorheic depression • Produces ~9% of India's total salt production; Ramsar site
Didwana, Pachpadra, Lunkaransar	**Rajasthan**	Inland saline playas	• Pachpadra produces high-grade sodium chloride (98% purity) used locally
Pushkar & Nakki Lake	**Rajasthan**	Pushkar: Sacred holy lake; Nakki: Mount Abu	• Nakki is highest lake in Rajasthan; legend says dug by gods' nails (*nakh*)
Pangong Tso	**Ladakh** (4,225 m elevation)	Endorheic saltwater lake	• **World's highest saltwater lake**; ~⅓ lies in India and ~⅔ in Tibet (China)
Tso Moriri	**Ladakh** (Rupshu valley)	High-altitude brackish lake	• Breeding ground for endangered Black-necked Crane and Bar-headed Goose; Ramsar site
Roopkund Lake	**Uttarakhand** (Chamoli, 5,029 m)	Glacial tarn lake	• Known as the **"Mystery Lake" / "Skeleton Lake"**; hundreds of human skeletons found
Bhimtal, Nainital, Naukuchiatal, Sattal	**Uttarakhand** (Kumaon)	Tectonic / fault freshwater lakes	• **Bhimtal** is the largest lake in Kumaon with a central island; Naukuchiatal has 9 corners
Suraj Tal & Chandra Tal	**Himachal Pradesh** (Lahaul & Spiti)	High-altitude glacial lakes	• Chandra Tal (source of Chandra river) and Suraj Tal (source of Bhaga river); join to form Chenab
Renuka Lake	**Himachal Pradesh** (Sirmaur)	Freshwater wetland lake	• Largest natural lake in Himachal Pradesh; shaped like a reclining woman; Ramsar site
Deepor Beel	**Assam** (Kamrup / Guwahati)	Freshwater oxbow lake	• Former channel of Brahmaputra; vital staging site for migratory waterbirds; Ramsar site
Rudrasagar Lake	**Tripura** (Melaghar)	Freshwater lowland lake	• Historic **Neermahal Palace** situated in the center of the lake; Ramsar site

16. Soils of India — ICAR Classification, Characteristics & Soil Health Master Card

A. ICAR 8 Soil Groups (% Share & Leading Characteristics)

1. **Alluvial Soil (~43.4% of total area — Largest)**:
2. **Genesis**: Deposited by Indus, Ganga, Brahmaputra river systems and coastal waves.
3. **Nutrient Profile**: **Rich in Potash and Lime**; severely **deficient in Nitrogen, Phosphorus, and Humus**.
4. **Classification**:
    - **Khadar**: New alluvium deposited by annual floods; fine loamy texture, highly fertile; light color.
    - **Bhangar**: Older alluvium situated above flood levels; clayey texture, less fertile; contains calcareous nodules (**Kankar**).
    - **Bhabar**: Narrow porous pebble-bed at foot of Shiwaliks where rivers disappear underground.
    - **Terai**: Marshy, waterlogged damp tract south of Bhabar with tall elephant grass; re-emerging streams.
5. **Crops**: Wheat, Rice, Sugarcane, Jute, Oilseeds, Maize, Pulses.

6. **Red & Yellow Soil (~18.6% — 2nd Largest Group)**:
7. **Genesis**: Weathering of ancient crystalline igneous (granites, gneisses) and metamorphic rocks in low-rainfall areas.
8. **Color Rationale**: Red color due to wide diffusion of **Iron (Ferric oxides)**; appears yellow when hydrated.
9. **Nutrient Profile**: **Deficient in Nitrogen, Phosphorus, Lime, and Humus**.
10. **Regions & Crops**: Tamil Nadu, Southern Karnataka, Telangana, Odisha, Chotanagpur, MP; ideal for millets, pulses, groundnut, tobacco.
11. **Black Soil / Regur / Tropical Chernozem (~15.2%)**:
12. **Genesis**: Weathering of basaltic lava rocks of the **Deccan Trap (Cretaceous lava flows)**.
13. **Distinctive Features**: Highly argillaceous (clayey >50%); swells and becomes sticky when wet; shrinks and develops wide deep cracks when dry (**"self-ploughing" capacity**); exceptional moisture-retention capability.
14. **Color**: Black color attributed to **Titaniferous Magnetite** and iron compounds.
15. **Nutrient Profile**: **Rich in Lime, Iron, Magnesia, Alumina, and Potash**; severely **deficient in Phosphorus, Nitrogen, and Organic Matter**.
16. **Regions & Crops**: Maharashtra, Gujarat, Malwa (MP), Northern Karnataka, Telangana; **Ideal for Cotton (Black Cotton Soil), Soyabean, Sugarcane, Citrus fruits**.
17. **Laterite Soil (~3.7%)**:
18. **Genesis**: Formed under conditions of **high temperature and alternating heavy wet and dry seasons with intense tropical rainfall**.
19. **Process**: Intense **leaching (desilication)** washes away silica and lime, leaving behind residual oxides of **Iron and Aluminium (Bauxite)**.
20. **Properties**: Highly acidic (pH 4.5–5.5); hardens like brick upon exposure to atmosphere (Latin *Later* = brick).
21. **Nutrient Profile**: Poor in Nitrogen, Phosphorus, Potash, Lime, and Humus.
22. **Regions & Crops**: Summits of Western Ghats (Kerala, Karnataka, Maharashtra), Eastern Ghats, Meghalaya plateau, Rajmahal Hills; **Premier soil for Plantation Crops: Tea, Coffee, Rubber, Cashewnut, Coconut**.
23. **Arid / Desert Soil (~4.4%)**:
24. **Properties**: Sandy to gravelly texture, saline in nature; low moisture and very low humus; phosphate normal but nitrogen deficient; with canal irrigation (Indira Gandhi Canal) produces bumper crops of wheat, barley, cotton.
25. **Saline & Alkaline Soils (Reh, Kallar, Usar, Rakar, Thur, Chopan) (~1.3%)**:
26. **Genesis**: Excessive canal irrigation in poorly drained arid/semi-arid regions causing high capillary action that pulls soluble sodium, calcium, and magnesium salts to topsoil.
27. **Reclamation**: Application of **Gypsum ($CaSO_4 \cdot 2H_2O$)** or Iron Pyrites to neutralize alkalinity; growing salt-tolerant green manure crops like **Dhaincha (*Sesbania*)**.
28. **Peaty and Marshy Soil (Kari) (~2%)**:
29. **Genesis**: Formed in humid regions with heavy rainfall and dense vegetation; high organic matter accumulation (40–50%).
30. **Properties**: Dark, heavy, highly acidic; found in **Kottayam and Alappuzha districts of Kerala (Kari lands)**, Sundarbans delta.
31. **Forest and Mountain Soil (~8.7%)**:
32. Immature loamy soil rich in humus from fallen leaf litter; acidic in higher alpine zones; terrace cultivation of apples, peaches, spices, tea.

17. Natural Vegetation & Forests of India (Champion & Seth / ISFR Key Indicators & Species)

A. Core Forest Statistics (India State of Forest Report - ISFR)

- **Total Forest & Tree Cover**: **~24.62%** of geographical area (Forest Cover: ~21.71%; Tree Cover: ~2.91%).
- **National Forest Policy (1988) Target**: **33%** of total geographical area (60% in mountainous/hilly regions).
- **State with Largest Forest Cover by Area**: **Madhya Pradesh** (1st) > Arunachal Pradesh (2nd) > Chhattisgarh (3rd) > Odisha (4th) > Maharashtra (5th).

- **State with Highest Forest Cover as % of Total Area**: **Mizoram** (84.5%) > Arunachal Pradesh (79.3%) > Meghalaya (76%) > Manipur (74.3%).
- **UT with Highest Forest Cover %**: **Lakshadweep** (90.3%) and **Andaman & Nicobar Islands** (81.7%).

B. Major Forest Types & Indicator Species (Match the Following Goldmine)

Forest Type	Annual Rainfall & Climate	Typical Regions	Dominant Commercial Tree Species (Prelims Match)
Tropical Wet Evergreen	**>250 cm**; Temp: 25–27°C; High humidity (>77%)	Western slopes of Western Ghats, NE Hills (Assam, Meghalaya), Andaman & Nicobar	• **Rosewood, Mahogany, Ebony, Aini, Cinchona (Quinine), Rubber, Wild Mango, Bamboo** • Dense multi-layered canopy; trees do not shed leaves simultaneously.
Tropical Semi-Evergreen	**200–250 cm**	Bordering wet evergreen belts in Western Ghats, Assam, Odisha	• **White Cedar, Hollock, Kadam, Kail, Champa**
Tropical Moist Deciduous	**100–200 cm**; distinct dry season	Eastern slopes of Western Ghats, Chotanagpur, foothills of Himalayas (Bhabar-Terai), Odisha, MP, UP	• **Teak (Sagan), Sal (Sakhu), Shisham, Sandalwood (Chandan), Mahua, Mulberry, Amla, Kusum, Semul, Bamboo** • Economically **most valuable timber forests of India**; shed leaves for 6–8 weeks in spring.
Tropical Dry Deciduous	**70–100 cm**; transitional between moist deciduous & thorn	Vast plains of UP, Bihar, MP, eastern Rajasthan, Deccan plateau	• **Tendu (Bidi wrapper), Palas / Dhak (*Flame of the Forest*), Amaltas, Bel, Khair, Axlewood (Dhaora)** • **MOST WIDESPREAD FOREST COVER IN INDIA (~41.8% of total forest area)**.
Tropical Thorn & Scrub	**<70 cm / <50 cm**	Rajasthan, SW Punjab, Haryana, Gujarat, rain-shadow leeward parts of Deccan	• **Acacia (Babool), Kikar, Ber, Wild Date Palm, Khair, Cacti, Euphorbia** • Deep roots, xerophytic thorns, fleshy stems to conserve water.
Montane Subtropical / Temperate	Altitude: **1,000 to 3,000 m**	Himalayas (J&K, HP, Uttarakhand, Sikkim) & Nilgiris/Anamalai (*Shola* forests)	• **1000–2000 m**: Broad-leaved **Oak and Chestnut** • **1500–3000 m**: Coniferous **Chir Pine** (resin & turpentine), **Deodar (*Cedrus deodara* — durable structural timber, western Himalayas), Blue Pine, Spruce, Silver Fir** • *Shola*: Stunted evergreen temperate grasslands/forests in Nilgiris and Anamalai.
Alpine & Sub-Alpine	Altitude: **>3,000 to 4,000 m**	High Himalayas above timber line up to snow line	• **Silver Fir, Juniper, Birch (Bhojpatra — ancient parchment bark), Rhododendron** • Alpine pastures called **Bugyals** in Uttarakhand and **Margs** in Kashmir (Gulmarg, Sonamarg).
Littoral / Mangrove / Tidal	Coastal saline estuaries, deltas, tidal creeks	Sundarbans (WB), Mahanadi delta (Bhitarkanika, Odisha), Godavari-Krishna deltas, Pichavaram (TN)	• **Sundari (*Heritiera fomes*), Goran, Kankra, Rhizophora, Sonneratia** • Halophytic adaptations: **Pneumatophores (breathing roots)**, stilt/prop roots, **vivipary** germination.

18. Shifting Cultivation (Jhum) Regional & Global Names Master Match-Sheet

A. Indian Regional Names for Shifting Cultivation

- **Jhum / Jhoom**: **North-Eastern India** (Assam, Meghalaya, Nagaland, Mizoram, Manipur, Arunachal Pradesh)
- **Podu / Dabi / Koman / Bhringa**: **Odisha** and Northern **Andhra Pradesh**
- **Bewar / Dahiya**: **Madhya Pradesh** (practiced by Baiga tribe in Bundelkhand/Baghelkhand)

- **Penda / Dipa**: **Bastar district (Chhattisgarh)** (practiced by Maria Gonds)
- **Valre / Waltre**: **South-Eastern Rajasthan** (Aravalli tribal belt)
- **Khil**: **Himalayan belt** (Himachal Pradesh, Uttarakhand)
- **Kuruwa / Khallu**: **Jharkhand** (Santhal Parganas)
- **Kumari**: **Western Ghats** (hilly tracts of Kerala and Karnataka)
- **Zabo**: **Nagaland** (indigenous forest-clearing + terraced pond water management)

B. International Names for Shifting Cultivation (Direct Exam Matches)

- **Milpa**: **Mexico and Central America**
- **Ladang**: **Indonesia and Malaysia**
- **Roca**: **Brazil** (Amazon basin)
- **Conuco**: **Venezuela** (Orinoco basin)
- **Taungya**: **Myanmar (Burma)**
- **Chena**: **Sri Lanka**
- **Ray**: **Vietnam and Laos**
- **Tamrai**: **Thailand**
- **Caingin**: **Philippines**
- **Masole**: **Congo Basin (Central Africa)**
- **Chitemene**: **Zambia, Zimbabwe and Malawi**

19. Agricultural Revolutions, Commodity Boards & Premier National Research Institutes

A. Agricultural Revolutions & Famous Pioneers

- **Green Revolution**: **Foodgrains (Wheat & Rice)** — Global Father: **Dr. Norman Borlaug** (Nobel Peace Prize 1970; semi-dwarf wheat at CIMMYT Mexico); Indian Father: **Dr. M.S. Swaminathan** (Bharat Ratna); Pantnagar University (1st agri university in India, 1960).
- **White Revolution (Operation Flood)**: **Milk & Dairy** — Father: **Dr. Verghese Kurien**; NDDB Anand, Gujarat (AMUL); launched in 1970.
- **Yellow Revolution**: **Oilseeds (Mustard & Sunflower)** — Pioneer: **Sam Pitroda** (Technology Mission on Oilseeds 1986).
- **Blue Revolution**: **Fish & Aquaculture** — Pioneers: **Dr. Hiralal Chaudhari** and **Dr. Arun Krishnan**.
- **Golden Revolution**: **Horticulture, Honey & Fruits** — Pioneer: **Nirpakh Tutej**.
- **Golden Fibre Revolution**: **Jute Production**.
- **Silver Revolution**: **Eggs & Poultry Production** — Pioneer: **Smt. Indira Gandhi**.
- **Silver Fibre Revolution**: **Cotton Production**.
- **Pink Revolution**: **Onion, Prawn / Shrimp & Pharmaceuticals** — Pioneer: **Durgesh Patel**.
- **Red Revolution**: **Meat and Tomato Production** — Pioneer: **Vishal Tewari**.
- **Grey Revolution**: **Chemical Fertilizers**.
- **Round Revolution**: **Potato Production**.
- **Brown Revolution**: **Leather, Cocoa & Non-Conventional Energy**.
- **Evergreen / Rainbow Revolution**: **Holistic sustainable agriculture** (coined by M.S. Swaminathan).

B. Commodity Boards Headquarters in India (Match Goldmine)

- **Tea Board of India**: **Kolkata** (West Bengal)
- **Coffee Board of India**: **Bengaluru** (Karnataka)
- **Rubber Board of India**: **Kottayam** (Kerala)
- **Spices Board of India**: **Kochi** (Kerala)
- **Tobacco Board of India**: **Guntur** (Andhra Pradesh)
- **National Jute Board**: **Kolkata** (West Bengal)

- **Central Silk Board**: **Bengaluru** (Karnataka)
- **Coir Board**: **Kochi** (Kerala)
- **National Fisheries Development Board (NFDB)**: **Hyderabad** (Telangana)

C. Premier National Research Institutes (Geography / Agriculture / Forestry)

- **Uttarakhand (UKPCS Top Triggers!)**:
- **Forest Research Institute (FRI)**: **Dehradun** (est. 1906, Greco-Roman architecture)
- **Indian Institute of Remote Sensing (IIRS)**: **Dehradun**
- **Wildlife Institute of India (WII)**: **Dehradun**
- **Wadia Institute of Himalayan Geology (WIHG)**: **Dehradun**
- **Indira Gandhi National Forest Academy (IGNFA)**: **Dehradun**
- **Govind Ballabh Pant Institute of Himalayan Environment & Development**: **Kosi-Katarmal, Almora**
- **Uttar Pradesh (UPPCS Top Triggers!)**:
- **Indian Institute of Sugarcane Research (IISR)**: **Lucknow**
- **National Sugar Institute (NSI)**: **Kanpur**
- **Indian Institute of Vegetable Research (IIVR)**: **Varanasi**
- **Indian Grassland and Fodder Research Institute (IGFRI)**: **Jhansi**
- **Central Institute for Research on Goats (CIRG)**: **Makhdoom, Mathura**
- **Central Institute for Subtropical Horticulture (CISH)**: **Rahmankhera, Lucknow**
- **Indian Institute of Pulses Research (IIPR)**: **Kanpur**
- **National Botanical Research Institute (NBRI)**: **Lucknow**
- **Central Drug Research Institute (CDRI)**: **Lucknow**
- **Central Institute of Medicinal and Aromatic Plants (CIMAP)**: **Lucknow**
- **All-India Premier Geographic & Agricultural Stations**:
- **Central Potato Research Institute (CPRI)**: **Shimla** (Himachal Pradesh)
- **National Rice Research Institute (NRRI / CRRI)**: **Cuttack** (Odisha)
- **Indian Agricultural Research Institute (IARI / Pusa)**: **New Delhi**
- **Central Arid Zone Research Institute (CAZRI)**: **Jodhpur** (Rajasthan)
- **National Dairy Research Institute (NDRI)**: **Karnal** (Haryana)
- **Central Soil Salinity Research Institute (CSSRI)**: **Karnal** (Haryana)
- **National Institute of Oceanography (NIO)**: **Dona Paula, Panaji** (Goa)
- **Indian National Centre for Ocean Information Services (INCOIS)**: **Hyderabad** (Telangana)
- **National Geophysical Research Institute (NGRI)**: **Hyderabad** (Telangana)
- **Central Plantation Crops Research Institute (CPCRI)**: **Kasaragod** (Kerala)
- **Central Tuber Crops Research Institute (CTCRI)**: **Thiruvananthapuram** (Kerala)
- **Central Tobacco Research Institute (CTRI)**: **Rajahmundry** (Andhra Pradesh)
- **Central Sheep and Wool Research Institute (CSWRI)**: **Avikanagar** (Rajasthan)
- **Central Institute for Research on Buffaloes (CIRB)**: **Hisar** (Haryana)

20. World Strategic Straits, Canals & Maritime Chokepoints Master Matrix

Strait / Canal	Water Bodies Connected	Landmasses Separated	High-Yield Prelims Exam Triggers
Strait of Malacca	**Andaman Sea (Indian Ocean) & South China Sea (Pacific Ocean)**	**Malay Peninsula & Sumatra (Indonesia)**	• **World's busiest shipping chokepoint**; shortest sea route between Persian Gulf/India and China/Japan

Strait / Canal	Water Bodies Connected	Landmasses Separated	High-Yield Prelims Exam Triggers
Strait of Hormuz	**Persian Gulf & Gulf of Oman (Arabian Sea)**	**Iran & Oman / UAE (Musandam Peninsula)**	• **World's most critical oil transit chokepoint** (~20% of global petroleum passes through it)
Bab-el-Mandeb	**Red Sea & Gulf of Aden (Indian Ocean)**	**Yemen (Arabia/Asia) & Djibouti / Eritrea (Horn of Africa)**	• Known as the **"Gate of Tears"**; connects Indian Ocean shipping to the Suez Canal
Suez Canal	**Mediterranean Sea & Red Sea**	**Sinai Peninsula & Mainland Egypt (Africa)**	• Opened in **1869**; designed by French engineer **Ferdinand de Lesseps** • Eliminates passage around Cape of Good Hope; Port Said (north) and Port Tewfik/Suez (south)
Strait of Gibraltar	**Atlantic Ocean & Mediterranean Sea**	**Spain (Europe) & Morocco (Africa)**	• Known as the **"Key to the Mediterranean"** / ancient **"Pillars of Hercules"**; Rock of Gibraltar (UK) on north
Bosporus (Bosphorus) Strait	**Black Sea & Sea of Marmara**	**European Istanbul & Asian Istanbul (Turkey)**	• World's narrowest strait used for international navigation; separates Europe from Asia
Dardanelles Strait	**Sea of Marmara & Aegean Sea (Mediterranean)**	**Gallipoli Peninsula (Europe) & Asian Turkey**	• Historic naval battle of Gallipoli (WWI); Turkish Straits system regulated by Montreux Convention
Bering Strait	**Chukchi Sea (Arctic Ocean) & Bering Sea (Pacific Ocean)**	**Russia (Siberia/Asia) & Alaska (USA/North America)**	• **International Date Line (IDL)** passes directly through this strait
Palk Strait	**Bay of Bengal & Palk Bay / Gulf of Mannar**	**Tamil Nadu (India) & Jaffna Peninsula (Sri Lanka)**	• Named after Robert Palk (Governor of Madras)
Strait of Dover	**English Channel & North Sea**	**England (Great Britain) & France (Mainland Europe)**	• Narrowest part of English Channel (34 km); Channel Tunnel (*Chunnel*) runs beneath it
Panama Canal	**Atlantic Ocean (Caribbean Sea) & Pacific Ocean**	**North America & South America (Isthmus of Panama)**	• Opened in **1914**; 82 km long; uses 3 sets of freshwater locks (Gatun, Pedro Miguel, Miraflores)
Kiel Canal	**North Sea & Baltic Sea**	**Jutland Peninsula (Germany)**	• Bypasses the Danish straits; world's busiest artificial waterway by vessel count
Sunda Strait	**Java Sea & Indian Ocean**	**Java and Sumatra (Indonesia)**	• **Krakatoa** volcanic island sits directly inside this strait
Lombok Strait	**Java Sea & Indian Ocean**	**Bali and Lombok (Indonesia)**	• Deeper alternative to Malacca for supertankers; marks the **Wallace Line** biogeographical boundary
Torres Strait	**Arafura Sea & Coral Sea (Pacific Ocean)**	**Cape York (Australia) & Papua New Guinea**	• Shallow tropical strait filled with coral reefs
Bass Strait	**Tasman Sea & Southern Ocean / Indian Ocean**	**Victoria (Mainland Australia) & Tasmania**	• King Island and Flinders Island lie in it
Cook Strait	**Tasman Sea & South Pacific Ocean**	**North Island & South Island (New Zealand)**	• Connects capital Wellington with South Island
Strait of Magellan	**Atlantic Ocean & Pacific Ocean**	**Mainland South America (Chile) & Tierra del Fuego**	• Natural passage discovered by Ferdinand Magellan (1520) avoiding turbulent Drake Passage

Strait / Canal	Water Bodies Connected	Landmasses Separated	High-Yield Prelims Exam Triggers
Drake Passage	**South Atlantic Ocean & South Pacific Ocean**	**Cape Horn (South America) & South Shetland Islands (Antarctica)**	• Roughest open water body on Earth; connects Atlantic to Southern Ocean
Florida Strait	**Gulf of Mexico & Atlantic Ocean**	**Florida Keys (USA) & Cuba**	• Conveys the powerful warm Gulf Stream northward

21. Major World Rivers, Drainage Features & Riverbank Cities Master Table

A. World Landmark Rivers & Unique Geographic Traits

River Name	Length & Basin	Source & Mouth	Landmark Exam Triggers & Geographic Superlatives
Nile River	**6,650 km** (Africa)	White Nile (Lake Victoria) + Blue Nile (Lake Tana) → **Mediterranean Sea**	• **LONGEST RIVER IN THE WORLD** • Confluence of White & Blue Nile at **Khartoum (Sudan)** • Forms classic **Arcuate Delta** in Egypt • **Aswan High Dam** creates **Lake Nasser** • Capitals on banks: **Cairo** (Egypt), **Khartoum** (Sudan)
Amazon River	**6,400 km** (South America)	Peruvian Andes (Nevado Mismi) → **Atlantic Ocean**	• **LARGEST RIVER IN THE WORLD BY WATER VOLUME & BASIN AREA** (~20% of world river discharge) • Flows across dense tropical rainforests (**Selvas** — "Lungs of the Earth") • Has **NO bridges** across its entire main stem • Spectacular tidal bore known as **Pororoca**
Yangtze River (Chang Jiang)	**6,300 km** (Asia / China)	Tibetan Plateau (Tangjula Mtns) → **East China Sea** (Shanghai)	• **LONGEST RIVER IN ASIA** (3rd longest in world); flows entirely in China • Hosts **Three Gorges Dam** (world's largest hydroelectric plant by capacity) • Cities: **Chongqing, Wuhan, Nanjing, Shanghai**
Mississippi–Missouri	**6,275 km** (North America)	Lake Itasca (Minnesota) → **Gulf of Mexico**	• Longest river system in North America; forms textbook **Bird's Foot Delta** • Cities: Minneapolis, St. Louis, Memphis, New Orleans
Danube River	**2,850 km** (Europe)	Black Forest (Germany) → **Black Sea**	• Flows through **10 countries** (more than any other river in the world) • Passes through **4 national capitals**: **Vienna** (Austria), **Bratislava** (Slovakia), **Budapest** (Hungary), **Belgrade** (Serbia)
Rhine River	**1,230 km** (Europe)	Swiss Alps (Lake Constance) → **North Sea**	• Flows through a **rift valley** between **Vosges Mountains (France)** and **Black Forest (Germany)** • **Busiest commercial inland waterway in Europe**; mouth at **Rotterdam Port** (Netherlands) • Basel, Strasbourg, Bonn, Cologne on its banks
Congo River (Zaire)	**4,700 km** (Central Africa)	Katanga Plateau → **Atlantic Ocean**	• **DEEPEST RIVER IN THE WORLD** (>220 m); 2nd largest discharge after Amazon • **CROSSES THE EQUATOR TWICE!** • Boyoma (Stanley) Falls and Livingstone Falls; **Kinshasa** and **Brazzaville** face each other across it
Limpopo River	**1,750 km** (Southern Africa)	Witwatersrand (South Africa) → **Indian Ocean** (Mozambique)	• **CROSSES THE TROPIC OF CAPRICORN TWICE!**

River Name	Length & Basin	Source & Mouth	Landmark Exam Triggers & Geographic Superlatives
Mahi River	**583 km** (India)	Vindhya Range (MP) → **Gulf of Khambhat**	• **CROSSES THE TROPIC OF CANCER TWICE!**
Volga River	**3,530 km** (Russia / Europe)	Valdai Hills → **Caspian Sea**	• **LONGEST RIVER IN EUROPE**; internal drainage into landlocked Caspian Sea; Volgograd (Stalingrad) on its bank
Amur River (Heilong Jiang)	**2,824 km** (Asia)	Confluence of Shilka & Argun → **Sea of Okhotsk**	• Forms the natural international border between **Russia and China** for >1,600 km
Mekong River	**4,350 km** (Southeast Asia)	Tibetan Plateau → **South China Sea**	• Known as the "Danube of the East"; flows through 6 countries: China, Myanmar, Laos, Thailand, Cambodia, Vietnam; **Vientiane** and **Phnom Penh** on banks
St. Lawrence River	**1,197 km** (North America)	Lake Ontario → **Gulf of St. Lawrence (Atlantic)**	• Natural outlet of the Great Lakes; forms part of USA–Canada international border; Montreal and Quebec City on its banks

B. Famous World Cities on Rivers (Direct Match List)

- **London**: **Thames** (UK)
- **Paris**: **Seine** (France)
- **Rome**: **Tiber** (Italy)
- **Berlin**: **Spree** (Germany)
- **Vienna, Budapest, Belgrade**: **Danube** (Austria, Hungary, Serbia)
- **Warsaw**: **Vistula** (Poland)
- **Kyiv (Kiev)**: **Dnieper** (Ukraine)
- **Cairo, Khartoum**: **Nile** (Egypt, Sudan)
- **Baghdad**: **Tigris** (Iraq)
- **Washington, D.C.**: **Potomac** (USA)
- **New York**: **Hudson** (USA)
- **New Orleans**: **Mississippi** (USA)
- **Buenos Aires**: **Río de la Plata** (Argentina)
- **Bangkok**: **Chao Phraya** (Thailand)
- **Yangon**: **Yangon River / Irrawaddy delta** (Myanmar)
- **Lahore**: **Ravi** (Pakistan)
- **Karachi**: **Indus** (Pakistan)

22. World Lakes, Inland Seas & Salinity Master Matrix

Lake Name	Location / Countries Bordering	Key Geographic Superlative	High-Yield Prelims Exam Triggers
Caspian Sea	Bordered by **5 Countries (mnemonic: TARIK)**: **T**urkmenistan, **A**zerbaijan, **R**ussia, **I**ran, **K**azakhstan	**LARGEST LAKE / INLAND WATER BODY IN THE WORLD**	• Surface area ~371,000 km²; slightly saline (~1.2%); largest endorheic basin • Baku (capital of Azerbaijan) situated on its western shore • Fed by Volga, Ural, and Kura rivers; has NO natural outlet
Lake Superior	**USA & Canada border** (Minnesota, Wisconsin, Michigan, Ontario)	**LARGEST FRESHWATER LAKE IN THE WORLD BY SURFACE AREA**	• Surface area ~82,100 km²; part of North American Great Lakes • Connected to Lake Huron by St. Mary's River and **Soo Locks**

Lake Name	Location / Countries Bordering	Key Geographic Superlative	High-Yield Prelims Exam Triggers
Lake Baikal	**Russia** (Siberia — Irkutsk & Buryatia)	**DEEPEST LAKE IN THE WORLD** (1,642 m)	• **Oldest freshwater lake in the world** (25–30 million years) • **Largest lake by volume of unfrozen freshwater (~20% of world's surface freshwater)** • Drained by only ONE river: the **Angara River** (tributary of Yenisey); UNESCO WHS
Lake Tanganyika	Shared by **Tanzania, DR Congo, Burundi, Zambia**	**LONGEST FRESHWATER LAKE IN THE WORLD** (673 km)	• 2nd deepest lake in the world (1,470 m); situated in East African Great Rift Valley
Lake Victoria	Shared by **Tanzania (49%), Uganda (45%), Kenya (6%)**	**LARGEST TROPICAL LAKE IN THE WORLD**	• 2nd largest freshwater lake by surface area (68,800 km^2) • **Equator passes directly through it** • Primary reservoir and source of the **White Nile River**
Lake Titicaca	**Andes Mountains** (Border of **Peru and Bolivia**)	**HIGHEST COMMERCIALLY NAVIGABLE LAKE IN THE WORLD**	• Elevation: **3,812 m** above sea level; high alpine lake • Indigenous Uros people live on artificial floating islands made of **Totora reeds**
Dead Sea	Bordered by **Jordan, Israel, and West Bank**	**LOWEST WATER BODY ON EARTH'S LAND SURFACE**	• Surface elevation: **~430 m below sea level** • Hypersaline terminal lake (~34% salinity — 10x oceans); high density allows humans to float effortlessly • Fed by Jordan River
Lake Van	**Eastern Turkey** (Anatolia)	**HIGHEST SALINITY AMONG MAJOR WATER BODIES (~330‰ / 33%)**	• Saline soda lake with no outlet; water does not freeze even at high elevation
Lake Assal	**Djibouti** (Afar Depression, Horn of Africa)	Hypersaline lake (~348‰ salinity)	• Lowest point in Africa (-155 m below sea level); 2nd highest salinity on Earth
Lake Maracaibo	**Venezuela** (South America)	Largest lake in South America	• Connected to Gulf of Venezuela by Tablazo Strait; rich petroleum basin; famous for **Catatumbo Lightning**
Aral Sea	Border of **Kazakhstan and Uzbekistan**	Catastrophic environmental shrinkage	• Formerly 4th largest lake; shrunken to <10% original size due to Soviet diversion of **Amu Darya and Syr Darya** rivers for cotton fields
Lake Balkhash	**Kazakhstan** (Central Asia)	Unique half-fresh, half-saline lake	• Western part is fresh water (fed by Ili River), while eastern part is saline water
Lake Chad	Shared by **Chad, Cameroon, Niger, Nigeria**	Endorheic Sahelian lake	• Crucial wetland in the Sahel; shrunken by >90% since 1960s

The Great Lakes of North America (Strict West-to-East Chronology)

- **Superior → Michigan → Huron → Erie → Ontario** (Mnemonic: **SMHEO** or **HOMES**).
- **Lake Michigan** is the **ONLY ONE of the 5 Great Lakes situated entirely within the United States** (the other four are shared between USA and Canada).
- **Niagara Falls** is located on the Niagara River between **Lake Erie and Lake Ontario**!
- **Welland Canal** bypasses Niagara Falls to connect Lake Erie and Lake Ontario for commercial shipping.

23. World Major Grasslands & Deserts Master Cheat-Sheet

A. Grasslands of the World (Tropical Savannas vs Temperate Steppes)

Grassland Name	Type (Temperate / Tropical)	Geographic Region & Countries	High-Yield Prelims Exam Triggers
Prairies	**Temperate** (Mid-latitude)	**North America (USA & Canada)**	• Known as the **"Granary of the World"** (massive commercial wheat belt) • Warm katabatic wind **Chinook** ("Snow Eater") melts winter snow on eastern slopes of Rockies.
Steppes	**Temperate** (Mid-latitude)	**Eurasia** (Ukraine, Southern Russia to Central Asia & Mongolia)	• Extensive temperate grassland plain; dark fertile **Chernozem** soils; premier wheat & grain farming.
Pampas	**Temperate** (Mid-latitude)	**South America (Argentina & Uruguay)**	• Fertile alluvial soils; rich in nutritious **Alfalfa** grass; extensive cattle ranching (**Estancias**) and cowboys (**Gauchos**); wheat cultivation.
Veld / Veldt	**Temperate** (Mid-latitude)	**South Africa** (High Veld & Bush Veld plateau)	• Renowned for rearing **Merino sheep** for high-grade wool and maize cultivation (Maize Triangle).
Downs	**Temperate** (Mid-latitude)	**Australia** (Murray–Darling River Basin)	• Rolling grassland plains; premier sheep-farming stations for wool and wheat growing.
Canterbury Plains	**Temperate** (Mid-latitude)	**New Zealand** (South Island)	• Rich pastoral plains for sheep rearing and dairy farming.
Puszta	**Temperate** (Mid-latitude)	**Hungary** (Danube River Basin)	• Traditional Hungarian horse-breeding and grain plain.
Savanna / Sudan Type	**Tropical**	**East & Central Africa** (Kenya, Tanzania, Sudan)	• Characterized by tall coarse grasses (elephant grass) and scattered umbrella-acacia / baobab trees • Known as **"Big Game Country"** (abundant herbivore & carnivore diversity); Serengeti; Masai pastoralists.
Llanos	**Tropical**	**Venezuela and Colombia** (Orinoco River Basin)	• Tropical savanna grassland with distinct wet and dry seasons; extensive cattle ranching.
Campos	**Tropical**	**Brazilian Highlands (Brazil)**	• Subtropical / tropical savanna plateau grasslands south of the Amazon forest.
Cerrado	**Tropical**	**Central Brazil** (Mato Grosso)	• Vast tropical savanna ecoregion; converted into world's largest soyabean and cattle production frontier.
Gran Chaco	**Tropical / Semi-arid**	**Paraguay, Bolivia, Northern Argentina**	• Scrub woodland and savanna; Quebracho tree (*axe-breaker*) source of tannin.

B. Major Deserts of the World & Origin Factors

Desert Name	Location & Continents	Type & Origin Factors	Unique Exam Superlative / Highlights
Sahara Desert	**North Africa** (11 countries)	**Subtropical Hot Desert**	• **LARGEST HOT DESERT ON EARTH** (~9.2 million km²); covers ~30% of Africa; Ergs (sand seas), Regs (stone plains), Hamadas (rock plateaus).
Atacama Desert	**Northern Chile & Southern Peru**	**Coastal Hyper-Arid Desert**	• **DRIEST NON-POLAR DESERT ON EARTH** • Formed by rain-shadow of high Andes + cold desiccating **Humboldt (Peru) Current** • Vast deposits of **Sodium Nitrate (Caliche)** and world-class copper mines.

Desert Name	Location & Continents	Type & Origin Factors	Unique Exam Superlative / Highlights
Namib Desert	**Namibia** (South-West Africa)	**Coastal Fog Desert**	• Formed by the cold **Benguela Current** offshore; ancient red sand dunes; home to ancient living fossil plant *Welwitschia mirabilis*.
Kalahari Desert	**Botswana, Namibia, South Africa**	**Semi-Arid Sandy Basin**	• Inhabited by the indigenous **San (Bushmen)** hunter-gatherers; vast red sand dunes (*Kgalagadi*).
Australian Deserts	**Central & Western Australia** (Great Victoria, Great Sandy, Gibson, Simpson)	**Subtropical Hot / Trade Wind Desert**	• Known as the **"Red Centre" / "Outback"**; formed by subtropical high-pressure belt + cold **West Australian Current**; Great Victoria is largest.
Gobi Desert	**Southern Mongolia & Northern China**	**Cold Continental Rain-Shadow Desert**	• Located in the rain-shadow of the Himalayas; extreme winter cold (-40°C); home to indigenous two-humped **Bactrian camels**.
Taklamakan Desert	**Xinjiang, Northwest China** (Tarim Basin)	**Cold Desert**	• Bounded by Tian Shan and Kunlun mountains; known as the "Sea of Death" / "Place of No Return".
Rub' al Khali	**Southern Arabian Peninsula** (Saudi Arabia, Oman, UAE, Yemen)	**Continuous Sand Desert (Erg)**	• Known as the **"Empty Quarter"**; **LARGEST CONTIGUOUS SAND DESERT IN THE WORLD** (~650,000 km²).
Patagonian Desert	**Southern Argentina** (South America)	**Cold Winter Rain-Shadow Desert**	• Rain-shadow of the Southern Andes; influenced by the cold **Falkland Current**; gravelly plateau.
Dasht-e Kavir & Dasht-e Lut	**Iran** (Central Plateau)	**Salt Deserts (Playa)**	• *Kavir* = great salt swamp; *Lut* recorded world's highest satellite land skin temperature (>70.7°C).
Mojave & Sonoran Deserts	**Southwestern USA (California, Arizona) & Mexico**	**Subtropical Hot Desert**	• Contains **Death Valley** (**Badwater Basin, -86 m** — lowest point in North America; hottest recorded air temp 56.7°C); giant **Saguaro cactus**.

24. World Volcanoes, Oceanic Trenches & Landmark Islands

A. Famous Volcanoes of the World (Direct Match Sheet)

Volcano Name	Country / Geographic Location	Type / Activity	High-Yield Prelims Exam Triggers
Mount Stromboli	**Lipari Islands / Sicily, Italy** (Tyrrhenian Sea)	Active Stratovolcano	• Known as the **"LIGHTHOUSE OF THE MEDITERRANEAN"** due to continuous glowing eruptions visible for miles.
Mount Etna	**Sicily, Italy**	Active Stratovolcano	• **Highest active volcano in Europe** (~3,357 m); continuously active; UNESCO World Heritage Site.
Mount Vesuvius	**Gulf of Naples, Italy**	Active / Dormant Stratovolcano	• Catastrophic Plinian eruption in **79 AD** completely buried and preserved the Roman cities of **Pompeii and Herculaneum**.
Mauna Loa & Kilauea	**Big Island of Hawaii, USA** (Pacific Ocean)	Active Shield Volcanoes	• **Mauna Loa** is the **world's largest subaerial shield volcano** by volume and area • **Kilauea** is among the world's most continuously active volcanoes; fluid basaltic lava flows.
Cotopaxi	**Andes Mountains, Ecuador** (South America)	Active Stratovolcano	• One of the world's highest active volcanoes (5,897 m); textbook symmetrical snow-capped cone.

Volcano Name	Country / Geographic Location	Type / Activity	High-Yield Prelims Exam Triggers
Ojos del Salado	**Andes Mountains (Border of Chile & Argentina)**	Active Stratovolcano	• **HIGHEST ACTIVE VOLCANO ON EARTH (6,893 m)**; 2nd highest peak in Western Hemisphere after Mount Aconcagua.
Mount Kilimanjaro	**Tanzania** (East Africa)	Dormant Stratovolcano	• **HIGHEST PEAK IN AFRICA (5,895 m)**; freestanding volcanic mountain with 3 volcanic cones: Kibo, Mawenzi, Shira; permanent snowcap near Equator.
Krakatoa (Krakatau)	**Sunda Strait (Between Java and Sumatra, Indonesia)**	Active Caldera Volcano	• Cataclysmic eruption in **1883** generated massive tsunamis and shockwaves heard 4,800 km away; modern growing cone is **Anak Krakatau**.
Mount Merapi & Mount Sinabung	**Java & Sumatra, Indonesia**	Highly Active Stratovolcanoes	• Merapi (*Mountain of Fire*) is the most active volcano in Indonesia; deadly pyroclastic density currents.
Mount Pinatubo & Mayon	**Luzon, Philippines**	Active Stratovolcanoes	• Pinatubo's massive **1991 eruption** injected huge SO2 aerosol clouds causing global temporary cooling of ~0.5°C • **Mayon** is celebrated for having the **world's most perfectly symmetrical volcanic cone**.
Mount Fuji (Fujisan)	**Honshu Island, Japan**	Dormant Stratovolcano	• Highest peak in Japan (3,776 m); sacred mountain; last erupted in 1707 (Hoei eruption).
Mount Erebus	**Ross Island, Antarctica**	Active Stratovolcano	• **SOUTHERNMOST ACTIVE VOLCANO ON EARTH**; features a long-lived persistent phonolitic lava lake.
Barren Island	**Andaman Sea, India** (~135 km NE of Port Blair)	**Active Volcano**	• **THE ONLY CONFIRMED ACTIVE VOLCANO IN SOUTH ASIA / INDIA**; erupted in 1991, 2005, and continues sporadic activity.
Narcondam Island	**Andaman Sea, India**	**Dormant Volcanic Island**	• Extinct/dormant andesite volcano; exclusive habitat of the endemic, endangered Narcondam Hornbill.

B. Deepest Oceanic Trenches of the World

- **Mariana Trench (Challenger Deep)**: **10,994 m** (Western Pacific Ocean near Guam) — **DEEPEST POINT ON PLANET EARTH**.
- **Tonga Trench (Horizon Deep)**: **10,882 m** (Southwest Pacific Ocean) — 2nd deepest trench on Earth.
- **Kuril–Kamchatka Trench**: **10,542 m** (Northwest Pacific Ocean).
- **Philippine Trench (Galathea Deep)**: **10,540 m** (Western Pacific Ocean).
- **Puerto Rico Trench (Milwaukee Deep)**: **8,376 m** (Atlantic Ocean) — **DEEPEST POINT IN THE ATLANTIC OCEAN**.
- **Java / Sunda Trench**: **7,450 m** (Eastern Indian Ocean off Sumatra/Java) — **DEEPEST POINT IN THE INDIAN OCEAN**.
- **Factorian Deep (South Sandwich Trench)**: **7,434 m** (Southern Ocean) — **DEEPEST POINT IN THE SOUTHERN OCEAN**.
- **Molloy Deep**: **5,550 m** (Fram Strait, Arctic Ocean) — **DEEPEST POINT IN THE ARCTIC OCEAN**.

25. World Major Industrial Regions & Specialised Manufacturing / Mining Centres

Region / City	Country	Primary Manufacturing / Mineral Specialization	High-Yield Prelims Exam Nickname / Triggers
Ruhr Valley & Saar Basin	**Germany**	**Coal mining, Iron and Steel, Heavy Engineering**	• Known as the **"Ruhrgebiet" / Industrial Heart of Europe**; Duisburg (world's largest inland port).

Region / City	Country	Primary Manufacturing / Mineral Specialization	High-Yield Prelims Exam Nickname / Triggers
Pittsburgh	**Pennsylvania, USA**	**Iron and Steel Industry**	• Celebrated as the **"STEEL CITY OF THE WORLD"**; confluence of Allegheny and Monongahela rivers.
Detroit	**Michigan, USA**	**Automobile Manufacturing**	• Celebrated as the **"AUTOMOBILE CAPITAL OF THE WORLD"** (Headquarters of Ford, GM, Chrysler).
Birmingham	**West Midlands, England, UK**	**Iron, Steel, Heavy Metal Hardware**	• Historic heart of the Industrial Revolution; central to the English **"Black Country"**.
Manchester & Lancashire	**North-West England, UK**	**Cotton Textile Manufacturing**	• World's premier historic cotton textile spinning cluster due to humid oceanic climate.
Katanga (Shaba) & Copperbelt	**DR Congo & Zambia** (Central Africa)	**Copper and Cobalt Extraction**	• World's richest mineral belt for battery-grade **Cobalt and high-grade Copper**.
Witwatersrand ("The Rand")	**Johannesburg, South Africa**	**Gold Mining**	• **World's largest gold-bearing reef formation**; produced ~40% of all gold mined in human history.
Kimberley	**Northern Cape, South Africa**	**Diamond Mining**	• Historic diamond capital; the **"Big Hole"**; origin of the geological term **Kimberlite pipe**.
Baku	**Azerbaijan** (Caspian Sea)	**Petroleum Extraction & Refining**	• Historic birthplace of modern commercial oil extraction; Nobel Brothers and Rothschild oil legacy.
Krivoy Rog & Donbass (Donets Basin)	**Ukraine**	**Iron Ore & Coking Coal**	• Krivoy Rog = massive iron-ore basin; Donbass = premier historic coking coalfield.
Magnitogorsk & Kuznetsk (Kuzbass)	**Russia** (Urals & Siberia)	**Iron & Steel Combine**	• Magnitogorsk built around Magnetic Mountain (iron ore); Kuznetsk is a giant Siberian coal basin.
Upper Silesia	**Poland**	**Coal Mining, Zinc & Lead, Heavy Industry**	• Industrial core of Central Europe around Katowice.
Chuquicamata	**Antofagasta, Atacama Desert, Chile**	**Copper Mining**	• **World's largest open-pit copper mine by volume** and second deepest open-pit mine on Earth.
Potosí (Cerro Rico)	**Bolivia** (Andes Mountains)	**Silver Mining**	• Historic "Rich Mountain" that bankrolled the Spanish Empire with vast silver wealth for centuries.
Kalgoorlie & Coolgardie	**Western Australia**	**Gold Mining**	• Historic Australian desert gold rush of 1890s; famous **"Golden Mile"** open-pit Super Pit gold mine.
Sudbury Basin	**Ontario, Canada**	**Nickel, Copper, Platinum Group Metals**	• Ancient 1.85-billion-year-old meteorite impact basin; premier global source of nickel.

Famous Indian Industrial Nicknames & City Specializations

- **Detroit of India**: **Chennai** (Tamil Nadu — automobile manufacturing hub of South Asia).
- **Manchester of India**: **Ahmedabad** (Gujarat — premier cotton textile hub).
- **Manchester of North India**: **Kanpur** (Uttar Pradesh — historic cotton mills and leather industry).
- **Manchester of South India**: **Coimbatore** (Tamil Nadu — cotton spinning mills).
- **Cottonopolis of India**: **Mumbai** (Maharashtra — historic concentration of textile mills).

- **Silicon Valley of India / Electronic City**: **Bengaluru** (Karnataka — premier IT & aerospace cluster).
- **Pittsburgh of India / Steel City**: **Jamshedpur / Tatanagar** (Jharkhand — TISCO established 1907 at Sakchi, confluence of Subarnarekha and Kharkai rivers).
- **Brass City (*Peetal Nagari*)**: **Moradabad** (Uttar Pradesh — world-renowned brass handicraft industry).
- **Glass City / Bangle City (*Suhag Nagari*)**: **Firozabad** (Uttar Pradesh — glass blowing and bangles).
- **Leather City of the World**: **Kanpur** (Uttar Pradesh).
- **Perfume Capital of India (*Itra Nagari*)**: **Kannauj** (Uttar Pradesh).
- **Weaver City / City of Handlooms**: **Panipat** (Haryana).
- **City of Locks**: **Aligarh** (Uttar Pradesh).
- **Sports Goods Hub**: **Meerut** (Uttar Pradesh) and **Jalandhar** (Punjab).

SUBJECT 06 · ENVIRONMENT & ECOLOGY

SUBJECT 06: ENVIRONMENT & ECOLOGY

पर्यावरण एवं पारिस्थितिकी — दैनिक उच्च-प्राथमिकता तथ्य

Environmental Legislation & Acts, Wildlife Projects, Biosphere Reserves, Ramsar Wetlands in UP & UK, International Conventions (UNFCCC, CBD, Ramsar), and IUCN Red List Status.

1. Core Environmental Legislation in India (Exact Years & Articles)

Legislation / Policy Name	Year of Enactment	Constitutional Backing / Key Mandate	Landmark Institutional Outcome
Indian Forest Act	**1927** (earlier 1865, 1878)	Consolidated forest laws; categorized into **Reserved, Protected, and Village Forests**	State Forest Departments
Wildlife (Protection) Act (WPA)	**1972** (9 Sept 1972)	Framework for protecting wild animals, birds, and plants; **6 original schedules restructured into 4 schedules in 2022 Amendment**	National Board for Wildlife (NBWL, chaired by PM); NTCA; Wildlife Crime Control Bureau (WCCB)
Water (Prevention and Control of Pollution) Act	**1974**	Prevention and abatement of water pollution; maintains wholesomeness of water	Established **Central Pollution Control Board (CPCB)** & State PCBs
Forest (Conservation) Act (FCA)	**1980** (25 Oct 1980)	Prohibits diversion of forest land for non-forest purposes without prior central approval	Forest clearance mechanism
Air (Prevention and Control of Pollution) Act	**1981**	Prevention, control, and abatement of air pollution; **Noise pollution included as an air pollutant by 1987 Amendment**	Empowered CPCB and SPCBs to set ambient air standards
Environment (Protection) Act (EPA)	**1986** (23 May 1986)	**Umbrella legislation** enacted under **Article 253** in the wake of the **Bhopal Gas Tragedy (Dec 1984)**	Empowered Union Govt to coordinate state actions, set emission norms; CRZ notifications; EIA rules
Public Liability Insurance Act	**1991**	Mandatory public insurance for providing immediate relief to victims of accidents occurring while handling hazardous substances	Environment Relief Fund
National Environment Tribunal Act	**1995**	Strict liability for damages arising out of any accident while handling hazardous substances	Precursor to NGT
Biological Diversity Act (BDA)	**2002**	Realized commitments under CBD (1992); 3-tier structure for conservation and equitable benefit sharing	**National Biodiversity Authority (NBA, Chennai)**, State Biodiversity Boards (SBB), Biodiversity Management Committees (BMC)
Forest Rights Act (FRA)	**2006**	Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act; vests forest land rights	Gram Sabha is the statutory authority to initiate rights recognition process

Legislation / Policy Name	Year of Enactment	Constitutional Backing / Key Mandate	Landmark Institutional Outcome
National Green Tribunal (NGT) Act	**2010** (18 Oct 2010)	Created dedicated environmental tribunal under **Article 21 (Right to a healthy environment)**; mandated disposal within **6 months**	**Principal Bench: New Delhi**; 4 Regional Benches: **Bhopal, Pune, Kolkata, Chennai**
CAMPA Act	**2016**	Compensatory Afforestation Fund Act; manages funds collected for diversion of forest land (90% to State, 10% to Centre)	National & State Compensatory Afforestation Funds

2. Environmental Movements in India (Pioneers & Locations)

Movement	Year & Place	Key Leaders / Activists	Core Environmental Objective / Strategy
Bishnoi Movement	**1730**, Khejarli village (Jodhpur, Rajasthan)	**Amrita Devi Bishnoi**	King Abhay Singh ordered felling of **Khejri trees** (*Prosopis cineraria*); Amrita Devi and her 3 daughters hugged trees; **363 Bishnois sacrificed their lives**; Amrita Devi Bishnoi Wildlife Protection Award instituted in memory.
Chipko Movement	**1973**, Reni village, Chamoli (**Uttarakhand**)	**Sunderlal Bahuguna**, **Chandi Prasad Bhatt** (Ramon Magsaysay awardee), **Gaura Devi** (*Chipko Mother*)	Protest against commercial logging of ash trees in Alaknanda basin; villagers hugged trees to prevent felling; coined slogan: "*Ecology is permanent economy*"; led to 15-year ban on green felling in UP/UK hills.
Appiko Movement	**1983**, Salkani, Uttara Kannada (Karnataka)	**Pandurang Hegde**	Southern version of Chipko; hugged trees in Western Ghats to save tropical forests from commercial monoculture plantations.
Silent Valley Movement	**1978**, Palakkad (Kerala)	Kerala Sastra Sahitya Parishad (**KSSP**), Romulus Whitaker	Protest against proposed **hydroelectric project on Kunthipuzha River** that would submerge the pristine moist evergreen rainforest habitat of the endangered **Lion-Tailed Macaque**; project scrapped; declared Silent Valley National Park (1984).
Narmada Bachao Andolan (NBA)	**1985**, Narmada Valley (Gujarat, MP, Maha)	**Medha Patkar**, **Baba Amte**	Mass movement against large dams on Narmada (particularly **Sardar Sarovar Dam**); demanded comprehensive rehabilitation, environmental impact assessment, and rights of displaced tribal villagers.
Jungle Bachao Andolan	**1982**, Singhbhum (Bihar / Jharkhand)	Local tribal communities	Tribals agitated against government scheme to replace indigenous, multipurpose **Sal forests** with commercially profitable **Teak (Sagwan)** plantations; known as "Greed Game Political Populism".
Tehri Dam Protest	1980s–1990s, Tehri (**Uttarakhand**)	**Sunderlal Bahuguna**	Agitation against the construction of the mega Tehri Dam on the seismic fault lines of Bhagirathi River; multiple hunger strikes.

3. Environmental Research Institutes & Organizations (Match Goldmine)

Institute / Organization	Location / Headquarters	Established Year / Key Function
Botanical Survey of India (BSI)	**Kolkata** (West Bengal)	**1890**; survey and conservation of plant resources of India
Zoological Survey of India (ZSI)	**Kolkata** (West Bengal)	**1916**; survey, exploration, and taxonomic research on animal diversity

Institute / Organization	Location / Headquarters	Established Year / Key Function
Forest Research Institute (FRI)	**Dehradun (Uttarakhand)**	**1906** (Imperial Forest Research Institute); Greco-Roman architectural marvel
Wildlife Institute of India (WII)	**Dehradun (Uttarakhand)**	**1982**; premier autonomous institution for wildlife research and management
National Environmental Engineering Research Institute (NEERI)	**Nagpur** (Maharashtra)	**1958**; pioneer in environmental monitoring, sewage, industrial pollution solutions (CSIR)
Central Pollution Control Board (CPCB)	**New Delhi**	**1974**; statutory organization under Water Act 1974; sets national air/water norms
National Biodiversity Authority (NBA)	**Chennai** (Tamil Nadu)	**2003**; statutory body under Biological Diversity Act 2002
Sálim Ali Centre for Ornithology & Natural History (SACON)	**Coimbatore** (Tamil Nadu)	Dedicated to conservation of avifauna and wetland ecology
Indian Institute of Forest Management (IIFM)	**Bhopal** (Madhya Pradesh)	**1982**; premier forestry management and planning institute
G.B. Pant National Institute of Himalayan Environment	**Kosi-Katarmal, Almora (Uttarakhand)**	**1988**; apex research body on sustainable development and ecology of Indian Himalayas
Central Arid Zone Research Institute (CAZRI)	**Jodhpur** (Rajasthan)	**1959**; research on desertification and arid land agriculture
National Disaster Management Authority (NDMA)	**New Delhi**	**2005**; apex body under Disaster Management Act 2005; **Chaired by the Prime Minister**

4. Indian Wildlife Conservation Projects (Chronological Ladder)

Project Name	Year Launched	Launch Location / Nodal Agency	Key Objective / Status / Prelims Fact
Project Tiger	**1 April 1973**	**Jim Corbett National Park (Uttarakhand)**	Central scheme; initiated with 9 reserves under Kailash Sankhala ("Tiger Man of India"); statutory backing via NTCA (2006); **58 Tiger Reserves** (Madhav TR, MP, is the 58th). Corbett has among the highest tiger densities.
Project Crocodile	**1975**	Hyderabad / Odisha (with UNDP & FAO aid)	Conservation of 3 crocodilian species: Gharial (*Gavialis gangeticus*), Mugger crocodile, and Saltwater crocodile (Bhitarkanika, Odisha)
Project Hangul	**1970**	Dachigam National Park (J&K)	Conservation of Kashmir Stag (*Cervus elaphus hanglu* - critically endangered) with IUCN/WWF aid
Project Rhino (IRV 2020)	**1987 / 2005**	Assam (Kaziranga, Manas, Orang, Pobitora)	Indian Rhino Vision 2020 aimed to attain a wild population of at least 3,000 Greater One-Horned Rhinos in Assam
Project Elephant	**February 1992**	Jharkhand (Singhbhum)	Centrally sponsored; protects elephant habitats and corridors; addresses human-elephant conflict; **Mike sites** (Monitoring of Illegal Killing of Elephants); campaign: *Haathi Mere Saathi*
Project Sea Turtle	**1999**	Odisha coast (Gahirmatha / Rushikulya)	Conservation of **Olive Ridley turtles** (*Lepidochelys olivacea*) during synchronized mass nesting (**Arribada**) with MoEFCC & UNDP
Project Snow Leopard	**2009**	High-altitude Himalayan states (Hemis, Gangotri, Nanda Devi)	Conservation of high-altitude ecosystem and the elusive Ghost of the Mountains (*Panthera uncia*)

Project Name	Year Launched	Launch Location / Nodal Agency	Key Objective / Status / Prelims Fact
Project Dolphin	**2020**	Ministry of Environment, Forest & Climate Change	Gangetic Dolphin (*Platanista gangetica*) declared **National Aquatic Animal on 5 October 2009** (5 Oct celebrated as **National Dolphin Day**); blind; uses echolocation; Vikramshila Sanctuary (Bihar)
Project Cheetah	**September 2022**	**Kuno National Park (Madhya Pradesh)**	Intercontinental translocation of African cheetahs from Namibia (8) and South Africa (12); Asiatic Cheetah declared extinct in India in **1952**

5. Ramsar Wetlands in Uttar Pradesh & Uttarakhand

- **Uttarakhand**:
- **Asan Conservation Reserve** (Dehradun; confluence of Asan and Yamuna; **first Ramsar of Uttarakhand, Oct 2020**).
- **Jhilmil Jheel Conservation Reserve** (Haridwar, **2005**) is the state's swamp-deer / barasingha wetland — **not** the Ramsar first (that is Asan).
- **Uttar Pradesh (12 Ramsar sites)** — do not stop at the older list of 10:
- **Bakhira Wildlife Sanctuary** (Sant Kabir Nagar).
- **Haiderpur Wetland** (Muzaffarnagar–Bijnor; Solani / Hastinapur WLS).
- **Nawabganj Bird Sanctuary** (*Chandra Shekhar Azad*, Unnao).
- **Parvati Arga Bird Sanctuary** (Gonda).
- **Saman Bird Sanctuary** (Mainpuri).
- **Samaspur Bird Sanctuary** (Rae Bareli).
- **Sandi Bird Sanctuary** (Hardoi).
- **Sarsai Nawar Jheel** (Etawah; Sarus Crane).
- **Sur Sarovar (Keetham Lake)** (Agra).
- **Upper Ganga River** (Brijghat to Narora).
- **Patna Bird Sanctuary** (Etah).
- **Shekha Jheel** (Aligarh).

> **Traps:** Sultanpur NP = **Haryana**. Rudrasagar = **Tripura**. Samaspur and Sarsai Nawar **are** in UP.

5A. Protected areas of UP & UK — Declared / still in force

PA	State	From (year)	High-yield
Hailey → Ramganga → Jim Corbett NP / TR	**Uttarakhand**	NP **1936** (Hailey); Project Tiger launch site **1973**	India's first National Park. Nainital–Pauri belt.
Rajaji NP / TR	**Uttarakhand**	NP **1983**; tiger reserve **2015**	Elephant corridor; Haridwar–Dehradun–Pauri
Nanda Devi NP	**Uttarakhand**	**1982** (BR **1988**; UNESCO mixed serial with VoF)	Chamoli; Nanda Devi peak
Valley of Flowers NP	**Uttarakhand**	**1982** (UNESCO **1988**, expanded **2005**)	Smallest UK NP; Chamoli
Gangotri NP	**Uttarakhand**	**1989**	Largest UK NP (~2390 sq km); Uttarkashi
Govind NP / WLS	**Uttarakhand**	WLS **1955**; NP later	Snow leopard belt; Uttarkashi
Dudhwa NP / TR	**Uttar Pradesh**	NP **1977**; TR **1987**	Lakhimpur Kheri Terai; rhino from **1984**; barasingha
Pilibhit TR	**Uttar Pradesh**	**2014**	Terai; tiger + woodland swamp deer

PA	State	From (year)	High-yield
Ranipur TR	**Uttar Pradesh**	**2022/2023**	Chitrakoot–Banda (Vindhyan); not Terai
Hastinapur WLS	**Uttar Pradesh**	**1986**	**Largest UP sanctuary**; Meerut–Bijnor belt; Haiderpur Ramsar inside
Katarniaghat WLS	**Uttar Pradesh**	**1975**	Terai Arc toward Nepal; gharial / dolphin

UP has **no** biosphere reserve and **no** marine PA. UK biosphere = **Nanda Devi (1988; UNESCO MAB 2004)**.

6. National Air Quality Index (AQI)

Launched in **October 2014** under Swachh Bharat Abhiyan by CPCB: - **Six Categories**: 1. *Good* (0–50) 2. *Satisfactory* (51–100) 3. *Moderately Polluted* (101–200) 4. *Poor* (201–300) 5. *Very Poor* (301–400) 6. *Severe* (401–500). - **Eight Monitored Pollutants (Frequently Tested in PCS!)**: 1. **Particulate Matter 10 (PM10)** 2. **Particulate Matter 2.5 (PM2.5)** 3. **Nitrogen Dioxide (NO2)** 4. **Sulphur Dioxide (SO2)** 5. **Carbon Monoxide (CO)** 6. **Ozone (O3)** 7. **Ammonia (NH3)** 8. **Lead (Pb)**. *(Notice: Carbon Dioxide (CO2) is NOT part of the AQI 8 pollutants!)*

7. The 18 Biosphere Reserves of India & 12 UNESCO MAB Reserves

No.	Biosphere Reserve	Year Set Up	State / Region	Part of UNESCO MAB (WNBR)?
1	**Nilgiri**	**1986**	Tamil Nadu, Kerala, Karnataka	**Yes (2000 — First in India)**
2	**Nanda Devi**	**1988**	**Uttarakhand** (Chamoli, Pithoragarh, Bageshwar)	**Yes (2004)**
3	**Nokrek**	**1988**	Meghalaya (Garo Hills)	**Yes (2009)**
4	**Great Nicobar**	**1989**	Andaman and Nicobar Islands	**Yes (2013)**
5	**Gulf of Mannar**	**1989**	Tamil Nadu (marine)	**Yes (2001)**
6	**Manas**	**1989**	Assam	No
7	**Sundarbans**	**1989**	West Bengal	**Yes (2001)**
8	**Similipal**	**1994**	Odisha (Mayurbhanj)	**Yes (2009)**
9	**Dibru-Saikhowa**	**1997**	Assam (*Smallest Biosphere Reserve*)	No
10	**Dehang-Debang**	**1998**	Arunachal Pradesh	No
11	**Pachmarhi**	**1999**	Madhya Pradesh (Satpura Range)	**Yes (2009)**
12	**Khangchendzonga**	**2000**	Sikkim	**Yes (2018)**
13	**Agasthyamalai**	**2001**	Kerala and Tamil Nadu	**Yes (2016)**
14	**Achanakmar-Amarkantak**	**2005**	Madhya Pradesh and Chhattisgarh	**Yes (2012)**
15	**Great Rann of Kutch**	**2008**	Gujarat (*Largest Biosphere Reserve*)	No
16	**Cold Desert**	**2009**	Himachal Pradesh (Pin Valley, Kibber, Chandratal)	No
17	**Seshachalam Hills**	**2010**	Andhra Pradesh (Tirupati)	No
18	**Panna**	**2011**	Madhya Pradesh	**Yes (2020)**

8. International Environmental Conventions & Protocols

Year	Convention / Agreement	Place / City	Primary Objective / Regulatory Scope
1971 (2 Feb)	**Ramsar Convention on Wetlands**	Ramsar (Iran)	Conservation and wise use of wetlands; entered into force 1975; **World Wetlands Day (2 Feb)**; India joined **1 Feb 1982**; Keoladeo & Chilika were first sites; **Montreux Record** (Keoladeo NP & Loktak Lake)
1972 (5–16 Jun)	**Stockholm Conference (UNCHE)**	Stockholm (Sweden)	UN Conference on the Human Environment; **Created UNEP (Nairobi, Kenya)**; celebrated as **World Environment Day (5 June)**
1973 (3 Mar)	**CITES (Washington Convention)**	Washington, D.C. (USA)	Convention on International Trade in Endangered Species of Wild Fauna and Flora; Appendix I, II, III; **World Wildlife Day (3 March)**
1979	**Bonn Convention (CMS)**	Bonn (Germany)	Convention on the Conservation of Migratory Species of Wild Animals
1982	**UNCLOS**	Montego Bay (Jamaica)	Law of the Sea: Territorial waters (12 nm), Contiguous zone (24 nm), Exclusive Economic Zone (200 nm)
1985	**Vienna Convention**	Vienna (Austria)	Framework convention for the protection of the stratospheric ozone layer
1987 (16 Sep)	**Montreal Protocol**	Montreal (Canada)	**Legally binding phase-out of Ozone Depleting Substances (ODS / CFCs)**; entered into force 1 Jan 1989; **World Ozone Day (16 Sept)**; **Kigali Amendment (2016)** phase out HFCs
1987	**Brundtland Report**	UN World Commission	Report titled ***Our Common Future***; formally defined **Sustainable Development**
1989	**Basel Convention**	Basel (Switzerland)	Control of transboundary movements of **Hazardous Wastes and their disposal**; entered into force 1992
1992 (3–14 Jun)	**Rio Earth Summit (UNCED)**	Rio de Janeiro (Brazil)	Landmark summit producing 3 Rio Conventions: 1. **UNFCCC** (Climate Change) 2. **CBD** (Biological Diversity) 3. **UNCCD** (Combating Desertification) Also produced: **Agenda 21**, Rio Declaration, and Forest Principles
1997 (11 Dec)	**Kyoto Protocol**	Kyoto (Japan)	UNFCCC COP-3; **Legally binding greenhouse gas emission reduction targets for Annex-I developed countries**; 6 greenhouse gases (CO2, CH4, N2O, HFCs, PFCs, SF6); Clean Development Mechanism (CDM); entered into force 2005; expired 2020
1998	**Rotterdam Convention**	Rotterdam (Netherlands)	Prior Informed Consent (PIC) procedure for certain hazardous chemicals and pesticides in international trade
2000	**Cartagena Protocol on Biosafety**	Montreal / Cartagena	Protocol to CBD governing the safe transfer, handling, and use of **Living Modified Organisms (LMOs)**
2001	**Stockholm Convention on POPs**	Stockholm (Sweden)	Elimination or restriction of **Persistent Organic Pollutants (POPs)** (original "Dirty Dozen" including DDT, Dioxins, PCBs)
2010	**Nagoya Protocol**	Nagoya (Japan)	Supplementary agreement to CBD on **Access to Genetic Resources and the Fair and Equitable Sharing of Benefits (ABS)**
2013	**Minamata Convention**	Kumamoto (Japan)	Global treaty to protect human health and the environment from adverse effects of **Mercury (Hg)**
2015 (12 Dec)	**Paris Agreement**	Paris (France)	UNFCCC COP-21; keep global temperature rise **well below 2°C** above pre-industrial levels, pursue efforts to limit rise to **1.5°C**; Nationally Determined Contributions (NDCs); entered into force **4 Nov 2016**

Year	Convention / Agreement	Place / City	Primary Objective / Regulatory Scope
2015	**Sustainable Development Goals (SDGs)**	UN General Assembly	**17 Goals, 169 Targets (2015–2030)**; SDG 1: No Poverty, SDG 2: Zero Hunger, SDG 3: Good Health, SDG 4: Quality Education, SDG 5: Gender Equality, SDG 6: Clean Water & Sanitation, SDG 13: Climate Action, SDG 14: Life Below Water, SDG 15: Life on Land
2022	**Kunming-Montreal GBF**	Montreal (CBD COP-15)	Global Biodiversity Framework; includes landmark **"30x30 target"** (protect at least 30% of planet's land and ocean by 2030)

9. Fundamental Ecological Principles & Laws

A. Lindeman's 10% Energy Law (1942)

- Formulated by **Raymond Lindeman**.
- Only **approximately 10% of the energy** entering any trophic level is transferred and organized into organic matter at the next higher trophic level.
- 90% of energy is lost as metabolic heat, respiration, and catabolism.
- Explains why food chains rarely exceed 4 to 5 trophic levels.

B. Ecological Pyramids (Charles Elton, 1927)

1. **Pyramid of Energy**:
2. **ALWAYS UPRIGHT, WITHOUT EXCEPTION!**
3. Energy decreases progressively at each successive trophic level according to thermodynamics.
4. **Pyramid of Biomass**:
5. **Terrestrial Ecosystems (Grassland, Forest)**: **UPRIGHT** (Producer biomass > Herbivore > Carnivore).
6. **Aquatic Ecosystems (Pond, Lake, Ocean)**: **INVERTED** (Phytoplankton standing crop biomass is much smaller than zooplankton and predatory fish).
7. **Pyramid of Numbers**:
8. **Grassland & Cropland**: **UPRIGHT** (Millions of grasses support fewer herbivores, and even fewer carnivores).
9. **Tree / Parasitic Ecosystem**: **INVERTED** (A single large tree supports hundreds of birds, which host thousands of parasites and hyper-parasites).
10. **Forest Ecosystem**: **Spindle-shaped** (Few large trees support many herbivores, which are eaten by fewer carnivores).

C. Bioaccumulation vs Biomagnification

- **Bioaccumulation**: Increase in the concentration of a chemical pollutant in an **individual organism over time** (rate of intake > rate of excretion).
- **Biomagnification**: Progressive increase in the concentration of persistent, non-biodegradable, fat-soluble toxins (e.g., **DDT, Mercury, PCBs**) at **successively higher trophic levels** of a food chain.
- *Classic example:* Phytoplankton (0.00005 ppm) → Zooplankton → Small Fish → Fish-eating birds / Falcons (25 ppm causing egg-shell thinning and population collapse).

D. Greenhouse Gases: Global Warming Potential (GWP, 100-Year Horizon)

Ranked from highest warming impact per molecule to lowest (relative to CO2 = 1):

SF6 (23,500) > PFCs (7,000–9,000) > HFCs (1,400–3,000) > N2O (273) > CH4 (28) > CO2 (1)

- **Water Vapour**: Most abundant natural greenhouse gas contributing greatest share to natural greenhouse effect (~60%), but has short atmospheric residence time.
- **Carbon Dioxide (CO2)**: Primary anthropogenic driver of global climate change due to massive volume and century-long residence time.
- **Methane (CH4)**: Major sources: Paddy/rice fields, ruminant livestock enteric fermentation, landfills, wetlands, coal mining.

10. Important Environmental Days (UPPCS Direct Match)

Date	Environmental Observance	Connected Milestone / Significance
2 February	**World Wetlands Day**	Signing of the Ramsar Convention on Wetlands (1971)
3 March	**World Wildlife Day**	Adoption of CITES (Washington Convention, 1973)
21 March	**International Day of Forests**	Promotes awareness of the importance of all types of forests
22 March	**World Water Day**	Focuses on the importance of freshwater
23 March	**World Meteorological Day**	Establishment of WMO (1950)
22 April	**Earth Day**	First held in 1970; global environmental movement
22 May	**International Day for Biological Diversity**	Adoption of the agreed text of the CBD (1992)
5 June	**World Environment Day**	Opening day of the historic Stockholm Conference (1972)
8 June	**World Oceans Day**	Concept originally proposed at Rio Earth Summit 1992
29 July	**International Tiger Day**	Instituted at the Saint Petersburg Tiger Summit (2010)
12 August	**World Elephant Day**	Dedicated to the preservation and protection of elephants
16 September	**World Ozone Day**	Signing of the Montreal Protocol (1987)
5 October	**National Dolphin Day (India)**	Declaration of Gangetic Dolphin as National Aquatic Animal (2009)
16 October	**World Food Day**	Founding of the FAO (1945)
5 December	**World Soil Day**	Promotes importance of healthy soil and sustainable soil management

11. Foundational Ecological Terms & Coining Scientists / Pioneers

Term / Ecological Concept	Coined By / Pioneer	Year	Core Definition / Exam Relevance
Ecology (*Oekologie*)	**Ernst Haeckel**	**1866**	Coined from Greek *oikos* (house/dwelling) + *logos* (study). Study of interactions between living organisms and their physical environment.
Ecosystem	**Arthur Tansley (A.G. Tansley)**	**1935**	The basic functional ecological unit consisting of biotic community together with its abiotic environment interacting as a system.
Biodiversity (Biological Diversity)	**Walter G. Rosen** (coined); popularized by **Edward O. Wilson (E.O. Wilson)**	**1985 / 1988**	Coined as contraction of "biological diversity". E.O. Wilson published landmark book *Biodiversity* (1988); known as "Father of Biodiversity".
Biodiversity Hotspots	**Norman Myers**	**1988**	Biogeographic regions characterized by exceptional levels of plant endemism and high levels of habitat loss (≥70% loss). Adopted by Conservation International.
Ecological Niche	**Joseph Grinnell** (habitat niche); refined by **Charles Elton** & **G.E. Hutchinson**	**1917 / 1927 / 1957**	The specific functional role and physical space occupied by an organism within its ecosystem (*address* + *profession of species*). Hutchinson introduced "fundamental vs realized niche".
Deep Ecology	**Arne Naess**	**1973**	Radical environmental philosophy rejecting anthropocentrism; advocates that all living beings have intrinsic moral value regardless of their utility to humans.

Term / Ecological Concept	Coined By / Pioneer	Year	Core Definition / Exam Relevance
Ecological Footprint	**William Rees & Mathis Wackernagel**	**1992**	Measure of human demand on Earth's ecosystems; the area of biologically productive land and water required to produce goods consumed and absorb waste.
Gaia Hypothesis	**James Lovelock & Lynn Margulis**	**1972**	Proposes that living organisms interact with their inorganic surroundings on Earth to form a synergistic and self-regulating complex system that helps maintain conditions for life.
Acid Rain	**Robert Angus Smith**	**1872**	Discovered connection between atmospheric pollution (SO2 & NOx from coal burning) and acid precipitation in Manchester, UK. Normal rain pH ~5.6; acid rain pH < 5.6.
Smog (Smoke + Fog)	**H.A. Des Voeux**	**1905**	Coined at the Public Health Congress in London to describe the deadly mixture of coal smoke and dense winter fog (**Classical / London Smog — reducing in nature, contains SO2**).
Photochemical Smog (Los Angeles Smog)	**Arie Haagen-Smit**	**1950s**	Sunlight-induced photochemical reaction between Nitrogen Oxides (NOx) and Volatile Organic Compounds (VOCs). **Oxidizing in nature; major components: Ozone (O3), PAN (Peroxyacetyl nitrate), and Aldehydes**.
Discovery of Antarctic Ozone Hole	**Joe Farman, Brian Gardiner, Jonathan Shanklin**	**May 1985**	British Antarctic Survey scientists discovered drastic springtime thinning of the stratospheric ozone layer over Halley Bay, Antarctica, caused by Chlorofluorocarbons (CFCs).
Biocoenosis	**Karl Möbius**	**1877**	An interacting community of different living species occupying a common biotope/habitat.
Father of Indian Ecology	**Prof. Ramdeo Misra**	1950s–1970s	Established pioneering ecological research and education in India at Banaras Hindu University (BHU), Varanasi.
Ecotone & Edge Effect	**Frederic Clements**	1905	**Ecotone**: Transition zone between two distinct biomes/communities (e.g., mangrove between marine & terrestrial; marshland between aquatic & dry land). **Edge Effect**: Phenomenon where species richness and population density are significantly greater in the ecotone than in either adjoining community (e.g., birds in forest edge).

12. Major Environmental Pollutants, Toxic Heavy Metals & Occupational Diseases

Toxic Pollutant / Metal	Source / Environmental Medium	Characteristic Disease / Syndrome	Clinical Pathology / Prelims Highlights
Methylmercury (Hg)	Industrial effluent discharged into coastal waters; biomagnified in fish	**Minamata Disease**	First discovered in **Minamata Bay (Kumamoto, Japan, 1956)**; caused by Chisso Corp chemical plant; severe neurological syndrome, ataxia, loss of motor control, sensory impairment, convulsions, congenital deformities.
Cadmium (Cd)	Mining runoff, zinc smelting, contaminated irrigation water for paddy fields	**Itai-Itai Disease ("Ouch-Ouch Disease")**	First discovered in **Toyama Prefecture (Japan, 1912)**; severe osteomalacia (extreme bone softening and painful skeletal fractures) and kidney failure; *Itai-Itai* means "it hurts, it hurts" in Japanese.

Toxic Pollutant / Metal	Source / Environmental Medium	Characteristic Disease / Syndrome	Clinical Pathology / Prelims Highlights
Nitrate (NO3-)	Runoff from nitrogenous chemical fertilizers in groundwater drinking water	**Blue Baby Syndrome (Methemoglobinemia)**	Nitrate converts to nitrite in infant gut; binds with infant hemoglobin to form **methemoglobin**, which cannot release oxygen to tissues; skin turns blue due to asphyxiation; fatal if untreated.
Arsenic (As)	Natural geochemical leaching into shallow groundwater aquifers	**Blackfoot Disease, Hyperkeratosis & Arsenicosis**	Severe peripheral vascular gangrene ("black foot"), skin hyper-pigmentation, keratosis, and bladder/skin cancer. Rampant in **Ganga-Brahmaputra alluvial plains: West Bengal, UP, Bihar, and Bangladesh**. Permissible limit: **0.01 mg/L** (WHO).
Fluoride (F-)	Over-extracted groundwater leaching from fluorapatite minerals	**Dental & Skeletal Fluorosis**	Permissible limit: **1.0 to 1.5 mg/L**. Excess causes mottling of teeth, calcification of ligaments, severe crippling skeletal deformity, and **Knock-Knee Syndrome (Genu Valgum)**. Widespread in Rajasthan, Telangana, UP.
Silica (SiO2 dust)	Inhalation of crystalline silica dust in stone quarries, agate cutting, sandblasting	**Silicosis**	Incurable occupational fibrotic lung disease causing permanent pulmonary scarring and progressive respiratory failure.
Asbestos Fibers	Asbestos cement manufacturing, building demolition, brake lining manufacture	**Asbestosis & Mesothelioma**	Asbestos fibers penetrate deep into lungs causing pulmonary fibrosis; strongly carcinogenic, inducing **malignant mesothelioma** (cancer of pleural lining).
Coal Dust	Underground coal mining and extraction	**Pneumoconiosis (Black Lung Disease / Anthracosis)**	Chronic inflammation and irreversible progressive massive fibrosis of lungs due to accumulated inhaled coal mine dust.
Cotton Dust	Cotton mills, textile spinning, and ginning factories	**Byssinosis ("Brown Lung Disease" / Monday Fever)**	Occupational asthma and chest tightness occurring on the first workday after a weekend break (*Monday disease*).
Sugarcane Bagasse	Inhalation of dry bagasse fibers during sugarcane processing & paper milling	**Bagassosis**	Hypersensitivity pneumonitis triggered by thermoactinomycetes in moldy bagasse.
Lead (Pb)	Lead-acid batteries, leaded gasoline (historic), paints, smelters, e-waste	**Plumbism / Saturnism**	Neurotoxic; causes severe cognitive and developmental impairment in children, anemia (inhibits heme synthesis), kidney damage, and characteristic **Burtonian blue line along gums**.
E-Waste Hazardous Elements	Discarded computers, mobile phones, printed circuit boards (PCBs)	Multi-organ toxicity	Lead (solder & CRTs), Cadmium (rechargeable Ni-Cd batteries & chip resistors), Mercury (switches & flat screens), Brominated Flame Retardants (BFRs - endocrine disruption).

13. Biodiversity Hotspots of India & Global Criteria

A. Global Criteria for Biodiversity Hotspot (Norman Myers, 1988)

To qualify as an international biodiversity hotspot, a biogeographic region must fulfill **TWO STRICT QUANTITATIVE CRITERIA**: 1. **High Endemism**: It must contain at least **1,500 species of vascular plants (>0.5% of the world's total) as endemics** (species found nowhere else on Earth). 2. **High Degree of Threat**: It must have lost at least **70% of its original**

primary native vegetation (i.e., has 30% or less of its original natural habitat remaining). - Globally, there are **36 recognized Biodiversity Hotspots** covering only 2.5% of Earth's land surface, but supporting over 50% of world's endemic plant species and 43% of endemic bird, mammal, reptile, and amphibian species.

B. The 4 Biodiversity Hotspots Covering India

Hotspot Name	Geographical Extent in India	Adjoining International Regions	Landmark Endemic Species / Key Features
1. Himalaya Hotspot	Covers the **entire Indian Himalayan region**: Jammu & Kashmir, Ladakh, Himachal Pradesh, **Uttarakhand**, Sikkim, and Arunachal Pradesh.	Pakistan, Nepal, Bhutan, Tibet / Southern China	Snow Leopard, Himalayan Musk Deer, Himalayan Tahr, Blue Sheep (*Bharal*), Red Panda; vast alpine and temperate rhododendron-oak ecosystems.
2. Indo-Burma Hotspot	Covers all of **North-East India south of the Brahmaputra River** (Assam plains/hills, Meghalaya, Nagaland, Manipur, Mizoram, Tripura) and the **Andaman Islands**.	Myanmar, Thailand, Laos, Vietnam, Cambodia, and Southern China	Hoolock Gibbon (India's only ape), Golden Langur, Stump-tailed Macaque; extraordinary diversity of orchids, ginger species, and freshwater turtles.
3. Western Ghats & Sri Lanka Hotspot	Covers the continuous mountain chain along the western coast across **Gujarat, Maharashtra, Goa, Karnataka, Kerala, and Tamil Nadu**.	Island nation of Sri Lanka	**Lion-Tailed Macaque** (Silent Valley), **Nilgiri Tahr** (Eravikulam), Nilgiri Langur, Malabar Gliding Frog; exceptional concentration of endemic amphibians (~80% endemic), reptiles, and flowering plants.
4. Sundaland Hotspot	Covers the **Nicobar Islands** of India (Great Nicobar, Little Nicobar).	Indonesia (Sumatra, Java, Borneo), Malaysia, Singapore, Brunei, and Southern Thailand	Nicobar Megapode (mound-building bird), Nicobar Tree Shrew, Saltwater Crocodile; rich coastal coral reefs and dense tropical rainforests.

Crucial Exam Trap: Andaman Islands fall under **Indo-Burma Hotspot**, while Nicobar Islands fall under **Sundaland Hotspot**!

SUBJECT 07 · POLITY & GOVERNANCE

SUBJECT 07: INDIAN POLITY & GOVERNANCE

भारतीय राजव्यवस्था एवं शासन — दैनिक उच्च-प्राथमिकता तथ्य

Borrowed Features, 25 Parts, 12 Schedules, High-Frequency Articles, Constitutional Offices (Ages/Tenures), Parliamentary Committees, Writs, and Landmark Supreme Court Judgments.

1. The **Constitution of India** is the lengthiest written constitution in the world; originally adopted on **26 November 1949** with **395 Articles, 22 Parts, and 8 Schedules**, it currently comprises **25 Parts and 12 Schedules** (with about 470 Articles in numerical continuity).
2. Under **Article 1**, India is described as a **'Union of States'** rather than a federation, signifying that the Indian federation is not the result of an agreement among States and that no State has the right to secede from the Union.
3. The **Preamble** declares India to be a **'Sovereign Socialist Secular Democratic Republic'**; the terms **'Socialist'**, **'Secular'**, and **'Integrity'** were added by the **42nd Constitutional Amendment Act, 1976** on the recommendation of the Swaran Singh Committee.
4. The Preamble secures to all citizens: **Justice** (Social, Economic, and Political — 3 types); **Liberty** (Thought, Expression, Belief, Faith, and Worship — 5 types); **Equality** (Status and Opportunity — 2 types); and promotes **Fraternity** assuring individual dignity and national unity/integrity.
5. In ***Kesavananda Bharati v. State of Kerala (1973)***, a 13-judge bench ruled (7:6) that the Preamble is an integral **part of the Constitution** (overruling the 1960 *Berubari Union* opinion) and formulated the **Basic Structure Doctrine**.
6. Major Constitutional Borrowings: **UK** provided Parliamentary Government, Rule of Law (Art. 14), Single Citizenship, Cabinet system, Prerogative Writs, Bicameralism, and the office of the Speaker.
7. **USA** provided **Fundamental Rights** (Part III), Independence of Judiciary, **Judicial Review** (Art. 13), **Impeachment of the President** (Art. 61), Removal of SC/HC Judges, and the office of the Vice-President.
8. **Ireland** provided the **Directive Principles of State Policy (DPSP)** (Part IV), the method of **Presidential Election** (PR-STV), and nomination of **12 members to Rajya Sabha** by the President.
9. **Canada** provided a Federation with a **strong Centre**, vesting of **Residuary Powers in the Centre** (Art. 248), appointment of State Governors by the Centre, and the **Advisory Jurisdiction of the Supreme Court** (Art. 143).
10. **Australia** provided the **Concurrent List** (7th Schedule), Freedom of Trade, Commerce and Intercourse (Part XIII), and the mechanism of **Joint Sitting of Parliament** (Art. 108).
11. **Germany (Weimar Constitution)** provided the provision for **suspension of Fundamental Rights during National Emergency** (Art. 359). Note: The emergency framework itself is largely drawn from the **Government of India Act, 1935**.
12. **USSR (Russia)** provided **Fundamental Duties** (Art. 51A, Part IVA) and the ideals of Social, Economic, and Political Justice in the Preamble; **France** provided the Republic and the ideals of Liberty, Equality, and Fraternity.
13. **South Africa** provided the procedure for **Amendment of the Constitution** (Art. 368) and the election of members of the Rajya Sabha by elected MLAs.
14. **Schedules Master Summary (TEARS OF OLD PM):** 1st (28 States & 8 UTs); 2nd (Salaries of President, Governors, Judges, CAG — *Ministers are NOT in 2nd Sch*); 3rd (Forms of Oaths — *President, VP, and Governors take oath under Arts. 60, 69, 159, NOT 3rd Sch*); 4th (Rajya Sabha seats: UP leads with 31 seats).
15. **Schedules 5 & 6:** The **5th Schedule** governs Scheduled Areas in 10 States; the **6th Schedule** provides Autonomous District Councils in **4 North-Eastern States: AMTM** (Assam, Meghalaya, Tripura, Mizoram; *Manipur is NOT included*).
16. **Schedules 7 to 12:** 7th (Union: 100, State: 61, Concurrent: 52 entries); 8th (22 Languages); 9th (Validation of Acts, added by 1st Amd 1951; post-1973 laws subject to judicial review per *Coelho case*); 10th (Anti-Defection Law, 52nd Amd 1985); 11th (Panchayats: **29 subjects**, 73rd Amd 1992); 12th (Municipalities: **18 subjects**, 74th Amd 1992).

17. **Eighth Schedule Language Additions:** Originally 14 languages; **Sindhi** added by 21st Amd (1967); **Konkani, Manipuri, Nepali** (KMN) added by 71st Amd (1992); **Bodo, Dogri, Maithili, Santhali** (BDMS) added by 92nd Amd (2003). *English and Bhojpuri are NOT in the 8th Schedule.*
18. **Citizenship (Articles 5–11, Part II):** India has **single citizenship**; the Citizenship Act of **1955** provides 5 modes of acquisition (Birth, Descent, Registration, Naturalisation, Incorporation) and 3 modes of loss (Renunciation, Termination, Deprivation).
19. **Fundamental Rights exclusively for Citizens:** Articles **15, 16, 19, 29, and 30** are available ONLY to Indian citizens; all other Fundamental Rights (Arts. 14, 20, 21, 21A, 22, 25, 26, 27, 28) apply to both citizens and foreigners (except enemy aliens).
20. **Article 12 & 13:** Article 12 defines 'State' for Part III purposes (including statutory and non-statutory authorities like LIC, ONGC, SAIL); Article 13 establishes **Judicial Review**, voiding any law violative of Fundamental Rights (*Doctrines of Severability and Eclipse*).
21. **Right to Equality (Articles 14–18):** Art. 14 (Equality before Law — UK negative concept; Equal Protection of Laws — US positive concept); Art. 15 (No discrimination *only* on grounds of religion, race, caste, sex, place of birth; 15(4) backward classes, 15(6) 10% EWS); Art. 16 (Equal opportunity in public employment); Art. 17 (Abolition of Untouchability — absolute right); Art. 18 (Abolition of Titles).
22. **Article 19 (Six Democratic Freedoms):** 19(1)(a) Speech & Expression; 19(1)(b) Peaceful assembly without arms; 19(1)© Associations/Unions/Cooperatives; 19(1)(d) Free movement throughout India; 19(1)(e) Reside and settle; 19(1)(g) Trade/Occupation. The right to property under 19(1)(f) was repealed by the **44th Amendment, 1978**.
23. **Article 20 (Three Criminal Safeguards):** Protection against **ex-post-facto criminal laws** (20(1)), protection against **Double Jeopardy** (20(2)), and protection against **Self-Incrimination** (20(3) — applies only to accused in criminal proceedings).
24. **Article 21 & 21A:** Article 21 guarantees protection of life and personal liberty; expanded in ***Maneka Gandhi (1978)*** to require 'Due Process of Law' (just, fair, and reasonable procedure) and in ***Puttaswamy (2017)*** to include the **Right to Privacy**. **Article 21A** (Right to Education ages 6–14) was added by the **86th Amendment, 2002**.
25. **Non-Suspendable Rights:** Under **Article 359** (as amended by 44th Amd 1978), the enforcement of **Articles 20 and 21 CANNOT BE SUSPENDED** even during a National Emergency.
26. **Prerogative Writs (Art. 32 vs Art. 226):** Dr. Ambedkar described **Article 32** as the 'Heart and Soul' of the Constitution. The Supreme Court issues writs only for Fundamental Rights, whereas the High Court under **Article 226** can issue writs for both Fundamental Rights and ordinary legal rights (making Art. 226 broader in scope).
27. **The Five Writs: Habeas Corpus** ('To have the body of', against both public and private entities); **Mandamus** ('We Command', against public officials for statutory duties; *cannot issue against President or Governor*); **Prohibition** (preventive against lower courts); **Certiorari** (quashes illegal orders of judicial/quasi-judicial/administrative bodies); **Quo-Warranto** ('By what authority', challenges illegal tenure in substantive public office).
28. **Key Directive Principles (Part IV):** Art. 39A (Equal Justice & Free Legal Aid — 42nd Amd); Art. 40 (Village Panchayats); Art. 43B (Cooperatives — 97th Amd); Art. 44 (Uniform Civil Code); Art. 45 (Early childhood care up to age 6 — 86th Amd); Art. 48A (Environment & Wildlife — 42nd Amd); Art. 50 (Separation of Judiciary from Executive).
29. **Fundamental Duties (Part IVA, Art. 51A):** Added by the **42nd Amendment, 1976** on the recommendation of the **Swaran Singh Committee**; non-justiciable; originally 10 duties; the **11th duty** (education of child aged 6–14 by parent/guardian) was added by the **86th Amendment, 2002**.
30. **Presidential Election (Articles 54 & 55):** Electoral College consists of: (1) Elected members of both Houses of Parliament; (2) Elected members of State Legislative Assemblies; (3) Elected members of Delhi and Puducherry Assemblies (70th Amd 1992). *Nominated MPs/MLAs and ALL Legislative Council members (MLCs) DO NOT VOTE.*
31. **Vice-Presidential Election (Article 66):** Electoral College consists of **ALL members of both Houses of Parliament (both Elected and Nominated)**. State MLAs and MLCs have NO role in the Vice-President's election.
32. **Impeachment of President (Article 61):** Quasi-judicial process initiated in either House on grounds of 'Violation of the Constitution'; requires 14 days' written notice signed by ¼th members; must be passed by a **special majority of not less than ⅔rds of the TOTAL MEMBERSHIP** of each House.
33. **Pardoning Power (Art. 72 vs Art. 161):** The **President (Art. 72)** can grant pardons in all cases including **Death Sentences** and **Court-Martial (military)** sentences; the **Governor (Art. 161)** has no power to pardon death sentences or

court-martial punishments.

34. **Ordinance Promulgation (Art. 123 vs Art. 213):** The President (Art. 123) and Governor (Art. 213) can issue ordinances only when the legislature is in recess; an ordinance ceases to operate **6 weeks** after the reassembly of the legislature (maximum duration is 6 months + 6 weeks).
35. **Council of Ministers Size (91st Amendment, 2003):** Total number of Ministers including the Prime Minister (or Chief Minister in States) shall not exceed **15% of the total strength** of the Lok Sabha (or State Legislative Assembly); in States, the minimum number of ministers cannot be less than **12** (Art. 164(1A)).
36. **Joint Sitting of Parliament (Article 108):** Summoned by the President to resolve deadlocks on Ordinary or Financial Bills; presided over by the **Speaker of Lok Sabha** (or Deputy Speaker / Deputy Chairman of Rajya Sabha; *Chairman of Rajya Sabha NEVER presides*). *NO joint sitting is permissible for Money Bills or Constitutional Amendment Bills*.
37. **Money Bill (Article 110):** Introduced only in Lok Sabha with prior presidential recommendation; the Speaker's decision that a bill is a Money Bill is final; Rajya Sabha cannot reject or amend it, and must return it within **14 days**.
38. **Comptroller and Auditor General (CAG, Article 148):** Appointed by the President; tenure is **6 years or up to 65 years of age**; removed on the same grounds and manner as a Supreme Court Judge; audits accounts of Union and States; reports submitted to PAC.
39. **Finance Commission (Article 280):** Quasi-judicial body constituted by the President every 5th year; consists of a **Chairman and 4 other members**; recommends devolution of net tax proceeds between Union and States.
40. **Election Commission of India (Article 324):** Consists of Chief Election Commissioner (CEC) and Election Commissioners; CEC is removed in the same manner and grounds as a Supreme Court Judge. *Elections to Panchayats and Municipalities are conducted by the State Election Commission (Arts. 243K & 243ZA), NOT by the ECI*.
41. **All-India Services (Article 312):** The **Rajya Sabha** has the exclusive constitutional power to pass a resolution by a **⅔rd majority of members present and voting** declaring it necessary in the national interest to create one or more new All-India Services.
42. **Emergency Provisions (Part XVIII): Art. 352** (National Emergency: War, External Aggression, Armed Rebellion — approved within 1 month by special majority); **Art. 356** (President's Rule: failure of constitutional machinery — approved within 2 months by simple majority; max 3 years); **Art. 360** (Financial Emergency — never imposed).
43. **Constitutional Amendment (Article 368, Part XX):** Requires Special Majority (majority of total membership + ⅔rd of members present and voting); federal provisions require additional ratification by legislatures of not less than half the States; the President **must give assent** to an amendment bill (24th Amd 1971).
44. **Warrant of Precedence (1979 Notification):** 1. President → 2. Vice-President → 3. Prime Minister → 4. Governors (within their States) → 5. Former Presidents → 6. CJI and Lok Sabha Speaker (Equal Rank) → 7. Union Cabinet Ministers, Chief Ministers (within States), Leaders of Opposition → 7A. Bharat Ratna awardees → 9. Supreme Court Judges → 9A. CEC, CAG, UPSC Chairman → 11. Attorney General for India, Cabinet Secretary.
45. **Uttar Pradesh Political Structure:** Vidhan Sabha has **403 seats** (84 reserved for SC, 2 for ST); Vidhan Parishad has **100 seats**; Lok Sabha representation is **80 seats** (17 SC, 0 ST); Rajya Sabha representation is **31 seats** (highest in the country).

1. Borrowed Features of the Indian Constitution (Country-Wise Master Table)

Source Country / Document	Borrowed Features & Principles
Government of India Act 1935 (*Major Source — >50% of text*)	Federal Scheme, Office of Governor, Judiciary structure, Public Service Commissions (UPSC/SPSC), Emergency provisions (administrative details), Administrative details.
United Kingdom (British Constitution)	Parliamentary form of government, Rule of Law (Art. 14), Legislative procedure, **Single Citizenship**, Cabinet system, Prerogative **Writs** (Art. 32/226), Parliamentary privileges (Art. 105), **Bicameralism**, Office of Speaker & Deputy Speaker.
United States (American Constitution)	**Fundamental Rights** (Part III), Independence of Judiciary, **Judicial Review** (Art. 13), **Impeachment of the President** (Art. 61), Removal of Supreme Court & High Court judges, Post of **Vice-President** (ex-officio Chairman of Rajya Sabha), Preamble phrasing ("We the People").

Source Country / Document	Borrowed Features & Principles
Ireland (Irish Constitution 1937)	**Directive Principles of State Policy (DPSP)** (Part IV), Nomination of 12 members to Rajya Sabha by President, Method of **Election of the President** (proportional representation by STV).
Canada	Federation with a **strong Centre**, Vesting of **Residuary Powers in the Centre** (Art. 248), **Appointment of State Governors by the Centre**, Advisory jurisdiction of the Supreme Court (Art. 143).
Australia	**Concurrent List** (7th Schedule), Freedom of Trade, Commerce, and Intercourse (Part XIII), **Joint Sitting of both Houses of Parliament** (Art. 108).
Germany (Weimar Constitution)	**Suspension of Fundamental Rights during National Emergency** (Art. 359). *(Note: Emergency itself is from GOI 1935, suspension of FR is from Weimar!)*
USSR / Russia (Soviet Constitution)	**Fundamental Duties** (Art. 51A, Part IVA - 42nd Amd 1976), Ideals of **Justice (Social, Economic, and Political)** in the Preamble, Five-Year Plans.
France (French Constitution)	Ideals of **Liberty, Equality, and Fraternity** in the Preamble; **Republic** character of the Indian State.
South Africa	Procedure for **Amendment of the Constitution** (Art. 368), **Election of members of the Rajya Sabha** (by elected MLAs via STV).
Japan	Procedure established by Law (*Vidhi dwara sthapit prakriya* - Art. 21).

2. Parts of the Indian Constitution (All 25 Parts Cheat-Sheet)

Originally 22 Parts in 1950; currently 25 Parts (counting sub-parts: IVA, IXA, IXB, XIVA; Part VII repealed).

Part	Roman	Subject Matter Covered	Article Span	Key Prelims Takeaway / Recurring Trap
Part I	**I**	**The Union and its Territory**	**Arts. 1–4**	Art. 1: "India, that is Bharat, shall be a Union of States"; Art. 3: Parliament alters state boundaries/names by simple majority
Part II	**II**	**Citizenship**	**Arts. 5–11**	Single citizenship (borrowed from UK); Citizenship Act 1955
Part III	**III**	**Fundamental Rights**	**Arts. 12–35**	**"Magna Carta of India"**; borrowed from US Bill of Rights; justifiable
Part IV	**IV**	**Directive Principles of State Policy (DPSP)**	**Arts. 36–51**	Borrowed from Ireland; non-justiciable; fundamental in the governance of the country
Part IVA	**IV-A**	**Fundamental Duties**	**Art. 51A**	Added by **42nd Amendment 1976** (Swaran Singh Committee); 10 originally, 11th added by **86th Amendment 2002**; non-justiciable
Part V	**V**	**The Union Government**	**Arts. 52–151**	Ch. 1: Executive (52–78), Ch. 2: Parliament (79–122), Ch. 3: President's Legislative Power (123), Ch. 4: Union Judiciary (124–147), Ch. 5: CAG (148–151)
Part VI	**VI**	**The State Governments**	**Arts. 152–237**	Governor (153–162), State Legislature (168–212), Governor's Ordinance (213), High Courts (214–232), Subordinate Courts (233–237)
Part VII	*VII*	*States in Part B of First Schedule*	*Art. 238*	**REPEALED** by **7th Constitutional Amendment Act 1956**
Part VIII	**VIII**	**The Union Territories**	**Arts. 239–242**	Administration of UTs by President through Administrator / Lieutenant Governor; Art. 239AA: Special status to NCT of Delhi (69th Amd 1991)
Part IX	**IX**	**The Panchayats**	**Arts. 243 to 243-O**	Inserted by **73rd Amendment 1992**; 3-tier Panchayati Raj; 11th Schedule (29 functional items)

Part	Roman	Subject Matter Covered	Article Span	Key Prelims Takeaway / Recurring Trap
Part IXA	**IX-A**	**The Municipalities**	**Arts. 243P to 243ZG**	Inserted by **74th Amendment 1992**; Urban Local Bodies; 12th Schedule (18 functional items)
Part IXB	**IX-B**	**The Co-operative Societies**	**Arts. 243ZH to 243ZT**	Inserted by **97th Amendment 2011** (partially struck down in 2021 for state cooperatives, holds for multi-state)
Part X	**X**	**The Scheduled and Tribal Areas**	**Arts. 244–244A**	5th & 6th Schedules administration
Part XI	**XI**	**Relations between the Union and the States**	**Arts. 245–263**	Ch. 1: Legislative Relations (245–255), Ch. 2: Administrative Relations (256–263); Art. 263: Inter-State Council. **Trap: NOT municipalities or tribunals!**
Part XII	**XII**	**Finance, Property, Contracts and Suits**	**Arts. 264–300A**	Consolidated Fund (266), Contingency Fund (267), GST Council (279A), Finance Commission (280); **Art. 300A**: Right to Property (Legal/Constitutional right, NOT fundamental right!)
Part XIII	**XIII**	**Trade, Commerce and Intercourse within India**	**Arts. 301–307**	Freedom of inter-state trade and commerce
Part XIV	**XIV**	**Services under the Union and the States**	**Arts. 308–323**	All India Services (Art. 312 created by Rajya Sabha); UPSC & State PSCs (Arts. 315–323)
Part XIVA	**XIV-A**	**Tribunals**	**Arts. 323A–323B**	Added by **42nd Amendment 1976** (Swaran Singh Committee); 323A: Administrative Tribunals (CAT/SAT), 323B: Other Tribunals
Part XV	**XV**	**Elections**	**Arts. 324–329A**	Art. 324: Election Commission of India; Art. 326: Universal Adult Suffrage
Part XVI	**XVI**	**Special Provisions Relating to Certain Classes**	**Arts. 330–342A**	Reservation of seats in Lok Sabha & State Assemblies (SC: 330/332; ST: 330/332); NCSC (338), NCST (338A), NCBC (338B)
Part XVII	**XVII**	**Official Language**	**Arts. 343–351**	Ch. 1: Official Language of Union (343 - Hindi in Devanagari script), Ch. 2: Regional Languages (345), Ch. 3: Judiciary (348), Ch. 4: Special Directives (350A: Primary instruction in mother tongue, 350B: Special Officer for Linguistic Minorities, 351: Promotion of Hindi)
Part XVIII	**XVIII**	**Emergency Provisions**	**Arts. 352–360**	Art. 352: National Emergency, Art. 356: President's Rule (State Emergency), Art. 360: Financial Emergency
Part XIX	**XIX**	**Miscellaneous**	**Arts. 361–367**	Art. 361: Immunity of President and Governors from criminal prosecution
Part XX	**XX**	**Amendment of the Constitution**	**Art. 368**	Constituent power of Parliament to amend the Constitution (borrowed from South Africa)
Part XXI	**XXI**	**Temporary, Transitional and Special Provisions**	**Arts. 369–392**	Special provisions for states: **Art. 371 (Maha & Guj), 371A (Nagaland), 371B (Assam), 371C (Manipur), 371D & E (Andhra/Telangana), 371F (Sikkim), 371G (Mizoram), 371H (Arunachal), 371-I (Goa), 371J (Karnataka - Hyderabad-Karnataka region)**
Part XXII	**XXII**	**Short Title, Commencement & Repeals**	**Arts. 393–395**	Short title: "Constitution of India"

3. All 12 Schedules (The Master Mnemonic: TEARS OF OLD PM)

Schedule	Mnemonic Letter	Subject Matter	Key Details / High-Yield Trap
1st	**T**	**Territories**	Names of the 28 States and 8 Union Territories and their territorial extents
2nd	**E**	**Emoluments & Salaries**	Salaries, allowances, and pensions of: President, Governors, Lok Sabha Speaker/Deputy Speaker, Rajya Sabha Chairman/Deputy Chairman, SC/HC Judges, CAG. **PM and Ministers are NOT in the 2nd Schedule!**
3rd	**A**	**Affirmations & Oaths**	Forms of Oaths for: Union/State Ministers, MPs/MLAs, SC/HC Judges, CAG. **President (Art. 60), Vice-President (Art. 69), and Governors (Art. 159) have oaths under separate articles, NOT in the 3rd Schedule!**
4th	**R**	**Rajya Sabha Seat Allocation**	Allocation of seats in the Council of States to States and UTs. **Uttar Pradesh has the highest: 31 seats; Uttarakhand has 3 seats.**
5th	**S**	**Scheduled Areas & Tribes**	Administration and control of Scheduled Areas and Scheduled Tribes in 10 states (excluding the 4 states of the 6th Schedule); Tribes Advisory Council
6th	**O**	**Other Tribal Areas**	Administration of Tribal Areas in **4 North-Eastern States: AMTM = Assam, Meghalaya, Tripura, Mizoram** (Autonomous District Councils; **Manipur is NOT in the 6th Schedule!**)
7th	**F**	**Federal Lists (Division of Powers)**	Three legislative lists: 1. **Union List**: Originally 97, now **100 subjects** 2. **State List**: Originally 66, now **61 subjects** 3. **Concurrent List**: Originally 47, now **52 subjects** (Education, Forests, Weights & Measures, Wild animals & birds, Administration of justice transferred by 42nd Amd)
8th	**O**	**Official Languages**	**22 recognized languages** (Originally 14): • 1967 (21st Amd): Added **Sindhi** • 1992 (71st Amd): Added **Konkani, Manipuri, Nepali** (KMN) • 2003 (92nd Amd): Added **Bodo, Dogri, Maithili, Santhali** (BDMS) **Traps: English and Bhojpuri are NOT in the 8th Schedule!**
9th	**L**	**Land Reforms & Validation**	Added by **1st Amendment 1951** to protect land reform laws from judicial review; **I.R. Coelho Case (2007)**: SC ruled laws placed in 9th Schedule post **24 April 1973** (Kesavananda judgment date) are **open to judicial review**
10th	**D**	**Defection (Anti-Defection Law)**	Added by **52nd Amendment 1985**; disqualification on ground of defection; decided by Presiding Officer; subject to judicial review (**Kihoto Hollohan 1992**); split exemption of ⅓ deleted by 91st Amd 2003; merger requires **⅔rd members**
11th	**P**	**Panchayati Raj**	Added by **73rd Amendment 1992**; specifies **29 functional subjects** within the purview of Panchayats (Article 243G)
12th	**M**	**Municipalities**	Added by **74th Amendment 1992**; specifies **18 functional subjects** within the purview of Municipalities (Article 243W)

4. Warrant of Precedence (Order of Precedence, 1979 Notification)

Frequently tested in arrange-order questions in UPPCS/UKPCS:

1. **Rank 1**: **President of India**
2. **Rank 2**: **Vice-President of India**
3. **Rank 3**: **Prime Minister of India**
4. **Rank 4**: **Governors of States** (within their respective States)

5. **Rank 5**: **Former Presidents**
6. *Rank 5A*: **Deputy Prime Minister** (if any)
7. **Rank 6**: **Chief Justice of India (CJI)** and **Speaker of Lok Sabha** (*Both have equal rank!*)
8. **Rank 7**: **Cabinet Ministers of the Union**, **Chief Ministers of States** (within respective states), Deputy Chairman of NITI Aayog (former Planning Commission), **Leaders of Opposition in Rajya Sabha and Lok Sabha**
9. *Rank 7A*: **Holders of Bharat Ratna**
10. **Rank 8**: Ambassadors Extraordinary & Plenipotentiary, Chief Ministers (outside their respective states), Governors (outside their respective states)
11. **Rank 9**: **Judges of the Supreme Court of India**
12. *Rank 9A*: **Chief Election Commissioner (CEC)**, **Comptroller and Auditor General (CAG)**, Chairperson of UPSC
13. **Rank 10**: Deputy Chairman of Rajya Sabha, Deputy Speaker of Lok Sabha, Deputy Chief Ministers, Ministers of State of the Union
14. **Rank 11**: **Attorney General for India**, Cabinet Secretary, Lieutenant Governors (within their UTs).

5. Parliamentary Terms, Motions & Working Procedure

Parliamentary Term	Core Meaning & Procedural Rules
Question Hour	First hour of every parliamentary sitting (11:00 AM to 12:00 PM): • **Starred Questions**: Oral answers required; **supplementary questions CAN be asked** (green paper). • **Unstarred Questions**: Written answers required; **NO supplementary questions allowed** (white paper). • **Short Notice Questions**: Relates to matter of urgent public importance; asked with less than 10 days notice; answered orally (light pink paper).
Zero Hour	**Unique Indian parliamentary innovation (introduced in 1962)**; starts immediately after Question Hour (12:00 PM); members raise urgent matters without prior notice; **not mentioned in Rules of Procedure**.
Adjournment Motion	Introduced in **LOK SABHA ONLY** to draw attention to a definite matter of urgent public importance; requires support of **at least 50 members** to be admitted; sitting adjourned for debate ≥ 2.5 hours; involves censure of government.
No-Confidence Motion	Moved in **LOK SABHA ONLY** under Rule 198; requires support of **at least 50 members**; no specific grounds needed; if passed, Council of Ministers must resign.
Censure Motion	Moved in Lok Sabha against specific ministers or entire council for specific policies; requires specific grounds; if passed, government is not obligated to resign immediately but must seek confidence.
Calling Attention Motion	Indian parliamentary innovation (since 1954); calling attention of a Minister to a matter of urgent public importance and seeking an authoritative statement. Mentioned in Rules of Procedure.
Cut Motions	Moved during Budget debate (Lok Sabha only): 1. **Disapproval of Policy Cut**: Amount reduced to **Re 1** (shows complete policy rejection). 2. **Economy Cut**: Amount reduced by a specific lump sum. 3. **Token Cut**: Amount reduced by **Rs 100** to express specific grievance.
Guillotine	When time allotted for discussion on demands for grants concludes, all remaining undiscussed demands are put to vote immediately without debate.
Lame-Duck Session	Refers to the last session of the existing Lok Sabha after a new Lok Sabha has been elected. Those existing MPs who failed to get re-elected are called "Lame Ducks".
Quorum	Minimum number of members required to be present to conduct a sitting: **1/10th of total membership** of the House (55 in Lok Sabha, 25 in Rajya Sabha) per **Article 100(3)**.

6. Parliamentary Committees (Must-Memorize Facts)

Committee Name	Total Strength	Lok Sabha Members	Rajya Sabha Members	Term	Chairperson Rules & Key Functions
Public Accounts Committee (PAC)	**22**	**15**	**7**	**1 year**	Set up first in **1921** (GOI Act 1919); scrutinizes **CAG Audit Reports**; elected by PR by STV; **Ministers CANNOT be members**; by convention since **1967**, Chairperson is appointed by the Speaker from the **Opposition**; known as **"Twin sister" of Estimates Committee** and **"Friend, Philosopher, and Guide"** of the CAG.
Estimates Committee	**30**	**30**	**0 (NO RS!)**	**1 year**	Set up in **1950** on recommendation of Finance Minister **John Mathai**; **LARGEST PARLIAMENTARY COMMITTEE**; all 30 members from Lok Sabha only; examines budget estimates and suggests alternative policies ("Continuous Economy Committee"); Chairperson appointed by Speaker from ruling party.
Committee on Public Undertakings (COPU)	**22**	**15**	**7**	**1 year**	Created in **1964** on recommendation of **Krishna Menon Committee**; examines reports of CAG on public undertakings; Chairperson appointed by Speaker from Lok Sabha members only.
Departmentally Related Standing Committees (DRSCs)	**31** (each)	**21**	**10**	**1 year**	Total **24 DRSCs** (16 work under Lok Sabha, 8 under Rajya Sabha); examine demands for grants, bills, and annual reports of respective ministries.

7. Constitutional Offices: Age Limits, Tenures & Appointments

Office / Position	Minimum Age Required	Retirement Age / Term of Office	Appointed By	Resignation Submitted To	Removal Procedure
President of India	**35 years**	**5 years** (eligible for re-election)	Electoral College (Elected MPs of LS/RS + Elected MLAs of States/UTs - Art. 54)	**Vice-President**	**Impeachment under Art. 61** (Quasi-judicial, 14 days notice, ⅔rd of total membership of both Houses; nominated MPs participate!)
Vice-President of India	**35 years**	**5 years**	Electoral College (All MPs of LS & RS, both elected & nominated - Art. 66; no MLAs!)	**President**	Resolution in Rajya Sabha by effective majority + agreed to by Lok Sabha (Art. 67)
Governor of a State	**35 years**	**5 years** (Holds office during **pleasure of President**)	President of India	**President**	Removed by President at any time without constitutional procedure
Prime Minister / Union Minister	**25 yrs** (LS) / **30 yrs** (RS)	No fixed tenure (during pleasure of President / confidence of LS)	President	**President**	Loss of majority in Lok Sabha
Member of Parliament (Lok Sabha)	**25 years**	**5 years** (unless dissolved earlier)	Direct Election (First-past-the-post)	**Speaker of Lok Sabha**	Disqualification under Art. 102 or 10th Schedule

Office / Position	Minimum Age Required	Retirement Age / Term of Office	Appointed By	Resignation Submitted To	Removal Procedure
Member of Parliament (Rajya Sabha)	**30 years**	**6 years** (Permanent body; **⅓rd members retire every 2 years**)	Indirect Election by elected MLAs (STV)	**Chairman of Rajya Sabha**	Disqualification under Art. 102 or 10th Schedule
Member of Legislative Assembly (MLA)	**25 years**	**5 years**	Direct Election	**Speaker of Assembly**	Disqualification under Art. 191
Member of Legislative Council (MLC)	**30 years**	**6 years** (⅓rd retire every 2 years)	Indirect (MLAs ⅓, Local bodies ⅓, Teachers 1/12, Grads 1/12, Gov nominates ⅙)	**Chairman of Council**	Disqualification under Art. 191
Panchayat / Municipality Member	**21 years**	**5 years**	Direct Election	Presiding Officer	Disqualification under state municipal/panchayat laws
Voter (Universal Adult Suffrage)	**18 years** (lowered from 21 by 61st Amd)	—	Enrolled in electoral roll	—	Ineligibility under Art. 326 (unsound mind, crime)
Supreme Court Judge / CJI	No min age	**65 years**	President (Collegium recommendation)	**President**	Removed by Presidential order supported by special majority of both Houses for proved misbehaviour or incapacity (Art. 124(4))
High Court Judge / Chief Justice	No min age	**62 years** (15th Amd raised from 60)	President (after consulting CJI, Governor, HC CJ)	**President**	Same manner and grounds as Supreme Court Judge
Comptroller & Auditor General (CAG)	No min age	**6 years or 65 years** (whichever earlier)	President	**President**	Same manner and grounds as Supreme Court Judge (Art. 148(1))
Chief Election Commissioner (CEC)	No min age	**6 years or 65 years**	President	**President**	Same manner and grounds as Supreme Court Judge (Art. 324(5))
Attorney General for India	Qualified as SC Judge	**No fixed term** (pleasure of President)	President	**President**	Removed by President at pleasure
Advocate General for the State	Qualified as HC Judge	**No fixed term** (pleasure of Governor)	Governor	**Governor**	Removed by Governor at pleasure
UPSC Chairman / Member	No min age	**6 years or 65 years**	President	**President**	Removed by President on ground of misbehaviour after inquiry by Supreme Court (Art. 317)
State PSC Chairman / Member	No min age	**6 years or 62 years**	**Governor appoints**	**Governor**	**REMOVED ONLY BY PRESIDENT** (Not Governor!) on SC inquiry under Art. 317

Office / Position	Minimum Age Required	Retirement Age / Term of Office	Appointed By	Resignation Submitted To	Removal Procedure
Central Vigilance Commissioner	No min age	**4 years or 65 years** (whichever earlier)	President on 3-member panel (PM, Home Min, LoP LS)	**President**	Removed by President on SC inquiry

8. High-Frequency Articles Must-Revise Daily

A. Fundamental Rights (Part III, Arts. 12–35)

- **Art. 12**: Definition of "State" (includes Union/State Governments, Parliament/Legislatures, Local authorities, other statutory authorities like LIC, ONGC).
- **Art. 13**: Laws inconsistent with or in derogation of Fundamental Rights shall be void (*Doctrine of Judicial Review*, *Doctrine of Severability*, *Doctrine of Eclipse*).
- **Art. 14**: Equality before law (British origin, negative concept) & Equal protection of the laws (American origin, positive concept).
- **Art. 15**: Prohibition of discrimination **ONLY** on grounds of religion, race, caste, sex, place of birth.
- *Art. 15(4)*: Special provisions for SC/ST/OBC (1st Amd 1951).
- *Art. 15(6)*: 10% EWS reservation in educational institutions (103rd Amd 2019).
- **Art. 16**: Equality of opportunity in matters of public employment.
- *Art. 16(4)*: Reservation for backward classes not adequately represented.
- *Art. 16(6)*: 10% EWS reservation in public employment (103rd Amd 2019).
- **Art. 17**: **Abolition of Untouchability** (Absolute right; enforced via Protection of Civil Rights Act 1955).
- **Art. 18**: Abolition of titles (Bharat Ratna/Padma awards are decorations, not titles — *Balaji Raghavan Case 1996*).
- **Art. 19**: Protection of **Six Democratic Freedoms**:
- *19(1)(a)*: Freedom of speech and expression (includes freedom of press, right to remain silent).
- *19(1)(b)*: Freedom to assemble peaceably and without arms.
- *19(1)©*: Freedom to form associations, unions, or co-operative societies (97th Amd).
- *19(1)(d)*: Freedom to move freely throughout the territory of India.
- *19(1)(e)*: Freedom to reside and settle in any part of the territory of India.
- *19(1)(g)*: Freedom to practice any profession, trade, or business.
- *(Note: 19(1)(f) Right to acquire, hold, and dispose of property was REPEALED by 44th Amd 1978).*
- **Art. 20**: Protection in respect of conviction for offences:
- *20(1)*: No ex-post-facto criminal law.
- *20(2)*: No double jeopardy (cannot be prosecuted twice for the same offence).
- *20(3)*: No self-incrimination (cannot be compelled to be a witness against oneself).
- **Art. 21**: **Protection of Life and Personal Liberty** (*Procedure Established by Law → Due Process of Law* via *Maneka Gandhi Case 1978*; encompasses Right to Privacy - *Puttaswamy 2017*, Right to clean environment, Right to livelihood, Right to sleep, Right to die with dignity / passive euthanasia - *Common Cause 2018*).
- **Art. 21A**: **Right to Free and Compulsory Education** for children aged 6 to 14 years (Inserted by **86th Amendment 2002**; RTE Act enacted in 2009, in force 1 April 2010).
- **Art. 22**: Protection against arrest and detention (production before magistrate within 24 hours; preventive detention max 3 months without advisory board approval).
- **Art. 23**: Prohibition of traffic in human beings and **Begar** (forced labour).
- **Art. 24**: Prohibition of employment of children **below 14 years in factories, mines**, or hazardous employment.
- **Arts. 25–28**: Freedom of Religion:
- *Art. 25*: Freedom of conscience, free profession, practice, and propagation of religion.

- *Art. 26*: Freedom to manage religious affairs.
- *Art. 27*: Freedom from payment of taxes for promotion of any particular religion.
- *Art. 28*: Freedom from attendance at religious instruction in state-funded educational institutions.
- **Arts. 29–30**: Cultural and Educational Rights of Minorities:
- *Art. 29*: Protection of language, script, and culture of minorities.
- *Art. 30*: Right of religious and linguistic minorities to establish and administer educational institutions.
- **Art. 32**: **Right to Constitutional Remedies**: Power of Supreme Court to issue 5 Writs (**Habeas Corpus, Mandamus, Prohibition, Certiorari, Quo-Warranto**); Dr. B.R. Ambedkar called Article 32 the **"Heart and Soul of the Constitution"**.

> **NON-SUSPENDABLE RIGHTS:** Under **Article 359 (as amended by 44th Amd 1978)**, the President **CANNOT suspend the enforcement of Articles 20 and 21** even during a National Emergency!

B. Directive Principles of State Policy (DPSP, Part IV, Arts. 36–51)

- **Art. 38**: State to secure a social order for the promotion of welfare of the people (minimize inequalities).
- **Art. 39(b) & ©**: Equitable distribution of material resources of the community to subserve common good (39b); prevention of concentration of wealth (39c) — *Articles 39(b) and 39© take precedence over Articles 14 and 19 per Art. 31C!*
- **Art. 39A**: **Equal justice and free legal aid** (Added by 42nd Amd 1976; Legal Services Authorities Act 1987 / NALSA).
- **Art. 40**: **Organization of Village Panchayats** (Gandhian principle).
- **Art. 41**: Right to work, to education, and to public assistance in cases of unemployment, old age, sickness.
- **Art. 42**: Provision for just and humane conditions of work and **Maternity relief**.
- **Art. 43**: Living wage and decent standard of life for workers.
- **Art. 43A**: **Participation of workers in management of industries** (Added by 42nd Amd 1976).
- **Art. 43B**: Promotion of **Co-operative societies** (Added by 97th Amd 2011).
- **Art. 44**: **Uniform Civil Code (UCC)** for all citizens (**Goa** had Portuguese civil code; **Uttarakhand became the first state in post-independence India to pass a UCC bill in 2024!**).
- **Art. 45**: Provision for early childhood care and education for all children until they complete the age of **6 years** (Amended by 86th Amd 2002).
- **Art. 48**: Organization of agriculture and animal husbandry; prohibiting slaughter of cows, calves, and milch cattle.
- **Art. 48A**: **Protection and improvement of environment, and safeguarding of forests and wildlife** (Added by 42nd Amd 1976).
- **Art. 50**: **Separation of Judiciary from the Executive** in the public services of the State.
- **Art. 51**: Promotion of international peace and security.

C. Union & State Executive / Legislature Comparisons

Constitutional Matter	Union Article (President / Parliament)	State Article (Governor / State Legislature)	Key Distinction / Comparison
Office of Head	**Art. 52**: President of India	**Art. 153**: Governor of a State	7th Amd 1956: Same person can be appointed Governor for two or more states
Executive Power	**Art. 53**: Vested in President	**Art. 154**: Vested in Governor	Governor is appointed by President (Canadian model); holds office during pleasure
Pardoning Power	**Art. 72**: President of India	**Art. 161**: Governor of a State	**President alone can pardon Death Sentences and Court Martial (Military) sentences.** Governor can suspend/remit, but cannot grant full pardon for death sentence
Council of Ministers	**Art. 74**: CoM to aid & advise	**Art. 163**: CoM to aid & advise	Governor has broader constitutional discretionary powers (Art. 163(2))

Constitutional Matter	Union Article (President / Parliament)	State Article (Governor / State Legislature)	Key Distinction / Comparison
Appointment of PM/CM	**Art. 75**: PM appointed by President	**Art. 164**: CM appointed by Governor	CoM size capped at **max 15%** of Lower House membership (91st Amd 2003); State min = 12
Top Legal Officer	**Art. 76**: **Attorney General for India**	**Art. 165**: **Advocate General for the State**	AG has right to speak in both Houses of Parliament but **NO right to vote** (Art. 88)
Joint Sitting	**Art. 108**: Joint Sitting of Parliament	**NO PROVISION** for Joint Sitting in State Legislature!	Summoned by President; presided over by **Lok Sabha Speaker** (simple majority)
Money Bill Definition	**Art. 110**: Money Bill in Parliament	**Art. 199**: Money Bill in State Assembly	Speaker's decision is final; Rajya Sabha / Legislative Council has only **14 days** to suggest
Annual Budget	**Art. 112**: Annual Financial Statement	**Art. 202**: Annual Financial Statement	Word "Budget" is nowhere used in the Constitution!
Ordinance Power	**Art. 123**: President	**Art. 213**: Governor	Can only be promulgated when legislature is not in session; max lifespan: **6 weeks from reassembly** (absolute maximum lifespan = 6 months + 6 weeks)
Assent to Bills	**Art. 111**: President's Assent	**Art. 200**: Governor's Assent	Governor can reserve bill for consideration of President (**Art. 201**)
Creation/Abolition of Council	—	**Art. 169**: Legislative Council	State Assembly passes resolution by special majority → Parliament approves by simple majority

9. Constitutional vs Statutory vs Executive Bodies

Body Name	Category	Governing Article / Parent Act	Head / Composition	Appointed By	Removal / Tenure
Election Commission of India (ECI)	**Constitutional**	**Art. 324**	Chief Election Commissioner + 2 Election Commissioners	President (per 2023 Act: panel of PM, LoP, Cabinet Min)	Same grounds and manner as Supreme Court Judge (6 yrs or 65 yrs)
Union Public Service Commission (UPSC)	**Constitutional**	**Arts. 315–323**	Chairman + Members (usually 9–11)	President	6 yrs or 65 yrs; removed by President on SC enquiry under Art. 317
Finance Commission	**Constitutional**	**Art. 280**	Chairman + 4 Members	President every 5th year	Recommends devolution of net tax proceeds between Union & States
Comptroller & Auditor General (CAG)	**Constitutional**	**Art. 148**	Single-member office	President	6 yrs or 65 yrs; removed like a Supreme Court Judge; **"Guardian of the Public Purse"**

Body Name	Category	Governing Article / Parent Act	Head / Composition	Appointed By	Removal / Tenure
Attorney General for India	**Constitutional**	**Art. 76**	Single-member office	President	Holds office **during the pleasure of the President**; must be qualified to be an SC Judge
Advocate General for the State	**Constitutional**	**Art. 165**	Single-member office	Governor	Holds office **during the pleasure of the Governor**; qualified to be an HC Judge
National Commission for SC (NCSC)	**Constitutional**	**Art. 338**	Chairperson, Vice-Chairperson + 3 Members	President	3 years tenure
National Commission for ST (NCST)	**Constitutional**	**Art. 338A**	Separated by 89th Amd 2003; Chairperson + 4	President	3 years tenure
National Commission for BC (NCBC)	**Constitutional**	**Art. 338B**	Accorded constitutional status by 102nd Amd 2018	President	3 years tenure
GST Council	**Constitutional**	**Art. 279A**	Chaired by Union Finance Minister; members: MoS Finance + State Finance Ministers	-	Voting weight: Centre = ⅓rd, States = ⅔rd; ¾th majority needed to pass decisions
Inter-State Council	**Constitutional**	**Art. 263**	Set up in 1990 (**Sarkaria Commission**); chaired by Prime Minister	-	CMs of all states + 6 Union Cabinet Ministers
National Human Rights Commission (NHRC)	**Statutory**	**Protection of Human Rights Act (PHRA) 1993**	Retired CJI or Judge of SC (Chairperson) + Members	President on panel (PM, Speaker, Home Min, LoP LS, LoP RS, Dy Chairman RS)	3 years or 70 years
Central Vigilance Commission (CVC)	**Statutory**	**CVC Act 2003** (set up 1964 on **Santhanam Committee**)	Central Vigilance Commissioner + max 2 Vigilance Commissioners	President on panel (PM, Home Minister, LoP LS)	**4 years or 65 years** (Unique tenure!)
Central Information Commission (CIC)	**Statutory**	**Right to Information (RTI) Act 2005**	Chief Information Commissioner + max 10 ICs	President on panel (PM, LoP LS, Union Cabinet Min)	Set by Central Govt or 65 yrs
Lokpal of India	**Statutory**	**Lokpal and Lokayuktas Act 2013**	Chairperson (CJI/SC Judge/eminent person) + max 8 Members	President on panel (PM, Speaker LS, LoP LS, CJI or nominee, eminent jurist)	5 years or 70 years

Body Name	Category	Governing Article / Parent Act	Head / Composition	Appointed By	Removal / Tenure
National Green Tribunal (NGT)	**Statutory**	**NGT Act 2010**	Chairperson (retired SC judge or HC CJ) + Judicial/Expert members	Central Govt in consultation with CJI	5 years; not eligible for reappointment
NITI Aayog	**Non-Constitutional (Executive Body)**	Created by **Union Cabinet Resolution on 1 Jan 2015** (replacing Planning Commission 1950)	Chairperson: Prime Minister; Vice-Chairperson + CEO + Ex-officio members	PM appoints	Think tank; promotes Cooperative & Competitive Federalism

10. Landmark Constitutional Amendments (The Complete PCS Sheet)

Amendment Act & Year	Prime Minister / Context	Landmark Provisions / Changes Made
1st (1951)	J.L. Nehru	Added **9th Schedule** to protect land reform laws from judicial review; placed reasonable restrictions on freedom of speech (public order, friendly relations with foreign states, incitement to an offence)
7th (1956)	J.L. Nehru	Implemented **States Reorganisation Act 1956** (Fazal Ali Commission); abolished Part A, B, C, D classification; created 14 States & 6 UTs; allowed common High Court for two or more states; allowed same person as Governor of multiple states
24th (1971)	Indira Gandhi	Affirmed Parliament's power to amend any part of the Constitution, including Fundamental Rights; made it **obligatory for the President to give assent** to a Constitutional Amendment Bill
42nd (1976)	Indira Gandhi (**Swaran Singh Committee**)	Known as the **"Mini-Constitution"**: 1. Preamble: Added words **"SOCIALIST, SECULAR, INTEGRITY"** 2. Added **Part IVA (Fundamental Duties, Art. 51A)** with 10 duties 3. Added **Part XIVA (Tribunals, Arts. 323A–323B)** 4. Added DPSPs: Arts. **39A** (Free legal aid), **43A** (Workers' participation), **48A** (Environment) 5. Transferred **5 subjects from State List to Concurrent List**: Education, Forests, Weights & Measures, Protection of Wild Animals & Birds, Administration of Justice
44th (1978)	Morarji Desai (Janata Party)	Undid emergency excesses of 42nd Amendment: 1. Replaced "Internal Disturbance" with **"Armed Rebellion"** in Art. 352 2. Mandated that President can declare National Emergency only on **written advice of the Cabinet** 3. **Right to Property deleted from Fundamental Rights** (Art. 19(1)(f) & Art. 31 deleted; made a legal right under **Art. 300A**) 4. Provided that **Articles 20 and 21 CANNOT be suspended** during emergency
52nd (1985)	Rajiv Gandhi	Added **10th Schedule (Anti-Defection Law)**; disqualified MPs/MLAs defecting from their political party
61st (1988)	Rajiv Gandhi	**Lowered voting age from 21 years to 18 years** for Lok Sabha and Legislative Assembly elections (amended **Art. 326**)
69th (1991)	P.V. Narasimha Rao	Conferred special status on Union Territory of Delhi as **National Capital Territory of Delhi (NCT)** with Legislative Assembly and 7-member Council of Ministers (inserted **Art. 239AA**)
73rd (1992)	P.V. Narasimha Rao	Accorded constitutional status to **Panchayati Raj Institutions (PRIs)**; inserted **Part IX (Arts. 243 to 243-O)** and **11th Schedule (29 subjects)**; 33% mandatory reservation for women

Amendment Act & Year	Prime Minister / Context	Landmark Provisions / Changes Made
74th (1992)	P.V. Narasimha Rao	Accorded constitutional status to **Urban Local Bodies (Municipalities)**; inserted **Part IXA (Arts. 243P to 243ZG)** and **12th Schedule (18 subjects)**
86th (2002)	A.B. Vajpayee	1. Inserted **Article 21A** (Right to free and compulsory education for children aged 6–14 as a Fundamental Right) 2. Substituted Article 45 (Early childhood care) 3. Added **11th Fundamental Duty** under **Art. 51A(k)**
91st (2003)	A.B. Vajpayee	1. Capped the size of the Council of Ministers to **maximum 15%** of the total strength of Lok Sabha / State Legislative Assembly (minimum 12 in states) 2. **Deleted ⅓rd split provision** from 10th Schedule (only ⅔rd merger protected)
99th (2014)	Narendra Modi	Established National Judicial Appointments Commission (NJAC); **struck down as unconstitutional by Supreme Court in 2015** (*Fourth Judges Case*), restoring the Collegium system
101st (2016)	Narendra Modi	Introduced **Goods and Services Tax (GST)**; inserted **Art. 246A, 269A, and 279A** (GST Council chaired by Union Finance Minister); in force from **1 July 2017**
102nd (2018)	Narendra Modi	Accorded constitutional status to the **National Commission for Backward Classes (NCBC)**; inserted **Articles 338B and 342A**
103rd (2019)	Narendra Modi	Provided **10% reservation for Economically Weaker Sections (EWS)** in educational admissions and civil posts; amended **Articles 15(6) and 16(6)**
104th (2020)	Narendra Modi	Extended reservation of seats for **SC and ST in Lok Sabha and State Assemblies for another 10 years (until 2030)**; **discontinued the nomination of 2 Anglo-Indian members** in Lok Sabha and 1 in Assemblies
105th (2021)	Narendra Modi	Clarified that State Governments maintain the power to prepare and maintain their own list of **Socially and Educationally Backward Classes (SEBCs / OBCs)**
106th (2023)	Narendra Modi	***Nari Shakti Vandan Adhiniyam***: Provides **33% reservation for women in Lok Sabha, State Legislative Assemblies, and NCT of Delhi Assembly** (inserted Arts. 330A, 332A, 239AA(2)(b)); effective after the next delimitation exercise

11. Panchayati Raj Committees Chronological Ladder

Committee Name	Year	Landmark Recommendation / Impact
Balwant Rai Mehta Committee	**1957**	Recommended **Three-Tier Panchayati Raj System**: 1. Gram Panchayat (Village) 2. Panchayat Samiti (Block - executive body) 3. Zilla Parishad (District - advisory body) **First implemented in Nagaur (Rajasthan) on 2 October 1959** by J.L. Nehru; second state was **Andhra Pradesh**
K. Santhanam Committee	**1963**	Panchayati Raj finances and budgetary powers
Ashok Mehta Committee	**1977** (Janata Govt)	Recommended **Two-Tier System**: 1. **Mandal Panchayat** (group of villages, pop. 15,000–20,000) 2. **Zilla Parishad** (District executive body) Advocated official participation of political parties in elections
G.V.K. Rao Committee	**1985** (Planning Commission)	Critiqued bureaucratic takeover, describing PRIs as **"Grass without Roots"**; recommended strengthening Zilla Parishad and creating post of District Development Commissioner (DDC)

Committee Name	Year	Landmark Recommendation / Impact
L.M. Singhvi Committee	**1986** (Rajiv Gandhi)	**FIRST TO STRONGLY RECOMMEND CONSTITUTIONAL RECOGNITION FOR PANCHAYATI RAJ**; advocated Nyaya Panchayats and financial autonomy
P.K. Thungon Committee	**1988**	Sub-committee of Parliamentary Consultative Committee; strongly urged **Constitutional status**, 3-tier system, fixed 5-year tenure, Zilla Parishad as main planning agency
V.N. Gadgil Committee	**1988**	Blueprint for the 73rd Amendment bill: 3 tiers, fixed 5-yr tenure, reservation for SC/ST/Women, State Election Commission, State Finance Commission

12. Landmark Supreme Court Judgments (Chronological Table)

Case Name & Year	Key Issue / Doctrine Established
Shankari Prasad Case (1951)	Parliament can amend any part of the Constitution, including Fundamental Rights under Article 368.
Golaknath Case (1967)	11-Judge bench; ruled that **Fundamental Rights are transcendental and sacrosanct**; Parliament CANNOT amend or take away Fundamental Rights.
Kesavananda Bharati Case (1973)	**13-Judge bench (largest ever, 7:6 verdict on 24 April 1973)**; upheld Parliament's power to amend the Constitution, BUT established the **"Basic Structure Doctrine"** — Parliament cannot alter the basic framework/features of the Constitution.
Indira Nehru Gandhi v. Raj Narain (1975)	Struck down 39th Amendment (Article 329A); held that **Free and Fair Elections and Judicial Review are part of the Basic Structure**.
Maneka Gandhi Case (1978)	Reinterpreted Article 21; replaced rigid "Procedure Established by Law" with **"Due Process of Law"**; procedure must be *just, fair, and reasonable*. Golden Triangle: Articles 14, 19, and 21 are interrelated.
Minerva Mills Case (1980)	Struck down clauses of 42nd Amendment; held that **limited amending power and judicial review are basic features**; Indian Constitution is founded on the bedrock of **balance between Fundamental Rights and DPSPs**.
Waman Rao Case (1981)	Reaffirmed Basic Structure Doctrine; set cutoff date of **24 April 1973** for applying Basic Structure scrutiny to the 9th Schedule.
Shah Bano Case (1985)	Upheld divorced Muslim woman's right to maintenance under Section 125 CrPC; urged enactment of a Uniform Civil Code.
Indira Sawhney Case (1992) (*Mandal Case*)	9-Judge bench; upheld 27% OBC reservation; introduced **"Creamy Layer"** exclusion; capped **total reservations at 50%**; no reservation in promotions.
S.R. Bommai Case (1994)	9-Judge bench; curbed misuse of **Article 356 (President's Rule)**; held that **Secularism is part of the Basic Structure**; floor test in Assembly is the only valid test of majority.
Vishaka v. State of Rajasthan (1997)	Formulated **Vishaka Guidelines** to prevent sexual harassment of women at workplaces (precursor to POSH Act 2013).
I.R. Coelho Case (2007)	9-Judge bench; held that all laws inserted into the **9th Schedule after 24 April 1973 are subject to judicial review** if they violate Fundamental Rights forming part of the Basic Structure.
Justice K.S. Puttaswamy Case (2017)	9-Judge bench unanimously held that the **Right to Privacy is a Fundamental Right** intrinsically protected under Article 21.
Navtej Singh Johar Case (2018)	Decriminalized consensual homosexual acts between adults by reading down **Section 377 of the Indian Penal Code (IPC)**.
Jarnail Singh Case (2018)	Held that the **"Creamy Layer" principle applies to SC/ST** when extending reservation in promotions.

Case Name & Year	Key Issue / Doctrine Established
Janki Janak v. UOI (2022)	Upheld the constitutional validity of the **103rd Constitutional Amendment (10% EWS reservation)** by 3:2 majority.

13. State Reorganization Chronology (Post-Independence India)

1. **1953**: **Andhra State** created (first linguistic state, separated Telugu-speaking areas of Madras after 56-day fast unto death of **Potti Sreeramulu**; Kurnool capital).
2. **1956**: **States Reorganisation Act 1956** (Fazal Ali Commission with H.N. Kunzru and K.M. Panikkar); created **14 States and 6 Union Territories**.
3. **1960**: **Bilingual Bombay State bifurcated into Maharashtra and Gujarat** (Gujarat became the 15th state).
4. **1961**: **Dadra & Nagar Haveli** (10th Amd) and **Goa, Daman & Diu** (12th Amd) integrated via Operation Vijay.
5. **1962**: **Pondicherry** (Puducherry) integrated via 14th Amd (ceded by French in 1954).
6. **1963**: **Nagaland** created as 16th state (incorporating Naga Hills-Tuensang Area).
7. **1966**: **Punjab bifurcated** into Punjab (Punjabi) and **Haryana** (17th state, Hindi) on recommendation of **Shah Commission**; **Chandigarh** made common UT.
8. **1971**: **Himachal Pradesh** elevated from UT to 18th State.
9. **1972**: North-Eastern reorganization: **Manipur (19th), Tripura (20th), and Meghalaya (21st)** became full states; Mizoram and Arunachal Pradesh created as UTs.
10. **1975**: **Sikkim**: 35th Amd 1974 made it "Associate State" (10th Schedule added); **36th Amendment 1975 made Sikkim 22nd full State of India**.
11. **1987**: **Mizoram (23rd)**, **Arunachal Pradesh (24th)**, and **Goa (25th state)** elevated to full statehood.
12. **2000 (November)**: Three new states carved out:
    - **Chhattisgarh (26th state)**: 1 Nov 2000 (carved from MP).
    - **Uttarakhand (Uttaranchal, 27th state)**: **9 Nov 2000** (carved from Uttar Pradesh, 13 hill districts; renamed Uttarakhand on 1 Jan 2007).
    - **Jharkhand (28th state)**: 15 Nov 2000 (carved from Bihar on Birsa Munda's birth anniversary).
13. **2014 (2 June)**: **Telangana (29th state)** carved out of Andhra Pradesh (Sri Krishna Committee).
14. **2019 (31 October)**: **Jammu & Kashmir Reorganisation Act 2019**: State of J&K bifurcated into **two Union Territories: J&K (with assembly) and Ladakh (without assembly)**; total count: **28 States and 8 Union Territories**.
15. **2020 (26 Jan)**: Merged **Dadra & Nagar Haveli and Daman & Diu** into a single Union Territory.

> **From–to (do not stop at the birth year): Uttarakhand** as a state runs **9 Nov 2000 → present** (name **Uttaranchal** till **1 Jan 2007**). **Jammu & Kashmir as a State** ran till **31 Oct 2019**; **Art. 370** operative **1950 → 5 Aug 2019**. Goa was a UT **1961–1987**, then the **25th state**.

14. UP & Uttarakhand State Legislature Quick Facts

State	Legislative Assembly (Vidhan Sabha)	Legislative Council (Vidhan Parishad)	Lok Sabha Seats	Rajya Sabha Seats	High Court & Benches
Uttar Pradesh	**403 seats** (SC: 84, ST: 2)	**100 seats** (Bicameral!)	**80 seats** (Highest in India; SC reserved: 17, ST: 0)	**31 seats** (Highest in India)	**Allahabad High Court** (Established 1866 at Agra, shifted to Allahabad 1869; Lucknow Bench created 1948)

State	Legislative Assembly (Vidhan Sabha)	Legislative Council (Vidhan Parishad)	Lok Sabha Seats	Rajya Sabha Seats	High Court & Benches
Uttarakhand	**70 seats** (SC: 13, ST: 2)	**Unicameral** (No Council)	**5 seats** (Tehri, Garhwal, Almora - SC, Haridwar, Nainital-US Nagar)	**3 seats**	**Uttarakhand High Court** (Established 9 Nov 2000 at Nainital; shifting to Haldwani approved)

15. The Five Types of Writs (Articles 32 & 226)

Power to issue prerogative writs: Supreme Court (Article 32 — only for Fundamental Rights) and High Courts (Article 226 — for Fundamental Rights AND ordinary legal rights; HC writ jurisdiction is wider than SC!)

Writ Name	Literal Latin Meaning	Purpose & Grounds for Issue	Against Whom Issued	Exceptions (Cannot be Issued Against)
Habeas Corpus (बन्दी प्रत्यक्षीकरण)	**"To have the body of"**	Remedy against **unlawful detention or illegal custody**; court orders the detaining authority to produce the detained person before it to examine legality of detention	**Both Public Authorities and Private Individuals**	1. Detention is lawful 2. Proceeding is for contempt of court/legislature 3. Detention is by competent court 4. Outside court's territorial jurisdiction
Mandamus (परमादेश)	**"We Command"**	Order issued to a public official/body directing them to **perform an official public duty** which they have failed or refused to perform	**Public officials, Public Corporations, Inferior Courts, Tribunals, Government**	1. **Private individuals or private bodies** 2. Discretionary duties (not mandatory) 3. To enforce private contracts 4. **President of India or State Governors** (Art. 361) 5. Chief Justice of HC acting in judicial capacity
Prohibition (प्रतिषेध)	**"To Forbid"**	Issued by a higher court to a lower court/tribunal to **prevent it from exceeding its jurisdiction** or usurping a jurisdiction not vested in it; preventive in nature	**Judicial and Quasi-Judicial Authorities only**	Administrative authorities, legislative bodies, and private individuals/bodies
Certiorari (उत्प्रेषण)	**"To be Certified" / "To be Informed"**	Issued by a higher court to a lower court/tribunal to **quash an order already passed** in excess of jurisdiction, lack of jurisdiction, or violation of natural justice; both **preventive and curative**	**Judicial, Quasi-Judicial, and Administrative Authorities** (SC expanded to administrative bodies in 1991)	Legislative bodies and private individuals/bodies
Quo-Warranto (अधिकार-पृच्छा)	**"By What Authority or Warrant"**	Issued to enquire into the **legality of the claim of a person to a public office**; prevents illegal usurpation of a substantive public office	**Substantive Public Offices of a permanent nature created by statute/Constitution**	Ministerial offices (PM/Minister), private offices. *(Can be sought by ANY interested person, not necessarily the aggrieved party!)*

16. Constituent Assembly Committees & Drafting Committee Members

A. Major Committees & Their Chairpersons (High-Frequency Match)

Committee Name	Chairperson	Scope & Responsibility
Drafting Committee (प्रारूप समिति)	**Dr. B.R. Ambedkar**	Prepared the draft of the Constitution; appointed on **29 August 1947**
Union Powers Committee	**Jawaharlal Nehru**	Defined the extent of Union legislative authority
Union Constitution Committee	**Jawaharlal Nehru**	Drafted structure of the Union Executive and Parliament
States Committee (Committee for Negotiating with States)	**Jawaharlal Nehru**	Negotiations with Princely States
Provincial Constitution Committee	**Sardar Vallabhbhai Patel**	Drafted structure of State Governors and Legislatures
Advisory Committee on Fundamental Rights, Minorities & Tribal Areas	**Sardar Vallabhbhai Patel**	Apex advisory body with 5 sub-committees: • *Fundamental Rights Sub-Committee*: **J.B. Kripalani** • *Minorities Sub-Committee*: **H.C. Mukherjee** • *North-East Frontier Tribal Areas*: **Gopinath Bardoloi** • *Excluded & Partially Excluded Areas*: **A.V. Thakkar**
Rules of Procedure Committee	**Dr. Rajendra Prasad**	Formulated rules for conduct of assembly business
Steering Committee	**Dr. Rajendra Prasad**	Steered legislative agenda and debates
Ad-hoc Committee on the National Flag	**Dr. Rajendra Prasad**	Adopted the National Flag on **22 July 1947**
Order of Business Committee	**Dr. K.M. Munshi**	Scheduled daily proceedings

B. The 7 Members of the Drafting Committee (Must-Memorize List)

Set up on 29 August 1947 under the chairmanship of Dr. B.R. Ambedkar:

1. **Dr. B.R. Ambedkar** (Chairman)
2. **N. Gopalaswami Ayyangar**
3. **Alladi Krishnaswamy Iyer**
4. **Dr. K.M. Munshi** (Only original Congress politician member)
5. **Syed Mohammad Saadulla** (Only Muslim League member)
6. **N. Madhava Rau** (Replaced **B.L. Mitter** who resigned due to ill-health)
7. **T.T. Krishnamachari** (Replaced **D.P. Khaitan** who died in 1948)

> **Constitutional Adviser**: **Sir B.N. Rau** (prepared original constitutional draft).
> **Chief Draftsman**: **S.N. Mukherjee**.
> **Calligrapher of English text**: **Prem Behari Narain Raizada** (flowery italic style).
> **Illuminator / Decorator**: **Nandalal Bose** and **Beohar Rammanohar Sinha** (Santiniketan artists).
> **Hindi Calligrapher**: **Vasant Krishan Vaidya** (decorated by Nandalal Bose).

17. Center-State Relations & Inter-State River Water Dispute Tribunals

A. Major Commissions on Center-State Relations

Commission / Committee	Year	Appointed By / State	Key Recommendations / Landmark Findings
Administrative Reforms Commission (ARC-I)	**1966**	Central Govt (Morarji Desai → K. Hanumanthaiah)	Recommended setting up of an **Inter-State Council under Article 263**; non-partisan Governors
Rajamannar Committee	**1969**	**Tamil Nadu Govt** (DMK, Dr. P.V. Rajamannar)	Strongly advocated State Autonomy; recommend immediate deletion of **Articles 356 and 365**; vesting residuary powers in States; abolishing All-India Services (IAS/IPS)
Anandpur Sahib Resolution	**1973**	Akali Dal (Punjab)	Demanded restriction of Central jurisdiction strictly to **Defence, Foreign Affairs, Currency, and Communications**
West Bengal Memorandum	**1977**	Left Front Govt (WB)	Demanded word "Union" be replaced by "Federation"; 75% of central revenues to states
Sarkaria Commission	**1983–1987**	Central Govt (Justice **R.S. Sarkaria**, B. Sivaraman, Dr. S.R. Sen)	**MOST INFLUENTIAL COMMISSION**: 1. **Article 356 must be used sparingly, only as a measure of last resort** 2. Institutionalize **Inter-State Council** under Art. 263 (established 1990) 3. Governor should be an eminent person from outside the state, detached from active politics, appointed after consulting CM 4. Opposed abolishing All-India Services, advocated strengthening them 5. Residuary legislative powers should remain with Centre (except taxation)
Punchhi Commission	**2007–2010**	Central Govt (Justice **M.M. Punchhi**)	1. Localized emergency (targeted under Art. 355/356 without dissolving state assembly) 2. Governor's tenure: fixed 5 years, removal only through impeachment by State Legislature 3. Governor should not be Chancellor of state universities

B. Inter-State River Water Disputes (Article 262)

Inter-State River Water Disputes Act 1956 empowers Central Govt to set up ad-hoc tribunals; jurisdiction of Supreme Court is barred:

Water Dispute Tribunal	Year Set Up	Riparian States Involved in Dispute
Krishna Water Dispute Tribunal I & II	1969 / 2004	Maharashtra, Karnataka, Andhra Pradesh, Telangana
Godavari Water Dispute Tribunal	1969	Maharashtra, Andhra Pradesh, MP, Chhattisgarh, Odisha, Karnataka
Narmada Water Dispute Tribunal	1969	Gujarat, Madhya Pradesh, Maharashtra, Rajasthan
Ravi & Beas Water Dispute Tribunal	1986	Punjab, Haryana, Rajasthan
Cauvery Water Dispute Tribunal	1990	Tamil Nadu, Karnataka, Kerala, Puducherry
Mahadayi (Mandovi) Tribunal	2010	Goa, Karnataka, Maharashtra
Vansadhara Water Dispute Tribunal	2010	Odisha and Andhra Pradesh
Mahanadi Water Dispute Tribunal	2018	Chhattisgarh and Odisha

SUBJECT 08 · ECONOMY

SUBJECT 08: INDIAN ECONOMY

भारतीय अर्थव्यवस्था — दैनिक उच्च-प्राथमिकता तथ्य

National Income Aggregates, 12 Five-Year Plans & Models, RBI Monetary Policy (Repo/SDF/CRR), CPI vs WPI, Fiscal Deficits, Finance Commission (15th/16th), and Poverty Committees.

1. National Income Aggregates & Macroeconomic Formulas

Concept / Metric	Formula / Technical Definition	Key Exam Traps & Prelims Triggers
GDP at Market Price	$GDP_{MP} = C + I + G + (X - M)$	Total market value of all final goods & services produced **within the domestic territory** in 1 year. Foreigners earning in India included; Indians earning abroad excluded.
GNP at Market Price	$GNP_{MP} = GDP_{MP} + NFIA$	Measures income earned by **nationals/citizens** regardless of geographical location. $NFIA$ = Net Factor Income from Abroad (Factor income from abroad $-$ Factor income paid abroad).
NNP at Market Price	$NNP_{MP} = GNP_{MP} - \text{Depreciation}$	Deducts consumption of fixed capital (depreciation). **Depreciation is also called Capital Consumption Allowance**.
NNP at Factor Cost (National Income)	$NNP_{FC} = NNP_{MP} - \text{Net Indirect Taxes (NIT)}$	Traditional official definition of **National Income**. Net Indirect Taxes (NIT) = Indirect Taxes $-$ Subsidies.
Gross Value Added (GVA)	$GVA_{Basic} = GVA_{Factor Cost} + \text{Production Taxes} - \text{Production Subsidies}$	Since 2015, India measures headline GDP as: **$GDP_{MP} = GVA_{Basic} + \text{Product Taxes} - \text{Product Subsidies}$**. Base year changed to **2011–12**.
Personal Disposable Income (PDI)	$PDI = \text{Personal Income} - \text{Direct Taxes} - \text{Miscellaneous Fees/Fines}$	Actual income available to households for consumption expenditure and savings.
Real GDP vs Nominal GDP	$\text{Real GDP} = \frac{\text{Nominal GDP}}{\text{GDP Deflator}} \times 100$	Real GDP is calculated at **constant base year prices** (eliminates inflation impact). Nominal GDP is at current prices.
GDP Deflator	$\frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$	Most comprehensive measure of inflation in an economy because it covers **all goods and services** produced, unlike CPI/WPI which use a fixed basket.

Examiner Trap: Transfer payments (old age pensions, unemployment allowance, student scholarships) are **excluded** from National Income because no corresponding production of goods or services occurs. Second-hand goods sales are also excluded (only broker commission is included).

2. Five-Year Plans (FYP) & NITI Aayog Master Matrix

Plan & Duration	Model / Architect	Core Objective / Strategy	Key Milestones & Exam Triggers
1st FYP (1951–56)	**Harrod-Domar Model**	Agriculture, irrigation, price stability	Bhakra-Nangal, Hirakud, Damodar Valley dams launched; Target 2.1%, Achieved 3.6%.

Plan & Duration	Model / Architect	Core Objective / Strategy	Key Milestones & Exam Triggers
2nd FYP (1956–61)	**P.C. Mahalanobis Model**	Rapid industrialisation, Heavy & Basic Industries	Steel plants at **Bhilai** (USSR), **Rourkela** (Germany), **Durgapur** (UK); Industrial Policy Resolution 1956.
3rd FYP (1961–66)	John Sandy & Sukhamoy Chakravarty	Self-reliant & self-generating economy	**Bokaro Steel Plant** (1964, USSR), FCI & CACP set up (1965). Plan failed due to 1962 China War, 1965 Pak War, 1965–66 Severe Drought.
Plan Holiday (1966–69)	Three Annual Plans	Green Revolution rollout	Devaluation of Indian Rupee (1966); High-Yielding Varieties (HYV) seeds introduced.
4th FYP (1969–74)	D.R. Gadgil Formula	Growth with stability & progressive self-reliance	**14 Major Commercial Banks Nationalised** (19 July 1969); Pokhran-I Nuclear Test (*Smiling Buddha*, 1974); 1971 Indo-Pak War.
5th FYP (1974–79)	D.P. Dhar Model	**Garibi Hatao** (Poverty Eradication) & Self-reliance	**20-Point Programme** (1975); Minimum Needs Programme (MNP); Terminated 1 year early in 1978 by Janata Govt.
Rolling Plan (1978–80)	Gunnar Myrdal concept	Annual revisions without fixed 5-year lock	Introduced by Janata Govt (Morarji Desai); ended when Congress returned in 1980.
6th FYP (1980–85)	Investment & Modernisation	Poverty eradication, economic & tech self-reliance	**NABARD established** (12 July 1982, Shivaraman Committee); **EXIM Bank** (1982); **6 more banks nationalised** (1980).
7th FYP (1985–90)	Pranab Mukherjee / Lakdawala	Food, Work & Productivity	Speed Post (1986); SEBI formed (1988); Jawahar Rozgar Yojana (JRY, 1989).
Annual Plans (1990–92)	Balance of Payments Crisis	Structural reforms & crisis stabilization	**LPG Reforms (1991)**: Liberalisation, Privatisation, Globalisation under P.V. Narasimha Rao & Dr. Manmohan Singh.
8th FYP (1992–97)	John W. Miller Model	Human Resource Development (employment, education)	India joined **WTO** (1 Jan 1995); National Stock Exchange (NSE) operationalised.
9th FYP (1997–2002)	Social Justice & Equality	Growth with Social Justice and Equity	50 years of Indian Independence; Asian Financial Crisis backdrop.
10th FYP (2002–07)	Target 8% GDP Growth	Doubling per capita income in 10 years, reduce poverty by 5%	National Rural Employment Guarantee Act (**NREGA**, 2005); Bharat Nirman.
11th FYP (2007–12)	C. Rangarajan / Montek Singh	**Faster and More Inclusive Growth**	Right to Education (RTE Act, 2009); National Rural Livelihoods Mission (NRLM).
12th FYP (2012–17)	Final Five-Year Plan	**Faster, More Inclusive and Sustainable Growth**	Terminated; replaced by NITI Aayog 15-Year Vision, 7-Year Strategy & 3-Year Action Agenda.

NITI Aayog (National Institution for Transforming India): Formed on **1 January 2015** replacing the Planning Commission (15 March 1950). Non-constitutional, non-statutory extra-constitutional think tank. Chairperson: **Prime Minister**. Governing Council comprises all State Chief Ministers, UT Lt Governors/CMs. Operates on **Bottom-Up** approach and **Cooperative Federalism** (unlike top-down Planning Commission).

3. Monetary Policy, RBI Instruments & Money Supply Aggregates

A. RBI Monetary Policy Instruments Master Table

Instrument	Nature / Category	Operational Mechanism & Definition
Cash Reserve Ratio (CRR)	Quantitative / Direct	Percentage of Net Demand and Time Liabilities (NDTL) commercial banks **must keep with RBI in cash**. No interest paid by RBI on CRR. Governed by RBI Act, 1934 (Sec 42).
Statutory Liquidity Ratio (SLR)	Quantitative / Direct	Percentage of NDTL commercial banks **must maintain with themselves** in approved liquid assets (Cash, Gold, Govt Securities). Governed by Banking Regulation Act, 1949 (Sec 24).
Repo Rate	Quantitative / Indirect	Rate at which RBI lends short-term money to commercial banks against **government securities as collateral** (repurchase agreement). Primary policy rate.
Reverse Repo Rate	Quantitative / Indirect	Rate at which RBI borrows short-term funds from commercial banks against government securities.
Standing Deposit Facility (SDF)	Quantitative / Liquidity	Introduced in April 2022 under amended RBI Act (Sec 17). Allows RBI to absorb excess liquidity from commercial banks **without providing any collateral (G-Secs)**. Replaced Reverse Repo as floor of LAF corridor.
Marginal Standing Facility (MSF)	Quantitative / Liquidity	Penal rate at which scheduled commercial banks can borrow overnight funds from RBI by dipping into their SLR quota (up to prescribed limit). Ceiling of LAF corridor.
Bank Rate	Quantitative / Long-term	Rate at which RBI buys or rediscounts bills of exchange or commercial papers; benchmark for penal interest rates on non-compliance with reserve requirements.
Open Market Operations (OMO)	Quantitative / Market	Outright purchase and sale of Government Securities (G-Secs) by RBI in open market to inject or absorb durable liquidity.
Moral Suasion, Credit Rationing, Margin Requirements	Qualitative / Selective	Directives to banks to channel credit to specific priority sectors, control speculative borrowing, and regulate loan-to-value (LTV) ratios.

B. Money Supply Measures (Monetary Aggregates)

- **Reserve Money (M_0 / High Powered Money):** Currency in Circulation $+$ Bankers' deposits with RBI $+$ 'Other' deposits with RBI.
- **Narrow Money (M_1):** Currency with Public $+$ Demand Deposits with Banking System (Current & Savings Account - CASA) $+$ 'Other' deposits with RBI. *(Most liquid aggregate).*
- **M_2:** $M_1 +$ Post Office Savings Bank deposits.
- **Broad Money (M_3):** $M_1 +$ Time Deposits (Fixed/Recurring Deposits) with the banking system. *(Most commonly used indicator of money supply in India).*
- **M_4:** $M_3 +$ All deposits with Post Office Savings Organisation (excluding National Savings Certificates / NSC).
- **Liquidity Hierarchy:** $M_1 > M_2 > M_3 > M_4$.

C. Monetary Policy Committee (MPC)

- Statutory body constituted under **Section 45ZB of RBI Act, 1934** (amended by Finance Act, 2016).
- **6 Members:** 3 from RBI (RBI Governor as ex-officio Chairperson, Deputy Governor in charge of monetary policy, 1 RBI officer) $+$ 3 external members appointed by Central Government.
- **Inflation Target:** Central Govt in consultation with RBI sets retail inflation target (CPI headline) at **4% with a tolerance band of $\pm 2\%$** (i.e., 2% to 6%).
- Meets at least **4 times a year**. Quorum is 4 members. Each member has 1 vote; RBI Governor has a **casting vote** in case of a tie.

4. Inflation Indices: CPI vs WPI Master Comparison

Parameter	Wholesale Price Index (WPI)	Consumer Price Index (CPI - Combined)
Publishing Authority	Office of Economic Adviser (DPIIT), Ministry of Commerce & Industry	National Statistical Office (NSO), Ministry of Statistics and Programme Implementation (MoSPI)
Base Year	**2011–12** (Revised from 2004–05 in 2017)	**2012**
Official Headline Anchor	Used for wholesale trade transactions	**Official headline anchor for RBI Monetary Policy** (since Urjit Patel Committee, 2014)
Services Included?	**NO services included** (Goods only)	**YES, Services are included** (Housing, Transport, Education, Medical, etc.)
Number of Commodities	**697 items**	448 items (Rural), 460 items (Urban)
Highest Weight Group	**Manufactured Products (64.23%)** > Primary Articles (22.62%) > Fuel & Power (13.15%)	**Food and Beverages (45.86%)** > Misc/Services (28.32%) > Housing (10.07%)
Price Point Measured	First point of bulk commercial transaction	Retail consumer purchase point

5. Public Finance, Deficits & Finance Commission

A. Deficit Formulas Master Sheet

1. **Fiscal Deficit (FD):** $\text{Total Expenditure} - \text{Total Receipts (excluding borrowings)}$. It represents the total borrowing requirements of the government from all sources in that year.
2. **Revenue Deficit (RD):** $\text{Revenue Expenditure} - \text{Revenue Receipts}$. Indicates that the government is borrowing to finance day-to-day consumption rather than capital asset creation.
3. **Effective Revenue Deficit (ERD):** $\text{Revenue Deficit} - \text{Grants for creation of capital assets}$. Introduced in Union Budget 2011–12.
4. **Primary Deficit (PD):** $\text{Fiscal Deficit} - \text{Interest Payments}$. Measures the current year's fiscal stance excluding the burden of past accumulated debt interest.

B. Finance Commission (Article 280) Master Matrix

- Quasi-judicial body constituted by the **President every 5 years** under Article 280.
- Recommends devolution of net proceeds of taxes between Union and States (Vertical devolution) and allocation among States (Horizontal devolution).
- **1st FC:** K.C. Neogy (1951)
- **13th FC:** Dr. Vijay Kelkar (recommending GST framework)
- **14th FC:** Dr. Y.V. Reddy (increased States' share sharply from 32% to **42%**)
- **15th FC: N.K. Singh** (2020–26): Devolution share set at **41%** (1% adjusted for newly created UTs of J&K and Ladakh). Criteria weights: Income Distance (45%), Population 2011 (15%), Area (15%), Forest & Ecology (10%), Demographic Performance (12.5%), Tax Effort (2.5%).
- **16th FC:** Constituted on 31 Dec 2023 under chairmanship of **Dr. Arvind Panagariya** (former Vice Chairman of NITI Aayog).

6. Banking & Financial Sector Milestones

- **1935 (1 April):** RBI established on recommendations of **Hilton Young Commission (1926)** under RBI Act, 1934. Initial capital ₹5 crore; nationalised on **1 January 1949**.

- **1955:** Imperial Bank of India (formed in 1921 by amalgamating Presidency Banks of Bengal, Bombay, Madras) nationalised and renamed **State Bank of India (SBI)** on recommendation of All India Rural Credit Survey Committee (AD Gorwala).
- **1969 (19 July): 14 Major Commercial Banks** (with deposits $\ge ₹50$ crore) nationalised by Indira Gandhi Govt via Banking Companies (Acquisition and Transfer of Undertakings) Ordinance.
- **1975 (2 October):** First Regional Rural Bank (**Prathama Bank**, Moradabad, UP) established on recommendations of **M. Narasimham Working Group**.
- **1980 (15 April): 6 More Commercial Banks** (with deposits $\ge ₹200$ crore) nationalised.
- **1982 (12 July): NABARD** established under NABARD Act, 1981 on recommendation of **B. Sivaraman Committee**.
- **1988:** SEBI set up as non-statutory body; granted statutory powers via **SEBI Act, 1992** on 30 January 1992.
- **Priority Sector Lending (PSL) Targets:** Total PSL target for Domestic Scheduled Commercial Banks and Foreign Banks with ≥ 20 branches is **40% of Adjusted Net Bank Credit (ANBC)**. Sub-targets: Agriculture (18%, within which 10% for Small & Marginal Farmers), Micro Enterprises (7.5%), Weaker Sections (12%). For Regional Rural Banks (RRBs) and Small Finance Banks (SFBs), total PSL target is **75% of ANBC**.

7. Agricultural Revolutions, Schemes & MSP

A. Colour Revolutions in Indian Agriculture

Colour Revolution	Associated Sector / Commodity	Pioneer / Father in India
Green Revolution	Food Grains (Wheat & Rice, HYV seeds)	**M.S. Swaminathan** (Norman Borlaug globally)
White Revolution (Operation Flood)	Milk & Dairy Development (1970)	**Dr. Verghese Kurien** (Amul / NDDB Anand)
Blue Revolution	Fisheries & Aquaculture	Dr. Hiralal Chaudhuri & Dr. Arun Krishnan
Yellow Revolution	Oilseeds (Mustard, Sunflower, Soyabean)	Sam Pitroda
Golden Revolution	Horticulture, Honey, Fruits	Nirpakh Tutej
Golden Fibre Revolution	**Jute** production	-
Silver Revolution	Eggs & Poultry	Indira Gandhi
Silver Fibre Revolution	**Cotton** production	-
Pink Revolution	Onion, Pharmaceuticals, Prawn	Durgesh Patel
Red Revolution	Meat & Tomato production	Vishal Tewari
Round Revolution	**Potato** production	-
Grey Revolution	Fertilizers	-
Brown Revolution	Leather, Non-conventional energy, Cocoa	-
Rainbow Revolution	Integrated development of all agri sectors	Announced in Agri Policy 2000

B. Minimum Support Price (MSP) Mechanism

- Announced by Central Government on recommendations of **CACP (Commission for Agricultural Costs and Prices)** before sowing season.
- Covers **22 mandated crops** $+$ Fair and Remunerative Price (FRP) for **Sugarcane**:
- 7 Cereals (Paddy, Wheat, Maize, Jowar, Bajra, Ragi, Barley)
- 5 Pulses (Gram, Arhar/Tur, Moong, Urad, Masur)
- 8 Oilseeds (Groundnut, Rapeseed/Mustard, Soyabean, Sesamum, Sunflower, Safflower, Nigerseed, Copra)
- Raw Cotton, Raw Jute, De-husked Coconut
- **Cost Concepts used by CACP:**
- $A2$: Direct paid-out costs on seeds, fertilizers, pesticides, hired labour, fuel, irrigation.

- $A2 + FL$: $A2 +$ Imputed value of unpaid family labour.
- $C2$: Comprehensive cost ($A2 + FL +$ rental value of owned land $+$ interest on owned capital assets).
- Central Govt formula since Union Budget 2018–19: MSP set at **at least 1.5 times (150%) of $A2 + FL$**.

8. Poverty Estimation Committees & Multidimensional Poverty

Committee & Year	Consumption Basket / Criteria Used	Poverty Line Benchmark (Monthly / Daily)
Y.K. Alagh (1979)	Minimum Nutritional Calorie intake	**2,400 kcal/day** in Rural areas; **2,100 kcal/day** in Urban areas.
D.T. Lakdawala (1993)	Calorie norms linked with state-specific price indices (CPI-AL for rural, CPI-IW for urban)	State-specific poverty lines; rejected uniform national basket.
Suresh Tendulkar (2009)	Shifted from calories to mixed reference basket including **Health, Education, Clothing, Footwear**	Set per capita daily consumption at **₹27.2 in rural** and **₹33.3 in urban** areas (₹816/month rural, ₹1000/month urban at 2011–12 prices). National poverty rate: **21.9%** (2011–12).
C. Rangarajan (2014)	Independent nutritional norm (2,155 kcal rural, 2,090 kcal urban + protein/fat) + non-food expenses	Set per capita daily consumption at **₹32 in rural** and **₹47 in urban** areas (₹972/month rural, ₹1407/month urban). National poverty rate: **29.5%** (2011–12).

Multidimensional Poverty Index (MPI) by NITI Aayog:
Follows Alkire-Foster methodology. **3 Dimensions and 12 Indicators** (each dimension weighted equally at $⅓$): 1. **Health (⅓):** Nutrition (⅙), Child & Adolescent Mortality (1/12), Maternal Health (1/12). 2. **Education (⅓):** Years of Schooling (⅙), School Attendance (⅙). 3. **Standard of Living (⅓):** Cooking Fuel, Sanitation, Drinking Water, Electricity, Housing, Assets, Bank Account (each 1/21).

9. International Financial Institutions & Global Economic Reports

Institution / Group	Established	Headquarters	Key Publications / Reports
International Monetary Fund (IMF)	1944 (Bretton Woods)	**Washington D.C.**	**World Economic Outlook (WEO)**, **Global Financial Stability Report (GFSR)**, Fiscal Monitor
World Bank (IBRD + IDA)	1944 (Bretton Woods)	**Washington D.C.**	**Global Economic Prospects (GEP)**, World Development Report (WDR)
World Economic Forum (WEF)	1971 (Klaus Schwab)	**Cologny/Geneva (Switzerland)**	**Global Competitiveness Report**, **Global Gender Gap Report**, Global Risks Report, Energy Transition Index
World Trade Organisation (WTO)	1 Jan 1995 (Marrakesh)	**Geneva (Switzerland)**	World Trade Report
Asian Development Bank (ADB)	1966	**Mandaluyong, Manila (Philippines)**	Asian Development Outlook (ADO)
Asian Infrastructure Investment Bank (AIIB)	2016	**Beijing (China)**	China largest shareholder, India 2nd largest shareholder
New Development Bank (NDB / BRICS Bank)	2014 (Fortaleza)	**Shanghai (China)**	First President: K.V. Kamath (India)
UNDP	1965	**New York (USA)**	**Human Development Report (HDR)** — HDI, IHDI, GII, MPI

Special Drawing Rights (SDR - Paper Gold): International reserve asset created by the IMF in 1969. The SDR basket includes **5 currencies**: US Dollar, Euro, Chinese Renminbi (Yuan - added in 2016), Japanese Yen, and British Pound Sterling.

SUBJECT 09 · CENSUS & URBANISATION

SUBJECT 09: CENSUS & URBANISATION (INDIA & UP)

जनगणना एवं नगरीकरण — दैनिक उच्च-प्राथमिकता तथ्य

Census 2011 National Indicators, State Top & Bottom Extremes, SC & ST Demographics, Religious Hierarchy, Uttar Pradesh District Extremes, and Urbanisation Rates in India & UP.

1. Indian Census History & Legal Framework

Milestone / Parameter	Year / Detail	Historical / Legal Context
First Census in Modern India	**1872**	Conducted non-synchronously under **Lord Mayo**.
First Synchronous Decennial Census	**1881**	Conducted across all India under Viceroy **Lord Ripon** (Census Commissioner: **W.C. Plowden**).
The Year of the Great Divide	**1921**	The only census decade in Indian history showing a **negative population growth rate (-0.31%)** due to 1918 Spanish Flu pandemic, plague, and severe famine.
First Census of Independent India	**1951**	Census Commissioner: **R.A. Gopalaswami**. Total population: 361 million; sex ratio: 946.
Census Act, 1948	Enacted under Sardar Vallabhbhai Patel	Permanent statutory basis; census figures are strictly confidential and inadmissible in courts as evidence.
Nodal Authority	Registrar General and Census Commissioner of India (RGCCI)	Under **Ministry of Home Affairs (MHA)**, New Delhi.
2011 Census Edition	**15th Census of India** (7th after Independence)	Motto: "*Our Census, Our Future*"; Mascot: Woman Enumerator; Commissioner: **Dr. C. Chandramouli**.

2. Census 2011 Master Matrix: National Indicators

Demographic Indicator	Census 2001	Census 2011	Absolute / Percentage Change
Total Population	1,028.7 million	**1,210.85 million** (~121.08 Cr)	$+182.1$ million (+17.7%)
Male Population	532.2 million (51.74%)	**623.27 million (51.54%)**	Male proportion decreased slightly
Female Population	496.5 million (48.26%)	**587.58 million (48.46%)**	Female proportion increased
Decadal Growth Rate (2001–11)	21.54%	**17.70%**	Decreased by **3.84 percentage points**
Annual Exponential Growth Rate	1.97%	**1.64%**	Decelerating trend
Population Density (persons/sq km)	325	**382**	Increased by **+57 persons/sq km**
Overall Sex Ratio (females/1000 males)	933	**943**	Improved by **+10 points**

Demographic Indicator	Census 2001	Census 2011	Absolute / Percentage Change
Child Sex Ratio (0–6 years)	927	**918**	**Declined by -9 points** (Critical alarm zone)
Child Population (0–6 years) share	15.93%	**13.59%**	Reflects falling fertility rates
Overall Literacy Rate	64.83%	**73.00%** (some official reports 74.04%)	Increased by **+8.17 percentage points**
Male Literacy Rate	75.26%	**80.90%** (or 82.14%)	Improved
Female Literacy Rate	53.67%	**64.60%** (or 65.46%)	Faster gain than males (+10.93%)
Gender Gap in Literacy	21.59%	**16.30%** (or 16.68%)	Narrowed by **5.29 percentage points**
Urban Population Share	27.81%	**31.16%**	Increased by $+3.35$ percentage points
Rural Population Share	72.19%	**68.84%**	Decreased proportionally

3. State & UT Rankings: Top & Bottom Extremes (Census 2011)

Demographic Parameter	Highest / Top 1 (State / UT)	Lowest / Bottom 1 (State / UT)
Total Population (States)	**Uttar Pradesh** (199.81 million, 16.51% of India)	**Sikkim** (610,577 persons)
Total Population (UTs)	**NCT Delhi** (16.78 million)	**Lakshadweep** (64,473 persons)
Decadal Growth Rate (States)	**Meghalaya** (27.95%) > Arunachal Pradesh (26.03%)	**Nagaland** (**-0.58%**, *only negative growth state*)
Decadal Growth Rate (UTs)	**Dadra & Nagar Haveli** (55.88%)	**Lakshadweep** (6.30%)
Population Density (States)	**Bihar** (1,106/sq km) > West Bengal (1,028) > Kerala (860) > UP (829)	**Arunachal Pradesh** (**17/sq km**)
Population Density (UTs)	**NCT Delhi** (11,320/sq km)	**Andaman & Nicobar Islands** (46/sq km)
Sex Ratio (States)	**Kerala** (1,084) > Tamil Nadu (996) > Andhra Pradesh (993)	**Haryana** (**879**) < Jammu & Kashmir (889)
Sex Ratio (UTs)	**Puducherry** (1,037)	**Daman & Diu** (**618**)
Child Sex Ratio (0–6 years)	**Mizoram** (971) > Meghalaya (970)	**Haryana** (**834**) < Punjab (846)
Overall Literacy Rate (States)	**Kerala** (94.00%) > Mizoram (91.33%) > Goa (88.70%)	**Bihar** (**61.80%**) < Arunachal (65.38%) < Rajasthan (66.11%)
Male Literacy Rate	**Kerala** (96.11%) > Mizoram (93.35%)	**Bihar** (**71.20%**)
Female Literacy Rate	**Kerala** (92.07%) > Mizoram (89.27%)	**Bihar** (**51.50%**) < Rajasthan (52.12%)
Highest Gender Gap in Literacy	**Rajasthan** (**27.1%** gap: Male 79.2% vs Female 52.1%)	**Meghalaya** (**3.1%** gap) < Kerala (4.04% gap)

4. Scheduled Castes (SC) and Scheduled Tribes (ST) Demographics

A. Scheduled Castes (SC) — Census 2011

- Total SC Population in India: **201.4 million** (**16.6%** of total population).
- **Highest SC Population (Absolute): Uttar Pradesh** (41.35 million, accounts for 20.5% of total Indian SCs) > West Bengal (21.46M) > Bihar (16.57M).
- **Highest SC Percentage (% of State Population): Punjab** (**31.9%**) > Himachal Pradesh (25.2%) > West Bengal (23.5%) > Uttar Pradesh (20.7%).
- **States/UTs with NO Scheduled Castes: Nagaland, Mizoram, Lakshadweep, Andaman & Nicobar Islands**.

B. Scheduled Tribes (ST) — Census 2011

- Total ST Population in India: **104.3 million** (**8.6%** of total population).
- **Highest ST Population (Absolute): Madhya Pradesh** (15.31 million, 14.7% of all Indian STs) > Maharashtra (10.51M) > Odisha (9.59M).
- **Highest ST Percentage (% of State/UT): Lakshadweep** (**94.8%**) and **Mizoram** (**94.4%**) > Nagaland (86.5%) > Meghalaya (86.1%).
- **States/UTs with NO Scheduled Tribes: Punjab, Haryana, Chandigarh, Delhi, Puducherry**.

5. Religious Demographics Hierarchy (Census 2011)

Religion	Total Population	% of India Population	Decadal Growth (2001–11)	Sex Ratio (2011)
Hindu	966.3 million	**79.80%**	16.76%	939
Muslim	172.2 million	**14.23%**	**24.60%** (highest among major)	951
Christian	27.8 million	**2.30%**	15.50%	**1,023** (highest sex ratio)
Sikh	20.8 million	**1.72%**	8.40%	903
Buddhist	8.4 million	**0.70%**	6.10%	965
Jain	4.5 million	**0.37%**	5.40%	954 (Highest literacy: 94.9%)
Other / Not Stated	10.8 million	0.88%	-	-

6. Uttar Pradesh Census 2011 District Extremes Master Sheet

Parameter in Uttar Pradesh	UP Overall Figure	Top / Highest District in UP	Bottom / Lowest District in UP
Total Population	**199,812,341** (16.51% of India)	**Prayagraj (Allahabad)** (5.95M) > Moradabad > Ghaziabad	**Mahoba** (875,958) < Chitrakoot < Hamirpur
Decadal Growth Rate	**20.23%** (2001–2011)	**Gautam Buddha Nagar** (**49.1%**) > Ghaziabad (41.3%)	**Kanpur Nagar** (**9.9%**) < Hamirpur (11.1%)
Population Density	**829 persons/sq km**	**Ghaziabad** (**3,971/sq km**) > Varanasi (2,395)	**Lalitpur** (**242/sq km**) < Sonbhadra (270)
Overall Sex Ratio	**912 females/1000 males**	**Jaunpur** (**1,024**) > Azamgarh (1,019) > Deoria (1,017)	**Gautam Buddha Nagar** (**851**) < Hamirpur/Kanpur (861)
Child Sex Ratio (0–6)	**902 females/1000 males**	**Balrampur** (**950**) > Sant Kabir Nagar (942)	**Baghpat** (**841**) < Gautam Buddha Nagar (843)

Parameter in Uttar Pradesh	UP Overall Figure	Top / Highest District in UP	Bottom / Lowest District in UP
Overall Literacy Rate	**67.68%**	**Auraiya** (**78.95%**) > Kanpur Nagar (79.65% urban / Auraiya overall) > Etawah	**Shrawasti** (**46.74%**) < Bahraich (49.36%) < Balrampur (49.51%)
Male Literacy Rate	**77.28%**	**Gautam Buddha Nagar** (**88.06%**) > Auraiya	**Shrawasti** (**57.16%**) < Bahraich
Female Literacy Rate	**57.18%**	**Kanpur Nagar** (**75.05%**) > Lucknow (71.54%)	**Shrawasti** (**34.78%**) < Balrampur (37.22%)
Lowest Gender Gap in Literacy	20.10% (Male 77.28% $-$ Female 57.18%)	**Kanpur Nagar** (Narrowest gender gap: 8.5%)	**Maharajganj** (Widest gender gap: 26.9%)
SC Population (Absolute)	**41.35 million** (20.70% of UP)	**Sitapur** (highest absolute SC pop)	**Baghpat** (lowest absolute SC pop)
SC Percentage (%)	20.70% of UP	**Kaushambi** (**34.72%**) > Sitapur	**Baghpat** (**11.44%**)
ST Population & %	**1.13 million** (0.57% of UP)	**Sonbhadra** (**20.67%** and highest absolute)	**Baghpat, Badaun, Ayodhya** (almost zero ST)

UP Urbanisation: 22.27% of UP population lives in urban areas; **77.73%** in rural areas.
Highest Urbanised District (%): **Ghaziabad (67.6%)** > Lucknow (66.2%) > Kanpur Nagar (65.8%).
Lowest Urbanised District (%): **Shrawasti (3.5%)** > Kushi Nagar (4.7%).
7 Million-Plus Cities in UP: Kanpur, Lucknow, Ghaziabad, Agra, Varanasi, Meerut, Prayagraj.

7. Uttarakhand Census 2011 Highlights

- **Total Population: 10,086,292** (~1.01 Crore, 0.83% of India).
- **Decadal Growth Rate: 18.81%** (Plains growing rapidly: US Nagar 33.4%, Haridwar 30.6%; Hill districts showing negative growth: **Almora -1.28%**, **Pauri Garhwal -1.41%** - ghost villages phenomenon).
- **Population Density: 189 persons/sq km** (Haridwar highest at 801; Uttarkashi lowest at 41).
- **Overall Sex Ratio: 963 females/1000 males** (Almora highest at 1,139; Haridwar lowest at 880).
- **Child Sex Ratio (0–6 years): 890** (Pithoragarh lowest at 816).
- **Overall Literacy Rate: 78.82%** (Male 87.40%, Female 70.01%). Highest literacy: **Dehradun (84.25%)**; Lowest: **Haridwar (73.43%)**.

SUBJECT 10 · UP SPECIAL GK

SUBJECT 10: UTTAR PRADESH SPECIAL GK

उत्तर प्रदेश विशेष सामान्य ज्ञान — दैनिक उच्च-प्राथमिकता तथ्य

Geographic Boundaries & 4 Extremes, Drainage & Rivers, Dudhwa NP & 4 Tiger Reserves, 10 Ramsar Sites, 9 Agro-Climatic Zones, Minerals, 1857 Revolt, Gharanas, and 75 Districts ODOP.

1. Uttar Pradesh Geographical Profile & Boundary Matrix

Geographical Dimension	Exact Specification & Measurement	High-Yield Prelims Anchor
Latitudinal Extent	$23^{\circ} 52'\text{ N}$ to $30^{\circ} 24'\text{ N}$	Latitudinal span: $6^{\circ} 32'$
Longitudinal Extent	$77^{\circ} 05'\text{ E}$ to $84^{\circ} 38'\text{ E}$	Longitudinal span: $7^{\circ} 33'$
Standard Time Meridian	**$82^{\circ} 30'\text{ E}$ (IST)** passes through **Mirzapur (near Prayagraj)**	Calculates Indian Standard Time (GMT $+5:30$)
Total Geographic Area	**240,928 sq km** (7.33% of India's total area)	**4th Largest State** by area (Rajasthan > MP > Maharashtra > UP)
East-to-West Length	**650 km**	Extreme length
North-to-South Width	**240 km**	Narrower width
Number of Neighboring Entities	**8 States + 1 UT (Delhi) + 1 International (Nepal)**	Shares borders with the **maximum number of states** in India

Boundary Touch-Points Master Grid

- **Northern Boundary:** Uttarakhand & Himachal Pradesh (Himachal touches only **Saharanpur** - shortest border).
- **North-Eastern Boundary: Nepal** (Touches **7 Districts:** Pilibhit, Lakhimpur Kheri, Bahraich, Shrawasti, Balrampur, Siddharthnagar, Maharajganj — 579 km border).
- **Western Boundary:** Haryana, Delhi (UT), Rajasthan.
- Touches Haryana: Saharanpur, Shamli, Baghpat, Gautam Buddha Nagar, Aligarh, Mathura (6 districts; separated by Yamuna river).
- Touches Delhi: Ghaziabad & Gautam Buddha Nagar.
- Touches Rajasthan: **Mathura & Agra** (2 districts).
- **Southern Boundary: Madhya Pradesh** (Touches **11 Districts:** Agra, Etawah, Jalaun, Jhansi, Lalitpur, Mahoba, Banda, Chitrakoot, Prayagraj, Mirzapur, Sonbhadra — longest inter-state border).
- **South-Eastern Boundary: Chhattisgarh** (Touches only **Sonbhadra**) & **Jharkhand** (Touches only **Sonbhadra**).
- **Eastern Boundary: Bihar** (Touches 7 Districts: Sonbhadra, Chandauli, Ghazipur, Ballia, Deoria, Kushinagar, Maharajganj).

Four District Extremes:
- **Northernmost District: Saharanpur** (surrounded by Haryana, HP, Uttarakhand)
- **Southernmost District: Sonbhadra** (**Only district in India bordering 4 states**: MP, Chhattisgarh, Jharkhand, Bihar)
- **Easternmost District: Ballia** (wedged into Bihar)
- **Westernmost District: Shamli**
- **Largest District by Area: Lakhimpur Kheri** (7,680 sq km)
- **Smallest District by Area: Hapur** (660 sq km)

2. Drainage System, Rivers, Canals & Wetlands of UP

A. Major Rivers of Uttar Pradesh

River	Entry Point in UP	Exit Point / Confluence	Total Districts in UP & Notes
Ganga	Enters at **Bijnor**	Exits UP at **Ballia** into Bihar	Traverses **28 districts** (~1,450 km in UP). Receives Ramganga at Kannauj, Yamuna at Prayagraj, Gomti at Saidpur (Ghazipur), Ghaghara at Chhapra (Bihar border).
Yamuna	Enters at **Saharanpur** (Faizabad village)	Joins Ganga at **Prayagraj (Triveni Sangam)**	Traverses **19 districts**. Receives Hindon at Noida, Chambal at Pachnada (Etawah), Betwa at Hamirpur, Ken at Chilla (Banda).
Ramganga	Enters at **Bijnor** (near Kalagarh)	Joins Ganga near **Kannauj**	Passes through Corbett National Park; flows via Moradabad, Bareilly.
Gomti	Originates at **Fulhar Lake (Panghaili Jheel), Pilibhit**	Joins Ganga near Saidpur, Ghazipur	**Only major Himalayan tributary originating in the plains of UP**. Lucknow, Sultanpur, Jaunpur situated on its banks.
Ghaghara (Saryu)	Enters near Kheri/Bahraich	Joins Ganga at Chhapra (Bihar)	Originates at Mapchachungo glacier (Tibet); called Karnali in Nepal, Saryu at Ayodhya.
Betwa (Vetravati)	Enters at **Lalitpur**	Joins Yamuna at **Hamirpur**	Originates in Vindhyan range (MP); Rajghat, Parichha, and Matatila dams built on it.
Ken (Karnavati)	Enters at **Banda**	Joins Yamuna at **Chilla (Banda)**	Ken-Betwa river interlinking project is India's first river link project.
Son	Enters and exits in **Sonbhadra**	Joins Ganga near Patna	Originates at Amarkantak plateau (MP).

B. Major Canals & Reservoirs

- **Sharda Canal:** Longest canal network in Uttar Pradesh (originates from Sharda river at Banbasa, India-Nepal border).
- **Upper Ganga Canal:** Constructed by Sir Proby Cautley (1854); originates at Haridwar (Bhimoda).
- **Govind Ballabh Pant Sagar: Largest artificial lake in India**, formed by the **Rihand Dam** on Rihand river at Pipri in **Sonbhadra**.
- **Parichha Dam & Matatila Dam:** Built on **Betwa river** (Jhansi / Lalitpur).
- **Keetham Lake (Sur Sarovar):** Ramsar site in **Agra**.

C. All 10 Ramsar Wetland Sites in Uttar Pradesh

1. **Upper Ganga River (Brijghat to Narora Stretch):** First Ramsar site in UP (designated 2005).
2. **Nawabganj Bird Sanctuary** (Unnao) - renamed Shahid Chandra Shekhar Azad Bird Sanctuary.
3. **Parvati Arga Bird Sanctuary** (Gonda) - oxbow lake.
4. **Saman Bird Sanctuary** (Mainpuri).
5. **Samaspur Bird Sanctuary** (Rae Bareli).
6. **Sandi Bird Sanctuary** (Hardoi).
7. **Sarsai Nawar Jheel** (Etawah) - sarus crane habitat.
8. **Sur Sarovar (Keetham Lake)** (Agra).
9. **Haiderpur Wetland** (Muzaffarnagar & Bijnor, designated 2021).
10. **Bakhira Wildlife Sanctuary** (Sant Kabir Nagar, designated 2022).

3. National Parks, Tiger Reserves & Wildlife Sanctuaries

- **Only National Park in UP: Dudhwa National Park** (Lakhimpur Kheri, established 1977). Famous for Swamp Deer (Barasingha) and One-horned Rhinoceros reintroduction.
- **Tiger Reserves in UP (4 Total):**
- **Dudhwa Tiger Reserve** (Lakhimpur Kheri - declared 1987)
- **Pilibhit Tiger Reserve** (Pilibhit - declared 2014; won TX2 global award)
- **Amangarh Tiger Reserve** (Bijnor - buffer zone of Jim Corbett NP retained in UP after UK partition)
- **Ranipur Tiger Reserve** (**Chitrakoot** - declared 2022 as UP's 4th and India's 53rd Tiger Reserve, in Bundelkhand region).
- **Elephant Reserve:** Terai Elephant Reserve (TER) established in Lakhimpur Kheri/Pilibhit belt (2022).
- **Hastinapur Wildlife Sanctuary: Largest wildlife sanctuary in UP** (spread across Meerut, Muzzafarnagar, Bijnor, Amroha, Hapur).
- **Mahavir Swami Wildlife Sanctuary: Smallest wildlife sanctuary in UP** (Lalitpur, ~5.4 sq km).
- **Kachhua (Turtle) Sanctuary:** Varanasi (now shifted to Bhadohi/Prayagraj stretch).
- **State Symbols:** Bird: **Sarus Crane** (*Grus antigone*); Animal: **Swamp Deer** (Barasingha); Tree: **Ashoka**; Flower: **Palash / Dhak** (*Butea monosperma* - Flame of the Forest); Aquatic Animal: **Gangetic Dolphin** (*Platanista gangetica*).

4. 9 Agro-Climatic Zones & Agriculture of UP

1. **Tarai Zone:** Bhabhar and Terai belt along foothills (high rainfall, sugarcane, paddy).
2. **Western Plain Zone:** Meerut division, high agricultural productivity (Green Revolution heartland).
3. **Mid-Western Plain Zone:** Bareilly & Moradabad divisions.
4. **South-Western Semi-Arid Zone:** Agra, Mathura, Aligarh (low rainfall, mustard, bajra).
5. **Central Plain Zone:** Lucknow, Kanpur, Ayodhya divisions (alluvial doab).
6. **Bundelkhand Zone:** 7 districts (Jhansi, Lalitpur, Jalaun, Hamirpur, Mahoba, Banda, Chitrakoot) — semi-arid, rocky terrain, pulses, oilseeds.
7. **North-Eastern Plain Zone:** Gorakhpur division (floods, sugarcane, rice).
8. **Eastern Plain Zone:** Varanasi, Azamgarh, Ghazipur, Jaunpur divisions.
9. **Vindhyan Zone:** Mirzapur, Sonbhadra, and southern Chandauli (plateau, red soil, coarse grains).

> **UP Agricultural Ranks in India:**
> - **Rank 1 in:** Total Foodgrain production, Wheat, Sugarcane, Potato, Milk, Mango, Guava, Ethanol production.
> - **Rank 2 in:** Rice (after West Bengal).

5. Minerals & Industrial Belts in Uttar Pradesh

- Uttar Pradesh is generally poor in metallic minerals; mineral wealth is concentrated in the southern plateau districts (Bundelkhand & Vindhyan region):
- **Uranium: Lalitpur** (only district in UP with Uranium occurrences)
- **Copper: Sonrai area, Lalitpur**
- **Limestone: Sonbhadra** (Gurma, Kanach, Kajrahat) & **Mirzapur** (supplies raw material for cement factories at Churk, Dalla, Chunar)
- **Glass Sand (Silica Sand): Shankargarh & Lohgara (Prayagraj)**, Chitrakoot, Banda (derived from disintegration of Vindhyan sandstones)
- **Bauxite (Aluminium ore): Banda, Chitrakoot, Chandauli** (supplies Renukoot Hindalco smelter)
- **Coal: Singrauli (Sonbhadra)** — power capital of India (NTPC plants)
- **Dolomite: Sonbhadra & Banda**
- **Pyrophyllite & Diaspore: Jhansi, Lalitpur, Mahoba, Hamirpur**
- **China Clay: Sonbhadra & Banda**

- **Gold:** Small placer deposits in **Sharda and Ramganga river sands**; Hardi village in Sonbhadra.

6. Revolt of 1857 in Uttar Pradesh: Centres & Leaders

Centre in UP	Indian Leader / Revolutionary	British Officer / Suppressor
Meerut	Sepoys of 3rd Native Cavalry (10 May 1857)	General Hewitt
Delhi / Western UP	General Bakht Khan (from Bareilly)	John Nicholson, Hudson
Kanpur	**Nana Saheb (Dhondu Pant)**, Tantia Tope (Ramchandra Pandurang), Azimullah Khan	Colin Campbell, Havelock
Lucknow	**Begum Hazrat Mahal** (on behalf of minor son Birjis Qadr)	Colin Campbell, Outram, Havelock
Jhansi	**Rani Laxmi Bai (Manikarnika)**	**Sir Hugh Rose** ("*Here lay the woman who was the only man among the rebels*")
Bareilly	**Khan Bahadur Khan**	Vincent Eyre, Colin Campbell
Faizabad (Ayodhya)	**Maulvi Ahmadullah Shah** (*Danka Shah*)	Campbell
Prayagraj (Allahabad)	**Maulvi Liaquat Ali**	Colonel James Neill
Jagdishpur / East UP	Kunwar Singh & Amar Singh	William Taylor, Vincent Eyre
Baghpat	**Shah Mal** (mobilized peasants of Baraut)	Dunlap
Mathura	Kadam Singh, Devi Singh	-
Fatehpur	Hikmatullah	-

7. Cultural Gharanas, Folk Music & Dances

- **Kathak:** Only Classical Dance of North India. Reached its zenith under **Nawab Wajid Ali Shah** of Awadh.
- **Lucknow Gharana exponents:** Thakur Prasad, Kalka-Bindadin Maharaj, Achchan Maharaj, Shambhu Maharaj, Lachhu Maharaj, **Pt. Birju Maharaj**.
- **Banaras Gharana exponents:** Sitara Devi, Pt. Kishan Maharaj (Tabla), Pt. Samta Prasad (Gudai Maharaj - Tabla), Bismillah Khan (Shehnai - Bharat Ratna 2001).
- **Kirana Gharana:** Founded by **Ustad Abdul Karim Khan & Abdul Wahid Khan** in Muzaffarnagar district (Bhimsen Joshi, Gangubai Hangal belonged to this tradition).
- **Atrali Gharana:** Aligarh.
- **Folk Music Genres:**
- **Rasiya:** Braj region (Mathura, Vrindavan) — themed on Radha-Krishna romance.
- **Kajri:** Mirzapur & Varanasi region — sung during monsoon / rainy season.
- **Birha:** Eastern UP (Bhojpuri belt) — heroic and narrative folklore.
- **Alha:** Bundelkhand region (Mahoba) — martial epic commemorating heroic deeds of Alha & Udal.
- **Chaiti & Hori:** Purvanchal & Braj region.
- **Folk Dances of UP:**
- **Charkula Dance: Braj region** — women balance a multi-tiered wooden pyramid with 108 oil lamps on their heads.
- **Karma Dance: Sonbhadra & Mirzapur** (performed by tribal communities like Kharwar).

- **Pai Danda & Diwari: Bundelkhand** (stick dance reminiscent of Raas Leela).
- **Nautanki:** Traditional folk theatre of UP (Hathras & Kanpur schools).

8. High-Yield One District One Product (ODOP) Master Match-List

District in UP	Official ODOP Product / Heritage Craft
Agra	Leather Goods & Footwear; Petha
Aligarh	**Locks and Hardware**
Amroha	**Musical Instruments (Dholak)**
Ayodhya	**Jaggery (Gud)**
Azamgarh	**Black Clay Pottery (Nizamabad)** — GI Tagged
Bhadohi	**Handmade Woollen Carpets (Dhurries)** — GI Tagged (*Carpet City*)
Chandauli	**Black Rice (Chak-Hao / Adamchini Rice)**
Farrukhabad	**Block Printing & Zari-Zardozi**
Firozabad	**Glassware & Glass Bangles** (*Suhag Nagari*)
Gorakhpur	**Terracotta Crafts** — GI Tagged
Kannauj	**Perfume (Attar) & Essential Oils** (*Grasse of the East*)
Kanpur Nagar	**Leather Products** (*Cawnpore Leather City*)
Lucknow	**Chikan Craft (Chikankari) & Zardozi** — GI Tagged
Mathura	Sanitary Fittings; Peda
Meerut	**Sports Goods** (*Sports Capital of India*)
Moradabad	**Metal Crafts / Brassware** (*Peetal Nagari*)
Muzaffarnagar	**Jaggery (Gud)**
Pilibhit	**Wooden Flute (Bansuri)** (*Bansuri Nagari*)
Pratapgarh	**Aonla (Indian Gooseberry) Products**
Saharanpur	**Wooden Carving Crafts**
Sambhal	**Horn & Bone Handicrafts**
Varanasi	**Banarasi Silk Sarees & Pink Meenakari**

SUBJECT 11 · SCIENCE & TECHNOLOGY

SUBJECT 11: GENERAL SCIENCE & TECHNOLOGY

सामान्य विज्ञान एवं प्रौद्योगिकी — दैनिक उच्च-प्राथमिकता तथ्य

Vitamins & Deficiencies, Pathogens & Diseases (Viral/Bacterial/Protozoan), Endocrine Hormones, Optics & Eye Defects, Commercial Chemicals, Alloys, ISRO Space Missions & DRDO Missiles.

1. Complete Vitamin Master Reference & Deficiency Diseases

Vitamin	Chemical / Scientific Name	Fat / Water Soluble	Key Deficiency Disease / Symptoms	Primary Natural Food Sources
Vitamin A	**Retinol**	Fat-soluble	**Night blindness (Nyctalopia)**, **Xerophthalmia** (dry cornea), Keratomalacia	Carrot, Papaya, Mango, Liver, Fish oil, Golden Rice
Vitamin B1	**Thiamine**	Water-soluble	**Beri-beri** (Peripheral neuropathy / cardiac failure), Wernicke-Korsakoff syndrome	Unpolished rice, yeast, whole grains (*lost in polished rice*)
Vitamin B2	**Riboflavin**	Water-soluble	**Cheilosis** (cracking at corners of mouth), Glossitis (red swollen tongue)	Milk, eggs, liver (*gives yellow colour to milk/whey*)
Vitamin B3	**Niacin (Nicotinic acid)**	Water-soluble	**Pellagra** (4Ds: Dermatitis, Diarrhea, Dementia, Death)	Yeast, meat, fish, legumes
Vitamin B5	**Pantothenic acid**	Water-soluble	Burning feet syndrome, fatigue	Mushroom, eggs, broccoli
Vitamin B6	**Pyridoxine**	Water-soluble	Peripheral neuropathy, microcytic anemia, convulsions in infants	Cereals, bananas, meat
Vitamin B7	**Biotin (Vitamin H)**	Water-soluble	Hair loss (alopecia), dermatitis	Egg yolk, nuts, intestinal flora (*avidin in raw egg binds biotin*)
Vitamin B9	**Folic Acid (Folate)**	Water-soluble	**Megaloblastic anemia**, Neural tube defects in fetuses (Spina bifida)	Green leafy vegetables, citrus fruits
Vitamin B12	**Cyanocobalamin**	Water-soluble	**Pernicious anemia**, demyelination of spinal cord (*contains Cobalt metal*)	Meat, fish, eggs, dairy (*Absent in plant foods*)
Vitamin C	**Ascorbic acid**	Water-soluble	**Scurvy** (bleeding gums, petechiae, impaired wound healing); heat-sensitive (*destroyed by cooking*)	Aonla (highest source), Guava, Citrus fruits (Lemon, Orange)
Vitamin D	**Calciferol** (D_2: Ergocalciferol, D_3: Cholecalciferol)	Fat-soluble	**Rickets** in children (bowed legs), **Osteomalacia** in adults (*hormone-like vitamin*)	Sunlight synthesis in skin, Cod liver oil
Vitamin E	**Tocopherol**	Fat-soluble	Infertility / reproductive failure, muscle weakness, hemolytic anemia (*Beauty Vitamin / Anti-oxidant*)	Wheat germ oil (richest), sunflower seeds, almonds
Vitamin K	**Phylloquinone** (K_1), Menaquinone (K_2)	Fat-soluble	**Defective blood clotting**, prolonged bleeding time (needed for synthesis of prothrombin in liver)	Spinach, kale, synthesized by human gut bacteria

Fat-Soluble vs Water-Soluble Mnemonic:
Fat-soluble: **K - E - D - A** (stored in liver and adipose tissue, hypervitaminosis possible).
Water-soluble: **B-complex and C** (excreted in urine, regular intake required).

2. Infectious Diseases & Causative Pathogens Master Matrix

A. Viral Diseases

Disease	Causative Virus	Mode of Transmission / Key Diagnostic Feature
COVID-19	SARS-CoV-2 (enveloped positive-sense RNA)	Respiratory droplets; RT-PCR test
Rabies (Hydrophobia)	Lyssavirus / Rhabdovirus (bullet-shaped RNA)	Bite of rabid dogs, bats; affects central nervous system
Poliomyelitis	Poliovirus (Enterovirus, single-stranded RNA)	Fecal-oral route; destroys anterior horn motor neurons of spinal cord; Oral Polio Vaccine (Sabin), Inactivated Polio Vaccine (Salk)
AIDS	Human Immunodeficiency Virus (HIV - Retrovirus with Reverse Transcriptase)	Blood, sexual, maternal; attacks **CD4+ Helper T-cells**; ELISA test, Western Blot confirmatory
Dengue (Break-bone fever)	Flavivirus (DEN-1 to DEN-4, ssRNA)	Transmitted by **Aedes aegypti** mosquito (day-biting); severe thrombocytopenia (platelet drop)
Chikungunya	Chikungunya virus (Alphavirus)	Transmitted by **Aedes aegypti** and *Aedes albopictus*; severe joint pain
Yellow Fever	Flavivirus	*Aedes aegypti* mosquito vector
Zika Virus	Flavivirus	*Aedes* mosquito; causes **Microcephaly** in newborns
Hepatitis B	Hepatitis B Virus (HBV - **DNA virus**, whereas Hep A, C, D, E are RNA)	Blood, body fluids; attacks liver; vaccine available
Smallpox	Variola virus (DNA virus)	Completely **eradicated from world in 1980** (WHO)
Measles, Mumps, Rubella	Morbillivirus, Paramyxovirus, Rubivirus	MMR vaccine

B. Bacterial Diseases

Disease	Causative Bacterium	Key Clinical Feature / Diagnostic Test
Tuberculosis (TB)	*Mycobacterium tuberculosis* (acid-fast bacillus)	Affects lungs; **Mantoux test**; **BCG vaccine** (Bacillus Calmette-Guérin)
Cholera	*Vibrio cholerae* (comma-shaped)	"*Rice water stools*", massive dehydration; fecal-oral route
Typhoid (Enteric fever)	*Salmonella typhi*	Step-ladder fever pattern, intestinal perforation; **Widal test**
Tetanus (Lockjaw)	*Clostridium tetani* (anaerobic, drumstick-shaped)	Neurotoxin tetanospasmin causes severe muscle spasms; ATS / Tetanus toxoid
Diphtheria	*Corynebacterium diphtheriae*	Grey pseudomembrane over throat / larynx; **Schick test**
Pertussis (Whooping Cough)	*Bordetella pertussis*	Paroxysmal coughing ending in whoop; DPT triple vaccine
Plague (Black Death)	*Yersinia pestis*	Transmitted by **Xenopsylla cheopis** (oriental rat flea); Bubonic, Pneumonic forms

Disease	Causative Bacterium	Key Clinical Feature / Diagnostic Test
Leprosy (Hansen's Disease)	*Mycobacterium leprae*	Hypopigmented skin patches with loss of sensation; Lepromin test
Anthrax	*Bacillus anthracis* (spore-forming)	Zoonotic disease of cattle; potential bioweapon
Syphilis & Gonorrhea	*Treponema pallidum* & *Neisseria gonorrhoeae*	Sexually transmitted bacterial infections

C. Protozoan & Other Pathogen Diseases

- **Malaria:** Caused by protozoan **Plasmodium** (*P. vivax, P. falciparum* [deadliest malignant tertian], *P. malariae, P. ovale*). Vector: **Female Anopheles mosquito**. Infective stage to humans: **Sporozoite**. Parasite multiplies in liver cells and destroys red blood cells (**RBCs**), releasing **hemozoin toxin** causing periodic shivering chills and high fever. Discovered by **Sir Ronald Ross** (in Secunderabad, India - Nobel Prize 1902). Treatment: Quinine (extracted from **Cinchona bark**), Artemisinin.
- **Kala-Azar (Visceral Leishmaniasis / Dum-Dum Fever):** Caused by *Leishmania donovani*. Vector: **Phlebotomus sandfly**. Enlarges spleen and liver (hepatosplenomegaly).
- **Sleeping Sickness (African Trypanosomiasis):** Caused by *Trypanosoma brucei*. Vector: **Tsetse fly** (*Glossina*).
- **Amoebic Dysentery:** Caused by *Entamoeba histolytica*.
- **Ringworm (Tinea):** Fungal infection (*Trichophyton, Microsporum, Epidermophyton*).
- **Filariasis / Elephantiasis:** Caused by parasitic nematode worm *Wuchereria bancrofti*. Transmitted by **Culex mosquito**. Blocks lymphatic vessels causing swelling of limbs/scrotum.

3. Human Physiology & Endocrine Hormones

Gland	Hormones Produced	Primary Function & Prelims Exam Triggers
Pituitary (Master Gland)	Growth Hormone (GH), TSH, ACTH, FSH, LH, Prolactin, **Oxytocin**, **Vasopressin (ADH)**	Regulated by Hypothalamus. Hypersecretion of GH causes **Gigantism / Acromegaly**; hyposecretion causes **Dwarfism**. **ADH (Antidiuretic Hormone)** controls water reabsorption in kidneys; deficiency causes **Diabetes Insipidus** (frequent urination without sugar). **Oxytocin** is the milk-ejection & birth hormone.
Thyroid	**Thyroxine (T_4)**, Triiodothyronine (T_3), Calcitonin	Largest endocrine gland; requires **Iodine**. Deficiency causes **Simple Goitre** (enlarged neck), **Cretinism** in infants, **Myxedema** in adults. Hyperthyroidism causes **Grave's Disease** (exophthalmic goitre).
Parathyroid	Parathyroid Hormone (PTH / Collip's hormone)	**Increases blood calcium levels** (hypercalcemic) by dissolving bone; acts antagonistically to thyroid calcitonin.
Pancreas (Islets of Langerhans)	**Insulin** (from β-cells), **Glucagon** (from α-cells), Somatostatin (from δ-cells)	**Insulin lowers blood glucose** by converting glucose to glycogen (glycogenesis). Deficiency causes **Diabetes Mellitus** (hyperglycemia, glycosuria). Discovered by Banting & Best (1921). Glucagon raises blood glucose.
Adrenal Gland	Adrenaline (Epinephrine) & Noradrenaline; Cortisol, Aldosterone	Adrenaline is the **"Emergency Hormone" / "Fight, Flight or Fright" (3F/4S hormone)**; increases heart rate, breathing, blood pressure.
Pineal Gland	**Melatonin**	Regulates circadian rhythm (24-hour sleep-wake cycle).

Blood Groups & ABO System: Discovered by **Karl Landsteiner** (1900).
- **Universal Donor: O^- (O Negative)** — has neither A nor B antigens and no Rh factor on RBCs.
- **Universal Recipient: AB^+ (AB Positive)** — has both A and B antigens and Rh factor; has no anti-A or anti-B antibodies in plasma.
- **Normal Blood Pressure: 120/80 mm Hg** (Systolic/Diastolic), measured by **Sphygmomanometer**. Blood pH: **7.35–7.45 (slightly basic)**.

4. Optics & Physics Principles (Key Phenomena & Eye Defects)

Phenomenon	Underlying Optical Principle	Exam Application / Examples
Mirage in deserts / Looming in cold	**Total Internal Reflection (TIR)**	Optical illusion; light travels from denser to rarer medium at angle $>$ critical angle.
Sparkling of Diamond	**Total Internal Reflection (TIR)**	Very small critical angle of diamond ($\approx 24.4^{\circ}$) due to high refractive index (≈ 2.42).
Endoscope / Optical Fibre Cables	**Total Internal Reflection (TIR)**	Zero signal loss communication; invented by Narinder Singh Kapany.
Blue colour of sky / Red colour of sunset	**Rayleigh Scattering of Light**	Scattering $\propto \frac{1}{\lambda^4}$. Blue light (shorter wavelength) scatters most; red light (longest wavelength) scatters least and reaches eyes at horizon.
Twinkling of Stars	**Atmospheric Refraction**	Continuously fluctuating refractive index of atmospheric air layers.
Coin appears raised / Stick bent in water	**Refraction of Light**	Light bends away from normal when entering air from water.
Rainbow Formation	**Dispersion $+$ Refraction $+$ Total Internal Reflection**	Water droplets act as tiny prisms. Primary rainbow: 2 refractions $+$ 1 internal reflection; Secondary rainbow: 2 refractions $+$ 2 internal reflections.
Soap bubble shining with brilliant colours	**Interference of Light**	Destructive and constructive interference of light reflected from upper and lower thin film surfaces.

Common Eye Defects & Corrective Lenses

1. **Myopia (Short-sightedness):** Nearby objects seen clearly, distant objects blurred. Image formed **in front of retina**. Corrected by **Concave Lens** (diverging lens).
2. **Hypermetropia (Long-sightedness):** Distant objects seen clearly, nearby objects blurred. Image formed **behind retina**. Corrected by **Convex Lens** (converging lens).
3. **Presbyopia (Old age vision):** Loss of accommodation power of ciliary muscles. Corrected by **Bifocal Lens** (upper part concave for distance, lower part convex for reading).
4. **Astigmatism:** Inability to focus simultaneously on horizontal and vertical lines due to irregular cornea curvature. Corrected by **Cylindrical Lens**.

5. Commercial Names of Common Chemical Compounds

Chemical / Common Commercial Name	Chemical Formula	Scientific / IUPAC Name	High-Yield Application
Baking Soda	NaHCO_3	Sodium hydrogen carbonate	Baking, antacid, fire extinguisher
Washing Soda	$\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$	Sodium carbonate decahydrate	Glass, soap, water softening

Chemical / Common Commercial Name	Chemical Formula	Scientific / IUPAC Name	High-Yield Application
Caustic Soda	NaOH	Sodium hydroxide	Soap manufacturing, petroleum refining
Caustic Potash	KOH	Potassium hydroxide	Soft soaps, alkaline batteries
Bleaching Powder	CaOCl_2	Calcium hypochlorite	Disinfectant for water, bleaching cotton
Quicklime	CaO	Calcium oxide	Cement manufacture, white washing
Slaked Lime	Ca(OH)_2	Calcium hydroxide	Mortar, neutralizing soil acidity
Plaster of Paris (POP)	$\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$	Calcium sulphate hemihydrate	Bone fractures, statues, false ceilings
Gypsum	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$	Calcium sulphate dihydrate	Retarder in cement setting, soil reclamation
Laughing Gas	N_2O	Nitrous oxide	Dental anaesthetic, propellant
Tear Gas	CCl_3NO_2 (Chloropicrin) / CS gas	2-chlorobenzalmalononitrile	Riot control
Heavy Water	D_2O	Deuterium oxide	Moderator in nuclear reactors
Dry Ice	Solid CO_2	Solid carbon dioxide	Refrigerant, sublimes without melting
Epsom Salt	$\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$	Magnesium sulphate heptahydrate	Laxative, bath salt
Blue Vitriol	$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$	Copper sulphate pentahydrate	Fungicide (Bordeaux mixture), electroplating
Green Vitriol	$\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$	Ferrous sulphate heptahydrate	Ink manufacturing
White Vitriol	$\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$	Zinc sulphate heptahydrate	Eye drops, pigments

6. Major Alloys & Their Compositions

Alloy	Major Constituents (%)	Primary Properties & Applications
Brass (Pital)	**Copper (60–70%) $+$ Zinc (30–40%)**	Utensils, musical instruments, cartridge cases (*No Tin*)
Bronze (Kansa)	**Copper (88%) $+$ Tin (12%)**	Statues, medals, coins, bells (*No Zinc*)
German Silver	**Copper (50%) $+$ Zinc (30%) $+$ Nickel (20%)**	**Contains ZERO Silver!** Utensils, artificial jewellery
Stainless Steel	**Iron $+$ Chromium (18%) $+$ Nickel (8%) $+$ Carbon (0.1–1%)**	Rust-resistant; Chromium gives corrosion resistance
Solder (Tanka)	**Lead (50%) $+$ Tin (50%)**	Low melting point; joining electrical wires

Alloy	Major Constituents (%)	Primary Properties & Applications
Duralumin	**Aluminium (95%) $+$ Copper (4%) $+$ Magnesium (0.5%) $+$ Manganese (0.5%)**	Light and strong; aircraft bodies, pressure cookers
Gun Metal	**Copper (88%) $+$ Tin (10%) $+$ Zinc (2%)**	Heavy machinery gears, gun barrels
Bell Metal	**Copper (78%) $+$ Tin (22%)**	High resonance for casting bells
Amalgam	**Mercury $+$ Any other metal** (except Iron, Platinum, Tungsten)	Dental fillings (Silver-Mercury amalgam); **Iron does not form amalgam**
Nichrome	**Nickel (60%) $+$ Chromium (40%)**	High resistance & high melting point; electric heater coils

7. Indian Space Programme (ISRO) & Defence Tech

- **ISRO Founded: 15 August 1969**; Headquarters: **Bengaluru**; Space Commission & Department of Space (DoS) formed in 1972.
- **Launch Port:** Satish Dhawan Space Centre (SDSC - SHAR), **Sriharikota (Andhra Pradesh)**, on Pulicat Lake barrier island. Second spaceport under construction at **Kulasekarapattinam (Tamil Nadu)** for SSLV launches.
- **Key Historic Missions:**
- **Aryabhata:** India's first satellite, launched on **19 April 1975** by Soviet Kosmos-3M rocket.
- **Chandrayaan-1:** Launched on **22 October 2008** via PSLV-C11; Moon Impact Probe detected water molecules (H_2O/OH) on lunar surface.
- **Mangalyaan (Mars Orbiter Mission - MOM):** Launched on **5 November 2013** via PSLV-C25; inserted into Martian orbit on **24 September 2014** on maiden attempt.
- **Chandrayaan-3:** Launched on **14 July 2023** via LVM3-M4; successfully achieved historic **soft landing near lunar South Pole** on **23 August 2023** (celebrated as **National Space Day**). Lander: *Vikram*, Rover: *Pragyan*. Landing site named **Shiv Shakti Point**.
- **Aditya-L1:** India's first solar observatory launched on **2 September 2023** via PSLV-C57; inserted into halo orbit around **Lagrange Point 1 (L1)** on 6 January 2024.
- **DRDO Missiles (Integrated Guided Missile Development Programme - IGMDP, 1983 - Dr. APJ Abdul Kalam):**
- **P - Prithvi:** Surface-to-Surface short-range ballistic missile (Liquid fuel).
- **A - Agni:** Surface-to-Surface medium-to-intercontinental ballistic missile (Agni I to V, Solid fuel, MIRV tech).
- **T - Trishul:** Surface-to-Air short-range missile.
- **N - Nag:** Anti-Tank guided missile ("*Fire and Forget*"); helicopter-launched version is **Helina / Dhruvastra**.
- **A - Akash:** Surface-to-Air medium-range missile with Rajendra multi-target phased array radar.
- **Nuclear Power Plants in India:**
- **Narora Atomic Power Station (NAPS): Bulandshahr, Uttar Pradesh** (PHWR, only nuclear plant in UP)
- **Tarapur:** Maharashtra (Oldest nuclear plant in India, 1969)
- **Rawatbhata:** Rajasthan (near Kota, Chambal river)
- **Kakrapar:** Gujarat (Tapi river)
- **Kaiga:** Karnataka (Kali river)
- **Kalpakkam (MAPS):** Tamil Nadu (Indira Gandhi Centre for Atomic Research)
- **Kudankulam:** Tamil Nadu (Largest nuclear power capacity in India, VVER Russian reactors)
- **Gorakhpur Nuclear Plant:** Under construction in **Fatehabad, Haryana** (Not in Gorakhpur, UP!)